

Optimal Trade Policies and Market Power in General Equilibrium Trade Models*

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Abstract

We derive optimal trade policies in a multi-country, multi-sector general-equilibrium (GE) model that encompasses a broad class of supply-side specifications. Unlike in partial-equilibrium models, the supply-side specification determines endogenous GE interdependencies across sectors and countries. In the two-country case, under our defined Generalized Independence of Irrelevant Alternatives (GIIA) Supply System, cross-sector interdependencies are simplified: although a policy in one sector affects supply in every sector, symmetric scaled cross-sector responses cancel from relative policy conditions. We obtain simple, explicit import-tariff and export-tax formulas that depend only on the targeted sector's structural parameters and relative market shares. With multiple countries, trade among foreign countries adds cross-country interdependence: the foreign trade network reallocates Home's aggregate sectoral net-import motive across foreign countries. We show that instead of depending solely on bilateral trade shares, network effects compress or amplify cross-country tariff differentials under different specifications. Using trade data, we show that ignoring these GE interdependencies can reduce Home's welfare gains.

Keywords: Optimal Policies, Tariffs, Export Taxes, General Equilibrium, GIIA Supply System, Imperfectly Substitutable Labor, Economies of Scale, Endogenous Self- and Cross- Export Supply Elasticities

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1 Introduction

A large body of research establishes that multi-country, multi-sector general equilibrium (GE) models are successful in accounting for global trade patterns, and are now standard tools for evaluating trade policies. Yet characterizing optimal trade policies within such frameworks remains challenging due to inherent complex interdependencies. When a tariff is imposed in one sector, it creates cascading effects: it changes demand and supply in both domestic and foreign markets, reallocates factors across sectors, and changes global prices in other sectors, which in turn reshape trade flows and determine optimal trade policy incentives in other countries and sectors.

The traditional optimal trade policy literature abstracts away from these interdependencies by using simple general equilibrium or partial equilibrium models, which reduces the problem to a single-foreign-country, single-good monopolist or monopsonist. This simplification makes the central mechanism transparent: optimal tariffs should equal the inverse of the self-elasticity of foreign export supply curves, reflecting the home country's buyer power.¹

Recent studies have advanced the analysis of optimal trade policy in modern trade models that emphasize general equilibrium (GE) effects. In these models, foreign export supply curves are equilibrium objects shaped by adjustments across sectors and countries, so the simple textbook formula no longer applies. The literature has been focused on two-country settings ([Costinot et al., 2015](#)), imposed approximations simplifying cross-country linkages ([Lashkaripour and Lugovskyy, 2023](#)), or relied on quantitative solutions ([Ossa, 2014](#)). What is still missing is a comprehensive characterization of optimal policy across the supply-side assumptions of existing trade models, and of how cross-sector and cross-country interdependencies shape those policies.

This paper fills the gap. We derive exact, explicit formulas for optimal policies in the full multi-country GE problem. The formulas hold in a unified framework with a generic supply system that nests existing supply-side assumption, including extended multi-sector Armington models, [Eaton and Kortum \(2002\)](#), and [Krugman \(1980\)](#) models, with different

¹See [Bagwell and Staiger \(1999\)](#), [Broda, Limao, and Weinstein \(2008\)](#) and [Feenstra \(2015\)](#).

economies of scale across sectors and labor market specifications ranging from perfectly mobile to fixed to imperfectly substitutable labor across sectors, as in the Ricardo-Roy model. We solve the problem of a home government who chooses domestic and trade policies to maximize welfare, accounting for their effects on consumption, production, trade flows, and prices. The economic rationales for these policies are classic: exploit terms-of-trade benefits and correct domestic market distortions, but the relevant supply responses are endogenous and interconnected. We use the Lagrange multiplier method to characterize Home's unilateral optimal policies.

Our first result provides exact and tractable formulas for the two-country case. We define a *Generalized Independence of Irrelevant Alternatives (GIIA) Supply System* across sectors:² after scaling by sector size and a sector-specific structural parameter η_j , cross-price supply responses are symmetric across sectors. The parameter η_j can be positive or negative. The class nests one-factor Ricardo-Roy models, generalized multi-sector Krugman models, and Armington and Eaton-Kortum specifications with different returns to scale.³

Specifically, in the two-country case with Cobb-Douglas demand across sectors and CES demand within sectors with elasticities ϵ_j (CD-CES demand), and a *GIIA supply system* with supply parameters η_j , optimal sectoral trade policies of Home (country 1) satisfy

$$\begin{aligned} \text{Import tariff: } 1 + \tau_j^m &= \frac{1}{\eta_j} \frac{M_{1,j}}{Y_{2,j}} + (1 + \bar{\tau}^m), \\ \text{Export tax: } 1 + \tau_j^x &= \frac{1 + \tau_j^d}{1 + \tau_j^m} \left(1 + \frac{1}{\epsilon_j \pi_{22,j}} \right), \end{aligned}$$

²Note that we call it *Generalized* IIA because it is not a demand-choice axiom as IIA, but an analogous cancellation property for scaled supply responses.

³The parameter η_j maps to primitives of the nested models. With imperfectly substitutable labor (elasticity κ) and production economies of scale (parameter ψ_j for sector j), it satisfies

$$\frac{1}{\eta_j} = \frac{1}{\kappa} - \psi_j \frac{\kappa - 1}{\kappa}.$$

A multi-sector Armington model corresponds to $\kappa = \infty$ and $\psi_j = 0$ and a Ricardo-Roy trade model to finite κ and $\psi_j = 0$. A multi-sector model with sectoral economies of scale ψ_j and perfectly mobile labor ($\kappa = \infty$), such as [Lashkaripour and Lugovskyy \(2023\)](#) and [Bartelme, Costinot, Donaldson, and Rodriguez-Clare \(2025\)](#), yields $1/\eta_j = -\psi_j$. The scale parameter ψ_j itself takes different forms across specifications. In a multi-sector Krugman, $\psi_j = 1/(\sigma_j - 1)$, where σ_j is the elasticity of substitution across varieties in sector j . In a multi-sector [Eaton and Kortum \(2001\)](#) model with endogenous technology, $\psi_j = 1/\theta$, where θ is the Fréchet shape parameter. And in a multi-sector Armington model with external economies of scale, ψ_j equals the sectoral scale elasticity directly.

with optimal domestic tax being $(1 + \bar{\tau}^d)^{\frac{\eta_j}{\eta_j - 1}}$.⁴ Here, $M_{1,j} = \beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}$ captures Home's net import in sector j , $Y_{2,j}$ is Foreign's sectoral income, ϵ_j is the trade elasticity, and $\pi_{22,j}$ is Foreign's expenditure share on its own goods. Tax neutrality leaves one domestic and one trade-policy normalization, represented by $\bar{\tau}^d$ and $\bar{\tau}^m$.

Three insights arise from our optimal policy formulas. First, relative tariffs across any two sectors depend on the relative magnitude of Home's net imports over foreign income scaled by sectoral supply parameters, reflecting Home's relative import market power across the two sectors. Under positive supply parameters, Home imposes higher tariffs in sectors where it is a larger buyer of foreign goods, and the tariff decreases with the sector's supply parameters. Second, conditional on the other taxes, export taxes are higher when trade demand is less elastic and Home has a larger share of foreign consumption, reflecting Home's seller power. Third, Home uses Pigouvian domestic taxes to fix domestic inefficiency across sectors.

Our formulas generalize the existing findings of uniform import tariffs under perfectly mobile labor across sectors. Absent economies of scale, η_j equals the elasticity of labor supply across sectors. With perfectly mobile labor, this elasticity is infinite, so $\eta_j = \infty$ and optimal tariff differences across sectors go to zero. This is because, with labor freely mobile across sectors, wages are equalized and technology is exogenously fixed, Home cannot change the relative price of any two imports and therefore uses uniform or zero tariffs. With imperfectly substitutable labor, η_j is finite, and Home imposes relatively higher tariffs in its net-import sectors. This occurs because decreased demand reduces foreign sectoral wages and hence supply prices in those sectors, thereby improving Home's terms of trade.

With alternative supply-side assumptions, such as Marshallian economies of scale, or endogenous technology,⁵ or a multi-sector Krugman model with scale parameters across sectors, $1/\eta_j$ equals the negative of the sectoral scale parameter. Home then imposes rel-

⁴Note that heterogeneous sectoral supply elasticities alone do not imply inefficiency: the η_j are the supply parameters in our definition of the GIIA Supply System, not the sectoral supply elasticities themselves. Differential domestic taxes are warranted only when the underlying mechanism creates a wedge between private and social marginal returns.

⁵Bai, Jin, and Lu (2025) extend the endogenous technology model in Eaton and Kortum (2001) to multiple sectors and study optimal dynamic trade policies with endogenous technology accumulation. The multi-sector versions of Krugman with identical scale and trade elasticity across sectors are isomorphic to Bai, Jin, and Lu (2025) at the steady state.

atively lower tariffs in sectors with net import shares. Lower tariffs increase demand for foreign goods, and reduce foreign producers' unit costs by increasing production scale, which lower foreign supply prices and improve Home's terms of trade.

A GIIA Supply System does not eliminate cross-sector linkages that arise from factor reallocation. Instead, it imposes symmetry on cross-price supply responses after scaling them by sector size and the relevant supply parameters. This symmetry makes the optimal-policy formulas tractable. When we compare the optimality conditions for two sectors, the scaled supply responses of all other sectors cancel. Relative import tariff formula therefore depends only on the two sectors' supply parameters and Home's net imports relative to Foreign income, rather than on the entire matrix of cross-sector supply elasticities. General-equilibrium effects have not disappeared. They remain embedded in equilibrium prices, incomes, and trade shares.

Crucially, the GIIA Supply System implies neither that a sector's partial supply elasticity nor that its own-price export-supply elasticity is fixed at η_j . This distinguishes our framework from partial-equilibrium models with exogenous, constant foreign supply elasticities. Cross-sector elasticities are generally nonzero; it is their scaled symmetry that simplifies the optimal-policy formulas despite cross-sector interdependencies.

An example of a non-GIIA Supply System helps to show how asymmetric interdependencies across sectors matter for optimal policies. Consider a multi-factor Ricardo-Roy model, where workers with different types select into different sectors based on comparative advantage. Price and hence wage changes in any two sectors can asymmetrically affect supply in a third sector, depending on the types of workers employed in the first two sectors and their likelihood of migrating to the third. This multi-factor reallocation creates asymmetric interdependencies across sectors, requiring the optimal tariff formula to incorporate all self- and cross-sector supply elasticities explicitly.

Our second result characterizes the exact multi-country problem. The multi-country framework adds interdependence across foreign countries. It differs from the two-country case in two key respects. First, GIIA restricts supply responses across sectors within a country, and this property does not extend across foreign countries. Factors are typically more mobile across sectors within a country than across countries, and expenditures are linked

through each country's representative household. Second, Home lacks policy instruments that directly control trade between two foreign countries. In contrast, in the two-country case, the home government can use sector-level import tariffs and export taxes to directly influence bilateral trade flows between Home and Foreign. In fact, when foreign countries do not trade with each other, or when they are identical, Home's optimal import tariffs and export taxes would again follow the same formulas based on bilateral trade flows between Home and each foreign country.

Because Home has no policy instrument that directly controls trade among foreign countries, it cannot independently control the resulting third-market adjustments. A foreign price change shifts third-market demand across foreign countries, thereby altering the residual export supplies that foreign countries offer to Home. The foreign trade network therefore generates endogenous self- and asymmetric cross-export-supply elasticities across countries and sectors. Under a general supply system, optimal tariffs depend on this entire elasticity matrix.

Under GIIA, the multi-country optimal tariff is

$$1 + \tau_{n,j}^m = \frac{1}{\eta_j} \frac{M_{1n,j} + \mathcal{N}_{n,j}}{Y_{n,j}} + (1 + \tau_n^m).$$

where $M_{1n,j}$ is Home's bilateral net import from country n in sector j , $\mathcal{N}_{n,j}$ captures additional effect of trade among foreign countries on countries' residual export supplies to Home, and τ_n^m is a country-specific component common across sectors. Because the network statistics $\mathcal{N}_{n,j}$ sum to zero across foreign countries within each sector, the foreign trade network reallocates Home's aggregate sectoral net-import motive across foreign countries without changing that aggregate motive.

Within each foreign country, GIIA continues to discipline tariff variation across sectors. Because τ_n^m is common across sectors within a country, the tariff difference between any two sectors depends explicitly only on their supply parameters and the corresponding sufficient statistics $(M_{1n,j} + \mathcal{N}_{n,j})/Y_{n,j}$. These statistics generally differ from Home's bilateral net-import shares because the network terms reallocate its aggregate sectoral net-import motive across foreign suppliers. GIIA thus preserves a tractable characterization of within-country

tariff differences even when the foreign trade network matters.

Across foreign countries within a sector, the supply specification determines whether this reallocation compresses or amplifies tariff differences. To isolate the mechanism, consider a sector in which foreign countries do not source from Home. This restriction shuts down the effect of Home's export taxes on foreign demand for other foreign countries' goods, which in turn affects their residual export supply to Home. The sign of $\mathcal{N}_{n,j}$ is then determined by whether country n 's tariff lies below or above a trade-network-weighted average of the tariffs on other foreign suppliers. If $\eta_j > 0$, the network term raises a tariff below the weighted average and lowers one above it, compressing cross-country tariff gaps relative to the bilateral benchmark. If $\eta_j < 0$, the same network term enters with the opposite sign and amplifies those gaps.

The intuition is a third-market substitution effect. Suppose $\eta_j > 0$ and the bilateral motive calls for a relatively high tariff on country n . The tariff lowers country n 's producer price, inducing other foreign countries to substitute toward its goods and leaving fewer goods available for export to Home. Country n 's residual export supply to Home is therefore more elastic than the bilateral export supply elasticity would suggest, which dampens the optimal tariff incentive. By the same logic, a foreign country facing a relatively low tariff loses third-market demand, which pushes its optimal tariff upward. The trade network thus compresses tariff differences across countries. When $\eta_j < 0$, scale effects reverse the price response and the same mechanism instead amplifies tariff differences. In that case, a subsidy to a relatively large net-import supplier expands demand and lowers its supply price. The lower price attracts additional third-market demand, which further reduces the country's residual export supply to Home. Export supply is therefore less elastic, which calls for a larger subsidy.

Ignoring cross-country interdependencies and simply applying bilateral trade policy formulas would reduce potential welfare gains. Moreover, since supply elasticities and imports are endogenous variables that respond to changes in Home's policies, using pre-tax bilateral trade shares and net import shares while ignoring how trade policies themselves alter these shares can generate welfare losses rather than gains. Through numerical examples, we demonstrate that these welfare costs become larger when foreign countries are

more asymmetric and engage in more extensive trade with each other.

We quantify optimal policies for the U.S. and China in a model with 20 countries and 13 sectors, using sectoral trade and production data. The explicit first-order conditions improve both accuracy and computational efficiency. We assume a positive supply parameter, coming from imperfectly mobile labor across sectors. Allowing for modest increasing returns to scale yields similar results. Optimal tariffs differ across countries and sectors, reflecting variation in Home's buyer power. In addition, cross-country tariff differences within a sector are compressed relative to the bilateral benchmark. To measure this compression, for each sector we regress optimal tariffs across foreign suppliers on Home's bilateral net imports divided by foreign sectoral income and scaled by the supply parameter. The bilateral formula implies a unit slope when its country-specific component is held fixed. In the quantitative model, every sector's slope is below one, indicating that tariff differences respond less than one-for-one to bilateral net-import share differences.

Two forces contribute to a limited tariff dispersion. First, the observed import shares and their cross-country-sector dispersion are small, consistent with high trade costs and limited heterogeneity in comparative advantage at this level of aggregation. Second, the endogenous network mechanism makes foreign residual export supplies more elastic: a high tariff on a large bilateral supplier redirects third-market demand toward that supplier and thereby moderates Home's tariff response. Because trade shares and export-supply elasticities also adjust to the policies, tariff dispersion falls further in equilibrium. Our welfare analysis confirms the importance of accounting for these endogenous effects. Policies using our formula, but based on *bilateral*, endogenous trade shares, yield smaller welfare gains than our optimal approach. Furthermore, policies using our formula, but with *bilateral observed* trade shares, yield about 81% of the optimal welfare gains for the U.S. and 77% for China, as these policies overlook endogenous changes in trade patterns and export supplies.

Our paper makes three contributions. First, we derive exact optimal policy formulas within a unified framework with general supply-side specifications, strengthening both theoretical clarity and computational efficiency. Second, we bridge the gap between traditional literature that relies on simplified general equilibrium or partial equilibrium models and

contemporary trade models that emphasize sectoral and cross-country interdependencies. Third, while quantitative models have recognized the importance of imperfectly mobile labor and extensively analyzed their trade implications, optimal trade policies within these frameworks remain largely unexplored. Our paper fills the gap by providing theoretical optimal taxation formulas for both general and specific cases of labor markets commonly used in quantitative trade models, and by quantifying their welfare impacts.

Literature Our approach of providing a general formula for optimal trade policies traces back to the classic work of [Dixit \(1985\)](#), which sets up the problem of optimal taxes in an open economy as a fictitious planning problem and derives the associated first-order conditions under a given foreign offer curve. However, Dixit’s framework does not deliver formulas that apply directly to contemporary trade models, in which the self- and cross-elasticities of foreign export supply are equilibrium objects and depend on the underlying model structure. We use the Lagrange multiplier method and derive optimal policies, trade shares, and prices from the home government’s first-order conditions and equilibrium conditions. This method allows us to derive optimal policies explicitly, ensuring accurate theoretical explanations and efficient computation.

Our paper is mostly related to the literature on optimal trade policies in GE models (see [Caliendo and Parro, 2022](#), for a survey). In particular, [Lashkaripour and Lugovskyy \(2023\)](#) (henceforth LL) characterizes optimal trade and industrial policies in a multi-country, multi-sector generalization of the [Krugman \(1980\)](#) model and quantifies the ineffectiveness of trade policy in correcting misallocation under scale economies. Their analytical formula, however, solve an approximated problem that fixes foreign relative wages and suppresses some higher-order cross-country linkages. Moreover, our comment, [Bai, Lu, and Wang \(2026\)](#), identifies omitted chain effects in their first-order conditions that affect their formulas even in the two-country case and overturn their conclusion that optimal tariffs are input-output blind.

We instead derive exact optimal policy formulas for the original problem. In our characterization, the full foreign trade network jointly determines Home’s market power and its country-sector-specific policies, and we explain theoretically how cross-sector and cross-

country interdependencies shape optimal policies. Finally, Our framework generalizes supply-side assumptions. LL's generalized Krugman model is one example within our GIIA supply system, which makes the sources and consequences of cross-sector interdependence transparent.

The incentives behind optimal policies in our framework combine terms of trade manipulation with the correction of domestic distortions, such as those arising from economies of scale. [Bartelme et al. \(2025\)](#) quantify the gains from optimal industrial policy in a small open economy with sector-specific economies of scale. [Adão, Becko, Costinot, and Donaldson \(2026a\)](#) estimate the welfare weights using observed global tariffs and find that countries only partially internalize the effects of their tariffs on trading partners' terms of trade. [Adão, Beirne, Costinot, and Donaldson \(2026b\)](#) provide direct evidence that increase trading partners' tariffs worsen a country's terms of trade.

To characterize transparently how GE forces shape optimal policies, we abstract from input-output linkages, cross-country technology diffusion, and incomplete markets. On input-output linkages, [Beshkar and Lashkaripour \(2020\)](#) derive optimal noncooperative trade policies in a neoclassical framework with global value chains, [Antras, Fort, Gutiérrez, and Tintelnot \(2024\)](#) provide an efficiency rationale for tariff escalation, and [Caliendo, Feenstra, Romalis, and Taylor \(2023\)](#) show that second-best tariffs on intermediate inputs are low. On diffusion, [Bai, Jin, Lu, and Wang \(2025\)](#) studies optimal dynamic trade policies with technology diffusion through imports, [Hémous, Lopot, Sampson, and Schärer \(2023\)](#) analyzes optimal patent policies for open economies, and [Santacreu \(2025\)](#) presents a quantitative theory of bilateral trade agreements that include intellectual property provisions. On incomplete markets, [Waugh \(2023\)](#) shows that they generate unequal gains from trade, and [Donald, Fukui, and Miyauchi \(2023\)](#) characterize constrained-efficient allocations in a dynamic spatial model with frictional migration.

The paper proceeds as follows: Section 2 extends the multi-country, multi-sector trade framework to incorporate general supply-side specifications. Section 3 derives the optimal policies; in particular, Section 3.2 works with a two-country case to focus on cross-sector interdependencies, Section 3.3 examines how trade networks and interdependencies in multi-country models affect the endogenous foreign export supply elasticities and, consequently,

the optimal policies. Section 4 quantifies the optimal unilateral policy for the U.S. and China across 20 countries, along with the Nash optimal policies and the associated welfare changes. Section 5 concludes.

2 Theoretical Framework

We study optimal trade and domestic policies in a multi-country, multi-sector general equilibrium model with a generic supply system that encompasses various supply-side specifications from existing models.

The world consists of N countries and J sectors, with Country 1 designated as Home. Each country n has a measure of $\bar{L}_n$ labor. All sectors utilize labor as the sole input. Consumers in all countries have the same utility function on final goods C given by $u(C) = C$. The final good is a Cobb-Douglas function over the consumptions of different sector $j \in J$ goods, $C_n = \prod_{j \in J} (C_{n,j})^{\beta_{n,j}}$, where the constant $\beta_{n,j}$ captures the consumption share of sector j in country n . Sector- j goods shipped from country i to country n are subject to an iceberg trade cost $d_{ni,j}$.

Within each sector, preference and technology assumptions can be those in [Eaton and Kortum \(2002\)](#) (EK) or Armington model, or a multi-sector model with external economies of scale, which is isomorphic to a generalized multi-sector [Krugman \(1980\)](#) model of internal economies of scale. As shown in [Kucheryavyi, Lyn, and Rodríguez-Clare \(2023\)](#), the equilibrium conditions across all these models can be characterized by a unified system of general equilibrium conditions. We use these CD-CES demand-side assumptions to focus on the effects of different supply-side specifications. We denote ϵ_j as the elasticity of bilateral trade flows with respect to bilateral trade costs in sector j . The set $\mathcal{S}$ represents general supply-side assumptions and is defined as $\mathcal{S} = \{(p_{n,j}, w_{n,j}, L_{n,j})\}$. For example, in a generalized multi-sector Krugman model, $\mathcal{S} = \{(p_{n,j}, w_{n,j}, L_{n,j}) : p_{n,j} = \bar{T}_{n,j}^{-1} w_{n,j} L_{n,j}^{1/(1-\sigma_j)}\}$ with $\bar{T}_{n,j}$ denoting the exogenous technology level.

In addition to these supply-side assumptions, we incorporate general labor market specifications that have been extensively studied in the trade literature. The labor market specification determines wages and labor in equilibrium. Let $w_{n,j}$ and $L_{n,j}$ be the wage and labor

in sector j of country n , and let Ω represent the equilibrium labor conditions. For example, when labor is perfectly mobile across sectors, we can define $\Omega = \{(w_{n,j}, L_{n,j}) : \sum_{j=1}^J L_{n,j} = \bar{L}_n, w_{n,j} = w_n\}$. Section 3.2 provides specific examples of Ω and the associated optimal policies, including extensions to multi-factor g with endowment $\bar{L}_{n,g}$.

Suppose that Home imposes unilateral trade and domestic policies, but foreign countries are passive with zero taxes. Specifically, the home government imposes country-sector-specific import tariffs $\tau_{n,j}^m$, export taxes $\tau_{n,j}^x$, and sector-specific domestic sales tax τ_j^d on its imports, exports, and domestic sales, respectively. Let P_n and x_n be the country n 's final goods price and expenditures, respectively, with $x_n = P_n C_n$.⁶ Below, we define the world market equilibrium under these Home's policies.

Definition 1 (World Market Equilibrium). *The world market equilibrium consists of an allocation of labor $\{L_{n,j}\}$, consumption $\{C_n\}$, expenditures $\{x_n\}$, consumer price indexes $\{P_n\}$, prices $\{p_{n,j}\}$ and wages $\{w_{n,j}\}$ such that consumers maximize utility, firms maximize profits, and the market clearing conditions hold, taking as given the home government's policies $\{\tau_{n,j}^m, \tau_{n,j}^x, \tau_j^d\}$:*

1. The labor market specifications $\Omega(\{w_{n,j}, L_{n,j}\})$ hold.
2. The supply-side assumptions $S(\{p_{n,j}, w_{n,j}, L_{n,j}\})$ hold.
3. Trade shares satisfy, for each sector j

$$\pi_{11,j} = \frac{(p_{1,j}(1 + \tau_j^d))^{-\epsilon_j}}{(p_{1,j}(1 + \tau_j^d))^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j}(1 + \tau_{i,j}^m)d_{1i,j})^{-\epsilon_j}}, \quad (1)$$

$$\pi_{1n,j} = \frac{(p_{n,j}(1 + \tau_{n,j}^m)d_{1n,j})^{-\epsilon_j}}{(p_{1,j}(1 + \tau_j^d))^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j}(1 + \tau_{i,j}^m)d_{1i,j})^{-\epsilon_j}}, \quad n \neq 1, \quad (2)$$

$$\pi_{n1,j} = \frac{(p_{1,j}(1 + \tau_{n,j}^x)d_{n1,j})^{-\epsilon_j}}{(p_{1,j}(1 + \tau_{n,j}^x)d_{n1,j})^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j}d_{ni,j})^{-\epsilon_j}}, \quad n \neq 1, \quad (3)$$

$$\pi_{nm,j} = \frac{(p_{m,j}d_{nm,j})^{-\epsilon_j}}{(p_{1,j}(1 + \tau_{n,j}^x)d_{n1,j})^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j}d_{ni,j})^{-\epsilon_j}}, \quad n \neq 1, m \neq 1. \quad (4)$$

⁶We assume perfect domestic financial market and C_n is the household's real consumption in country n .

4. Expenditures are given by

$$x_1 = \sum_{j=1}^J Y_{1,j} + \sum_{j=1}^J \sum_{i \neq 1}^N \beta_{i,j} \frac{\tau_{i,j}^x}{1 + \tau_{i,j}^x} \pi_{i1,j} x_i + \sum_{j=1}^J \beta_{1,j} \sum_{i \neq 1}^N \frac{\tau_{i,j}^m}{1 + \tau_{i,j}^m} \pi_{1i,j} x_1 + \sum_{j=1}^J \beta_{1,j} \frac{\tau_j^d}{1 + \tau_j^d} \pi_{11,j} x_1, (\gamma_x) \quad (5)$$

$$x_n = \sum_{j=1}^J Y_{n,j}, \quad n \neq 1, \quad (6)$$

where $Y_{n,j} = w_{n,j} L_{n,j}$ is the income in sector j of country n .

5. Consumer price indexes are given by

$$P_1 = \prod_{j=1}^J \left[(p_{1,j}(1 + \tau_j^d))^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j}(1 + \tau_{i,j}^m) d_{1i,j})^{-\epsilon_j} \right]^{-\frac{\beta_{1,j}}{\epsilon_j}}, \quad (7)$$

$$P_n = \prod_{j=1}^J \left[(p_{1,j}(1 + \tau_{n,j}^x) d_{n1,j})^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j} d_{ni,j})^{-\epsilon_j} \right]^{-\frac{\beta_{n,j}}{\epsilon_j}}, \quad n \neq 1. \quad (8)$$

6. Goods market clearing conditions, for each sector j

$$\beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} x_1 = Y_{1,j} - \sum_{i \neq 1}^N \beta_{i,j} \frac{1}{1 + \tau_{i,j}^x} \pi_{i1,j} x_i, \quad (\gamma_{1,j}) \quad (9)$$

$$\beta_{1,j} \frac{1}{1 + \tau_{n,j}^m} \pi_{1n,j} x_1 = Y_{n,j} - \sum_{i \neq 1}^N \beta_{i,j} \pi_{in,j} x_i, \quad (\gamma_{n,j}), \quad n \neq 1. \quad (10)$$

Without loss of generality, we normalize the price of Home's final goods, $P_1 = 1$. Note that in the world market equilibrium, the market-clearing conditions and the definition of expenditures imply that the balanced trade conditions hold for each country. The equilibrium also satisfies a neutrality property with respect to Home's domestic taxes, import tariffs, and export taxes.

Lemma 1 (Tax Neutrality). *Consider two sets of Home government's trade policies $\Gamma = \{(1 + \tau_{n,j}^m, 1 + \tau_{n,j}^x, 1 + \tau_j^d) : \forall n \neq 1, j\}$ and $\check{\Gamma} = \left\{ \left(\lambda(1 + \tau_{n,j}^m), \frac{1 + \tau_{n,j}^x}{\mu}, \lambda \frac{1 + \tau_j^d}{\mu} \right) : \forall n \neq 1, j \right\}$ for any constants $\mu > 0$ and $\lambda > 0$. When the labor market allocation in Ω is homogeneous of degree zero in wages, and prices in $\mathbb{S}$ are homogeneous of degree one in wages, the welfare and allocations of the*

world market equilibrium under Γ and $\check{\Gamma}$ are the same.

Proof. Appendix A.

This lemma establishes tax neutrality,⁷ in the unilateral problem if the supply-side specifications have homogeneity properties, implying multiple equivalent policy combinations. Because the lemma leaves two degrees of freedom, λ and μ , we can set the domestic tax in one sector and one trade policy for a given country and sector to zero without changing the allocation or welfare. For example, we can set sector 1's domestic tax to zero, $\tau_1^d = 0$, along with the export tax on sector 1 to country 2, $\tau_{2,1}^x = 0$, and scale the remaining taxes accordingly.

3 Optimal Unilateral Policies

This section examines the unilateral policies of the home country, assuming that foreign governments remain passive. We consider country-sector-specific export taxes, import tariffs, and sector-specific domestic taxes to address the externalities arising from terms of trade and potential domestic market frictions.

3.1 General Formula

We begin by defining Home government's problem and then characterize the optimal policies in general form.

Definition 2 (Home Problem). *Home government's problem is to choose country-sector-specific trade policies $\{\tau_{n,j}^x, \tau_{n,j}^m\}$ and sector-specific domestic policies $\{\tau_j^d\}$, for $\forall j, n \neq 1$, to maximize domestic consumers' consumption, $\max x_1/P_1$, subject to the world market equilibrium given by Definition 1.*

We solve the home government's optimization problem in Definition 2 using the Lagrange multiplier method through a two-step approach. In the first step, we derive each

⁷Lerner (1936) establishes the equivalence between import tariff and export tax in a two-country, two-good neoclassical economy. Lashkaripour and Lugovskyy (2023) derive a related tax-neutrality result in a generalized multi-industry, multi-country Krugman model. Costinot and Werning (2019) provide a more general treatment with arbitrary numbers of countries, goods, firms, and households, under perfect or imperfect competition, and state the conditions under which tax neutrality continues to hold.

country's supply curves $Y_{n,j}(\{p_{n,k}\})$ where $Y_{n,j}$ represents the sectoral income as $Y_{n,j} = w_{n,j}L_{n,j}$. This derivation combines the supply-side assumption $S(\{p_{n,j}, w_{n,j}, L_{n,j}\})$ with the labor market specification $\Omega(\{w_{n,j}, L_{n,j}\})$. We can then express the equilibrium variables, including $\{w_{n,j}, L_{n,j}\}$, as functions of sectoral prices $\{p_{n,k}\}$ and policy instruments. In the second step, we substitute $Y_{n,j}(\{p_{n,k}\})$ for sectoral income, the supply-side assumption, and the labor market specification, and then take first-order conditions with respect to expenditures, sectoral prices, and policy instruments. Details see Appendix B.

This approach provides a unified framework for characterizing optimal policies by first deriving supply curves that encapsulate the underlying supply structure. A key advantage is its generality: One can readily apply the resulting general optimal formulas to any specification of S and Ω with $Y_{n,j} = w_{n,j}L_{n,j}$. The policy formulas obtained through this approach also offer clear economic intuition, with interpretations built on supply elasticities and trade shares that reflect the home country's market power.⁸

While the supply-curve approach shares similarity with the classic literature, our endogenous supply curves capture important cross-sector linkages, with sectoral income $Y_{n,j}$ depending not only on own-sector prices $p_{n,j}$ but also on prices across all other sectors $p_{n,k}$. These interdependencies arise from the general equilibrium nature of factor allocation and production technologies with returns to scale. As a result, policy interventions in one sector generate spillover effects that propagate throughout the economy.

According to Walras' law, one equilibrium condition is redundant; thus, we drop Home's expenditure constraint from the optimization problem and only consider the goods market clearing conditions. Let $\gamma_{1,j}$ and $\gamma_{n,j}$ be the multipliers on the goods market clearing conditions (9) and (10), respectively, with each constraint written as Home demand equal to residual export supply.⁹

Proposition 1 (Optimal Policies and Multipliers). *Irrespective of supply-side specifications,*

⁸An alternative approach involves directly taking first-order conditions with respect to $\{w_{n,j}, L_{n,j}\}$ and imposing multipliers on the labor market specifications Ω . Both approaches yield identical optimal policies and equilibrium outcomes. We demonstrate the equivalence of these two approaches across three examples in Appendix C.

⁹Equivalently, if we use Home's budget constraint, one of the market-clearing conditions becomes redundant. By utilizing Home's budget constraint and a multiplier on the constraint, we can eliminate one of the goods market-clearing conditions by setting $\gamma_{2,j} = 0$. The same optimal policy formulas can be attained. We provide an example of the proof of Home's optimal policies in the two-country case in Appendix L.

optimal policies in sector j take the form of

$$1 + \tau_j^d = \gamma_{1,j}, \quad 1 + \tau_{n,j}^m = \gamma_{n,j}, \quad \frac{1 + \tau_{n,j}^x}{1 + \tau_j^d} = \frac{1 + \epsilon_j(1 - \pi_{n1,j})}{\sum_{i \neq 1}^N (1 + \tau_{i,j}^m) \epsilon_j \pi_{ni,j}}, \quad n \neq 1. \quad (11)$$

Proof. Appendix E.

The first two equalities have a direct decentralization interpretation. To see this, rewrite the left-hand side of the goods market clearing condition (10) as

$$\beta_{1,j} \frac{1}{1 + \tau_{n,j}^m} \pi_{1n,j} x_1 = p_{n,j} d_{1n,j} C_{1n,j},$$

which states that Home's expenditure on country n 's sector- j good, $p_{n,j} d_{1n,j} C_{1n,j}$. Home government's first-order condition with respect to $C_{1n,j}$ implies $U_{1n,j} = \gamma_{n,j} p_{n,j} d_{1n,j}$ where $U_{1n,j}$ denotes the marginal utility. Thus $\gamma_{n,j}$ is the shadow value relative to the supply price. Home consumer, by contrast, values the same good at its tariff-inclusive price, $U_{1n,j} = (1 + \tau_{n,j}^m) p_{n,j} d_{1n,j}$. The tariff is therefore the instrument to make consumers face the shadow value: $1 + \tau_{n,j}^m = \gamma_{n,j}$. The same argument applied to Home's own good gives $1 + \tau_j^d = \gamma_{1,j}$, so the multiplier on domestic market clearing is the optimal gross domestic tax.

The third equality describes the additional wedge on exports $(1 + \tau_{n,j}^x)/(1 + \tau_j^d)$, relative to the domestic-policy applied to Home production. This additional wedge reflects Home's monopoly power in destination n . Consider first a case in which domestic taxes and import tariffs are all zero. Then, the optimal export tax simplifies to $\frac{1 + \epsilon_j(1 - \pi_{n1,j})}{\epsilon_j(1 - \pi_{n1,j})}$. Thus, Home sets a higher export tax on destinations where $\pi_{n1,j}$, country n 's expenditure share on Home's sector j goods, and hence Home's seller power, is larger. In general, the export tax formula in (11) also contains Home's import tariffs because an export tax on sales to country n shifts country n 's demand toward goods supplied by country i , thereby changing country i 's residual export supply to Home and the value of Home's tariff on those imports. We defer the detailed economic interpretation of these interactions until the two- and multi-country cases below.

Our goal in the following sections is to unpack these multipliers. They capture two reasons why the home government uses taxes: to manipulate terms of trade and to address domestic distortions. Terms-of-trade gains come from Home's ability to change import and

export prices by affecting the demand that Foreign faces and the supply of Home goods to Foreign. In other words, these gains depend on the Home country's buyer power in the foreign goods market and its monopoly power as an exporter. Home's buyer power in country n is governed by the elasticity of the residual export supply that country n offers to Home, which is endogenous because sectoral supply and demand from all other destinations respond to prices.

Section 3.2 isolates cross-sector interdependencies due to factor allocation and returns to scale in a two-country setting. For models featuring symmetric cross-elasticities in foreign supply curves, we show that relative import tariffs depend only on sectoral supply parameters and Home's relative net-import shares across sectors. Section 3.3 extends the analysis to multiple countries, introducing both cross-sector and cross-country interdependencies. The foreign trade network reallocates Home's aggregate sectoral net-import motive across foreign countries. We show that, rather than depending solely on bilateral net import share, network effects compress or amplify cross-country tariff differentials under different specifications.

3.2 Cross-Sector Interdependencies and Optimal Policies

Here, we isolate the impact of cross-sector interdependencies on optimal policies by analyzing the case with two countries, Home and Foreign.

The multipliers $\gamma_{2,j}$ on Foreign's goods market clearing condition depend on Foreign's supply curves $Y_{2,j}(\{p_{2,k}\})$ and its domestic demand $\beta_{2,j}\pi_{22,j}x_2$, which is affected by Home's exports to Foreign. Accordingly, we define Foreign's supply elasticity and sectoral net import as a share in foreign sectoral income. Together, these two components characterize optimal tariffs in the two-country case.

Definition 3. Matrix Λ_n consists of scaled partial supply elasticities in country n , with the j th-row- i th-column entry given by $\frac{\partial \ln Y_{n,i}}{\partial \ln p_{n,j}} \frac{Y_{n,i}}{Y_{n,j}}$. Home's sectoral net import as a share in foreign income in sector j is $M_{1,j}/Y_{2,j}$, where $M_{1,j} \equiv \beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}$.

Proposition 2 (Two-Country General Optimal Policy Formula). Home government's unilat-

eral optimal policy satisfies

$$\text{Domestic tax vector: } (\Lambda_1 - I)(\mathbf{1} + \boldsymbol{\tau}^d) = 0, \quad (12)$$

$$\text{Import tariff vector: } \Lambda_2(\boldsymbol{\tau}^m + \mathbf{1} + \Phi_2 \mathbf{1}) = \left(\frac{M_{1,j}}{Y_{2,j}} \right)_{j=1}^J, \quad (13)$$

$$\text{Export tax: } 1 + \tau_j^x = \frac{1 + \tau_j^d}{1 + \tau_j^m} \left(1 + \frac{1}{\epsilon_j \pi_{22,j}} \right), \quad \forall j, \quad (14)$$

and the equilibrium conditions (1)-(10), where $\boldsymbol{\tau}^d$ and $\boldsymbol{\tau}^m$ are vectors of domestic taxes and import tariffs across sectors, respectively. I is an identity matrix. $\left(\frac{M_{1,j}}{Y_{2,j}} \right)_{j=1}^J$ denotes the $J \times 1$ column vector whose j th element is $M_{1,j}/Y_{2,j}$. Φ_2 is common across all sectors.¹⁰

Proof. Appendix B.

First, domestic taxes relate directly to Home's scaled partial supply elasticities matrix Λ_1 , which depend on Home's own supply-side and labor market specifications. Second, with two countries, import tariffs depend directly on Foreign's matrix Λ_2 and how important Home's purchases are to Foreign, $M_{1,j}/Y_{2,j}$. Lastly, export taxes follow directly from Proposition 1 when there are two countries: given the other taxes, they depend on the trade elasticity and the share of foreign spending on Home exports. All taxes are jointly determined with equilibrium allocations and prices satisfying conditions (1)-(10).

To demonstrate how various forces influence endogenous supply and demand and optimal policies, we focus below on a class of models featuring a supply system with symmetric cross-elasticity effects.

3.2.1 Optimal Policies under GIIA Supply System

Here, we study a class of models with *Generalized Independence of Irrelevant Alternatives (GIIA) Supply System*, as defined below. This system offers two key advantages. First, many standard trade models fall within this system. Second, this system yields analytical policy formulas, even considering general equilibrium effects and interlinkages across sectors.

Definition 4 (GIIA Supply System). *We define a Generalized Independence of Irrelevant Alternatives*

¹⁰ $\Phi_2 = -\sum_{s=1}^J (\beta_{2,s} \frac{1+\tau_s^d}{1+\tau_s^m} \pi_{21,s} + (1 + \tau_s^m) \beta_{2,s} \pi_{22,s})$.

natives Supply System if for any three sectors s, i, j ,

$$\frac{\partial \ln Y_s}{\partial \ln p_j} \frac{Y_s}{Y_j} \frac{1}{\eta_j} - \frac{\partial \ln Y_s}{\partial \ln p_i} \frac{Y_s}{Y_i} \frac{1}{\eta_i} = \begin{cases} 0 & \text{if } s \neq i, j, \\ 1 & \text{if } s = j, \\ -1 & \text{if } s = i, \end{cases} \quad (15)$$

where η_j is a sector-specific constant related to supply elasticity.

Lemma 2 (CES Sufficiency). *Constant returns to scale production with CES labor supply across sectors yields a supply system that satisfies GIIA (Definition 4).*

Rather than assuming a specific functional form, we impose a GIIA Supply System. This restriction requires that, for any two sectors, all other sectors affect their relative weighted supply responses symmetrically. As a result, third sectors cancel out of relative policy comparisons despite general equilibrium interactions. A CES labor supply across sectors satisfies this restriction, but the results do not rely on CES per se.

Under a GIIA Supply System, the partial elasticities of a sector's income with respect to prices in any other two sectors (scaled by the size and elasticity parameters of the two sectors) are the same. Additionally, the difference between a sector's own-price supply elasticity (scaled by its size and supply parameter) and its cross-price elasticity relative to any other sector (scaled by that sector's size and supply parameter) equals one, i.e., $\frac{\partial \ln Y_j}{\partial \ln p_j} \frac{1}{\eta_j} - \frac{\partial \ln Y_j}{\partial \ln p_i} \frac{Y_j}{Y_i} \frac{1}{\eta_i} = 1$.

The GIIA Supply System encompasses many quantitative trade models, including extended multi-sector Armington, Eaton-Kortum, and different economies of scale across sectors like Armington with external economies of scale or the generalized Krugman model. It also accommodates various labor market specifications: perfectly mobile labor, fixed labor, and imperfectly substitutable labor across sectors as in the one-factor Ricardo-Roy model.

Proposition 3 (Two-country Optimal Policies). *Under Cobb-Douglas and CES demand with demand elasticity ϵ_j and a GIIA Supply System with supply parameter η_j , the home government's*

unilateral optimal policies satisfy the following conditions for any sector j ,

$$\text{Domestic tax: } 1 + \tau_j^d = (1 + \bar{\tau}^d) \frac{\eta_j}{\eta_j - 1}, \quad (16)$$

$$\text{Import tariff: } 1 + \tau_j^m = \frac{1}{\eta_j} \frac{M_{1,j}}{Y_{2,j}} + (1 + \bar{\tau}^m), \quad (17)$$

$$\text{Export tax: } 1 + \tau_j^x = \frac{1 + \tau_j^d}{1 + \tau_j^m} \left(1 + \frac{1}{\epsilon_j \pi_{22,j}} \right). \quad (18)$$

and the equilibrium conditions (1)-(10).¹¹

Proof. Appendix B.

Given the tax neutrality established in Lemma 1, one domestic tax level and one trade tax level remain indeterminate. Thus, the uniform shifter $\bar{\tau}^d$ in equation (16) can take any arbitrary value, and one trade tax can be normalized without changing the real allocations and welfare impacts of optimal policies.

This proposition demonstrates that under a GIIA Supply System, Home imposes differential domestic taxes for all sectors if the parameters η_j are heterogeneous. Heterogeneity here refers to the sector-specific constants that govern the scaled supply responses as in our definition of GIIA, and in general, heterogeneity in sectoral supply elasticities alone does not imply inefficiency. When η_j is identical across sectors, Home imposes uniform domestic taxes, or equivalently zero domestic taxes by choosing $\bar{\tau}^d$.

The import tariff of any sector depends only on Home's net import relative to foreign sectoral income, the sector's supply parameters η_j , which can be positive or negative, and a sector-common component $\bar{\tau}^m$. A larger Home net import spending relative to Foreign income, $\frac{M_{1,j}}{Y_{2,j}}$, reflects greater Home buyer power and thus justifies a higher import tariff when $\eta_j > 0$, or a larger import subsidy when $\eta_j < 0$. Importantly, the relative import tariff across sectors j and i , $\tau_j^m - \tau_i^m = M_{1,j}/(\eta_j Y_{2,j}) - M_{1,i}/(\eta_i Y_{2,i})$, does not explicitly depend on Home's net import share in other sectors. Furthermore, instead of depending on the entire matrix Λ_n , where each sector's self-elasticity is not a constant and cross-elasticities are not zero, the relative import tariffs only depend on the parameters η_j and η_i .

Given other taxes, sector j 's export tax is higher if the demand elasticity ϵ_j is smaller, i.e.,

¹¹In the special case of $\eta_j = 1$ for some j , it is the fixed labor case and domestic taxes can be any value.

the demand is less elastic, and if $\pi_{22,j}$ is lower, which implies a greater foreign demand for Home goods. The schedule of export taxes is consistent with [Costinot et al. \(2015\)](#), which shows that export taxes reflect the home government's incentive to manipulate relative prices in its favor by exploiting its export market power.

In the following, we present three examples of the GIIA Supply System and explain the mechanisms. The first focuses on labor supply and allocation across sectors. The second examines models with different returns to scale across sectors. The third explores different returns to scale across sectors combined with labor market specifications.

Example 1: One-Factor Ricardo-Roy Model (imperfectly substitutable inputs) We adopt the framework in [Galle, Rodríguez-Clare, and Yi \(2023\)](#), which combines the multi-sector EK model in [Costinot, Donaldson, and Komunjer \(2012\)](#) and the Roy model as in [Lagakos and Waugh \(2013\)](#) and [Hsieh, Hurst, Jones, and Klenow \(2019\)](#). We set up the general multi-factor setup and show that when there is only one factor, the model belongs to a GIIA Supply System. See Appendix [C.1](#) for a proof.

There are G types of workers in each country. The total number of type- g workers in country n is fixed at $\bar{L}_{n,g}$. A type- g worker in country n draws an efficiency unit z_j in sector j from a Fréchet distribution $F_{n,j,g}$ with a shape parameter κ and a scale parameter $A_{n,j,g}$. A worker with the vector $\mathbf{z} = (z_1, z_2, \dots, z_J)$ chooses to work in a sector that gives the highest income based on her productivity and wage in that sector. Let $\Xi_{n,j}$ be the set of workers choosing sector j , $\Xi_{n,j} \equiv \{\mathbf{z}: w_{n,j}z_j \geq w_{n,k}z_k \text{ for all } k\}$.

In equilibrium, the share of type- g workers that enter sector j in country n is given by

$$\lambda_{n,j,g} = \frac{A_{n,j,g}(w_{n,j})^\kappa}{\sum_{s=1}^J A_{n,s,g}(w_{n,s})^\kappa}.$$

The sum of type- g efficiency units supplied to sector j is $L_{n,j,g} = \frac{[\sum_{s=1}^J A_{n,s,g}(w_{n,s})^\kappa]^{\frac{1}{\kappa}}}{w_{n,j}} \lambda_{n,j,g} \bar{L}_{n,g}$, and the total efficiency labor working in sector j in country n is given by $L_{n,j} = \sum_{g=1}^G L_{n,j,g}$.¹²

¹²The sum of type- g efficiency units supplied to sector j in country n is $L_{n,j,g} \equiv L_{n,g} \int_{\Xi_{n,j}} z_j dF_{n,g}(\mathbf{z}) = \frac{\zeta_n [\sum_{s=1}^J A_{n,s,g}(w_{n,s})^\kappa]^{\frac{1}{\kappa}}}{w_{n,j}} \lambda_{n,j,g} \bar{L}_{n,g}$, where $F_{n,g}(\mathbf{z})$ is the joint probability distribution of $\mathbf{z}$ for type- g workers across sectors in country n , and $\zeta_n \equiv \Gamma(1 - 1/\kappa)$. For notation simplification, we scale $A_{n,s,g}$ by ζ_n^κ , i.e., setting $A_{n,s,g}$

Thus, the labor market specification Ω satisfies

$$\Omega = \left\{ (w_{n,j}, L_{n,j}) : w_{n,j} L_{n,j} = \sum_{g=1}^G \left[\frac{A_{n,j,g} (w_{n,j})^\kappa}{W_{n,g}^\kappa} W_{n,g} \bar{L}_{n,g} \right], W_{n,g} = \left[\sum_{s=1}^J A_{n,s,g} (w_{n,s})^\kappa \right]^{\frac{1}{\kappa}} \right\}. \quad (19)$$

With perfect competition in the goods market, price equals marginal cost in each sector. The supply-side assumption S therefore specifies that for any country n and sector j :

$$p_{n,j} = \frac{w_{n,j}}{\bar{T}_{n,j}}. \quad (20)$$

Consider the case with only one type of worker ($G = 1$).¹³ We can derive supply curves by combining the labor market specification Ω (Eq.19) with the supply-side assumption S (Eq.20) and the definition of income $Y_{n,j} = w_{n,j} L_{n,j}$. Eliminating wages and labor $\{w_{n,j}, L_{n,j}\}$ yields the following supply curves:

$$Y_{n,j}(\{p_{n,k}\}) = A_{n,j} (p_{n,j} \bar{T}_{n,j})^\kappa \left[\sum_{s=1}^J A_{n,s} (p_{n,s} \bar{T}_{n,s})^\kappa \right]^{\frac{1-\kappa}{\kappa}} \bar{L}_n.$$

The corresponding element of the scaled partial supply elasticity matrix Λ_n is given by

$$\frac{\partial \ln(Y_{n,s})}{\partial \ln(p_{n,j})} \frac{Y_{n,s}}{Y_{n,j}} = I_{s=j}(\kappa) - (\kappa - 1) \lambda_{n,s}, \quad (21)$$

where $I_{s=j}(\kappa)$ is an indicator function that equals κ when $s = j$ and 0 otherwise. $\lambda_{n,j} = Y_{n,j}/Y_n$ is the share of income in sector j produced in country n .

We can derive the optimal tariff formula in two equivalent ways. In the first, we start from the general optimality condition in Proposition 2. Written sector by sector rather than in matrix form, the condition reads

$$\sum_{s=1}^J (\tau_s^m + 1 + \Phi_2) \frac{\partial \ln(Y_{2,s})}{\partial \ln(p_{2,j})} \frac{Y_{2,s}}{Y_{2,j}} = \frac{M_{1,j}}{Y_{2,j}}, \quad \forall j. \quad (22)$$

to $\zeta_n^\kappa A_{n,s,g}$.

¹³The one-factor Ricardo-Roy model is equivalent to a CES labor market characterized by constant elasticity of substitution of labor across sectors. See Appendix C.4 for details.

Substituting the scaled partial supply elasticities from equation (21), dividing equation (22) for each sector j by κ , and differencing across sectors i and j yields the relative tariffs

$$\tau_j^m - \tau_i^m = \frac{1}{\kappa} \frac{M_{1,j}}{Y_{2,j}} - \frac{1}{\kappa} \frac{M_{1,i}}{Y_{2,i}}.$$

Alternatively, we can demonstrate that this model's supply system, which features a one-factor Ricardo-Roy structure and perfect goods market competition, satisfies the GIIA Supply System requirements. Hence, the optimal policies satisfy the formula in Proposition 3. Using the elements from equation (21), the matrix Λ_n takes the form

$$\Lambda_n = \begin{pmatrix} \kappa - (\kappa - 1)\lambda_{n,1} & \dots & -(\kappa - 1)\lambda_{n,j} & \dots & -(\kappa - 1)\lambda_{n,J} \\ \dots & \dots & \dots & \dots & \dots \\ -(\kappa - 1)\lambda_{n,1} & \dots & \kappa - (\kappa - 1)\lambda_{n,j} & \dots & -(\kappa - 1)\lambda_{n,J} \\ \dots & \dots & \dots & \dots & \dots \\ -(\kappa - 1)\lambda_{n,1} & \dots & -(\kappa - 1)\lambda_{n,j} & \dots & \kappa - (\kappa - 1)\lambda_{n,J} \end{pmatrix}. \quad (23)$$

When we take the difference between any two rows of Λ_n (say i and j), all entries except columns j and i are eliminated, leaving κ in column j and $-\kappa$ in column i . The GIIA Supply System, as defined in Definition 4, is clearly satisfied with the constant parameter $\eta_i = \eta_j = \kappa$. This enables direct application of Proposition 3 to determine optimal policies. Given that η_j is constant across sectors, optimal domestic taxes are uniform, reflecting an efficient domestic market.

The salient symmetry feature of the partial supply elasticity matrix Λ_n produces the powerful tariff formula in (17). The cross-elasticities are symmetric, meaning any two sectors' prices have symmetric impacts on a third sector. This symmetry implies that third-sector supply responses do not show up explicitly in the optimal relative tariffs between any two sectors, even when cross-elasticities are non-zero. In this model, relative foreign supply curves are upward-sloping with an elasticity of $\kappa > 0$ (the labor substitutability parameter). To induce relatively lower foreign prices in a given sector, the home government must raise tariffs in that sector, which reduces domestic demand for foreign goods. Since

labor is not perfectly substitutable across sectors, this demand reduction lowers wages and thus prices in the corresponding foreign sector, improving Home's terms of trade. Furthermore, Proposition 3 demonstrates that the home government should increase tariffs more in sectors where Home has larger import shares in foreign income, as Home has greater buyer power in these sectors.

Two important special cases merit attention. When $\kappa = \infty$, the Ricardo-Roy model reduces to the standard specification of perfectly mobile labor across sectors. Proposition 3 then implies that $\tau_j^m - \tau_i^m = 0$ for any two sectors i and j , generating uniform optimal tariffs across all sectors. When $\kappa = 1$, the model becomes equivalent to the sector-specific labor case, and optimal relative tariffs equal the relative net import shares of Home across any two sectors.

Example 2: Generalized Multi-Sector Krugman (different scale elasticities across sectors)

We introduce increasing returns to scale through a generalized multi-sector Krugman with different scale elasticities across sectors. Appendix C.2 presents this model, proves it belongs to the GIIA Supply System, and also provides a model-specific proof that yields the same optimal policy formula as in Proposition 3.

In this case, we assume perfectly mobile labor across sectors. The labor market specification Ω is characterized by equal wages across sectors, $w_{n,j} = w_n$ for any j , and labor market clearing $\sum_j L_{n,j} = \bar{L}_n$. The supply-side assumption S incorporates increasing returns to scale with

$$p_{n,j} = \bar{T}_{n,j}^{-1} L_{n,j}^{-\psi_j} w_n, \quad (24)$$

for each country n and sector j . Here, the scale elasticity ψ_j relates to the elasticity of substitution σ_j between varieties within a sector j through the relationship $\psi_j = 1/(\sigma_j - 1)$. Combining Ω and S, the supply curves in country n are given by

$$Y_{n,j} = w_n^{(1+\psi_j)/\psi_j} (p_{n,j} \bar{T}_{n,j})^{-1/\psi_j}, \quad \text{where } w_n \text{ satisfies } \sum_{j=1}^J (p_{n,j} \bar{T}_{n,j})^{-\frac{1}{\psi_j}} w_n^{\frac{1}{\psi_j}} = \bar{L}_n. \quad (25)$$

Using the implicit function theorem, we derive the scaled partial supply elasticities:

$$\frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,j}} \frac{Y_{n,s}}{Y_{n,j}} = I_{s=j} \left(-\frac{1}{\psi_j} \right) + \frac{1}{\psi_j} \frac{(1 + \frac{1}{\psi_s}) Y_{n,s}}{\sum_{k=1}^J \frac{1}{\psi_k} Y_{n,k}}.$$

This model satisfies the requirements of the GIIA Supply System with $1/\eta_j = -\psi_j$.

We can then apply Proposition 3 to get optimal policies. In particular, optimal tariffs satisfy

$$\tau_j^m - \tau_k^m = -\psi_j \frac{M_{1,j}}{Y_{2,j}} + \psi_k \frac{M_{1,k}}{Y_{2,k}}.$$

Increasing returns to scale generate a downward-sloping relative supply curve. In this scenario, to achieve lower prices in a foreign sector, the home government reduces import tariffs in that sector, thereby increasing domestic demand for foreign goods. This higher demand incentivizes foreign producers to allocate more labor to that sector. Due to increasing returns to scale, Home consumers benefit from lower import prices. Home government reduces tariffs more in sectors where Home has larger net import shares in foreign income. Since the scale elasticity ψ_j is sector-specific and introduces inefficiency in the product market, optimal domestic taxes are no longer uniform across sectors.

Note that this generalized multi-sector Krugman framework is isomorphic to multi-sector Eaton-Kortum and Armington models with Marshallian economies of scale or endogenous technology but different returns to scale across sectors.¹⁴ Our optimal policy formulas therefore apply to these models.

Example 3: Differential Scale Elasticities and Imperfectly Mobile Labor We now extend the generalized multi-sector Krugman model with imperfectly mobile labor. This example combines imperfectly mobile labor across sectors in Example 1, and the increasing

¹⁴See Bartelme et al. (2025) for a trade framework with Marshallian externalities. Kucheryavyi, Lyn, and Rodríguez-Clare (2023) show the isomorphic gravity equations for models with varying returns to scale, including the generalized multi-sector Krugman model, multi-sector Eaton-Kortum and Armington models with external economies of scale. Eaton and Kortum (2001) allow for endogenous technology accumulation, where total labor is divided into two components: production labor and researchers. Labor moves freely across production and research, and technology endogenously depends on the number of researchers. Bai, Jin, and Lu (2025) extend this framework to a multi-sector setting and study the optimal dynamic trade policies. In steady-state equilibrium, the ratio of production labor to researchers remains constant, allowing technology to be expressed as a function of the sector's labor as in equation (A.77).

returns to scale in Example 2. This model belongs to the GIIA Supply System.¹⁵

Specifically, the labor market specification Ω follows equation (19), and the supply-side assumption follows equation (24). Using the implicit function theorem, we can obtain scaled partial supply elasticities and show that these elasticities satisfy the requirement of the GIIA Supply System, where parameters η_j incorporate both the scale elasticity and labor market parameter:

$$\frac{1}{\eta_j} = \frac{1}{\kappa} - \psi_j \frac{\kappa - 1}{\kappa}. \quad (26)$$

We apply Proposition 3 to determine the optimal policies. Whether the foreign relative supply curve in sector j is upward- or downward-sloping depends on the sign of the parameter η_j , which influences whether the home government imposes a relatively higher or lower tariff on this sector.

In sum, the two-country case with a GIIA Supply System yields a clear formula showing that Home's relative optimal tariffs depend on its buyer power across foreign sectors. By emphasizing buyer power, this formula echoes traditional optimal tariff theory while differing significantly from a partial equilibrium setup that uses exogenous, constant foreign export elasticities or overlooks sectoral linkages. The key distinction lies in how our formula captures interdependencies across sectors through general equilibrium effects. Although the formula appears sector-independent at first glance, all underlying components, such as trade shares, prices, and trade policies, are jointly determined. Traditional formulations abstract away from these connections, treating foreign sectors in isolation. In the following subsection, we highlight the importance of *interdependence across sectors* by an example of general supply system.

3.2.2 Optimal Policies under General Supply System

When a model does not belong to the GIIA Supply System, interdependence across sectors shows up explicitly: the cross-elasticities of foreign scaled supply curves are not symmetric, and the relative import tariffs across any two sectors depend not only on Home's relative net import shares of these two sectors but also on those of other sectors. Thus, the import

¹⁵Appendix C.3 also provides a model-specific proof that derives the same optimal policy formulas stated in Proposition 3.

tariff formula depends on a matrix of foreign partial supply elasticities and a vector of Home's net import shares, as shown in Proposition 2.

One example of a non-GIIA Supply System is the multi-factor Ricardo-Roy model, as laid out in Example 1. In this case, the supply-side assumption S is specified by equation (20) and Ω is defined as equation (19). Combining these two yields the following supply curves:

$$Y_{n,j} = (p_{n,j} \bar{T}_{n,j})^\kappa \sum_{g=1}^G A_{n,j,g} \left[\sum_{s=1}^J A_{n,s,g} (p_{n,s} \bar{T}_{n,s})^\kappa \right]^{\frac{1-\kappa}{\kappa}} \bar{L}_{n,g}$$

The corresponding element of the scaled partial supply elasticity matrix Λ_n is given by

$$\frac{\partial \ln(Y_{n,s})}{\partial \ln(p_{n,j})} \frac{Y_{n,s}}{Y_{n,j}} = I_{s=j}(\kappa) - (\kappa - 1) \frac{\sum_g \lambda_{n,s,g} \lambda_{n,j,g} W_{n,g} \bar{L}_{n,g}}{Y_{n,j}}$$

where $\lambda_{n,j,g} = Y_{n,j,g} / (W_{n,g} \bar{L}_{n,g})$ is the share of type- g workers that enter sector j in country n . In the matrix format, Λ_n takes the form of

$$\Lambda_2 = \begin{pmatrix} \kappa - (\kappa - 1) \frac{\sum_g \lambda_{2,1,g} \lambda_{2,1,g} W_{2,g} \bar{L}_{2,g}}{Y_{2,1}} & \dots & -(\kappa - 1) \frac{\sum_g \lambda_{2,1,g} \lambda_{2,j,g} W_{2,g} \bar{L}_{2,g}}{Y_{2,1}} & \dots & -(\kappa - 1) \frac{\sum_g \lambda_{2,1,g} \lambda_{2,J,g} W_{2,g} \bar{L}_{2,g}}{Y_{2,1}} \\ \dots & \dots & \dots & \dots & \dots \\ -(\kappa - 1) \frac{\sum_g \lambda_{2,j,g} \lambda_{2,1,g} W_{2,g} \bar{L}_{2,g}}{Y_{2,j}} & \dots & \kappa - (\kappa - 1) \frac{\sum_g \lambda_{2,j,g} \lambda_{2,j,g} W_{2,g} \bar{L}_{2,g}}{Y_{2,j}} & \dots & -(\kappa - 1) \frac{\sum_g \lambda_{2,j,g} \lambda_{2,J,g} W_{2,g} \bar{L}_{2,g}}{Y_{2,j}} \\ \dots & \dots & \dots & \dots & \dots \\ -(\kappa - 1) \frac{\sum_g \lambda_{2,J,g} \lambda_{2,1,g} W_{2,g} \bar{L}_{2,g}}{Y_{2,J}} & \dots & -(\kappa - 1) \frac{\sum_g \lambda_{2,J,g} \lambda_{2,j,g} W_{2,g} \bar{L}_{2,g}}{Y_{2,J}} & \dots & \kappa - (\kappa - 1) \frac{\sum_g \lambda_{2,J,g} \lambda_{2,J,g} W_{2,g} \bar{L}_{2,g}}{Y_{2,J}} \end{pmatrix}.$$

This general Ricardo-Roy model fails to meet the GIIA requirements. Foreign sector s 's supply, represented by $Y_{2,s}$, is affected by the sector j 's price $p_{2,j}$ through the expression $-(\kappa - 1) \sum_g \frac{\lambda_{2,j,g} W_{2,g} \bar{L}_{2,g}}{Y_{2,j}} \lambda_{2,s,g}$, and affected by sector J 's price $p_{2,J}$ through $-(\kappa - 1) \sum_g \frac{\lambda_{2,J,g} W_{2,g} \bar{L}_{2,g}}{Y_{2,J}} \lambda_{2,s,g}$. In general, these cross-elasticities are asymmetrically influenced by endogenous variables and cannot be canceled out. Thus, the multi-factor Ricardo-Roy model is not a GIIA Supply System.

The Home government internalizes the impact of its import tariffs on sectoral wages, which in turn influence worker allocation across sectors. Whether an increase in sector j 's wage in Foreign reduces labor and supply in the foreign sector i depends on the types

of workers in sector j , $\frac{\lambda_{2,j,g} W_{2,g} \bar{L}_{2,g}}{Y_{2,j}}$, and how likely they are to enter sector i , i.e., $\lambda_{2,i,g}$. The covariance between $\frac{\lambda_{2,j,g} W_{2,g} \bar{L}_{2,g}}{Y_{2,j}}$ and $\lambda_{2,i,g}$ indicates how likely type- g workers within sector j apply to sector i , and a higher covariance results in greater elasticity, meaning that changes in sector j 's wage will have a more pronounced effect on sector i 's income due to the increased likelihood of worker mobility between the two sectors. The covariance between sector j and i can differ from that between sector J and i . Thus, the impact of price changes of sector j and sector J on the third sector i is different. These interdependencies show up explicitly in optimal policies.

We can directly apply Proposition 2 for optimal policies. In addition, even if the condition for the GIIA Supply System is not met, we can still prove that the home government does not use any domestic taxes, i.e., $\tau_j^d = 0$ for any sector j . The reason is that the Ricardo-Roy model is efficient and has no market frictions. See Appendix D for details.

3.3 Cross-Country Interdependencies and Optimal Policies

A multi-country framework differs from the two-country case in two key respects. First, Home lacks policy instruments that directly control trade between foreign countries. A price change in one foreign country-sector redirects third-market demand and changes the residual export supplies that all foreign countries offer to Home. Thus even under GIIA, the trade-network effect appears in the tariff formula.

Second, although GIIA continues to simplify cross-sector supply responses within each country, it no longer simplifies the tariff formula as fully as in the two-country case. There, GIIA simplifies the formula because of its cross-sector symmetry assumption and because aggregate demand within the single foreign country is identical across sectors. With multiple countries, factors are typically less mobile across countries than across sectors within a country, and aggregate demand differs across countries. Both differences matter for the optimal tariff formula. As a result, the tariff formula includes an additional trade-network term and a country-specific term.

This section focuses on import tariffs. Optimal domestic taxes continue to satisfy (12), and export taxes remain given by Proposition 1. We first state the general tariff charac-

terization, then impose GIIA to obtain a compact sufficient-statistic formula, illustrate the underlying trade-network effect, and finally characterize tariff convergence and divergence.

3.3.1 Optimal Tariffs and Export Supply Elasticity Matrix

Define country n 's residual export supply to Home in sector j as

$$E_{1n,j} \equiv Y_{n,j} - \sum_{i \neq 1}^N \beta_{i,j} \pi_{in,j} x_i. \quad (27)$$

This is the right-hand side of the goods market clearing condition (10). It depends on both country n 's supply $Y_{n,j}$ and demand from all non-Home destinations $\sum_{i \neq 1}^N \beta_{i,j} \pi_{in,j} x_i$. Let bold symbols stack their scalar counterparts across the J sectors, for example $\mathbf{E}_{1n} = [E_{1n,1}, \dots, E_{1n,J}]'$ and $\circ$ denotes element-wise multiplication.

The key object for optimal tariff is the export supply elasticity matrix dE , the block matrix with (n, i) block

$$[dE]_{ni} \equiv \frac{\partial E_{1i}}{\partial \ln \mathbf{p}_n} \circ \frac{1}{Y_n}, \quad n, i \neq 1. \quad (28)$$

Each block measures how country i 's residual export supply to Home responds to country n 's prices, scaled by country n 's output. We also define $\mathbf{b}_n \equiv \beta_1 x_1 \circ \pi_{1n} \circ \frac{1}{Y_n} - \sum_{k=1}^J (1 + \tau_k^d) \frac{\partial E_{11,k}}{\partial \ln \mathbf{p}_n} \circ \frac{1}{Y_n}$, which collects Home's demand for country n 's goods and the response of Home's own market clearing to country n 's prices, weighted by domestic taxes.

Proposition 4 (General Multi-Country Tariff Characterization). *Home's optimal gross tariffs satisfy*

$$dE \begin{pmatrix} 1 + \tau_2^m \\ \vdots \\ 1 + \tau_N^m \end{pmatrix} = \begin{pmatrix} \mathbf{b}_2 \\ \vdots \\ \mathbf{b}_N \end{pmatrix}. \quad (29)$$

Proof. Appendix E.

By Proposition 1, the import tariff equals the multiplier $\gamma_{n,j}$ on the goods market clearing condition (10): $1 + \tau_{n,j}^m = \gamma_{n,j}$. These multipliers are tied to the elasticities of residual export supply dE and Proposition 4 follows. Here is the intuition. A tariff on country n 's sector- j good reduces Home's demand for it and affects $p_{n,j}$. Foreign production, expenditure, and

third-market substitution then adjust, changing the residual export supplies of all foreign countries. Home's buyer power is its ability to move the supply price: the less elastic the residual export supply, the further the price changes and the more Home gains.

Hence the optimal tariffs are jointly determined by the entire endogenous export-supply system. Three forces move them across countries and sectors. The first is the supply responses within the exporting country. The second is the trade-network effect. A tariff on one country's good moves prices in every foreign market and reallocates trade among countries, and this reallocation can dampen or amplify Home's bilateral buyer power. The third is an income effect. Price changes affect foreign incomes and hence demand across countries.

These forces show up in the elements of dE . Its diagonal blocks collect a supplier's own residual export-supply responses:

$$\frac{\partial E_{1n,i}}{\partial \ln p_{n,j}} \frac{1}{Y_{n,j}} = \underbrace{\frac{\partial \ln Y_{n,i}}{\partial \ln p_{n,j}} \frac{Y_{n,i}}{Y_{n,j}}}_{\text{supply elasticity}} - \underbrace{\beta_{n,i} \pi_{nm,i} \frac{\partial \sum_{\ell=1}^J Y_{n,\ell}}{\partial \ln p_{n,j}} \frac{1}{Y_{n,j}}}_{\text{income effect}} + \underbrace{I_{i=j} \epsilon_j \sum_{h \neq 1}^N (1 - \pi_{hn,j}) s_{hn,j}}_{\text{trade-network effect}}, \quad (30)$$

where $s_{hn,j} \equiv \pi_{hm,j} \beta_{h,j} x_h / Y_{n,j}$ is the share of country-sector (n, j) 's income earned from sales to destination h . The own elasticity consists of the supply elasticity, the income effect, and the trade-network effect, which works through trade shares. As $p_{n,j}$ increases, each importer h reduces its demand for goods from (n, j) , which increases (n, j) 's export supply to Home.

In a two-country model, these are the only channels, so that $dE = [dE]_{22}$. With multiple countries, a price change in one country also alters the residual export supply of every other country to Home. These indirect effects appear in the off-diagonal blocks:

$$\frac{\partial E_{1m,i}}{\partial \ln p_{n,j}} \frac{1}{Y_{n,j}} = - \underbrace{\beta_{n,i} \pi_{nm,i} \frac{\partial \sum_{\ell=1}^J Y_{n,\ell}}{\partial \ln p_{n,j}} \frac{1}{Y_{n,j}}}_{\text{income effect}} - \underbrace{I_{i=j} \epsilon_j \sum_{h \neq 1}^N \pi_{hm,j} s_{hn,j}}_{\text{trade-network effect}}. \quad (31)$$

The trade-network term is what carries cross-country interdependence within sector j . The term depends on the covariance between $\pi_{hm,j}$ (country h 's import share from m) and

$s_{hn,j}$ (how important h is in buying goods from n). The economic mechanism operates as follows: when country n 's prices rise, other countries reduce their imports from n and increase their purchases from alternative suppliers, such as country m . This substitution effect reduces m 's supply to Home. Higher covariance amplifies the elasticity, meaning that m 's export to Home is more sensitive to price changes elsewhere. This covariance is shaped by the correlation between productivity and trade costs, and trade networks across countries. Finally, the income effect captures the distinct channel through which the price change alters country n 's income and hence demand for country m 's goods in sector i , which in turn affects m 's supply to Home.

3.3.2 Optimal Tariffs under GIIA

We now impose GIIA. Its scaled symmetry collapses the general elasticity system into a bilateral net-import component and a trade-network adjustment.

Proposition 5 (Optimal Tariffs under GIIA). *Under a GIIA Supply System with parameters η_j and multiple countries:*

1. Home's unilateral optimal tariff on country n in sector j satisfies

$$1 + \tau_{n,j}^m = \frac{1}{\eta_j} \frac{M_{1n,j} + \mathcal{N}_{n,j}}{Y_{n,j}} + (1 + \tau_n^m), \quad (32)$$

where

$$M_{1n,j} \equiv \pi_{1n,j} \beta_{1,j} x_1 - \frac{1 + \tau_j^d}{1 + \tau_{n,j}^x} \pi_{n1,j} \beta_{n,j} x_n$$

is Home's bilateral net import with n . τ_n^m is a country-specific component common across sectors.¹⁶ The network statistic $\mathcal{N}_{n,j}$ is

$$\mathcal{N}_{n,j} \equiv \frac{1 + \tau_j^d}{1 + \tau_{n,j}^x} \beta_{n,j} \pi_{n1,j} x_n + \sum_{h \neq 1}^N \beta_{h,j} \pi_{hn,j} x_h \left[(\epsilon_j + 1) \frac{1 + \tau_j^d}{1 + \tau_{h,j}^x} - \epsilon_j (1 + \tau_{n,j}^m) \right], \quad (33)$$

¹⁶ $\tau_n^m = -(\Phi_n + 1) - \sum_{k=1}^J (\Phi_n + 1 + \tau_{n,k}^m) \left[\frac{\partial \ln Y_{n,k}}{\partial \ln p_{n,0}} \frac{Y_{n,k}}{Y_{n,0}} \frac{1}{\eta_0} - I_{k=0} \right]$, where 0 can be any sector, and $\Phi_n = -\sum_{i \neq 1}^N \sum_{k=1}^J \beta_{n,k} (1 + \tau_{i,k}^m) \pi_{ni,k} - \sum_{k=1}^J \beta_{n,k} (1 + \tau_k^d) \frac{\tau_{n1,k}}{1 + \tau_{n,k}^x}$ summarizes income effects.

and satisfies

$$\sum_{n \neq 1}^N \mathcal{N}_{n,j} = 0. \quad (34)$$

Consequently,

$$\sum_{n \neq 1}^N \frac{Y_{n,j}}{\sum_{i \neq 1}^N Y_{i,j}} (\tau_{n,j}^m - \tau_n^m) = \frac{1}{\eta_j} \frac{M_{1,j}}{\sum_{i \neq 1}^N Y_{i,j}}, \quad (35)$$

where $M_{1,j} = \sum_{n \neq 1}^N M_{1n,j}$ is Home's aggregate net import in sector j .

2. If foreign countries are identical or do not trade with one another, $\mathcal{N}_{n,j} = 0$, and optimal policies reduce to

$$1 + \tau_{n,j}^m = \frac{1}{\eta_j} \frac{M_{1n,j}}{Y_{n,j}} + (1 + \tau_n^m), \quad (36)$$

$$1 + \tau_{n,j}^x = \frac{1 + \tau_j^d}{1 + \tau_{n,j}^m} \left(1 + \frac{1}{\epsilon_j(1 - \pi_{n1,j})} \right). \quad (37)$$

Proof. Appendix F.

Equation (32) retains the two-country bilateral motive $M_{1n,j}/Y_{n,j}$ but adds $\mathcal{N}_{n,j}/Y_{n,j}$, which captures how trade among foreign countries changes countries' residual export supplies to Home. In particular, $\mathcal{N}_{n,j}$ in equation (33) are reduced from the following equation:

$$\mathcal{N}_{n,j} \equiv \frac{1 + \tau_j^d}{1 + \tau_{n,j}^x} \beta_{n,j} \pi_{n1,j} x_n + \sum_{h \neq 1}^N \beta_{h,j} x_h \left[\frac{\gamma_{1,j}}{1 + \tau_{h,j}^x} \frac{\partial \pi_{h1,j}}{\partial \ln p_{n,j}} + \sum_{g \neq 1}^N \gamma_{g,j} \frac{\partial \pi_{hg,j}}{\partial \ln p_{n,j}} \right], \quad (38)$$

reflecting the responses of trade shares in foreign countries. Because the network statistics add up to zero within each sector, they reallocate Home's aggregate sectoral net-import motive across foreign suppliers rather than changing its aggregate value. Accordingly, equation (35) determines the foreign-income-weighted average of the tariff components $\tau_{n,j}^m - \tau_n^m$ from Home's aggregate net import in sector j . When $\eta_j > 0$, Home applies higher weighted average tariffs to sectors where it has a larger net-import share. This tariff pattern reflects Home's incentive to reduce foreign prices in sectors where Home has greater buyer

power.

GIIA again disciplines variation across sectors within a foreign country. Within a foreign country, τ_n^m is common across sectors, and cross-sector tariff differences depend only on η_j and the network-adjusted sufficient statistic $(M_{1n,j} + \mathcal{N}_{n,j})/Y_{n,j}$. These sufficient statistics generally differ from bilateral net-import ratios. If foreign countries are identical or do not trade with one another, $\mathcal{N}_{n,j} = 0$ and the bilateral formulas reemerge.

Conditional Convergence & Divergence The network statistic $\mathcal{N}_{n,j}$ makes the tariff on a country-sector pair deviate from the bilateral net import $M_{1n,j}$. Depending on the sign of the supply parameters, it can compress the bilateral trade share effect, leading tariffs to converge across countries within a sector, or amplify it, leading tariffs to diverge. These effects are present in the general framework, but they are intertwined with general equilibrium forces since Home is not only a buyer but also an exporter, and its own exports move foreign income and demands.

To isolate the network effect, we consider a simplified scenario where Home does not export in one sector. Home is then a pure buyer in that sector, and the remaining interdependence across foreign countries reflects only the third-market demand for foreign goods. In this case, we can show that tariffs converge or diverge across countries according to the sign of η_j .

For each country n , define the weighted average tariff on other foreign countries as $\bar{\tau}_{-n,j}^m \equiv \sum_{i \neq 1,n} \omega_{in,j} \tau_{i,j}^m$, where the weight is given by $\omega_{in,j} \equiv \frac{\sum_{g \neq 1} \beta_{g,j} x_g \pi_{gn,j} \pi_{gi,j}}{\sum_{\tilde{i} \neq 1,n} \sum_{\tilde{g} \neq 1} \beta_{\tilde{g},j} x_{\tilde{g}} \pi_{\tilde{g}n,j} \pi_{\tilde{g}\tilde{i},j}}$.

Lemma 3. *Under a GIIA Supply System with parameters η_j and multiple countries, consider a sector j in which no foreign country sources from Home; that is, $d_{g1,j} \rightarrow \infty$ for every $g \neq 1$. Then*

$$\tau_{n,j}^m \begin{matrix} \leq \\ \geq \end{matrix} \bar{\tau}_{-n,j}^m \iff \mathcal{N}_{n,j} \begin{matrix} \geq \\ \leq \end{matrix} 0. \quad (39)$$

Thus, relative to its bilateral benchmark, when $\eta_j > 0$, the network term raises a tariff below the weighted average and lowers a tariff above it; when $\eta_j < 0$, these directions are reversed.

Proof. Appendix G.

In this case, the network statistic becomes

$$\mathcal{N}_{n,j} = \epsilon_j \left(\sum_{g \neq 1} \sum_{i \neq 1,n} \beta_{g,j} x_g \pi_{gn,j} \pi_{gi,j} \right) (\bar{\tau}_{-n,j}^m - \tau_{n,j}^m). \quad (40)$$

Substituting this expression into the optimal tariff formula (32) yields Lemma 3. In the three-country case, the weighted average tariff reduces to the tariff of the other foreign country. In particular,

$$\mathcal{N}_{2,j} = \epsilon_j \left(\sum_{g \neq 1} \beta_{g,j} x_g \pi_{g2,j} \pi_{g3,j} \right) (\tau_{3,j}^m - \tau_{2,j}^m), \quad \mathcal{N}_{3,j} = -\mathcal{N}_{2,j}. \quad (41)$$

Hence, if $\tau_{2,j}^m > \tau_{3,j}^m$, then $\mathcal{N}_{2,j} < 0 < \mathcal{N}_{3,j}$. For $\eta_j > 0$, the network lowers country 2's tariff relative to its bilateral benchmark and raises country 3's tariff. For $\eta_j < 0$, the directions are reversed. The same argument applies with countries 2 and 3 interchanged when $\tau_{2,j}^m < \tau_{3,j}^m$. Equivalently, the tariff gap satisfies

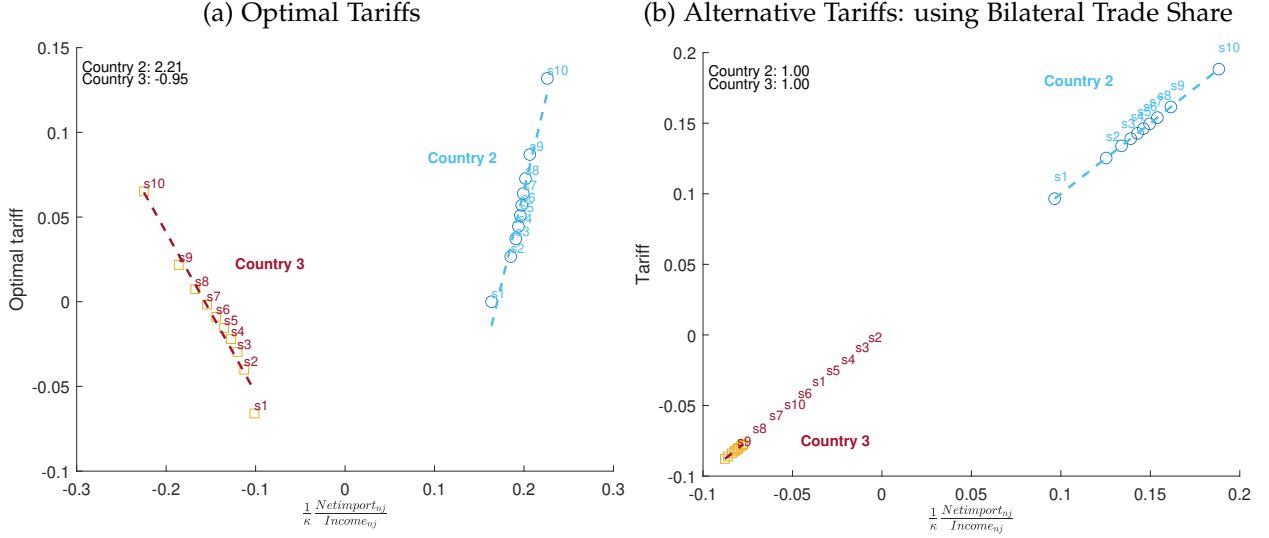
$$\tau_{2,j}^m - \tau_{3,j}^m = \frac{\frac{1}{\eta_j} \left(\frac{M_{12,j}}{Y_{2,j}} - \frac{M_{13,j}}{Y_{3,j}} \right) + \tau_2^m - \tau_3^m}{1 + \frac{\epsilon_j}{\eta_j} \left(\sum_{g \neq 1} \beta_{g,j} x_g \pi_{g2,j} \pi_{g3,j} \right) \left(\frac{1}{Y_{2,j}} + \frac{1}{Y_{3,j}} \right)}. \quad (42)$$

Equation (42) gives a precise convergence benchmark. When $\eta_j > 0$, the foreign trade network makes the denominator exceed one and compresses the absolute tariff gap relative to the counterfactual without that network; when $\eta_j < 0$, it amplifies the gap, provided the denominator remains positive. Convergence in this sense does not imply that all countries face the same tariff in sector j . First, it is compared with no trade network benchmark, and country-sector tariffs remain heterogeneous because bilateral net-import shares reflect each country's technology and comparative advantage, and second, because the country-specific components τ_n^m differ.

Taken together, the export-supply-elasticity matrix and Proposition 5 establish that optimal tariffs are jointly determined across foreign countries and generally cannot be expressed using bilateral net imports alone. Lemma 3 isolates a sufficient condition for a

sharp convergence or divergence result. The numerical example below relaxes Lemma 3's assumption, allowing Home to export in all sectors. Even under this general case, the same network force can generate tariff convergence across countries under a positive supply parameter.

Figure 1: Effects of Trade Network Between Foreign Countries



Note: Each panel plots Home's import tariff against its after-tax bilateral net import from a country-sector, divided by that country-sector's income and scaled by $1/\kappa$. Panel (a) reports the optimal tariffs. Panel (b) reports the first alternative policy: tariffs and export taxes are jointly determined by (36)-(37) using endogenous equilibrium trade shares, with Home treating each foreign country independently. The dashed lines are fitted lines, and the numbers in the upper-left corner denote their slopes.

Numerical Example Consider three countries and ten sectors.¹⁷ Country 2 has a comparative advantage in sector 10, relative to Home (country 1), while country 3 has a comparative advantage in sector 1. The technology parameters are: $T_1 = [50, \dots, 1]$, $T_2 = [1, \dots, 50]$, and $T_3 = T_1 \circ r$, where the ten elements of r decrease uniformly from 2 to 0.5. Due to asymmetric trade costs (listed in Case 4 of Table A-1 in Appendix H), Home imports significantly less from country 3 than country 2, while also exporting less to country 2.

Figure 1 plots tariffs against after-tax Home net import shares from each country-sector relative to foreign country-sector income, normalized by κ . Panel (a) presents the optimal tariffs calculated using the optimal policies from (29) and the equilibrium conditions. A

¹⁷In all numerical examples, $\epsilon = 4$, $\kappa = 1.1$, $\bar{L}_n = 1, \forall n$. $A_{n,j} = 1, \forall n, j$.

key takeaway is that although country 3 lacks a comparative advantage in sector 10, Home nevertheless imposes a higher tariff on imports of sector 10 from country 3.

The mechanism runs through the third market. Home's large sector-10 imports from country 2 create a strong bilateral tariff motive. The resulting fall in country 2's sector-10 price induces country 3 to substitute toward country 2's good, reducing the quantity country 2 supplies to Home. This additional response makes country 2's residual export supply more elastic and moderates its optimal tariff. At the same time, the cross-country response raises the tariff applied to country 3. The two tariffs therefore converge. Home thus exploits its monopsony power as a large buyer in sector 10, using tariffs to depress foreign prices and improve its terms of trade across all suppliers in that sector.

Panel (b) isolates the impact of trade network effects by analyzing an alternative policy: Home treats countries 2 and 3 as two independent bilateral problems, when determining policies. Specifically, import tariffs and export taxes are given by setting $\mathcal{N}_{n,j} = 0$ and applying the bilateral formulas (36)–(37), even though countries 2 and 3 trade with one another. Thus, the trade shares, prices, expenditures, tariffs, and export taxes are endogenous under the alternative policy formulas. This exercise nevertheless omits two multi-country objects: the network statistic $\mathcal{N}_{n,j}$ and the relative country-specific tariff component $\tau_2^m - \tau_3^m$.¹⁸

According to equation (36), the slope between tariffs and after-tax Home net import shares should be equal to one, as shown in Panel (b). Within each foreign country, the ranking of sectoral tariffs aligns with Home's after-tax net import share in that sector. Given the opposing comparative advantages of countries 2 and 3 relative to Home, the sectoral tariff rankings are largely reversed across the two countries when only bilateral relationships are considered. However, the optimal tariff rankings align with Home's overall sectoral net imports. Ignoring trade networks and interdependence leads to a smaller welfare gain: 2.51%, only 76% of the gain (3.31%) achieved with optimal policies (Table A-1, Case 4, Column 2).

¹⁸Tax neutrality permits one normalization in the joint multi-country problem. Treating each foreign country as an independent bilateral problem instead normalizes the country-specific component separately for each country, and different ways of normalizations generate different results. In this implementation we use $\tau_{n,j}^m = \kappa^{-1} M_{1n,j} / Y_{n,j}$. Although cross-sector tariff differences within each country remain consistent with (36), this alternative procedure cannot recover the economically relevant cross-country difference $\tau_2^m - \tau_3^m$.

Note that while the formulas (36) and (37) may not be optimal in this numerical example, the resulting trade shares, prices, and expenditures are still from the equilibrium induced by these formulas. The tariffs and export taxes are consistent with the equilibrium following the alternative formulas. Thus, the policies largely capture the endogenous direct effect $\frac{\partial E_{1n}}{\partial \ln p_n}$, the self-elasticities, and changes in imports. When prices change, they trigger sectoral reallocations and alter other foreign countries' imports from a specific country-sector, which in turn affects its export supply to the Home country. All these are incorporated into the equilibrium induced by the policies prescribed by (36) and (37).

The second alternative also jointly calculates import tariffs and export taxes from (36)–(37), but evaluates both formulas using pre-tax bilateral trade shares and net import shares. It therefore omits not only the network statistic and the relative country-specific tariff component, but also the endogenous adjustment of the sufficient statistics used to set policy. After these tariffs and export taxes are calculated, welfare is evaluated in the equilibrium induced by them. In the example, this procedure produces a 5.09% welfare loss rather than a gain.

Appendix H provides additional numerical examples illustrating how technology and trade costs shape tariff heterogeneity and convergence, as well as the welfare consequences of ignoring endogenous trade-network effects and GE effects.

4 Quantitative Analysis

In this section, we quantify the optimal policies and calculate the welfare changes associated with these policies in the multi-country case satisfying the GIIA Supply System with constant returns to scale, where $\eta_j = \kappa$.¹⁹ We study both unilateral and Nash optimal trade policies. We begin by extending the exact hat method, using optimal policy formulas and equilibrium conditions, to compute the counterfactual equilibrium under these optimal policies.

¹⁹The quantitative results are very similar when we additionally allow for increasing returns to scale, where $\frac{1}{\eta_j} = \frac{1}{\kappa} - \psi_j \frac{\kappa-1}{\kappa}$. We set $\psi_j = 0.21$ in all sectors, corresponding to an average estimated scale elasticity across manufacturing sectors in [Bartelme et al. \(2025\)](#).

4.1 Computing Welfare with Exact Hat Method

To compute the welfare changes after trade policies, we employ the exact hat method. Let variables without ‘prime’ denote the observed variables, which include the trade matrix $\{\pi_{ni,j}\}$ and sectoral income $\{w_{n,j}L_{n,j}\}$. Variables denoted with ‘prime’ represent counterfactuals after implementing the optimal policies, and variables with ‘hat’ denote the ratios of prime variables to the observed ones. We describe below how welfare changes are computed under unilateral and Nash optimal policies.

Given data on trade shares and sectoral income, $\{\pi_{ni,j}, w_{n,j}L_{n,j}\}$, along with parameter values $\{\epsilon_j, \kappa, \beta_{n,j}\}$, the counterfactual equilibrium is obtained by jointly solving the exact-hat versions of the trade-share equations, price indexes, expenditure equations, goods-market clearing conditions, and labor-market conditions. For Home’s unilateral optimal policy, we combine the counterfactual equilibrium conditions with Home’s optimal import-tariff and export-tax conditions, given by equations (29) and (11).²⁰ Appendix J.1 reports the complete system of equations for the unilateral-policy counterfactual.

We also consider Nash optimal policies where all countries are able to retaliate against the policies of other countries. Country i ’s policy is defined as $\{\tau_{ni,j}^x, \tau_{in,j}^m\}$, where $\tau_{ni,j}^x$ denotes the export tax imposed by country i on country n in sector j , and $\tau_{in,j}^m$ represents the import tariff imposed by country i on country n in sector j . Given the policies of other countries, country i ’s government chooses $\{\tau_{ni,j}^x, \tau_{in,j}^m\}$ to maximize domestic consumers’ consumption, $\max x_i/P_i$, subject to world market equilibrium and the policies of other countries. Appendix I derives the Nash optimal policies. Appendix J.2 lists the counterfactual equilibrium conditions under Nash optimal policies using the exact hat method and the formulas for optimal policies. Once the counterfactual equilibrium is obtained, the welfare change for each country is calculated as $\hat{C}_n = \frac{\hat{x}_n}{\hat{P}_n}$.

In summary, this procedure enables us to calculate optimal policies and equilibrium changes using bilateral trade and sector-level income data, without the need to explicitly recover technology $\bar{T}_{n,j}$, fundamental productivity $A_{n,j}$, or trade costs $d_{ni,j}$. We adapt the standard exact hat method but incorporate labor market specifications with imperfect labor

²⁰As we have proven in Appendix F.4, under the multi-country case we consider, a country implementing optimal policies will not impose any domestic taxes because the domestic market is efficient. Therefore, domestic taxes will not be included from now on.

substitution as in Ricardo-Roy, and our first-order conditions (FOCs) for optimal policies. With explicit FOCs for optimal policies, our method provides notable gains in accuracy and computational efficiency.²¹

4.2 Data and Measurement

To conduct the analysis, we need sector-level data on gross production and bilateral trade for each country. Data on domestic and international production and expenditure are from the World Input-Output Database (WIOD) 2016 Release, which covers 56 sectors, 43 countries, and an aggregate for the rest of the world for the period 2000-2014. The 56 sectors are classified according to International Standard Industrial Classification Revision 4 (ISIC Rev. 4). We utilize data from 2014. Given the absence of certain sector data for specific countries, we aggregate some of the 56 sectors and focus on $J = 13$ manufacturing sectors, similar to [Lashkaripour and Lugovskyy \(2023\)](#). See Table A-2 in the appendix for the list of aggregated sectors. We retain data for the top 19 countries by GDP in 2014.²² The 19 largest countries include the U.S., China, Japan, Germany, the United Kingdom, France, Brazil, Italy, the Russian Federation, India, Canada, the Republic of Korea, Australia, Spain, Mexico, Turkey, the Netherlands, Indonesia, and Switzerland. All other countries are aggregated into the rest of the world (ROW).

Two parameters are common across countries and sectors. We set the trade elasticity to $\epsilon = 4$, consistent with the estimation of [Simonovska and Waugh \(2014\)](#), and the GIIA parameter $\kappa = 1.5$ following [Galle, Rodríguez-Clare, and Yi \(2023\)](#). The sector-level expenditure shares for each country n are based on the following calculations:

$$\beta_{n,j} = \frac{\sum_{i=1}^N X_{ni,j}}{\sum_{k=1}^J \sum_{i=1}^N X_{ni,k}},$$

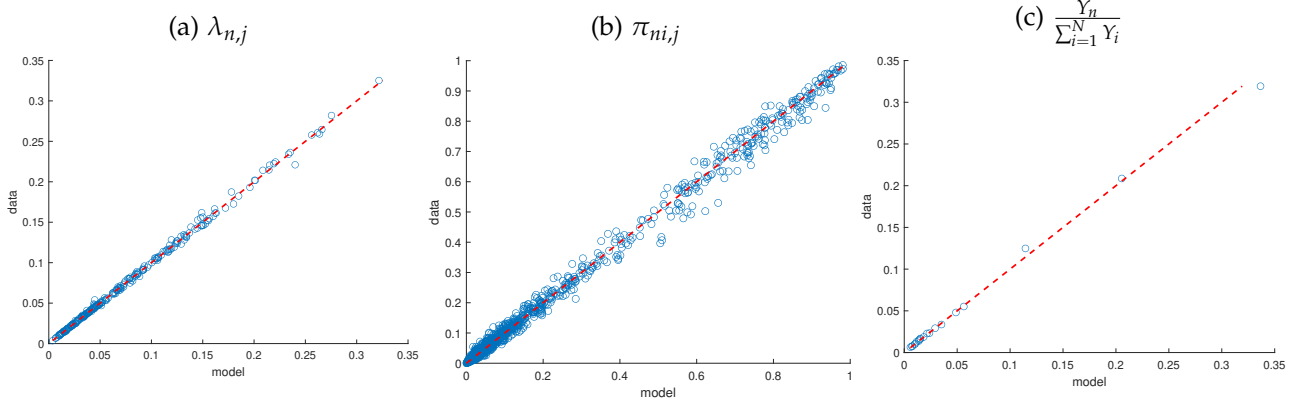
where $X_{ni,j}$ is the trade flow from country i to country n in sector j .

To ensure that the data satisfy the world market equilibrium conditions described in

²¹As noted by [Lashkaripour and Lugovskyy \(2023\)](#), without FOCs, we would have to rely on numerical optimization, which can become increasingly difficult and unstable to implement when dealing with many policy variables with multiple sectors and countries.

²²Nominal GDP data are from the World Bank's World Development Indicators (WDI) database. Saudi Arabia is excluded due to the lack of data in the WIOD database.

Figure 2: Model and Data Comparison



Note: This figure shows the relationship between the data and the model's income shares $\lambda_{n,j}$, trade shares $\pi_{ni,j}$, and the country's share of global income $\frac{Y_n}{\sum_{i=1}^N Y_i}$. The red dashed line represents the 45-degree line.

equations (1)-(10), we first apply the hat method under the assumption of zero export taxes and import tariffs, and balanced trade²³, and then update the trade shares and sectoral income values to $\{\pi'_{ni,j}, w'_{n,j} L'_{n,j}\}$ based on the system of equations detailed in the Appendix J.1. Figure 2 compares the model with data in three endogenous variables: the share of sector j income in country n , denoted by $\lambda_{n,j}$, the trade share from country i to country n in sector j , $\pi_{ni,j}$, and the share of country n 's income in global income $Y_n / \sum_{i=1}^N Y_i$. The model captures the data well, with all the dots aligning around the 45-degree line.

4.3 Unilateral Optimal Policies of the U.S.

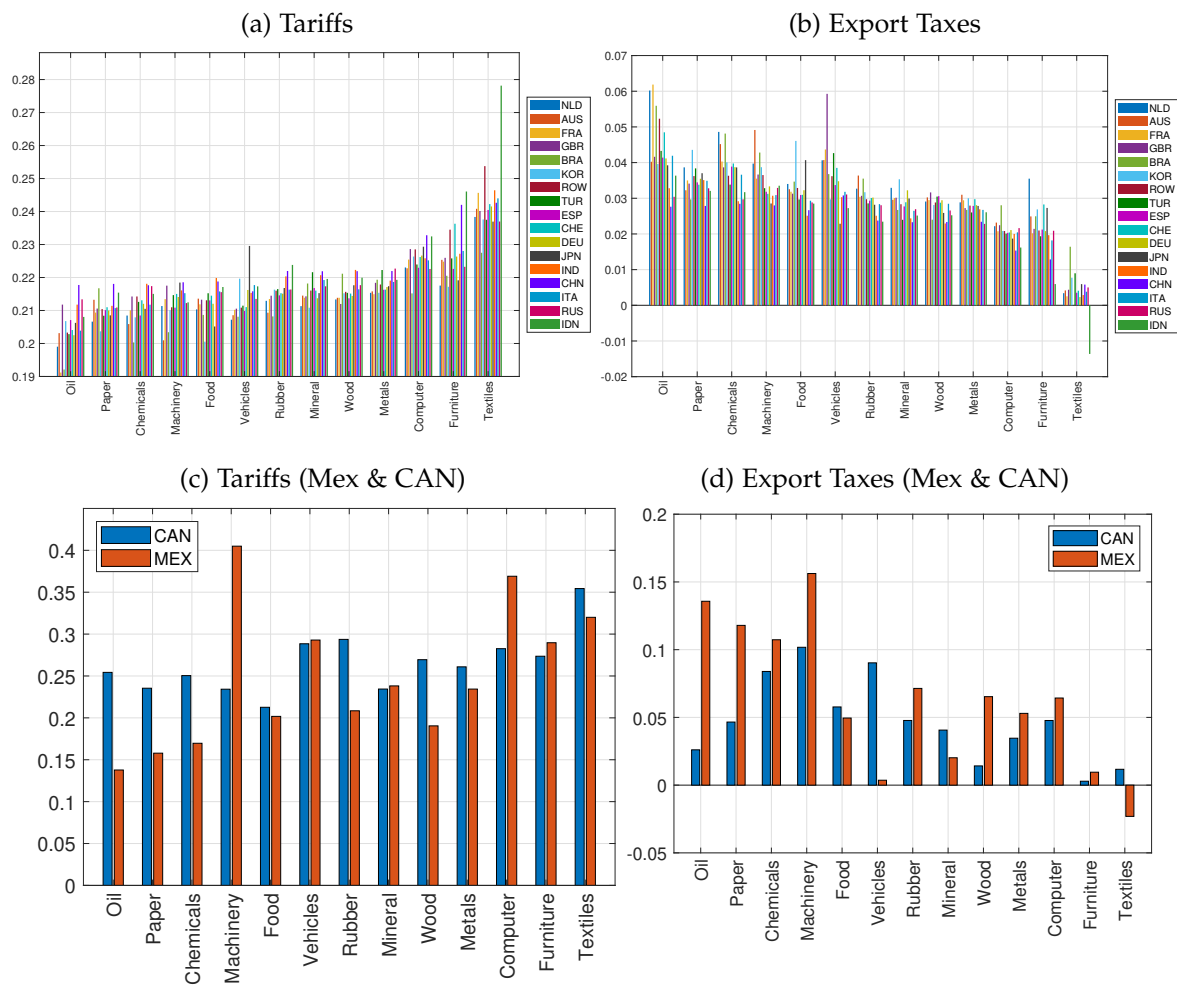
Figure 3 plots the U.S. unilateral optimal policies.²⁴ We rank sectors in ascending order based on the U.S. sectoral net imports relative to world sectoral income before policy implementation. For example, relative to other sectors, the U.S. oil sector has the highest share of net exports, whereas the U.S. textiles sector is imported on net. Following the tax neutrality principle established in Lemma 1, we normalize the U.S. export tax on textiles shipped to ROW to zero.

²³Our model features an equilibrium with balanced trade. In the data, trade is not balanced. This implies that the welfare gains computed include both the welfare changes from eliminating trade imbalances and those from implementing optimal policies. To separate them, we first eliminate the imbalances. We then take the new equilibrium to be our private equilibrium observations, which are used to calculate the optimal policies and welfare changes. The results show additional changes brought about by optimal policies.

²⁴Optimal tariffs and export taxes on Canada and Mexico are separately shown because their means and standard deviations are much higher than those of other countries.

The first feature is that export taxes and import tariffs are heterogeneous across sectors and countries. Second, tariffs imposed by the U.S. on other countries are generally higher in its importing sectors, while export taxes are higher in U.S. exporting sectors. As shown in columns (1) and (3) of Table 1, on average, U.S. optimal import tariffs are lowest in the oil sector, at 20.41%, and highest in the textiles sector, at 25.29%, with an average rate of 22.17%. The oil sector faces the highest export tax at 4.72%, while the textiles sector has the lowest export tax at 0.29%. The average export tax rate across all sectors is 3.19%.

Figure 3: U.S. Unilateral Optimal Policies



Note: This figure plots U.S. unilateral optimal policies. The sectors are ranked in ascending order based on the U.S. sectoral net imports relative to world sectoral income before the policy. Countries are ranked in ascending order based on the U.S. total net import share relative to each country's total income before the policy. Optimal tariffs and export taxes on Canada and Mexico are separately shown because their means and standard deviations are much higher than those of other countries.

Table 1: U.S. Policies Across Sectors: Optimal and Alternative

Sectors	Optimal				Endogenous Bilateral Share			
	Tariff (%)		Export tax (%)		Tariff (%)		Export tax (%)	
	Mean	Std	Mean	Std	Mean	Std	Mean	Std
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Oil	20.41	2.01	4.72	2.30	21.03	4.81	4.37	5.20
Paper	20.94	1.37	3.98	1.89	21.72	4.50	3.49	5.00
Chemicals	21.15	1.38	4.39	1.90	21.53	4.28	4.21	4.87
Machinery	22.39	4.32	4.48	3.06	22.44	3.50	3.84	4.65
Food	21.23	0.50	3.44	0.81	21.98	3.94	2.92	3.83
Vehicles	22.18	2.42	3.71	1.63	22.34	3.12	3.61	3.33
Rubber	21.95	1.79	3.27	1.05	22.60	3.45	2.77	3.75
Mineral	21.86	0.68	2.86	0.45	22.51	3.28	2.41	3.14
Wood	21.79	1.38	2.91	0.94	22.53	4.00	2.45	3.81
Metals	22.12	1.03	2.92	0.62	22.97	3.20	2.26	3.28
Computer	23.63	3.40	2.44	1.14	23.50	3.64	2.32	2.89
Furniture	23.29	1.85	2.01	0.77	23.53	1.75	1.94	1.64
Textiles	25.29	3.10	0.29	0.83	25.33	0.81	0.00	0.00
Avg.	22.17	1.94	3.19	1.34	22.62	3.41	2.81	3.49

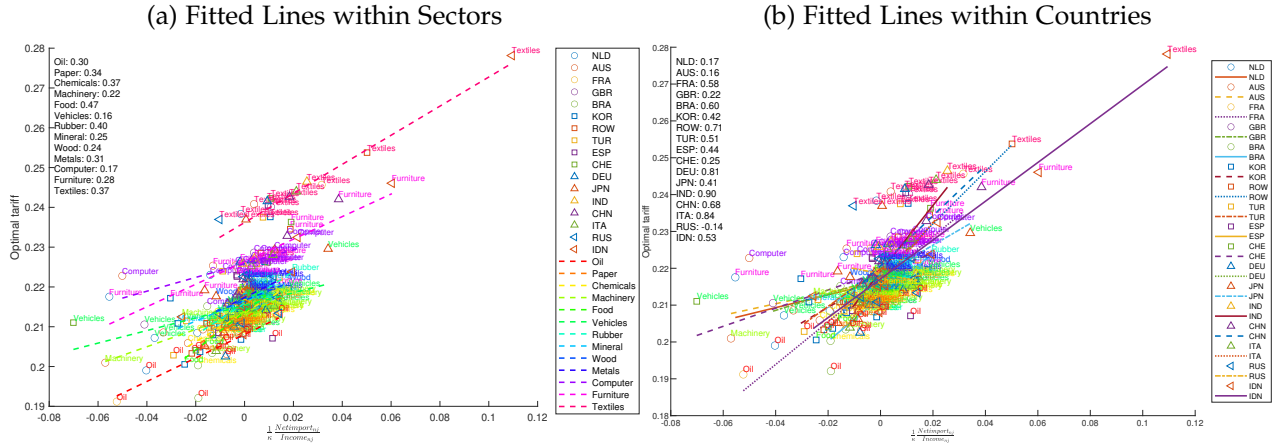
Note: This table summarizes the U.S. policies (mean and standard deviation) for 13 sectors under two scenarios: unilateral optimal policies and alternative policies using formulas (36) and (37) with endogenous bilateral trade share. The sectors are ranked in ascending order based on the U.S. sectoral net imports relative to world sectoral income before the policy. The last row represents the average of the entire column.

Table 2: U.S. Optimal Policies Across Countries: Optimal and Alternative

Cty	Optimal					Endogenous Bilateral Share				
	Tariff (%)			Export tax (%)		Tariff (%)			Export tax (%)	
	Mean (1)	Std (2)	Slope (3)	Mean (4)	Std (5)	Mean (6)	Std (7)	Slope (8)	Mean (9)	Std (10)
NLD	21.35	0.91	0.17	3.43	1.28	23.56	1.55	1.00	1.98	1.90
AUS	21.44	1.01	0.16	3.23	1.09	23.46	1.89	1.00	1.84	1.97
FRA	21.52	1.18	0.58	3.16	1.32	21.68	1.61	1.00	3.23	1.81
GBR	21.78	0.84	0.22	3.15	1.21	23.99	1.13	1.00	1.44	1.61
BRA	21.23	0.95	0.60	3.31	1.04	24.04	1.32	1.00	1.13	1.38
KOR	21.44	0.94	0.42	3.22	0.99	23.11	1.36	1.00	2.03	1.28
ROW	21.89	1.26	0.71	2.98	1.16	20.32	1.67	1.00	4.34	1.59
TUR	21.75	0.84	0.51	2.91	0.93	23.70	1.22	1.00	1.41	1.28
ESP	21.59	0.85	0.44	2.86	0.92	24.05	0.52	1.00	0.97	0.54
CHE	21.82	1.02	0.25	3.02	1.00	23.62	1.80	1.00	1.57	1.72
DEU	21.71	0.93	0.81	2.92	0.96	24.09	0.55	1.00	1.07	0.50
JPN	21.78	0.91	0.41	2.85	0.91	24.79	1.21	1.00	0.52	1.02
IND	22.11	0.86	0.90	2.50	0.78	22.70	0.78	1.00	2.02	0.72
CHN	22.40	0.88	0.68	2.29	0.70	23.63	1.09	1.00	1.24	0.85
ITA	21.87	0.93	0.84	2.75	0.91	23.26	0.87	1.00	1.62	0.80
CAN	26.50	3.46	0.32	4.66	2.94	21.46	3.77	1.00	7.64	4.40
RUS	21.84	0.66	-0.14	2.60	0.71	26.08	0.72	1.00	-0.77	0.61
IDN	22.50	1.79	0.53	2.29	1.30	15.29	3.41	1.00	8.62	3.09
MEX	24.73	7.94	0.67	6.39	5.18	16.88	9.25	1.00	11.55	7.70
Avg.	22.17	1.48	0.48	3.19	1.33	22.62	1.88	1.00	2.81	1.83

Note: This table summarizes the U.S. import tariffs (mean, standard deviation, and tariff-import slope) and export taxes (mean and standard deviation) for the other 19 countries under two scenarios: unilateral optimal policies and alternative policies using formulas (36) and (37) with endogenous bilateral trade share. The column "Slope" represents tariff-import slope, which is the regression coefficient of U.S. optimal tariffs on each country's sectors with respect to the after-tax U.S. net import share relative to that country-sector income in different sectors, normalized by κ . Countries are ranked in ascending order based on the U.S. total net import share relative to each country's total income before the policy. The last row represents the average of the entire column.

Figure 4: U.S. Unilateral Optimal Tariffs and Net Import Share



Note: This figure shows the relationship between U.S. unilateral optimal tariffs and U.S. net import share, normalized by κ . The net import share is defined as after-tax U.S. net imports from a certain country-sector over the foreign country-sector income. Panel (a) plots separate fitted regression lines for each sector, and panel (b) presents separate fitted regression lines for each country. Canada and Mexico are excluded from the figure because their means and standard deviations are much higher than those of other countries.

Third, the average import tariffs vary substantially across countries, as demonstrated in column (1) of Table 2, where countries are ranked in ascending order based on the U.S. total net import share relative to each country's total income before the policy. The U.S. imposes the highest import tariffs on Canada (26.50%) and Mexico (24.73%), while the lowest tariffs are levied on Brazil (21.23%) and the Netherlands (21.35%), with an average rate of 22.17%, as shown in column (1). The export taxes across countries have a mean of 3.19% and, for countries other than Mexico and Canada, range from 2.29% to 3.43%. The export taxes on Mexico and Canada are 6.39% and 4.66%, respectively, as shown in column (4).

Fourth, optimal tariffs depend not only on U.S. market power over individual country-sector pairs but also on overall U.S. market power within a sector, reflecting the interdependency among countries. Within sectors, there is dispersion in tariffs across countries, as illustrated by the scatter points in Figure 4.²⁵ Columns (2) and (4) of Table 1 quantify this variation by reporting the standard deviations of the trade policies within each sector. Consider the textiles sector, where import tariffs range from 22.7% for Brazil to 35.4% for

²⁵Figure 4 excludes Canada and Mexico because optimal tariff values on these two countries are substantially larger than those for other countries, as shown in Figure 3(c) and (d). This reflects their large trade volumes and correspondingly low trade costs with the U.S., which gives the U.S. significantly greater market power in these two countries. Including Canada and Mexico in Figure 4 would compress the figure's scale and obscure tariff patterns in other trading partners.

Canada, yielding a standard deviation of 3.10%. The U.S. tends to impose higher tariffs on countries where it has a larger net import share relative to their income. This relationship explains why optimal U.S. tariffs are higher on Indonesia in the textiles sector and on Japan in vehicles, compared to other countries within those respective sectors.

However, in addition to U.S. market power over individual country-sector pairs, interdependence among countries also shapes the tariffs imposed. Figure 4(a) plots the fitted regression line of tariffs against bilateral after-tax U.S. net import shares relative to each country's income for each sector (normalized by κ). We refer to the slope of this relationship as the *tariff-import slope*. The textiles line has the highest intercept, reflecting that import tariffs on textiles exceed those of other sectors across most countries. Furthermore, all sector-specific slopes are below one, ranging from 0.16 for vehicles to 0.47 for food. Thus, cross-country tariff differences within a sector respond less than one-for-one to differences in bilateral net import shares. This compression relative to the unit-slope bilateral benchmark is consistent with the within-sector convergence mechanism in Lemma 3. Column (3) of Table 2 presents, for each country, the regression coefficients of optimal sectoral tariffs with respect to the bilateral after-tax U.S. net import share relative to that country's income. These coefficients are consistently below 1, averaging 0.48. Figure 4(b) illustrates how these regression slopes differ across countries while remaining below 1.

As explained in the theoretical results in Section 3.3.1, the tariff-import slope in optimal policies deviates from 1 due to cross-country interdependency. Even though the U.S. does not import textiles from Russia, it imposes higher tariffs because the cross-country trade network means that relative wage changes affect demand and supply. Without tariffs on Russia's textiles, the U.S. would import relatively more from Russia, and relative wage changes would increase Russia's imports from Indonesia and China, further reducing their export supplies to the U.S. Therefore, optimal tariffs within a sector tend to converge across countries due to these interdependencies.

Alternative policies. We now consider a scenario where the U.S. does not account for interdependence across countries. The import tariff and export tax are determined by equations (36) and (37), but the trade shares and prices are endogenously determined

in equilibrium given these trade policy formulas. Columns (5)-(8) of Table 1 show the cross-sector differences under the U.S. “Endogenous Bilateral Share” policies. Compared to the “Optimal” case, both tariffs and export taxes are more dispersed, with the standard deviation in most sectors higher than that of the “Optimal” case.

Columns (6)-(10) of Table 2 illustrate the cross-country differences under these alternative U.S. policies. In this case, the regression coefficients of U.S. tariffs on each country’s sectors with respect to the U.S. after-tax net import share relative to that country-sector income (normalized by κ) in different sectors are 1, which is shown in column (8). These columns exhibit significantly different patterns compared to the “Optimal”: optimal policies depend less on bilateral imports and have smaller standard deviations on average, again highlighting the impact of the trade network among countries on optimal policies.

Welfare. The first column of Table 5 presents the welfare changes resulting from the unilateral optimal policies implemented by the U.S. The results indicate that the U.S. experiences welfare gains of approximately 1.3%, while other countries suffer welfare losses, as the U.S.’s unilateral policies improve its own terms of trade while worsening those of other countries. The U.S. welfare gains under optimal policies are larger than those under the “Endogenous Bilateral Share” and “Observed Bilateral Share” scenarios, as shown in columns (2) and (3).

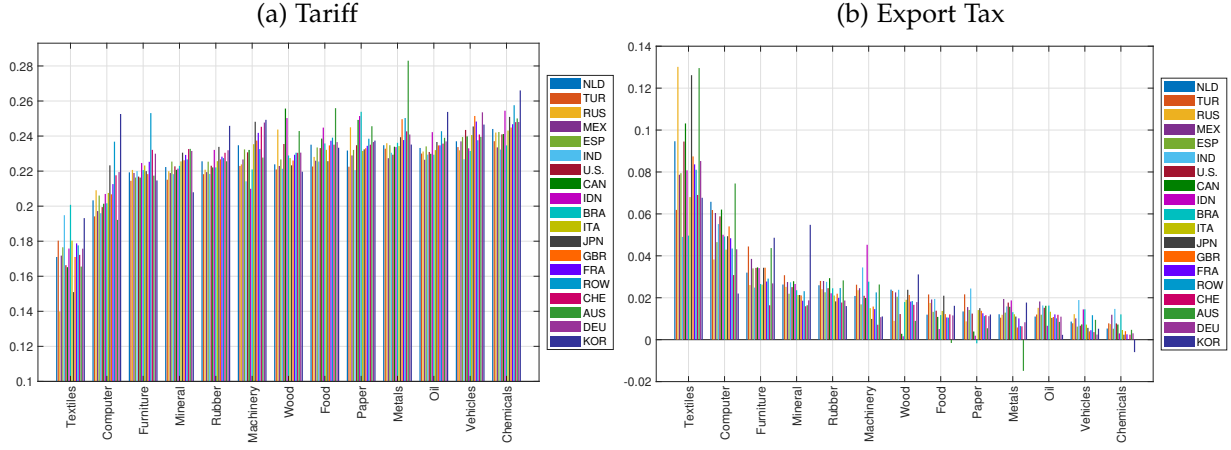
4.4 Unilateral Optimal Policies of China

Now we turn to the case when China imposes unilateral policies. Figure 5 shows China’s optimal policies across countries and sectors, with sectors ranked in ascending order based on China’s sectoral net imports over world sectoral income. China’s export tax on chemicals shipped to ROW is normalized to zero. As with our U.S. analysis, we summarize these results in two ways: Table 3 reports average trade policies by sector, while Table 4 shows average policies by country, along with the tariff-import slopes. Figure 6 illustrates these tariff-import relationships and interdependency through scatter plots of optimal tariffs against China’s net import shares.²⁶ Panel (a) displays sector-specific fitted lines, and

²⁶Russia is excluded from Figure 6 because China’s net import share from it in the textiles sector (−0.76) is an extreme outlier compared to other countries. See Figure 5 for tariffs on Russia.

panel (b) shows country-specific fitted lines.

Figure 5: China's Unilateral Optimal Policies



Note: This figure plots China's unilateral optimal policies. The sectors are ranked in ascending order based on China's sectoral net imports relative to world sectoral income before the policy. Countries are ranked in ascending order based on China's total net import share relative to each country's total income before the policy.

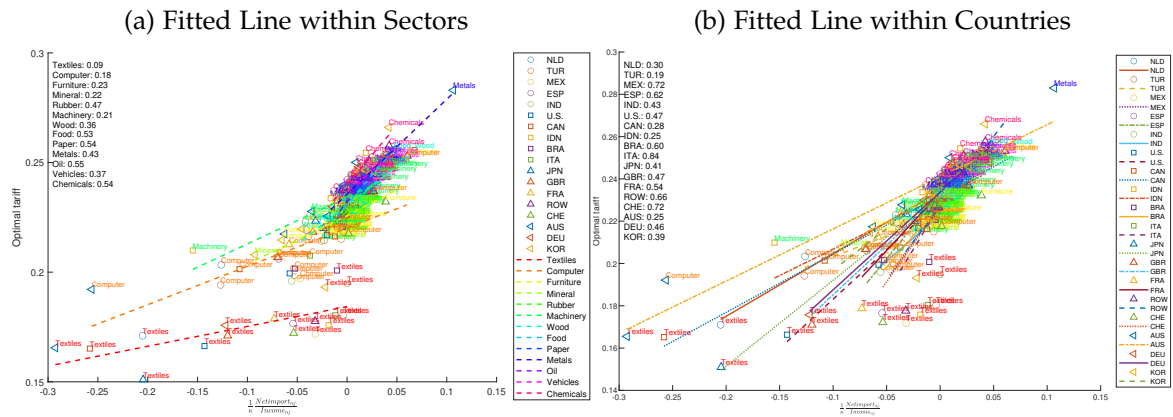
China's optimal trade policies follow similar principles to those in the U.S. First, both optimal export taxes and import tariffs are heterogeneous across sectors and countries. Second, China tends to impose higher tariffs on sectors where it is a net importer and higher export taxes on sectors where it is a net exporter, as shown in columns (1) and (3) of Table 3. Across sectors, China's optimal import tariffs range from 17.41% in textiles (where China is a major exporter) to 24.51% in chemicals (where China is a net importer), with an average of 22.69%. Export taxes show the reverse pattern, reaching 8.53% in textiles while falling to 0.53% in chemicals, with an average of 2.43%.

Third, cross-country interdependencies shape these policies greatly. Figure 6(a) is consistent with the convergence mechanism within sectors. All sector-specific slopes are below one, ranging from 0.09 for Textiles to 0.55 for Oil. Column (3) of Table 4 and Figure 6(b) show that tariff-import slopes are consistently smaller than 1, indicating that optimal trade policies toward individual countries reflect broader sectoral considerations rather than purely bilateral trade flows. For example, China imposes relatively high tariffs on chemical imports from Mexico despite its small imports from Mexico in this sector. China also imposes lower tariffs on textiles for most countries.

Alternative policies. Similarly, we consider a scenario where China does not account for interdependence across countries. The import tariff and export tax are determined by equations (36) and (37). Columns (5)-(8) of Table 3 illustrate the cross-sector differences under the policies implemented by China, “Endogenous Bilateral Share”. Compared with the “Optimal” case, most sectors exhibit a significantly higher standard deviation, reflecting less similarity in tariffs and export taxes across countries. Columns (6)-(10) of Table 4 illustrate the cross-country differences under the policies implemented by China, “Endogenous Bilateral Share”. In this case, the tariff-import slopes are 1 by construction. These columns exhibit significantly different patterns compared to the optimal case. The standard deviations of import tariffs and export taxes in most countries are larger than those in the optimal case, highlighting the impact of trade networks among countries on optimal policies.

Column (4) of Table 5 illustrates the welfare changes resulting from China’s unilateral optimal policies. China benefits from welfare gains, while most other countries experience welfare losses. China’s welfare gains are greater than those under the “Endogenous Bilateral Share” and “Observed Bilateral Share” scenarios, as shown in columns (5) and (6).

Figure 6: China’s Unilateral Optimal Tariffs and Net Import Share



Note: This figure shows the relationship between China’s unilateral optimal tariffs and China’s net import share, normalized by κ . The net import share is defined as China’s after-tax net-imports from a certain country-sector over the foreign country-sector income. Panel (a) plots separate fitted regression lines for each sector, and panel (b) presents separate fitted regression lines for each country.

Table 3: China's Optimal Policies: Optimal and Alternative

Sectors	Optimal				Endogenous Bilateral Share			
	Tariff (%)		Export tax (%)		Tariff (%)		Export tax (%)	
	Mean	Std	Mean	Std	Mean	Std	Mean	Std
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Textiles	17.41	1.37	8.53	2.32	15.58	8.18	10.97	9.48
Computer	20.97	1.47	5.04	1.22	19.70	3.73	6.77	3.87
Furniture	22.22	0.86	3.25	0.75	23.07	2.45	2.86	2.39
Mineral	22.37	0.62	2.49	0.81	23.81	2.41	1.38	2.52
Rubber	22.65	0.64	2.33	0.39	24.12	1.12	1.16	1.05
Machinery	23.23	1.09	2.07	0.90	23.84	3.10	1.84	3.02
Wood	23.10	0.99	1.75	0.74	24.52	2.86	0.73	2.48
Food	23.44	0.74	1.32	0.54	25.07	1.85	0.04	1.55
Paper	23.66	0.88	1.17	0.60	25.26	1.95	-0.07	1.55
Metals	23.91	1.19	1.11	0.73	25.17	2.51	0.23	1.96
Oil	23.49	0.62	1.21	0.37	24.86	0.91	0.17	0.74
Vehicles	23.99	0.69	0.84	0.42	25.19	1.72	0.05	1.43
Chemicals	24.51	0.82	0.53	0.45	25.37	0.19	0.00	0.00
Avg.	22.69	0.92	2.43	0.79	23.50	2.54	2.01	2.46

Note: This table summarizes China's policies (mean and standard deviation) for 13 sectors under two scenarios: unilateral optimal policies and alternative policies using formulas (36) and (37) with endogenous bilateral trade share. The sectors are ranked in ascending order based on China's sectoral net imports relative to world sectoral income before the policy. The last row represents the average of the entire column.

Table 4: China's Optimal Policies: Optimal and Alternative

Cty	Optimal					Endogenous Bilateral Share				
	Tariff (%)			Export tax (%)		Tariff (%)			Export tax (%)	
	Mean (1)	Std (2)	Slope (3)	Mean (4)	Std (5)	Mean (6)	Std (7)	Slope (8)	Mean (9)	Std (10)
NLD	22.43	1.83	0.30	2.71	2.46	22.37	4.51	1.00	3.26	5.03
TUR	21.88	1.53	0.19	2.75	1.77	25.16	2.74	1.00	0.69	3.14
RUS	22.25	2.59	0.12	2.69	3.10	22.38	8.70	1.00	3.35	10.24
MEX	21.96	1.66	0.72	2.88	1.91	24.39	1.58	1.00	1.20	2.08
ESP	22.50	1.68	0.62	2.35	1.94	23.85	2.23	1.00	1.48	2.50
IND	21.91	1.13	0.43	2.71	1.20	25.32	1.53	1.00	0.14	1.68
U.S.	22.36	1.98	0.47	2.65	2.37	23.20	3.59	1.00	2.16	3.95
CAN	22.66	2.25	0.28	2.62	2.71	23.16	5.71	1.00	2.45	6.04
IDN	22.93	2.12	0.25	2.39	2.27	23.36	3.81	1.00	2.32	3.95
BRA	22.61	1.36	0.60	2.19	1.39	25.41	1.87	1.00	0.04	1.86
ITA	22.61	1.57	0.84	2.14	1.63	24.58	1.21	1.00	0.70	1.35
JPN	22.77	2.41	0.41	2.68	3.11	22.41	4.45	1.00	3.22	5.20
GBR	22.78	2.02	0.47	2.36	2.26	23.56	3.70	1.00	1.90	3.92
FRA	22.96	1.72	0.54	2.18	2.10	23.36	2.68	1.00	1.96	3.01
ROW	23.47	1.88	0.66	2.25	2.01	22.40	2.22	1.00	3.01	2.02
CHE	23.06	1.84	0.72	1.71	1.66	23.80	2.09	1.00	1.18	1.90
AUS	23.21	2.80	0.25	2.61	3.69	22.66	7.32	1.00	3.47	8.03
DEU	23.22	1.85	0.46	2.07	2.13	23.86	3.38	1.00	1.59	3.54
KOR	23.50	2.00	0.39	2.30	2.09	21.36	3.50	1.00	4.07	3.47
Avg.	22.69	1.91	0.46	2.43	2.20	23.50	3.52	1.00	2.01	3.84

Note: This table summarizes China's tariffs (mean, standard deviation, and tariff-import slope) and export taxes (mean and standard deviation) for the other 19 countries under two scenarios: unilateral optimal policies and alternative policies using formulas (36) and (37) with endogenous bilateral trade share. The column "Slope" represents tariff-import slope, which is the regression coefficient of China's optimal tariffs on each country's sectors with respect to China's after-tax net-import share relative to that country-sector income in different sectors, normalized by κ . Countries are ranked in ascending order based on China's total net import share relative to each country's total income before the policy. The last row represents the average of the entire column.

Table 5: Welfare Changes

	U.S. Unilateral Policy (%)			China's Unilateral Policy (%)			Nash Policy (%)
	Optimal	Endogenous Bilateral Share	Observed Bilateral Share	Optimal	Endogenous Bilateral Share	Observed Bilateral Share	Optimal
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
U.S.	1.305	1.252	1.051	-0.388	-0.414	-0.523	-0.905
NLD	-0.381	-0.359	-0.452	-0.737	-0.858	-1.197	-8.061
AUS	-0.286	-0.308	-0.378	-1.400	-1.731	-2.086	-4.185
FRA	-0.266	-0.281	-0.288	-0.340	-0.360	-0.418	-4.659
GBR	-0.520	-0.532	-0.577	-0.504	-0.558	-0.709	-3.310
BRA	-0.214	-0.207	-0.223	-0.197	-0.196	-0.221	-0.981
KOR	-0.253	-0.250	-0.265	-0.849	-0.863	-0.935	-1.958
ROW	-0.260	-0.272	-0.274	-0.738	-0.714	-0.709	-1.299
TUR	-0.165	-0.156	-0.178	0.045	0.089	0.103	-3.799
ESP	-0.052	-0.044	-0.035	-0.204	-0.202	-0.252	-2.228
CHE	-0.272	-0.277	-0.283	-0.148	-0.152	-0.162	-3.591
DEU	-0.213	-0.216	-0.191	-0.447	-0.485	-0.543	-3.277
JPN	-0.237	-0.246	-0.247	-0.677	-0.716	-0.852	-1.546
IND	-0.078	-0.077	-0.076	-0.117	-0.097	-0.089	-0.709
CHN	-0.077	-0.078	-0.076	0.566	0.553	0.433	-0.313
ITA	-0.126	-0.128	-0.119	-0.070	-0.048	-0.001	-1.737
CAN	-4.858	-4.733	-5.262	-0.555	-0.637	-0.844	-5.578
RUS	-0.020	-0.030	-0.029	-0.834	-1.175	-1.567	-2.931
IDN	-0.167	-0.132	-0.222	-0.361	-0.400	-0.399	-2.395
MEX	-4.829	-4.473	-5.279	-0.150	-0.131	-0.107	-5.442

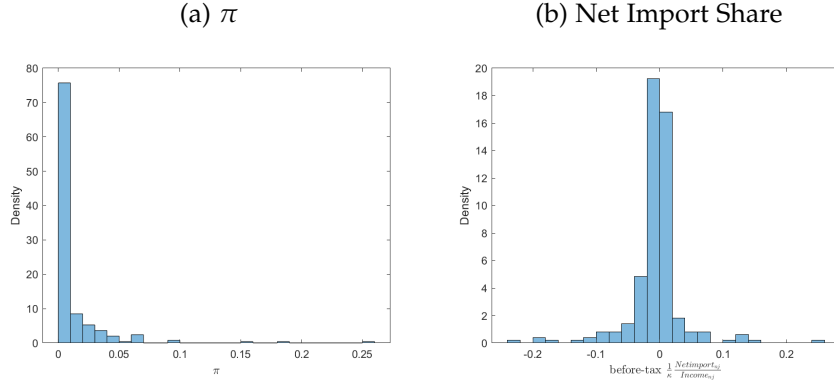
Note: The welfare change is calculated as the percentage change relative to real consumption (x_n/P_n) before the policy. Countries except the U.S. are ranked in ascending order based on the U.S. total net import share relative to each country's total income before the policy. The welfare change for each country under the unilateral policies of the U.S. and China is evaluated across three scenarios: optimal policies, alternative policies using formulas (36) and (37) with endogenous bilateral trade shares, and alternative policies using formulas (36) and (37) with pre-tax observed bilateral trade shares and net import shares. The last column presents each country's welfare change under the optimal Nash policy.

4.5 Discussions and Nash Policies

Welfare discussion The quantitative exercise illustrates that, given the underlying fundamentals, optimal policies are positively related to Home's overall net imports, with convergence across countries. The welfare change is not substantial, which is typical, as the data suggest high trade costs. In addition, the welfare differences arising from using the bilateral (endogenous) trade share are small. In our theoretical and numerical examples, we demonstrate that these welfare differences can vary significantly across different scenarios. Using our formula with pre-tax observed bilateral trade shares and net import shares results in a much larger gap.

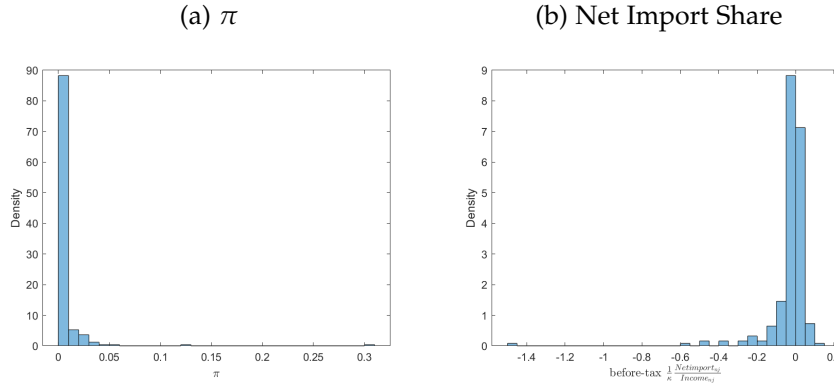
For the two-digit sectors and the 20 countries we analyzed, the following observations contribute to the small differences: The level and standard deviation of the U.S. or China's import expenditure share across country-sector pairs are very small (see Figure 7(a) and 8(a)), and the pre-tax net import share also exhibits very small standard deviations (see Figure 7(b) and 8(b)). This indicates high trade costs and limited heterogeneity across two-digit sectors and foreign countries. Although the optimal tariff-import slope is not equal to 1 for all countries, optimal tariffs remain positively correlated with net import shares for most countries. This suggests that foreign countries have similar comparative advantages relative to the U.S., indicating limited heterogeneity across foreign countries for the U.S. Hence, using our formulas with bilateral endogenous trade shares, which partly account for the endogenous self-elasticities, largely captures the gains. Using our formula with pre-tax observed bilateral trade shares and net import shares results in a larger gap: the welfare change is about 81% of the optimal for the U.S. and 77% for China, as these policies overlook the endogenous changes in export supplies in response to the policies.

Figure 7: Distribution of U.S. Trade and Bilateral Net Import Share



Note: Panel (a) shows the distribution of U.S. before-tax import share for all country-sector pairs. Panel (b) shows the distribution of U.S. before-tax net import share in foreign income, normalized by κ , for all country-sector pairs.

Figure 8: Distribution of China's Trade and Bilateral Net Import Share



Note: Panel (a) shows the distribution of China's before-tax import share for all country-sector pairs. Panel (b) shows the distribution of China's before-tax net import share in foreign income, normalized by κ , for all country-sector pairs.

Nash Optimal Policies We analyze the scenario in which the governments of all countries implement policies that result in a Nash equilibrium. In Appendix K, Tables A-3–A-6 show the U.S. and China's optimal policies across countries and sectors under the Nash equilibrium.²⁷

As in the unilateral-policy case, Nash tariffs and export taxes are heterogeneous across sectors and countries. The average Nash tariff and export tax are 22.15% and 2.98% for the U.S., and 22.47% and 2.55% for China. U.S. Nash import tariffs are, on average, lowest in

²⁷To compare with the unilateral optimal policies, we use the same normalizations as unilateral when the U.S. and China implement optimal policies in the Nash equilibrium.

the oil sector and highest in the textiles sector, preserving the cross-sector pattern observed under unilateral policies.

In general, Nash policies are much more complicated due to foreign countries' endogenous market power and the interdependencies in their foreign trading partners. In a two-country world with a sectoral comparative advantage, trade will be significantly reduced by Nash policies. As differential market power across sectors diminishes, the standard deviation of Home's tariffs across sectors would also decrease, compared with unilateral policies. However, with multiple countries, the third-country policies would affect trade between Home and a foreign country. This may lead to an increase in the dispersion of imports across foreign sectors, meaning that the standard deviation of tariffs across sectors for a foreign country may not necessarily decrease, compared with unilateral policies.

The welfare outcomes under Nash optimal policies are presented in the last column of Table 5. Although unilateral policies yield higher welfare gains for Home, countries in a Nash equilibrium face retaliation, resulting in lower welfare for all countries. Countries that face significant additional reductions in welfare, for example, European countries, are those countries that trade extensively with countries other than the U.S. and China.

5 Conclusion

In this paper, we provide a comprehensive understanding of optimal trade policies within a general equilibrium context. We derive explicit formulas for a unified setup incorporating general supply-side specifications that not only enhance theoretical clarity but also streamline the calculation of optimal taxes. The explicit formula for a two-country model with the GIIA Supply System establishes a direct link to market power, offering novel insights into optimal policies in general equilibrium.

We also examine specifications beyond the GIIA Supply System framework. Our analysis reveals how sectoral interdependencies and trade network effects across countries influence endogenous foreign (self- and cross-) export supply elasticities and shape optimal policies. We quantify these impacts for unilateral policies by the U.S. and China within a 20-country framework, along with the Nash optimal policies.

In addition to assumptions on production technology, other factors can also influence trade and optimal trade policies. These factors include preferences, market structure, and political and strategic considerations. While an extensive literature has examined some of these factors, our paper specifically concentrates on the implications of supply-side assumptions and the general equilibrium interdependencies across sectors and countries for trade policies. Using a unified framework, we aim to provide a clear and explicit understanding of how different model specifications matter in optimal taxation.

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ONLINE APPENDIX TO “OPTIMAL TRADE POLICIES AND MARKET POWER IN GENERAL EQUILIBRIUM TRADE MODELS”

BY YAN BAI, DAN LU, AND HANXI WANG

This appendix is organized as follows.

- A. Proof for Lemma 1: Tax Neutrality
- B. Proof for Proposition 2 and 3: General Formula for Two-Country Optimal Policies and Optimal Policies under GIIA Supply System
- C. Optimal Policies under GIIA Supply System: Examples
 - C.1. One-factor Ricardo-Roy Model
 - C.2. Generalized Multi-sector Krugman Model
 - C.3. Differential Scale Elasticities and Imperfectly Mobile Labor
 - C.4. Proof for Lemma 2: CES Labor Market
- D. Non-GIIA Supply System: Multi-factor Ricardo-Roy Model
- E. Proof for Proposition 1 and 4: General Formula for Multi-Country Optimal Policies
- F. Proof for Proposition 5: Multi-Country Optimal Tariffs under GIIA Supply System
- G. Proof for Lemma 3: Convergence & Divergence
- H. Numerical Examples: Heterogeneity and Convergence in Optimal Policies
- I. Nash Optimal Policies
- J. Quantifying the Consequences of Optimal Policies
- K. Data and Additional Quantitative Results
- L. Alternative Proof for Proposition 2 and 3

A Proof for Lemma 1: Tax Neutrality

This section presents the Home government’s problem and proves tax neutrality.

Home's problem Let $\tau_{n,j}^x$ denote the export tax imposed by country 1 (Home) on country n in sector j . Similarly, $\tau_{n,j}^m$ represents the import tariff imposed by Home on country n in sector j , and τ_j^d represents the domestic tax imposed by Home in sector j . Home government chooses $\{\tau_{n,j}^x, \tau_{n,j}^m, \tau_j^d\}$ for $n \neq 1$ to solve the following problem:

$$\max \frac{x_1}{P_1},$$

subject to world market equilibrium characterized by the following constraints:

The labor market specifications $\Omega(\{w_{n,j}, L_{n,j}\})$ hold.

The supply-side assumptions $\mathcal{S}(\{p_{n,j}, w_{n,j}, L_{n,j}\})$ hold.

Expenditures are given by

$$\begin{aligned} x_1 &= \sum_{j=1}^J Y_{1,j} + \sum_{j=1}^J \sum_{i \neq 1}^N \beta_{i,j} \frac{\tau_{i,j}^x}{1 + \tau_{i,j}^x} \pi_{i1,j} x_i + \sum_{j=1}^J \beta_{1,j} \sum_{i \neq 1}^N \frac{\tau_{i,j}^m}{1 + \tau_{i,j}^m} \pi_{1i,j} x_1 + \sum_{j=1}^J \beta_{1,j} \frac{\tau_j^d}{1 + \tau_j^d} \pi_{11,j} x_1, \quad (\gamma_x) \\ x_n &= \sum_{j=1}^J Y_{n,j}, \quad n \neq 1, \end{aligned}$$

where $Y_{n,j} = w_{n,j} L_{n,j}$ is the income in sector j of country n .

Goods market clearing conditions, for each j

$$\begin{aligned} \beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} x_1 &= E_{11,j} \equiv Y_{1,j} - \sum_{i \neq 1}^N \beta_{i,j} \frac{1}{1 + \tau_{i,j}^x} \pi_{i1,j} x_i, \quad (\gamma_{1,j}) \\ \beta_{1,j} \frac{1}{1 + \tau_{n,j}^m} \pi_{1n,j} x_1 &= E_{1n,j} \equiv Y_{n,j} - \sum_{i \neq 1}^N \beta_{i,j} \pi_{in,j} x_i, \quad (\gamma_{n,j}), \quad n \neq 1 \end{aligned}$$

where we define $E_{11,j}$ as the supply of Home to its domestic market in sector j , and $E_{1n,j}$ as the export supply of country n to Home in sector j .

Consumer prices indexes are given by

$$\begin{aligned} P_1 &= \prod_{j=1}^J \left[(p_{1,j}(1 + \tau_j^d))^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j}(1 + \tau_{i,j}^m) d_{1i,j})^{-\epsilon_j} \right]^{-\frac{\beta_{1,j}}{\epsilon_j}}, \\ P_n &= \prod_{j=1}^J \left[(p_{1,j}(1 + \tau_{n,j}^x) d_{n1,j})^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j} d_{ni,j})^{-\epsilon_j} \right]^{-\frac{\beta_{n,j}}{\epsilon_j}}, \quad n \neq 1. \end{aligned}$$

Trade shares for each sector j are given by:

$$\begin{aligned}\pi_{11,j} &= \frac{(p_{1,j}(1 + \tau_j^d))^{-\epsilon_j}}{(p_{1,j}(1 + \tau_j^d))^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j}(1 + \tau_{i,j}^m) d_{1i,j})^{-\epsilon_j}}, \\ \pi_{1n,j} &= \frac{(p_{n,j}(1 + \tau_{n,j}^m) d_{1n,j})^{-\epsilon_j}}{(p_{1,j}(1 + \tau_j^d))^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j}(1 + \tau_{i,j}^m) d_{1i,j})^{-\epsilon_j}}, \quad n \neq 1, \\ \pi_{n1,j} &= \frac{(p_{1,j}(1 + \tau_{n,j}^x) d_{n1,j})^{-\epsilon_j}}{(p_{1,j}(1 + \tau_{n,j}^x) d_{n1,j})^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j} d_{ni,j})^{-\epsilon_j}}, \quad n \neq 1, \\ \pi_{nm,j} &= \frac{(p_{m,j} d_{nm,j})^{-\epsilon_j}}{(p_{1,j}(1 + \tau_{n,j}^x) d_{n1,j})^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j} d_{ni,j})^{-\epsilon_j}}, \quad n \neq 1.\end{aligned}$$

Tax neutrality Given $\Gamma = \{(\tau_{n,j}^m + 1, \tau_{n,j}^x + 1, \tau_j^d + 1) : \forall j, n \neq 1\}$ and $\check{\Gamma} = \{(\check{\tau}_{n,j}^m + 1, \check{\tau}_{n,j}^x + 1, \check{\tau}_j^d + 1) : \forall j, n \neq 1\}$, let $\mathcal{E}(\Gamma)$ denote the set of $\{\pi_{1m,j}, \pi_{nm,j}, \frac{w_{1,j}}{P_1}, \frac{w_{n,j}}{P_n}, \frac{x_1}{P_1}, \frac{x_n}{P_n}\}$ that form an equilibrium. We say that from Γ to $\check{\Gamma}$ is neutral if $\mathcal{E}(\Gamma) = \mathcal{E}(\check{\Gamma})$. This captures neutrality because the equilibrium allocations and welfare under Γ and $\check{\Gamma}$ are the same.

Assume $1 + \check{\tau}_{n,j}^m = \lambda(1 + \tau_{n,j}^m)$, $1 + \check{\tau}_{n,j}^x = \frac{1 + \tau_{n,j}^x}{\mu}$, $1 + \check{\tau}_j^d = \lambda \frac{1 + \tau_j^d}{\mu}$ for any constant $\mu > 0$ and $\lambda > 0$. We guess the allocations $\{\check{\pi}_{1m,j}, \check{\pi}_{nm,j}, \frac{\check{w}_{1,j}}{\check{P}_1}, \frac{\check{w}_{n,j}}{\check{P}_n}, \frac{\check{x}_1}{\check{P}_1}, \frac{\check{x}_n}{\check{P}_n}\}$ in the new equilibrium are the same as allocations in the old equilibrium with $\check{\pi}_{1m,j} = \pi_{1m,j}$, $\check{\pi}_{nm,j} = \pi_{nm,j}$, $\check{P}_1 = P_1$, $\check{P}_n = \frac{P_n}{\lambda}$, $\check{w}_{1,j} = \mu \frac{w_{1,j}}{\lambda}$, $\check{w}_{n,j} = \frac{w_{n,j}}{\lambda}$, $\check{x}_1 = x_1$, $\check{x}_n = \frac{x_n}{\lambda}$ for $n \neq 1$. When the labor market specification in Ω is homogeneous of degree zero on wages, $\check{L}_{n,j} = L_{n,j}$, and prices $\$$ are homogeneous of degree one on wages, $\frac{\check{p}_{n,j}}{p_{n,j}} = \frac{\check{w}_{n,j}}{w_{n,j}}$. We then verify that all equilibrium conditions hold under the new equilibrium.

$$\begin{aligned}\check{x}_1 &= \frac{\mu}{\lambda} \sum_{j=1}^J Y_{1,j} + \sum_{i \neq 1}^N \sum_{j=1}^J (1 - \frac{\mu}{1 + \tau_{i,j}^x}) \beta_{i,j} \pi_{i1,j} x_i \frac{1}{\lambda} + \sum_{j=1}^J \beta_{1,j} \sum_{i \neq 1}^N (1 - \frac{1}{\lambda(1 + \tau_{i,j}^m)}) \pi_{1i,j} x_i \\ &+ \sum_{j=1}^J \beta_{1,j} (1 - \frac{\mu}{\lambda(1 + \tau_j^d)}) \pi_{11,j} x_1 \\ &= \frac{\mu}{\lambda} \sum_{j=1}^J \beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} x_1 + \frac{\mu}{\lambda} \sum_{i \neq 1}^N \beta_{i,j} \frac{1}{1 + \tau_{i,j}^x} \pi_{i1,j} x_i + \sum_{i \neq 1}^N \sum_{j=1}^J \beta_{i,j} (1 - \frac{\mu}{1 + \tau_{i,j}^x}) \pi_{i1,j} \frac{x_i}{\lambda} \\ &+ \sum_{j=1}^J \beta_{1,j} \sum_{i \neq 1}^N (1 - \frac{1}{\lambda(1 + \tau_{i,j}^m)}) \pi_{1i,j} x_i + \sum_{j=1}^J \beta_{1,j} (1 - \frac{\mu}{\lambda(1 + \tau_j^d)}) \pi_{11,j} x_1 \\ &= \sum_{i \neq 1}^N \sum_{j=1}^J \beta_{i,j} \pi_{i1,j} \frac{x_i}{\lambda} + \sum_{j=1}^J \beta_{1,j} \sum_{i \neq 1}^N (1 - \frac{1}{\lambda(1 + \tau_{i,j}^m)}) \pi_{1i,j} x_i + \sum_{j=1}^J \beta_{1,j} \pi_{11,j} x_1 \\ &= \sum_{i \neq 1}^N \sum_{j=1}^J \beta_{1,j} \pi_{1i,j} x_i + \sum_{j=1}^J \beta_{1,j} \pi_{11,j} x_1 = x_1,\end{aligned}$$

$$\begin{aligned}
\check{x}_n &= \sum_{j=1}^J \frac{1}{\lambda} Y_{n,j} = \frac{x_n}{\lambda}, \quad n \neq 1 \\
\check{Y}_{1,j} &= \check{w}_{1,j} L_{1,j} = \frac{\mu}{\lambda} w_{1,j} L_{1,j} = \frac{\mu}{\lambda} Y_{1,j}, \\
\check{Y}_{n,j} &= \check{w}_{n,j} L_{n,j} = \frac{1}{\lambda} w_{n,j} L_{n,j} = \frac{1}{\lambda} Y_{n,j}, \quad n \neq 1 \\
\check{Y}_{1,j} &= \frac{\mu}{\lambda(1+\tau_j^d)} \beta_{1,j} \pi_{11,j} x_1 + \sum_{i \neq 1}^N \beta_{i,j} \frac{\mu}{1+\tau_{i,j}^x} \pi_{i1,j} \frac{x_i}{\lambda} = \frac{\mu}{\lambda} Y_{1,j}, \\
\check{Y}_{n,j} &= \beta_{1,j} \frac{1}{\lambda(1+\tau_{n,j}^m)} \pi_{1n,j} x_1 + \sum_{i \neq 1}^N \beta_{i,j} \pi_{in,j} \frac{x_i}{\lambda} = \frac{1}{\lambda} Y_{n,j}, \quad n \neq 1 \\
\check{P}_1 &= \prod_{j=1}^J \left[\left(\frac{\mu}{\lambda} p_{1,j} \frac{\lambda}{\mu} (1+\tau_j^d) \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \left(\frac{p_{i,j}}{\lambda} \lambda (1+\tau_{i,j}^m) d_{1i,j} \right)^{-\epsilon_j} \right]^{-\frac{\beta_{1,j}}{\epsilon_j}} = P_1 = 1, \\
\check{P}_n &= \prod_{j=1}^J \left[\left(\frac{\mu}{\lambda} p_{1,j} \frac{1+\tau_{n,j}^x}{\mu} d_{n1,j} \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \left(\frac{p_{i,j}}{\lambda} d_{ni,j} \right)^{-\epsilon_j} \right]^{-\frac{\beta_{n,j}}{\epsilon_j}} = \frac{P_n}{\lambda}, \quad n \neq 1 \\
\check{\pi}_{11,j} &= \frac{\left(\frac{\mu}{\lambda} p_{1,j} \frac{\lambda}{\mu} (1+\tau_j^d) \right)^{-\epsilon_j}}{\left(\frac{\mu}{\lambda} p_{1,j} \frac{\lambda}{\mu} (1+\tau_j^d) \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \left(\frac{p_{i,j}}{\lambda} \lambda (1+\tau_{i,j}^m) d_{1i,j} \right)^{-\epsilon_j}} = \pi_{11,j}, \\
\check{\pi}_{n1,j} &= \frac{\left(\frac{\mu}{\lambda} p_{1,j} \frac{1+\tau_{n,j}^x}{\mu} d_{n1,j} \right)^{-\epsilon_j}}{\left(\frac{\mu}{\lambda} p_{1,j} \frac{1+\tau_{n,j}^x}{\mu} d_{n1,j} \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \left(\frac{p_{i,j}}{\lambda} d_{ni,j} \right)^{-\epsilon_j}} = \pi_{n1,j}, \quad n \neq 1
\end{aligned}$$

The same allocations satisfy the equilibrium conditions under $\{\check{\tau}_{n,j}^m + 1, \check{\tau}_{n,j}^x + 1, \check{\tau}_j^d + 1\}$. Hence, we proved that the allocations and welfare are the same under $\{\tau_{n,j}^m + 1, \tau_{n,j}^x + 1, \tau_j^d + 1\}$ and $\{\check{\tau}_{n,j}^m + 1, \check{\tau}_{n,j}^x + 1, \check{\tau}_j^d + 1\}$.

The Home government can manipulate $2J(N-1) + J$ prices. We can first normalize one domestic tax to zero, for example, $\check{\tau}_1^d = 0$ for sector 1. For any τ_1^d , we can always choose λ and μ , such that $\mu/\lambda = 1 + \tau_1^d$ to get $\check{\tau}_1^d = \lambda(1 + \tau_1^d)/\mu - 1 = 0$ and scale other taxes accordingly. The welfare and allocation are equivalent. Second, we can set an import tariff from one specific country and sector to zero, for example, $\check{\tau}_{2,1}^m = 0$ by setting $\lambda = 1/(\tau_{2,1}^m + 1)$ for any $\tau_{2,1}^m$ and adjusting other taxes accordingly to achieve the same allocation. Alternatively, one of the export taxes, $\check{\tau}_{2,1}^x$, can be normalized to zero by setting $\mu = 1 + \tau_{2,1}^x$ for any $\tau_{2,1}^x$. In either case, it suffices for the government to use only the $2J(N-1) + J - 2$ instruments to implement the same allocation. In our optimization framework, we normalize one domestic tax and one trade tax to uniquely determine the level of all remaining taxes. $\square$

B Proof for Proposition 2 and 3: General Formula for Two-Country Optimal Policies and Optimal Policies under GIIA Supply System

This section presents the Home government's problem in the two-country case. We first present the maximization problem, then prove the optimal policies under our defined GIIA Supply System and the general framework, as shown in Propositions 2 and 3.

Home government chooses policies $\{\tau_j^d, \tau_j^x, \tau_j^m\}$ to maximize domestic households' consumption x_1/P_1 subject to the world market equilibrium defined in Definition 1. We solve this optimization problem with the Lagrangian multiplier through a two-step approach.

In the first step, we derive supply curves $Y_{n,j}(\{p_{n,k}\})$ for each country-sector pair using the supply-side assumptions $\mathcal{S}$ and labor market specifications Ω . We can then express the equilibrium variables, including $\{w_{n,j}, L_{n,j}\}$, as functions of sectoral prices $\{p_{n,j}\}$ and policy instruments. In the second step, we take first-order conditions with respect to expenditures x_1 , sectoral prices $\{p_{1,j}, p_{2,j}\}$, and policy instruments, imposing multipliers on the goods-market clearing conditions while treating income $Y_{n,j}$ as endogenous functions of $\{p_{n,k}\}$ through the derived supply curves.

An alternative approach involves taking first-order conditions directly with respect to $\{w_{n,j}, L_{n,j}, p_{n,j}\}$ while imposing additional multipliers on the supply-side assumptions $\mathcal{S}$ and labor market specifications Ω . Both approaches yield identical optimal policies and equilibrium outcomes. However, our two-step approach provides a unified framework for characterizing optimal policies by first deriving supply curves that encapsulate the underlying supply structure. The policy formulas obtained through this approach also offer a clear economic intuition and interpretation. We demonstrate the equivalence of these two approaches across three examples in Appendix C.

Step 1: Derive supply curves $Y_{n,j}(\{p_{n,k}\})$ Income $Y_{n,j}$ in each sector is defined as $Y_{n,j} = w_{n,j}L_{n,j}$. We derive the supply curves $Y_{n,j}(\{p_{n,k}\})$ for sector j in country n by combining the supply-side assumptions $\mathcal{S}$ and the labor market specifications Ω . In each country, the set $\mathcal{S}$ specifies J equations that link sectoral prices, wages, and labor $\{p_{n,j}, w_{n,j}, L_{n,j}\}$ both within and across sectors, while the set Ω provides an additional J equations governing labor market conditions.

For each country n , we have $4J$ variables $\{Y_{n,j}, p_{n,j}, w_{n,j}, L_{n,j}\}$ that must satisfy the following $3J$

conditions,

$$\begin{aligned}
[\text{income definition}] \quad & Y_{n,j} = w_{n,j} L_{n,j}, \quad \text{for all } j \\
[\text{supply-side assumption}] \quad & S(\{p_{n,j}, w_{n,j}, L_{n,j}\}), \\
[\text{labor market specification}] \quad & \Omega(\{w_{n,j}, L_{n,j}\}).
\end{aligned}$$

This system of $3J$ equations in $4J$ unknowns yields J degrees of freedom, which generate the supply curves linking sectoral income $Y_{n,j}$ with prices across all sectors $\{p_{n,k}\}$. From these supply curves, we can construct a matrix of partial supply elasticities Λ_n , where each element represents the weighted elasticity $\frac{\partial \ln Y_{n,k}}{\partial \ln p_{n,j}} \frac{Y_{n,k}}{Y_{n,j}}$.

The derived supply curves have incorporated both the supply-side assumptions S and the labor market specifications Ω . This allows us to simplify our optimization problem by eliminating the explicit consideration of conditions S and Ω along with wages and labor $\{w_{n,j}, L_{n,j}\}$. Instead, we can focus on the reduced system involving only $\{Y_{n,j}, p_{n,k}\}$ and the derived supply relationships.

Example: Consider a generalized Krugman model where each sector has identical scale elasticity. The supply-side assumption S specifies $p_{n,j} = \bar{T}_{n,j}^{-1} w_{n,j} L_{n,j}^{1/(1-\sigma)}$. The labor market specification Ω assumes perfectly mobile labor across sectors, which generates one labor constraint $\sum_{j=1}^J L_{n,j} = \bar{L}_n$ and wage equalization conditions $w_{n,j} = w_n$ for all sectors. The complete system of $3J$ conditions for country n consists of

$$\begin{aligned}
[\text{definition of income}] \quad & Y_{n,j} = w_{n,j} L_{n,j}, \\
[\text{supply-side assumption}] \quad & p_{n,j} = \bar{T}_{n,j}^{-1} w_{n,j} L_{n,j}^{1/(1-\sigma)}, \\
[\text{labor market specification}] \quad & w_{n,j} = w_n, \\
& \sum_{j=1}^J L_{n,j} = \bar{L}_n.
\end{aligned}$$

To derive the supply curves, we eliminate the wage and labor variables $w_{n,j}$ and $L_{n,j}$ from this system, and obtain the following supply curves

$$Y_{n,j} = \left(\frac{1}{\bar{L}_n} \sum_{k=1}^J p_{n,k}^{1-\sigma} \bar{T}_{n,k}^{1-\sigma} \right)^{\frac{\sigma}{1-\sigma}} p_{n,j}^{1-\sigma} \bar{T}_{n,j}^{1-\sigma}.$$

This expression demonstrates that sectoral income $Y_{n,j}$ depends not only on its own prices $p_{n,j}$ but also on prices across all other sectors $p_{n,k}$.

Step 2: Derive optimal policies We now replace the conditions $\mathbb{S}$ and Ω with the derived supply curves $Y_{n,j}(\{p_{n,k}\})$. Note that foreign prices do not affect home supply curves directly.

In the two-country case, the Home government choose $\{x_1, p_{1,j}, p_{2,j}, \tau_j^d, \tau_j^x, \tau_j^m\}$ to solve the following problem:

$$\max_{\{\tau_j^d, \tau_j^x, \tau_j^m, x_1, p_{1,j}, p_{2,j}\}} \frac{x_1}{P_1},$$

subject to world market equilibrium characterized by the following constraints:

Home expenditures

$$x_1 = \sum_{j=1}^J Y_{1,j}(\{p_{1,k}\}) + \sum_{j=1}^J \beta_{2,j} \frac{\tau_j^x}{1 + \tau_j^x} \pi_{21,j} x_2 + \sum_{j=1}^J \beta_{1,j} \frac{\tau_j^m}{1 + \tau_j^m} \pi_{12,j} x_1 + \sum_{j=1}^J \beta_{1,j} \frac{\tau_j^d}{1 + \tau_j^d} \pi_{11,j} x_1, \quad (\gamma_x) \quad (\text{A.1})$$

Goods market clearing conditions

$$\beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} x_1 = Y_{1,j}(\{p_{1,k}\}) - \beta_{2,j} \frac{1}{1 + \tau_j^x} \pi_{21,j} x_2, \quad (\gamma_{1,j}, \quad J) \quad (\text{A.2})$$

$$\beta_{1,j} \frac{1}{1 + \tau_j^m} \pi_{12,j} x_1 = Y_{2,j}(\{p_{2,k}\}) - \beta_{2,j} \pi_{22,j} x_2, \quad (\gamma_{2,j}, \quad J) \quad (\text{A.3})$$

where $\{x_2, P_1, P_2\}$ and trade shares are given by

$$x_2 = \sum_{j=1}^J Y_{2,j}(\{p_{2,k}\}), \quad (\text{A.4})$$

$$\pi_{11,j} = \frac{(p_{1,j}(1 + \tau_j^d))^{-\epsilon_j}}{(p_{1,j}(1 + \tau_j^d))^{-\epsilon_j} + (p_{2,j}(1 + \tau_j^m)d_{12,j})^{-\epsilon_j}}, \quad (\text{A.5})$$

$$\pi_{12,j} = \frac{(p_{2,j}(1 + \tau_j^m)d_{12,j})^{-\epsilon_j}}{(p_{1,j}(1 + \tau_j^d))^{-\epsilon_j} + (p_{2,j}(1 + \tau_j^m)d_{12,j})^{-\epsilon_j}}, \quad (\text{A.6})$$

$$\pi_{21,j} = \frac{(p_{1,j}(1 + \tau_j^x)d_{21,j})^{-\epsilon_j}}{(p_{1,j}(1 + \tau_j^x)d_{21,j})^{-\epsilon_j} + (p_{2,j})^{-\epsilon_j}}, \quad (\text{A.7})$$

$$\pi_{22,j} = \frac{(p_{2,j})^{-\epsilon_j}}{(p_{1,j}(1 + \tau_j^x)d_{21,j})^{-\epsilon_j} + (p_{2,j})^{-\epsilon_j}}, \quad (\text{A.8})$$

$$P_1 = \prod_{j=1}^J \left[(p_{1,j}(1 + \tau_j^d))^{-\epsilon_j} + (p_{2,j}(1 + \tau_j^m)d_{12,j})^{-\epsilon_j} \right]^{-\frac{\beta_{1,j}}{\epsilon_j}} = 1, \quad (\text{A.9})$$

$$P_2 = \prod_{j=1}^J \left[(p_{1,j}(1 + \tau_j^x)d_{21,j})^{-\epsilon_j} + p_{2,j}^{-\epsilon_j} \right]^{-\frac{\beta_{2,j}}{\epsilon_j}}. \quad (\text{A.10})$$

One of the equilibrium conditions in (A.1), (A.2) and (A.3) is redundant, according to Walras' law, thus, we drop the expenditure constraint in optimization and set $\gamma_x = 0$. Or, equivalently, if we use Home's income equation, one of the market-clearing conditions is redundant.

FOC over x_1

$$1 - \sum_{j=1}^J \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} - \sum_{j=1}^J \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \pi_{12,j} = 0.$$

FOC over export tax τ_j^x

$$\gamma_{1,j} \beta_{2,j} \frac{\pi_{21,j} x_2}{(1 + \tau_j^x)^2} - \gamma_{1,j} \beta_{2,j} \frac{1}{1 + \tau_j^x} \frac{\partial \pi_{21,j}}{\partial \tau_j^x} x_2 - \gamma_{2,j} \beta_{2,j} \frac{\partial \pi_{22,j}}{\partial \tau_j^x} x_2 = 0.$$

After plugging in derivatives and simplifying, the FOC over τ_j^x becomes

$$1 + \tau_j^x = \frac{\gamma_{1,j}(\epsilon_j \pi_{22,j} + 1)}{\gamma_{2,j} \epsilon_j \pi_{22,j}}. \quad (\text{A.11})$$

FOC over import tariff τ_j^m

$$- \beta_{1,j} x_1 \frac{\pi_{12,j}}{1 + \tau_j^m} - \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \frac{\partial \pi_{11,j}}{\partial \tau_j^m} x_1 + \gamma_{2,j} \beta_{1,j} \frac{\pi_{12,j}}{(1 + \tau_j^m)^2} x_1 - \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \frac{\partial \pi_{12,j}}{\partial \tau_j^m} x_1 = 0.$$

Plugging in derivatives and simplifying, the FOC over τ_j^m becomes

$$1 + \tau_j^m = \frac{\gamma_{2,j} (1 + \epsilon_j \pi_{11,j})}{\gamma_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j \pi_{11,j} + 1}. \quad (\text{A.12})$$

FOC over domestic tax τ_j^d

$$- \beta_{1,j} x_1 \frac{\pi_{11,j}}{1 + \tau_j^d} + \gamma_{1,j} \frac{1}{(1 + \tau_j^d)^2} \beta_{1,j} \pi_{11,j} x_1 - \gamma_{1,j} \frac{1}{1 + \tau_j^d} \beta_{1,j} \frac{\partial \pi_{11,j}}{\partial \tau_j^d} x_1 - \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \frac{\partial \pi_{12,j}}{\partial \tau_j^d} x_1 = 0.$$

Plugging in derivatives and simplifying, the FOC over τ_j^d becomes

$$1 + \tau_j^d = \frac{\gamma_{1,j} (1 + \epsilon_j \pi_{12,j})}{\gamma_{2,j} \frac{1}{1 + \tau_j^m} \epsilon_j \pi_{12,j} + 1}. \quad (\text{A.13})$$

Combining the optimal tariff (A.12) and domestic tax (A.13), we can get

$$1 + \tau_j^m = \gamma_{2,j}, \quad 1 + \tau_j^d = \gamma_{1,j}. \quad (\text{A.14})$$

Combining the above two equations with equation (A.11), we get

$$1 + \tau_j^x = \frac{1 + \tau_j^d}{1 + \tau_j^m} \left(1 + \frac{1}{\epsilon_j \pi_{22,j}} \right). \quad (\text{A.15})$$

FOC over $p_{1,j}$

$$\begin{aligned} & -\beta_{1,j} x_1 \pi_{11,j} + \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j \pi_{11,j} \pi_{12,j} x_1 + \gamma_{1,j} \beta_{2,j} \frac{1}{1 + \tau_j^x} \epsilon_j \pi_{21,j} \pi_{22,j} x_2 \\ & - \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \epsilon_j \pi_{12,j} \pi_{11,j} x_1 - \gamma_{2,j} \beta_{2,j} \epsilon_j \pi_{22,j} \pi_{21,j} x_2 + \sum_{k=1}^J \gamma_{1,k} \frac{\partial Y_{1,k}}{\partial p_{1,j}} p_{1,j} = 0. \end{aligned} \quad (\text{A.16})$$

FOC over $p_{2,j}$

$$\begin{aligned} & -\beta_{1,j} x_1 \pi_{12,j} - \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j \pi_{11,j} \pi_{12,j} x_1 - \gamma_{1,j} \beta_{2,j} \frac{1}{1 + \tau_j^x} \epsilon_j \pi_{21,j} \pi_{22,j} x_2 \\ & + \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \epsilon_j \pi_{12,j} \pi_{11,j} x_1 + \gamma_{2,j} \beta_{2,j} \epsilon_j \pi_{22,j} \pi_{21,j} x_2 + \sum_{k=1}^J (\Phi_2 + \gamma_{2,k}) \frac{\partial Y_{2,k}}{\partial p_{2,j}} p_{2,j} = 0, \end{aligned} \quad (\text{A.17})$$

where $\Phi_2 = -\sum_{s=1}^J (\gamma_{1,s} \beta_{2,s} \frac{1}{1 + \tau_s^x} \pi_{21,s} + \gamma_{2,s} \beta_{2,s} \pi_{22,s})$.

B.1 Proof of Optimal Domestic Tax Formula

Combining FOCs over $p_{1,j}$ (A.16), τ_j^x (A.15), and τ_j^d, τ_j^m (A.14), we get

$$\begin{aligned} & -\beta_{1,j} x_1 \pi_{11,j} + \beta_{2,j} \left(\frac{1 + \tau_j^d}{1 + \tau_j^x} - (1 + \tau_j^m) \right) \epsilon_j \pi_{21,j} \pi_{22,j} x_2 + \sum_{k=1}^J (1 + \tau_k^d) \frac{\partial Y_{1,k}}{\partial p_{1,j}} p_{1,j} = 0 \\ \Rightarrow & -\beta_{1,j} x_1 \pi_{11,j} - \beta_{2,j} \frac{1 + \tau_j^d}{1 + \tau_j^x} \pi_{21,j} x_2 + \sum_{k=1}^J (1 + \tau_k^d) \frac{\partial Y_{1,k}}{\partial p_{1,j}} p_{1,j} = 0 \\ \Rightarrow & \sum_{k=1}^J (1 + \tau_k^d) \frac{\partial Y_{1,k}}{\partial p_{1,j}} p_{1,j} - (1 + \tau_j^d) Y_{1,j} = 0 \\ \Rightarrow & \sum_{k=1}^J (1 + \tau_k^d) \frac{\partial \ln(Y_{1,k})}{\partial \ln(p_{1,j})} \frac{Y_{1,k}}{Y_{1,j}} - (1 + \tau_j^d) = 0. \end{aligned} \quad (\text{A.18})$$

Hence, domestic taxes $\tau^d = [\tau_1^d, \dots, \tau_j^d, \dots, \tau_J^d]'$ for J sectors satisfy

$$(\Lambda_1 - I)(1 + \tau^d) = 0,$$

where element of matrix Λ_n at row j and column k is given by $\frac{\partial \ln(Y_{n,k})}{\partial \ln(p_{n,j})} \frac{Y_{n,k}}{Y_{n,j}}$, and

$$\Lambda_1 - I = \begin{pmatrix} \frac{\partial \ln(Y_{1,1})}{\partial \ln(p_{1,1})} \frac{Y_{1,1}}{Y_{1,1}} - 1 & \dots & \frac{\partial \ln(Y_{1,j})}{\partial \ln(p_{1,1})} \frac{Y_{1,j}}{Y_{1,1}} & \dots & \frac{\partial \ln(Y_{1,J})}{\partial \ln(p_{1,1})} \frac{Y_{1,J}}{Y_{1,1}} \\ \dots & \dots & \dots & \dots & \dots \\ \frac{\partial \ln(Y_{1,1})}{\partial \ln(p_{1,j})} \frac{Y_{1,1}}{Y_{1,j}} & \dots & \frac{\partial \ln(Y_{1,j})}{\partial \ln(p_{1,j})} \frac{Y_{1,j}}{Y_{1,j}} - 1 & \dots & \frac{\partial \ln(Y_{1,J})}{\partial \ln(p_{1,j})} \frac{Y_{1,J}}{Y_{1,j}} \\ \dots & \dots & \dots & \dots & \dots \\ \frac{\partial \ln(Y_{1,1})}{\partial \ln(p_{1,J})} \frac{Y_{1,1}}{Y_{1,J}} & \dots & \frac{\partial \ln(Y_{1,j})}{\partial \ln(p_{1,J})} \frac{Y_{1,j}}{Y_{1,J}} & \dots & \frac{\partial \ln(Y_{1,J})}{\partial \ln(p_{1,J})} \frac{Y_{1,J}}{Y_{1,J}} - 1 \end{pmatrix}.$$

Divide equation (A.18) by η_j . For any sector k and j , optimal domestic taxes satisfy

$$\sum_{s=1}^J \frac{\partial \ln Y_{1,s}}{\partial \ln p_{1,j}} \frac{Y_{1,s}}{Y_{1,j}} (1 + \tau_s^d) \frac{1}{\eta_j} = (1 + \tau_j^d) \frac{1}{\eta_j}, \quad (\text{A.19})$$

$$\sum_{s=1}^J \frac{\partial \ln Y_{1,s}}{\partial \ln p_{1,k}} \frac{Y_{1,s}}{Y_{1,k}} (1 + \tau_s^d) \frac{1}{\eta_k} = (1 + \tau_k^d) \frac{1}{\eta_k}, \quad (\text{A.20})$$

where η_j is the partial supply elasticity parameter in sector j . Using the definition of GIIA Supply System in Definition 4, and subtracting (A.20) from (A.19), we have

$$(1 + \tau_j^d) - (1 + \tau_k^d) = (1 + \tau_j^d) \frac{1}{\eta_j} - (1 + \tau_k^d) \frac{1}{\eta_k} \Rightarrow \frac{1 + \tau_j^d}{1 + \tau_k^d} = \frac{\frac{\eta_j}{\eta_j - 1}}{\frac{\eta_k}{\eta_k - 1}}.$$

According to the tax neutrality shown in Lemma 1, we can set $1 + \tau_j^d = (1 + \bar{\tau}^d) \frac{\eta_j}{\eta_j - 1}, \forall j$. The uniform shifter $\bar{\tau}^d$ can be assigned any arbitrary value, and one of the trade policies can be normalized, without changing the real allocations and welfare impacts of optimal policies.

B.2 Proof of Optimal Import Tariff Formula

Using FOC over $p_{2,j}$ (A.17), τ_j^x (A.15), and τ_j^d, τ_j^m (A.14), we get for each sector j

$$\begin{aligned} \sum_{k=1}^J (\Phi_2 + 1 + \tau_k^m) \left[\frac{\partial \ln Y_{2,k}}{\partial \ln p_{2,j}} Y_{2,k} \right] &= \beta_{1,j} \pi_{12,j} x_1 + \left[\frac{1 + \tau_j^d}{1 + \tau_j^x} - (1 + \tau_j^m) \right] \beta_{2,j} \pi_{21,j} \pi_{22,j} x_2 \\ \Rightarrow \sum_{k=1}^J (\Phi_2 + 1 + \tau_k^m) \left[\frac{\partial \ln Y_{2,k}}{\partial \ln p_{2,j}} \frac{Y_{2,k}}{Y_{2,j}} \right] &= \frac{\beta_{1,j} \pi_{12,j} x_1}{Y_{2,j}} - \frac{1 + \tau_j^d}{1 + \tau_j^x} \beta_{2,j} \pi_{21,j} \frac{x_2}{Y_{2,j}} \\ \Rightarrow \sum_{k=1}^J (\Phi_2 + 1 + \tau_k^m) \left[\frac{\partial \ln Y_{2,k}}{\partial \ln p_{2,j}} \frac{Y_{2,k}}{Y_{2,j}} \right] &= \frac{\beta_{1,j} \pi_{12,j} x_1}{Y_{2,j}} - (1 + \tau_j^d) \frac{Y_{1,j}}{Y_{2,j}} + \frac{\beta_{1,j} \pi_{11,j} x_1}{Y_{2,j}}. \end{aligned} \quad (\text{A.21})$$

The above equation becomes

$$\sum_{k=1}^J (\Phi_2 + 1 + \tau_k^m) \left[\frac{\partial \ln Y_{2,k}}{\partial \ln p_{2,j}} \frac{Y_{2,k}}{Y_{2,j}} \right] = \frac{\beta_{1,j} x_1 - (1 + \tau_j^d) Y_{1,j}}{Y_{2,j}}, \quad (\text{A.22})$$

where Φ_2 is common across all sectors.

Let $\tau^m = [\tau_1^m, \dots, \tau_j^m, \dots, \tau_J^m]'$ denote the import tariff vector. We can write equation (A.22) in matrix form as $\Lambda_2(\tau^m + \Phi_2 \mathbf{1} + \mathbf{1}) = \left(\frac{M_{1,j}}{Y_{2,j}} \right)_{j=1}^J$, where Λ_2 is given by

$$\Lambda_2 = \begin{pmatrix} \frac{\partial \ln(Y_{2,1})}{\partial \ln(p_{2,1})} \frac{Y_{2,1}}{Y_{2,1}} & \dots & \frac{\partial \ln(Y_{2,j})}{\partial \ln(p_{2,1})} \frac{Y_{2,j}}{Y_{2,1}} & \dots & \frac{\partial \ln(Y_{2,J})}{\partial \ln(p_{2,1})} \frac{Y_{2,J}}{Y_{2,1}} \\ \dots & \dots & \dots & \dots & \dots \\ \frac{\partial \ln(Y_{2,1})}{\partial \ln(p_{2,j})} \frac{Y_{2,1}}{Y_{2,j}} & \dots & \frac{\partial \ln(Y_{2,j})}{\partial \ln(p_{2,j})} \frac{Y_{2,j}}{Y_{2,j}} & \dots & \frac{\partial \ln(Y_{2,J})}{\partial \ln(p_{2,j})} \frac{Y_{2,J}}{Y_{2,j}} \\ \dots & \dots & \dots & \dots & \dots \\ \frac{\partial \ln(Y_{2,1})}{\partial \ln(p_{2,J})} \frac{Y_{2,1}}{Y_{2,J}} & \dots & \frac{\partial \ln(Y_{2,j})}{\partial \ln(p_{2,J})} \frac{Y_{2,j}}{Y_{2,J}} & \dots & \frac{\partial \ln(Y_{2,J})}{\partial \ln(p_{2,J})} \frac{Y_{2,J}}{Y_{2,J}} \end{pmatrix}. \quad (\text{A.23})$$

Divide equation (A.22) by η_j . For any sector k and j , optimal import tariffs satisfy

$$\sum_{s=1}^J \frac{\partial \ln Y_{2,s}}{\partial \ln p_{2,j}} \frac{Y_{2,s}}{Y_{2,j}} (\tau_s^m + \Phi_2 + 1) \frac{1}{\eta_j} = \frac{1}{\eta_j} \frac{M_{1,j}}{Y_{2,j}}, \quad (\text{A.24})$$

$$\sum_{s=1}^J \frac{\partial \ln Y_{2,s}}{\partial \ln p_{2,k}} \frac{Y_{2,s}}{Y_{2,k}} (\tau_s^m + \Phi_2 + 1) \frac{1}{\eta_k} = \frac{1}{\eta_k} \frac{M_{1,k}}{Y_{2,k}}. \quad (\text{A.25})$$

Using the definition of GIIA Supply System in Definition 4, and subtracting (A.25) from (A.24), we

have

$$\begin{aligned}
& \left(\frac{\partial \ln Y_{2,j}}{\partial \ln p_{2,j}} \frac{1}{\eta_j} - \frac{\partial \ln Y_{2,j}}{\partial \ln p_{2,k}} \frac{Y_{2,j}}{Y_{2,k}} \frac{1}{\eta_k} \right) (\tau_j^m + \Phi_2 + 1) + \left(\frac{\partial \ln Y_{2,k}}{\partial \ln p_{2,j}} \frac{Y_{2,k}}{Y_{2,j}} \frac{1}{\eta_j} - \frac{\partial \ln Y_{2,k}}{\partial \ln p_{2,k}} \frac{1}{\eta_k} \right) (\tau_k^m + \Phi_2 + 1) \\
&= \frac{1}{\eta_j} \frac{M_{1,j}}{Y_{2,j}} - \frac{1}{\eta_k} \frac{M_{1,k}}{Y_{2,k}} \\
&\Rightarrow \tau_j^m - \tau_k^m = \frac{1}{\eta_j} \frac{M_{1,j}}{Y_{2,j}} - \frac{1}{\eta_k} \frac{M_{1,k}}{Y_{2,k}},
\end{aligned}$$

Since this relationship holds for every pair of sectors, there exists a sector-common endogenous term $\bar{\tau}^m$ such that

$$1 + \tau_j^m = \frac{1}{\eta_j} \frac{M_{1,j}}{Y_{2,j}} + (1 + \bar{\tau}^m), \quad \forall j.$$

The GIIA restriction implies that the jk th element of Λ_n , after scaling by $1/\eta_j$, can be written as

$$\frac{\Lambda_{n,jk}}{\eta_j} \equiv \frac{\partial \ln Y_{n,k}}{\partial \ln p_{n,j}} \frac{Y_{n,k}}{Y_{n,j}} \frac{1}{\eta_j} = I_{k=j} + a_{n,k}. \quad (\text{A.26})$$

Hence,

$$\bar{\tau}^m = -(\Phi_2 + 1) - \sum_{k=1}^J (\Phi_2 + 1 + \tau_k^m) a_{2,k}.$$

Equivalently, for any reference sector s ,

$$\bar{\tau}^m = -(\Phi_2 + 1) - \sum_{k=1}^J (\Phi_2 + 1 + \tau_k^m) \left[\frac{\partial \ln Y_{2,k}}{\partial \ln p_{2,s}} \frac{Y_{2,k}}{Y_{2,s}} \frac{1}{\eta_s} - I_{k=s} \right].$$

C Optimal Policies under GIIA Supply System: Examples

This section presents three examples of GIIA Supply System and derives their optimal policies. The three models include a one-factor Ricardo-Roy model (Section C.1), a generalized multi-sector Krugman model (Section C.2), and a model with sector-specific scale elasticities and imperfectly mobile labor (Section C.3). In addition, Section C.4 provides a proof that the labor market specification in a CES labor market is equivalent to that in the one-factor Ricardo-Roy model, and examines two special cases: perfectly mobile labor across sectors and immobile/sector-specific labor.

We use two methods to characterize optimal policies and demonstrate that both yield the same tax formulas.

In the first method, we formulate the Home government's problem and the world market equilibrium based on the original equilibrium conditions of each model. In each case, we use the explicit

functional forms of S and Ω specific to each case and solve $\{p_{n,j}\}$ as functions of wages and labor $\{w_{n,j}, L_{n,j}\}$ using the model-specific supply function $S(\{p_{n,j}, w_{n,j}, L_{n,j}\})$. We then take first-order conditions with respect to $\{w_{n,j}, L_{n,j}\}$ and impose multipliers on each model's labor market conditions Ω . This method differs from our general approach in two ways: first, it relies on specific functional forms of S and Ω rather than treating them generically, and second, it takes derivatives directly with respect to $\{w_{n,j}, L_{n,j}\}$ rather than with respect to derived supply curves while treating wages and labor as functions of sectoral prices $\{p_{n,j}\}$.

In the second method, we prove that each model satisfies the GIIA Supply System requirement and apply the formulas from Proposition 3 directly. Both methods yield identical tax formulas, confirming the validity of our general approach.

C.1 One-factor Ricardo-Roy Model

In Section C.1.1, we formulate the Home government's problem and the world market equilibrium based on the original equilibrium conditions of the one-factor Ricardo-Roy model. We solve $\{p_{n,j}\}$ as functions of wages and labor $\{w_{n,j}, L_{n,j}\}$ using the model-specific supply function $S(\{p_{n,j}, w_{n,j}, L_{n,j}\})$. We then directly take first-order conditions with respect to $\{w_{n,j}, L_{n,j}\}$ to derive optimal policies in this model. In Section C.1.2, we show that this model is a GIIA Supply System and employ matrix operations to establish a connection with Proposition 3.

C.1.1 Method 1: Model Specific Proof

In the one-factor Ricardo-Roy model, due to the constant returns to scale, prices are proportional to wages. The supply-side assumption S satisfies

$$p_{n,j} = \frac{w_{n,j}}{\bar{T}_{n,j}}. \quad (\text{A.27})$$

where $\bar{T}_{n,j}$ denotes the exogenous technology level beyond the economies of scale in the sector j of country n . In the following market equilibrium constraints, we substitute $p_{n,j}$ with $w_{n,j}$.

Applying the labor market specification Ω (Eq.19) with one-factor $G = 1$, we obtain the one-factor Ricardo-Roy model Ω satisfying

$$w_{n,j}L_{n,j} = A_{n,j}\left(\frac{w_{n,j}}{W_n}\right)^\kappa W_n \bar{L}_n, \quad W_n = \left[\sum_{j=1}^J A_{n,j}w_{n,j}^\kappa\right]^{\frac{1}{\kappa}}. \quad (\text{A.28})$$

Using this condition, we can eliminate $L_{n,j}$ from the optimization system and substitute

$$L_{n,j} = A_{n,j}(w_{n,j})^{\kappa-1} \left[\sum_{j=1}^J A_{n,j} w_{n,j}^{\kappa} \right]^{\frac{1-\kappa}{\kappa}} \bar{L}_n.$$

Now Home government chooses $\{\tau_j^d, \tau_j^x, \tau_j^m, x_1, w_{1,j}, w_{2,j}\}$ to solve the following problem:

$$\max \frac{x_1}{P_1},$$

subject to world market equilibrium conditions

$$\begin{aligned} x_1 = & \sum_{j=1}^J A_{1,j}(w_{1,j})^{\kappa} \left[\sum_{j=1}^J A_{1,j} w_{1,j}^{\kappa} \right]^{\frac{1-\kappa}{\kappa}} \bar{L}_1 + \sum_{j=1}^J \beta_{2,j} \frac{\tau_j^x}{1 + \tau_j^x} \pi_{21,j} x_2 + \sum_{j=1}^J \beta_{1,j} \frac{\tau_j^m}{1 + \tau_j^m} \pi_{12,j} x_1 \\ & + \sum_{j=1}^J \beta_{1,j} \frac{\tau_j^d}{1 + \tau_j^d} \pi_{11,j} x_1, \quad (\gamma_x) \end{aligned} \quad (\text{A.29})$$

$$\beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} x_1 = A_{1,j}(w_{1,j})^{\kappa} \left[\sum_{j=1}^J A_{1,j} w_{1,j}^{\kappa} \right]^{\frac{1-\kappa}{\kappa}} \bar{L}_1 - \beta_{2,j} \frac{1}{1 + \tau_j^x} \pi_{21,j} x_2, \quad (\gamma_{1,j}, \quad J) \quad (\text{A.30})$$

$$\beta_{1,j} \frac{1}{1 + \tau_j^m} \pi_{12,j} x_1 = A_{2,j}(w_{2,j})^{\kappa} \left[\sum_{j=1}^J A_{2,j} w_{2,j}^{\kappa} \right]^{\frac{1-\kappa}{\kappa}} \bar{L}_2 - \beta_{2,j} \pi_{22,j} x_2, \quad (\gamma_{2,j}, \quad J) \quad (\text{A.31})$$

where $\{x_2, P_1, P_2\}$ and trade shares are given by

$$x_2 = \sum_{j=1}^J A_{2,j} (w_{2,j})^\kappa \left[\sum_{j=1}^J A_{2,j} w_{2,j}^\kappa \right]^{\frac{1-\kappa}{\kappa}} \bar{L}_n, \quad (\text{A.32})$$

$$\pi_{11,j} = \frac{\bar{T}_{1,j}^{\epsilon_j} (w_{1,j} (1 + \tau_j^d))^{-\epsilon_j}}{\bar{T}_{1,j}^{\epsilon_j} (w_{1,j} (1 + \tau_j^d))^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} (w_{2,j} (1 + \tau_j^m) d_{12,j})^{-\epsilon_j}}, \quad (\text{A.33})$$

$$\pi_{12,j} = \frac{\bar{T}_{2,j}^{\epsilon_j} (w_{2,j} (1 + \tau_j^m) d_{12,j})^{-\epsilon_j}}{\bar{T}_{1,j}^{\epsilon_j} (w_{1,j} (1 + \tau_j^d))^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} (w_{2,j} (1 + \tau_j^m) d_{12,j})^{-\epsilon_j}}, \quad (\text{A.34})$$

$$\pi_{21,j} = \frac{\bar{T}_{1,j}^{\epsilon_j} (w_{1,j} (1 + \tau_j^x) d_{21,j})^{-\epsilon_j}}{\bar{T}_{1,j}^{\epsilon_j} (w_{1,j} (1 + \tau_j^x) d_{21,j})^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} (w_{2,j})^{-\epsilon_j}}, \quad (\text{A.35})$$

$$\pi_{22,j} = \frac{\bar{T}_{2,j}^{\epsilon_j} (w_{2,j})^{-\epsilon_j}}{\bar{T}_{1,j}^{\epsilon_j} (w_{1,j} (1 + \tau_j^x) d_{21,j})^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} (w_{2,j})^{-\epsilon_j}}, \quad (\text{A.36})$$

$$P_1 = \prod_{j=1}^J \left[\bar{T}_{1,j}^{\epsilon_j} (w_{1,j} (1 + \tau_j^d))^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} (w_{2,j} (1 + \tau_j^m) d_{12,j})^{-\epsilon_j} \right]^{-\frac{\beta_{1,j}}{\epsilon_j}} = 1, \quad (\text{A.37})$$

$$P_2 = \prod_{j=1}^J \left[\bar{T}_{1,j}^{\epsilon_j} (w_{1,j} (1 + \tau_j^x) d_{21,j})^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} (w_{2,j})^{-\epsilon_j} \right]^{-\frac{\beta_{2,j}}{\epsilon_j}}. \quad (\text{A.38})$$

One of the equilibrium conditions in (A.29), (A.30) and (A.31) is redundant, according to Walras' law, thus, we drop the expenditure constraint in optimization and set $\gamma_x = 0$. Or, equivalently, if we use Home's budget constraint, one of the market-clearing conditions is redundant.

The FOC over $\tau_j^d, \tau_j^m, \tau_j^x$ are the same as in Appendix B. We can get the optimal policies:

$$1 + \tau_j^m = \gamma_{2,j}, \quad 1 + \tau_j^d = \gamma_{1,j}. \quad (\text{A.39})$$

$$1 + \tau_j^x = \frac{1 + \tau_j^d}{1 + \tau_j^m} \left(1 + \frac{1}{\epsilon_j \pi_{22,j}} \right). \quad (\text{A.40})$$

FOC over x_1

$$1 - \sum_{j=1}^J \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} - \sum_{j=1}^J \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \pi_{12,j} = 0.$$

FOC over $w_{1,j}$

$$\begin{aligned}
& -\beta_{1,j}x_1\pi_{11,j} + \gamma_{1,j}\beta_{1,j}\frac{1}{1+\tau_j^d}\epsilon_j\pi_{11,j}\pi_{12,j}x_1 + \gamma_{1,j}\beta_{2,j}\frac{1}{1+\tau_j^x}\epsilon_j\pi_{21,j}\pi_{22,j}x_2 - \gamma_{2,j}\beta_{1,j}\frac{1}{1+\tau_j^m}\epsilon_j\pi_{12,j}\pi_{11,j}x_1 \\
& - \gamma_{2,j}\beta_{2,j}\epsilon_j\pi_{22,j}\pi_{21,j}x_2 + \kappa\gamma_{1,j}w_{1,j}L_{1,j} - \sum_{s=1}^J \gamma_{1,s}(\kappa-1)\lambda_{1,s}w_{1,j}L_{1,j} = 0,
\end{aligned} \tag{A.41}$$

where the share of sector j income in country n as $\lambda_{n,j} = Y_{n,j}/Y_n$ and $Y_{n,j} = w_{n,j}L_{n,j}$, $Y_n = \sum_{j=1}^J Y_{n,j}$.

FOC over $w_{2,j}$

$$\begin{aligned}
& -\beta_{1,j}x_1\pi_{12,j} - \gamma_{1,j}\beta_{1,j}\frac{1}{1+\tau_j^d}\epsilon_j\pi_{11,j}\pi_{12,j}x_1 - \gamma_{1,j}\beta_{2,j}\frac{1}{1+\tau_j^x}\epsilon_j\pi_{21,j}\pi_{22,j}x_2 + \gamma_{2,j}\beta_{1,j}\frac{1}{1+\tau_j^m}\epsilon_j\pi_{12,j}\pi_{11,j}x_1 \\
& + \gamma_{2,j}\beta_{2,j}\epsilon_j\pi_{22,j}\pi_{21,j}x_2 + \kappa(\Phi_2 + \gamma_{2,j})w_{2,j}L_{2,j} - \sum_{s=1}^J (\Phi_2 + \gamma_{2,s})(\kappa-1)\lambda_{2,s}w_{2,j}L_{2,j} = 0,
\end{aligned} \tag{A.42}$$

where $\Phi_2 = -\sum_{s=1}^J (\gamma_{1,s}\beta_{2,s}\frac{1}{1+\tau_s^x}\pi_{21,s} + \gamma_{2,s}\beta_{2,s}\pi_{22,s})$.

Combining FOCs over $w_{1,j}$ (A.41), τ_j^x (A.40), and τ_j^d, τ_j^m (A.39), we get

$$\begin{aligned}
& -\beta_{1,j}x_1\pi_{11,j} + \beta_{2,j}\left(\frac{1+\tau_j^d}{1+\tau_j^x} - (1+\tau_j^m)\right)\epsilon_j\pi_{21,j}\pi_{22,j}x_2 + \kappa\gamma_{1,j}w_{1,j}L_{1,j} - \sum_{s=1}^J \gamma_{1,s}(\kappa-1)\lambda_{1,s}w_{1,j}L_{1,j} = 0 \\
\Rightarrow & -\beta_{1,j}x_1\pi_{11,j} - \beta_{2,j}\frac{1+\tau_j^d}{1+\tau_j^x}\pi_{21,j}x_2 + \kappa\gamma_{1,j}w_{1,j}L_{1,j} - \sum_{s=1}^J \gamma_{1,s}(\kappa-1)\lambda_{1,s}w_{1,j}L_{1,j} = 0 \\
\Rightarrow & (\kappa-1)(1+\tau_j^d)w_{1,j}L_{1,j} - \sum_{s=1}^J (1+\tau_s^d)(\kappa-1)\lambda_{1,s}w_{1,j}L_{1,j} = 0 \\
\Rightarrow & 1+\tau_j^d = \sum_{s=1}^J (1+\tau_s^d)\lambda_{1,s}.
\end{aligned}$$

Thus, the optimal domestic tax is uniform across sectors.

Combining FOCs over $w_{2,j}$ (A.42), τ_j^x (A.40), and τ_j^d, τ_j^m (A.39), we get

$$\begin{aligned}
& \kappa(\Phi_2 + \gamma_{2,j})Y_{2,j} - \sum_{s=1}^J (\Phi_2 + \gamma_{2,s})(\kappa - 1)\lambda_{2,s}Y_{2,s} = \beta_{1,j}\pi_{12,j}x_1 + \left[\frac{1 + \tau_j^d}{1 + \tau_j^x} - (1 + \tau_j^m) \right] \beta_{2,j}\epsilon_j\pi_{21,j}\pi_{22,j}x_2, \\
& \Rightarrow \kappa(\Phi_2 + 1 + \tau_j^m) - \sum_{s=1}^J (\Phi_2 + 1 + \tau_s^m)(\kappa - 1)\lambda_{2,s} = \frac{\beta_{1,j}\pi_{12,j}x_1}{Y_{2,j}} - \frac{1 + \tau_j^d}{1 + \tau_j^x} \beta_{2,j}\pi_{21,j} \frac{x_2}{Y_{2,j}} \\
& \Rightarrow \kappa(\Phi_2 + 1 + \tau_j^m) - \sum_{s=1}^J (\Phi_2 + 1 + \tau_s^m)(\kappa - 1)\lambda_{2,s} = \frac{\beta_{1,j}\pi_{12,j}x_1}{Y_{2,j}} - (1 + \tau_j^d) \frac{Y_{1,j}}{Y_{2,j}} + \frac{\beta_{1,j}\pi_{11,j}x_1}{Y_{2,j}} \\
& \Rightarrow (\Phi_2 + 1 + \tau_j^m) - \sum_{s=1}^J (\Phi_2 + 1 + \tau_s^m) \frac{\kappa - 1}{\kappa} \lambda_{2,s} = \frac{1}{\kappa} \frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}}. \tag{A.43}
\end{aligned}$$

Finally, we take the difference of (A.43) between sectors j and k and derive the formula for optimal tariffs

$$\tau_j^m - \tau_k^m = \frac{1}{\kappa} \left[\frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}} - \frac{\beta_{1,k}x_1 - (1 + \tau_k^d)Y_{1,k}}{Y_{2,k}} \right]. \tag{A.44}$$

C.1.2 Method 2: Applying Proposition 3

We first prove that the one-factor Ricardo-Roy is a GIIA Supply System, then apply the optimal policy formulas from Proposition 3 using the structural parameters from this model.

We can derive supply curves by combining the labor market specification Ω (Eq.A.28) with the supply-side assumption S (Eq.A.27) and the definition of income $Y_{n,j} = w_{n,j}L_{n,j}$. Eliminating the wage and labor $\{w_{n,j}, L_{n,j}\}$ yields the following supply curves:

$$Y_{n,j} = A_{n,j} (p_{n,j}\bar{T}_{n,j})^\kappa \left[\sum_{s=1}^J A_{n,s} (p_{n,s}\bar{T}_{n,s})^\kappa \right]^{\frac{1-\kappa}{\kappa}} \bar{L}_n.$$

The corresponding element of the scaled partial supply elasticity matrix Λ_n is given by

$$\frac{\partial \ln(Y_{n,s})}{\partial \ln(p_{n,j})} \frac{Y_{n,s}}{Y_{n,j}} = I_{s=j}(\kappa) - (\kappa - 1)\lambda_{n,s},$$

where $I_{s=j}(\kappa)$ is an indicator function that equals κ when $s = j$ and 0 otherwise, $\lambda_{n,j} = Y_{n,j}/Y_n$ is the share of income in sector j produced in country n .

The scaled partial supply elasticity matrix Λ_n of country n takes the form of

$$\Lambda_n = \begin{pmatrix} \kappa - (\kappa - 1)\lambda_{n,1} & \dots & -(\kappa - 1)\lambda_{n,j} & \dots & -(\kappa - 1)\lambda_{n,J} \\ \dots & \dots & \dots & \dots & \dots \\ -(\kappa - 1)\lambda_{n,1} & \dots & \kappa - (\kappa - 1)\lambda_{n,j} & \dots & -(\kappa - 1)\lambda_{n,J} \\ \dots & \dots & \dots & \dots & \dots \\ -(\kappa - 1)\lambda_{n,1} & \dots & -(\kappa - 1)\lambda_{n,j} & \dots & \kappa - (\kappa - 1)\lambda_{n,J} \end{pmatrix}. \quad (\text{A.45})$$

This matrix is invertible since

$$\det(\Lambda_n) = \kappa^J \left[1 - \frac{\kappa - 1}{\kappa} \lambda_n^T \mathbf{1} \right] = \kappa^J \left(1 - \frac{\kappa - 1}{\kappa} \right) = \kappa^{J-1} \neq 0.$$

Obviously, this model satisfies the requirement of the GIIA Supply System. The partial elasticities of a sector's income with respect to price (scaled by the two sectors' size and elasticity parameter) satisfy $\frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,j}} \frac{Y_{n,s}}{Y_{n,j}} \frac{1}{\eta_j} - \frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,i}} \frac{Y_{n,s}}{Y_{n,i}} \frac{1}{\eta_i} = 0$ for $s \neq i$ or j , and $\frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,j}} \frac{Y_{n,s}}{Y_{n,j}} \frac{1}{\eta_j} - \frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,i}} \frac{Y_{n,s}}{Y_{n,i}} \frac{1}{\eta_i} = 1$ for $s = j$, where the parameter $\eta_j = \kappa$.

Proof of optimal domestic taxes According to Proposition 2, optimal domestic taxes satisfy

$$(\Lambda_1 - I)(1 + \tau^d) = 0.$$

Consider two rows, j th and k th of (A.45),

$$\begin{aligned} -(\kappa - 1) \sum_{s=1}^J \lambda_{2,s} (1 + \tau_s^d) + (\kappa - 1)(1 + \tau_j^d) &= 0, \\ -(\kappa - 1) \sum_{s=1}^J \lambda_{2,s} (1 + \tau_s^d) + (\kappa - 1)(1 + \tau_k^d) &= 0. \end{aligned}$$

The difference of the two implies

$$(\kappa - 1)(\tau_j^d - \tau_k^d) = 0.$$

This indicates $\tau_j^d = \tau_k^d$ for any k, j . Thus, the domestic tax is uniform across all sectors.

Proof of optimal import tariffs According to Proposition 2, the optimal tariffs satisfy

$$\sum_{k=1}^J \tau_k^m \frac{\partial \ln(Y_{2,k})}{\partial \ln(p_{2,j})} \frac{Y_{2,k}}{Y_{2,j}} = \frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}}, \forall j$$

Consider two rows, j th and k th of (A.45)

$$\begin{aligned} -(\kappa - 1) \sum_{s=1}^J \lambda_{2,s} \tau_s^m + \kappa \tau_j^m &= \frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}}, \\ -(\kappa - 1) \sum_{s=1}^J \lambda_{2,s} \tau_s^m + \kappa \tau_k^m &= \frac{\beta_{1,k}x_1 - (1 + \tau_k^d)Y_{1,k}}{Y_{2,k}}. \end{aligned}$$

The difference of the two implies

$$\begin{aligned} \kappa(\tau_j^m - \tau_k^m) &= \frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}} - \frac{\beta_{1,k}x_1 - (1 + \tau_k^d)Y_{1,k}}{Y_{2,k}}, \\ \Rightarrow \tau_j^m - \tau_k^m &= \frac{1}{\kappa} \left[\frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}} - \frac{\beta_{1,k}x_1 - (1 + \tau_k^d)Y_{1,k}}{Y_{2,k}} \right]. \end{aligned}$$

The optimal policies align with Proposition 3 under the GIIA Supply System, where the parameter $\eta_j = \kappa$. Furthermore, this formula is the same as (A.44) that we derived directly using the Lagrange method. $\square$

C.2 Generalized Multi-sector Krugman Model

In Section C.2.1, we formulate the Home government's problem and the world market equilibrium based on the original equilibrium conditions of the generalized multi-sector Krugman model. We solve $\{p_{n,j}\}$ as functions of wages and labor $\{w_{n,j}, L_{n,j}\}$ using the model-specific supply function $S(\{p_{n,j}, w_{n,j}, L_{n,j}\})$. We then directly take first-order conditions with respect to $\{w_{n,j}, L_{n,j}\}$ and impose multipliers on the labor market specifications Ω to derive optimal policies in this model. In Section C.2.2, we show that this model is a GIIA Supply System and employ matrix operations to establish a connection with Proposition 3.

Furthermore, in Section C.2.3, we demonstrate that the multi-sector Armington and Eaton and Kortum (2002) model with external economies of scale, and model with endogenous technology but different returns to scale across sectors are isomorphic to this model.

C.2.1 Method 1: Model Specific Proof

In the generalized multi-sector Krugman model, labor is perfectly mobile, which indicates $w_{n,j} = w_n$ for any j . Due to the increasing returns to scale, prices are affected by both wages and labor. The supply-side assumption S satisfies

$$p_{n,j} = \frac{w_n}{\bar{T}_{n,j} L_{n,j}^{\psi_j}}, \quad (\text{A.46})$$

where $\bar{T}_{n,j}$ denotes the exogenous technology level beyond the economies of scale in sector j of country n and $\psi_j = 1/(\sigma_j - 1)$ is the scale elasticity in sector j . In the following market equilibrium constraints, we substitute $p_{n,j}$ with w_n and $L_{n,j}$.

Home government solves the following problem:

$$\max_{\{\tau_j^d, \tau_j^x, \tau_j^m, x_1, w_1, w_2, L_{1,j}, L_{2,j}\}} \frac{x_1}{P_1},$$

subject to world market equilibrium characterized by the following constraints:

$$x_1 = \sum_{j=1}^J w_1 L_{1,j} + \sum_{j=1}^J \beta_{2,j} \frac{\tau_j^x}{1 + \tau_j^x} \pi_{21,j} x_2 + \sum_{j=1}^J \beta_{1,j} \frac{\tau_j^m}{1 + \tau_j^m} \pi_{12,j} x_1 + \sum_{j=1}^J \beta_{1,j} \frac{\tau_j^d}{1 + \tau_j^d} \pi_{11,j} x_1, \quad (\gamma_x) \quad (\text{A.47})$$

$$\beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} x_1 = w_1 L_{1,j} - \beta_{2,j} \frac{1}{1 + \tau_j^x} \pi_{21,j} x_2, \quad (\gamma_{1,j}, \quad J) \quad (\text{A.48})$$

$$\beta_{1,j} \frac{1}{1 + \tau_j^m} \pi_{12,j} x_1 = w_2 L_{2,j} - \beta_{2,j} \pi_{22,j} x_2, \quad (\gamma_{2,j}, \quad J) \quad (\text{A.49})$$

The labor market specifications are

$$\sum_{j=1}^J L_{1,j} = \bar{L}_1, \quad (\gamma_{L1}) \quad (\text{A.50})$$

$$\sum_{j=1}^J L_{2,j} = \bar{L}_2, \quad (\gamma_{L2}) \quad (\text{A.51})$$

$\{x_2, P_1, P_2\}$ and trade shares are given by

$$x_2 = \sum_{j=1}^J w_2 L_{2,j}, \quad (\text{A.52})$$

$$\pi_{11,j} = \frac{\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_1 (1 + \tau_j^d))^{-\epsilon_j}}{\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_1 (1 + \tau_j^d))^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} (w_2 (1 + \tau_j^m) d_{12,j})^{-\epsilon_j}}, \quad (\text{A.53})$$

$$\pi_{12,j} = \frac{\bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} (w_2 (1 + \tau_j^m) d_{12,j})^{-\epsilon_j}}{\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_1 (1 + \tau_j^d))^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} (w_2 (1 + \tau_j^m) d_{12,j})^{-\epsilon_j}}, \quad (\text{A.54})$$

$$\pi_{21,j} = \frac{\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_1 (1 + \tau_j^x) d_{21,j})^{-\epsilon_j}}{\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_1 (1 + \tau_j^x) d_{21,j})^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} (w_2)^{-\epsilon_j}}, \quad (\text{A.55})$$

$$\pi_{22,j} = \frac{\bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} (w_2)^{-\epsilon_j}}{\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_1 (1 + \tau_j^x) d_{21,j})^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} (w_2)^{-\epsilon_j}}, \quad (\text{A.56})$$

$$P_1 = \prod_{j=1}^J \left[\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_1 (1 + \tau_j^d))^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} (w_2 (1 + \tau_j^m) d_{12,j})^{-\epsilon_j} \right]^{-\frac{\beta_{1,j}}{\epsilon_j}} = 1, \quad (\text{A.57})$$

$$P_2 = \prod_{j=1}^J \left[\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_1 (1 + \tau_j^x) d_{21,j})^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} w_2^{-\epsilon_j} \right]^{-\frac{\beta_{2,j}}{\epsilon_j}}. \quad (\text{A.58})$$

One of the equilibrium conditions in (A.47), (A.48) and (A.49) is redundant, according to Walras' law, thus, we drop the expenditure constraint in optimization and set $\gamma_x = 0$. Or, equivalently, if we use Home's budget constraint, one of the market-clearing conditions is redundant. The derivation of the market equilibrium constraints is provided in Appendix C.2.3.

The FOC over $\tau_j^d, \tau_j^m, \tau_j^x$ are the same as in Appendix B. We can get the optimal policies:

$$1 + \tau_j^m = \gamma_{2,j}, \quad 1 + \tau_j^d = \gamma_{1,j}. \quad (\text{A.59})$$

$$1 + \tau_j^x = \frac{1 + \tau_j^d}{1 + \tau_j^m} \left(1 + \frac{1}{\epsilon_j \pi_{22,j}} \right). \quad (\text{A.60})$$

FOC over x_1

$$1 - \sum_{j=1}^J \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} - \sum_{j=1}^J \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \pi_{12,j} = 0.$$

FOC over w_1

$$\begin{aligned}
& - \sum_{j=1}^J \beta_{1,j} x_1 \pi_{11,j} + \sum_{j=1}^J \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j \pi_{11,j} \pi_{12,j} x_1 + \sum_{j=1}^J \gamma_{1,j} \beta_{2,j} \frac{1}{1 + \tau_j^x} \epsilon_j \pi_{21,j} \pi_{22,j} x_2 \\
& - \sum_{j=1}^J \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \epsilon_j \pi_{12,j} \pi_{11,j} x_1 - \sum_{j=1}^J \gamma_{2,j} \beta_{2,j} \epsilon_j \pi_{22,j} \pi_{21,j} x_2 + \sum_{j=1}^J \gamma_{1,j} w_1 L_{1,j} = 0. \tag{A.61}
\end{aligned}$$

FOC over w_2

$$\begin{aligned}
& - \sum_{j=1}^J \beta_{1,j} x_1 \pi_{12,j} - \sum_{j=1}^J \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j \pi_{11,j} \pi_{12,j} x_1 - \sum_{j=1}^J \gamma_{1,j} \beta_{2,j} \frac{1}{1 + \tau_j^x} \epsilon_j \pi_{21,j} \pi_{22,j} x_2 \\
& + \sum_{j=1}^J \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \epsilon_j \pi_{12,j} \pi_{11,j} x_1 + \sum_{j=1}^J \gamma_{2,j} \beta_{2,j} \epsilon_j \pi_{22,j} \pi_{21,j} x_2 + \sum_{j=1}^J (\Phi_2 + \gamma_{2,j}) w_2 L_{2,j} = 0, \tag{A.62}
\end{aligned}$$

where $\Phi_2 = - \sum_{s=1}^J (\gamma_{1,s} \beta_{2,s} \frac{1}{1 + \tau_s^x} \pi_{21,s} + \gamma_{2,s} \beta_{2,s} \pi_{22,s})$.

FOC over $L_{1,j}$

$$\begin{aligned}
& \psi_j \beta_{1,j} x_1 \pi_{11,j} - \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j \psi_j \pi_{11,j} \pi_{12,j} x_1 - \gamma_{1,j} \beta_{2,j} \frac{1}{1 + \tau_j^x} \epsilon_j \psi_j \pi_{21,j} \pi_{22,j} x_2 \\
& + \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \epsilon_j \psi_j \pi_{12,j} \pi_{11,j} x_1 + \gamma_{2,j} \beta_{2,j} \epsilon_j \psi_j \pi_{22,j} \pi_{21,j} x_2 + \gamma_{1,j} w_1 L_{1,j} - \gamma_{L1} L_{1,j} = 0. \tag{A.63}
\end{aligned}$$

FOC over $L_{2,j}$

$$\begin{aligned}
& \psi_j \beta_{1,j} x_1 \pi_{12,j} + \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j \psi_j \pi_{11,j} \pi_{12,j} x_1 + \gamma_{1,j} \beta_{2,j} \frac{1}{1 + \tau_j^x} \epsilon_j \psi_j \pi_{21,j} \pi_{22,j} x_2 \\
& - \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \epsilon_j \psi_j \pi_{12,j} \pi_{11,j} x_1 - \gamma_{2,j} \beta_{2,j} \epsilon_j \psi_j \pi_{22,j} \pi_{21,j} x_2 + (\Phi_2 + \gamma_{2,j}) w_2 L_{2,j} - \gamma_{L2} L_{2,j} = 0. \tag{A.64}
\end{aligned}$$

Combining FOCs over $L_{1,j}$ (A.63), τ_j^x (A.60), and τ_j^d, τ_j^m (A.59), we get

$$\begin{aligned}
& \psi_j \beta_{1,j} x_1 \pi_{11,j} - \beta_{2,j} \left(\frac{1 + \tau_j^d}{1 + \tau_j^x} - (1 + \tau_j^m) \right) \epsilon_j \psi_j \pi_{21,j} \pi_{22,j} x_2 + \gamma_{1,j} w_1 L_{1,j} - \gamma_{L1} L_{1,j} = 0 \\
& \Rightarrow \psi_j \beta_{1,j} x_1 \pi_{11,j} + \beta_{2,j} \frac{1 + \tau_j^d}{1 + \tau_j^x} \psi_j \pi_{21,j} x_2 + \gamma_{1,j} w_1 L_{1,j} - \gamma_{L1} L_{1,j} = 0 \\
& \Rightarrow \gamma_{1,j} w_1 L_{1,j} - \gamma_{L1} L_{1,j} + \psi_j (1 + \tau_j^d) w_1 L_{1,j} = 0 \\
& \Rightarrow (\psi_j + 1) (1 + \tau_j^d) = \gamma_{L1} \frac{1}{w_1}.
\end{aligned}$$

Hence, for any sectors j and k , the domestic taxes satisfy

$$\frac{1 + \tau_j^d}{1 + \tau_k^d} = \frac{1 + \psi_k}{1 + \psi_j}.$$

Combining FOCs over $L_{2,j}$ (A.64), τ_j^x (A.60), and τ_j^d, τ_j^m (A.59), we get

$$\begin{aligned} & -(\Phi_2 + \gamma_{2,j})w_2L_{2,j} + \gamma_{L2}L_{2,j} = \psi_j\beta_{1,j}\pi_{12,j}x_1 + \left[\frac{1 + \tau_j^d}{1 + \tau_j^x} - (1 + \tau_j^m) \right] \beta_{2,j}\epsilon_j\psi_j\pi_{21,j}\pi_{22,j}x_2 \\ \Rightarrow & -(\Phi_2 + 1 + \tau_j^m) + \gamma_{L2}\frac{1}{w_2} = \psi_j\frac{\beta_{1,j}\pi_{12,j}x_1}{Y_{2,j}} - \frac{1 + \tau_j^d}{1 + \tau_j^x}\beta_{2,j}\psi_j\pi_{21,j}\frac{x_2}{Y_{2,j}} \\ \Rightarrow & -(\Phi_2 + 1 + \tau_j^m) + \gamma_{L2}\frac{1}{w_2} = \psi_j\frac{\beta_{1,j}\pi_{12,j}x_1}{Y_{2,j}} - \psi_j(1 + \tau_j^d)\frac{Y_{1,j}}{Y_{2,j}} + \psi_j\frac{\beta_{1,j}\pi_{11,j}x_1}{Y_{2,j}} \\ \Rightarrow & -(\Phi_2 + 1 + \tau_j^m) + \gamma_{L2}\frac{1}{w_2} = \psi_j\frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}}. \end{aligned} \quad (\text{A.65})$$

Finally, we take the difference of (A.65) between sectors j and k and derive the formula for optimal tariffs

$$\tau_j^m - \tau_k^m = -\psi_j\frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}} + \psi_k\frac{\beta_{1,k}x_1 - (1 + \tau_k^d)Y_{1,k}}{Y_{2,k}}, \quad (\text{A.66})$$

where $Y_{n,j} = w_nL_{n,j}$ and $\psi_j = 1/(\sigma_j - 1)$.

C.2.2 Method 2: Applying Proposition 3

We first prove that the generalized multi-sector Krugman is a GIIA Supply System, then apply the optimal policy formulas from Proposition 3 using the structural parameters from this model.

We begin by deriving $L_{n,j}$ as a function of $p_{n,j}$ and w_n , from equation (A.46), then summing them up to derive w_n as a function of $\bar{L}_n, p_{n,j}$ and other sectors' prices. Then, we use the implicit function

theorem to obtain $\frac{\partial \ln w_n}{\partial \ln p_{n,j}}$.

$$\begin{aligned}
p_{n,j} &= \bar{T}_{n,j}^{-1} L_{n,j}^{-\psi_j} w_n \\
\Rightarrow L_{n,j} &= \left(p_{n,j} \bar{T}_{n,j} w_n^{-1} \right)^{-\frac{1}{\psi_j}} \\
\Rightarrow \bar{L}_n &= \sum_{j=1}^J \left(p_{n,j} \bar{T}_{n,j} \right)^{-\frac{1}{\psi_j}} w_n^{\frac{1}{\psi_j}} \\
\Rightarrow \sum_{j=1}^J \frac{1}{\psi_j} \left(p_{n,j} \bar{T}_{n,j} \right)^{-\frac{1}{\psi_j}} w_n^{\frac{1}{\psi_j}} \frac{\partial \ln w_n}{\partial \ln p_{n,j}} &= \frac{1}{\psi_j} \left(p_{n,j} \bar{T}_{n,j} \right)^{-\frac{1}{\psi_j}} w_n^{\frac{1}{\psi_j}} \\
\Rightarrow \frac{\partial \ln w_n}{\partial \ln p_{n,j}} &= \frac{\frac{1}{\psi_j} \left(p_{n,j} \bar{T}_{n,j} \right)^{-\frac{1}{\psi_j}} w_n^{\frac{1}{\psi_j}}}{\sum_{s=1}^J \frac{1}{\psi_s} \left(p_{n,s} \bar{T}_{n,s} \right)^{-\frac{1}{\psi_s}} w_n^{\frac{1}{\psi_s}}} = \frac{\frac{1}{\psi_j} Y_{n,j}}{\sum_{s=1}^J \frac{1}{\psi_s} Y_{n,s}}. \tag{A.67}
\end{aligned}$$

Since the sectoral income is $Y_{n,j} = w_n L_{n,j}$, the supply curves in country n are given by,

$$Y_{n,j} = w_n^{\frac{1+\psi_j}{\psi_j}} \left(p_{n,j} \bar{T}_{n,j} \right)^{-\frac{1}{\psi_j}}, \quad \text{where } w_n \text{ satisfies } \sum_{j=1}^J \left(p_{n,j} \bar{T}_{n,j} \right)^{-\frac{1}{\psi_j}} w_n^{\frac{1}{\psi_j}} = \bar{L}_n. \tag{A.68}$$

We then derive scaled partial supply elasticities $\frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,j}} \frac{Y_{n,s}}{Y_{n,j}}$ in Λ_n by incorporating (A.67) and (A.68).

$$\begin{aligned}
\frac{\partial \ln Y_{n,k}}{\partial \ln p_{n,j}} \frac{Y_{n,k}}{Y_{n,j}} &= I_{k=j} \left(-\frac{1}{\psi_j} \right) + \left(1 + \frac{1}{\psi_k} \right) \frac{Y_{n,k}}{Y_{n,j}} \frac{\partial \ln w_n}{\partial \ln p_{n,j}} \\
&= I_{k=j} \left(-\frac{1}{\psi_j} \right) + \left(1 + \frac{1}{\psi_k} \right) \frac{\frac{1}{\psi_j} Y_{n,k}}{\sum_{s=1}^J \frac{1}{\psi_s} Y_{n,s}} \\
&= I_{k=j} \left(-\frac{1}{\psi_j} \right) + \frac{1}{\psi_j} \frac{\left(1 + \frac{1}{\psi_k} \right) Y_{n,k}}{\sum_{s=1}^J \frac{1}{\psi_s} Y_{n,s}}, \tag{A.69}
\end{aligned}$$

where $I_{s=j} \left(-\frac{1}{\psi_j} \right)$ is an indicator function that equals $-\frac{1}{\psi_j}$ when $s = j$, and 0 otherwise.

The partial supply elasticity matrix Λ_n of country n takes the form of

$$\Lambda_n = \begin{pmatrix} -\frac{1}{\psi_1} \left(1 - \frac{(1 + \frac{1}{\psi_1})Y_{n,1}}{\sum_{s=1}^J \frac{1}{\psi_s} Y_{n,s}}\right) & \dots & \frac{1}{\psi_1} \frac{(1 + \frac{1}{\psi_j})Y_{n,j}}{\sum_{s=1}^J \frac{1}{\psi_s} Y_{n,s}} & \dots & \frac{1}{\psi_1} \frac{(1 + \frac{1}{\psi_J})Y_{n,J}}{\sum_{s=1}^J \frac{1}{\psi_s} Y_{n,s}} \\ \dots & \dots & \dots & \dots & \dots \\ \frac{1}{\psi_j} \frac{(1 + \frac{1}{\psi_1})Y_{n,1}}{\sum_{s=1}^J \frac{1}{\psi_s} Y_{n,s}} & \dots & -\frac{1}{\psi_j} \left(1 - \frac{(1 + \frac{1}{\psi_j})Y_{n,j}}{\sum_{s=1}^J \frac{1}{\psi_s} Y_{n,s}}\right) & \dots & \frac{1}{\psi_j} \frac{(1 + \frac{1}{\psi_J})Y_{n,J}}{\sum_{s=1}^J \frac{1}{\psi_s} Y_{n,s}} \\ \dots & \dots & \dots & \dots & \dots \\ \frac{1}{\psi_J} \frac{(1 + \frac{1}{\psi_1})Y_{n,1}}{\sum_{s=1}^J \frac{1}{\psi_s} Y_{n,s}} & \dots & \frac{1}{\psi_J} \frac{(1 + \frac{1}{\psi_j})Y_{n,j}}{\sum_{s=1}^J \frac{1}{\psi_s} Y_{n,s}} & \dots & -\frac{1}{\psi_J} \left(1 - \frac{(1 + \frac{1}{\psi_J})Y_{n,J}}{\sum_{s=1}^J \frac{1}{\psi_s} Y_{n,s}}\right) \end{pmatrix}. \quad (\text{A.70})$$

This matrix is invertible since

$$\det(\Lambda_n) = (-1)^{J+1} \left(\prod_{j=1}^J \frac{1}{\psi_j} \right) \frac{\sum_j Y_{n,j}}{\sum_j Y_{n,j} / \psi_j} \neq 0.$$

The formula for Λ_n shows that this model satisfies the GIIA Supply System in Definition 4. The partial elasticities of a sector's income with respect to price (scaled by the two sectors' size and elasticity parameter) satisfy $\frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,j}} \frac{Y_{n,s}}{Y_{n,j}} \frac{1}{\eta_j} - \frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,i}} \frac{Y_{n,s}}{Y_{n,i}} \frac{1}{\eta_i} = 0$ for $s \neq i$ or j , and $\frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,j}} \frac{Y_{n,s}}{Y_{n,j}} \frac{1}{\eta_j} - \frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,i}} \frac{Y_{n,s}}{Y_{n,i}} \frac{1}{\eta_i} = 1$ for $s = j$, where the parameter

$$\frac{1}{\eta_j} = -\psi_j = \frac{1}{1 - \sigma_j}.$$

Proof of optimal domestic taxes According to Proposition 2, optimal domestic taxes satisfy

$$(\Lambda_1 - I)(1 + \tau^d) = 0.$$

Consider two rows, j th and k th of (A.70)

$$-\frac{1}{\psi_j}(1 + \tau_j^d) + \frac{1}{\psi_j} \sum_{s=1}^J \frac{(1 + \frac{1}{\psi_s})Y_{n,s}}{\sum_{i=1}^J \frac{1}{\psi_i} Y_{n,i}} (1 + \tau_s^d) = (1 + \tau_j^d), \quad (\text{A.71})$$

$$-\frac{1}{\psi_k}(1 + \tau_k^d) + \frac{1}{\psi_k} \sum_{s=1}^J \frac{(1 + \frac{1}{\psi_s})Y_{n,s}}{\sum_{i=1}^J \frac{1}{\psi_i} Y_{n,i}} (1 + \tau_s^d) = (1 + \tau_k^d). \quad (\text{A.72})$$

Multiplying (A.71) and (A.72) by $-\psi_j$ and $-\psi_k$ respectively, and subtracting (A.72) from (A.71), we have

$$\begin{aligned} (1 + \tau_j^d) - (1 + \tau_k^d) &= -(1 + \tau_j^d)\psi_j + (1 + \tau_k^d)\psi_k \\ \Rightarrow \frac{1 + \tau_j^d}{1 + \tau_k^d} &= \frac{1 + \psi_k}{1 + \psi_j} = \frac{\frac{\eta_j}{\eta_j - 1}}{\frac{\eta_k}{\eta_k - 1}}, \end{aligned}$$

where $\frac{1}{\eta_j} = -\psi_j = \frac{1}{1 - \sigma_j}$ and ψ_j is the scale elasticity in sector j . When ψ_j is identical across sectors, we get the uniform domestic tax across sectors.

Proof of optimal import tariffs According to Proposition 2, the optimal tariffs satisfy

$$\sum_{k=1}^J \tau_k^m \frac{\partial \ln(Y_{2,k})}{\partial \ln(p_{2,j})} \frac{Y_{2,k}}{Y_{2,j}} = \frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}}, \forall j$$

Consider two rows, j th and k th of (A.70),

$$-\frac{1}{\psi_j} \tau_j^m + \frac{1}{\psi_j} \sum_{s=1}^J \frac{(1 + \frac{1}{\psi_s})Y_{n,s}}{\sum_{i=1}^J \frac{1}{\psi_i} Y_{n,i}} \tau_s^m = \frac{M_{1,j}}{Y_{2,j}}, \quad (\text{A.73})$$

$$-\frac{1}{\psi_k} \tau_k^m + \frac{1}{\psi_k} \sum_{s=1}^J \frac{(1 + \frac{1}{\psi_s})Y_{n,s}}{\sum_{i=1}^J \frac{1}{\psi_i} Y_{n,i}} \tau_s^m = \frac{M_{1,k}}{Y_{2,k}}, \quad (\text{A.74})$$

Multiplying (A.73) and (A.74) by $-\psi_j$ and $-\psi_k$ respectively, and subtracting (A.74) from (A.73), we have

$$\begin{aligned} \tau_j^m - \tau_k^m &= -\psi_j \frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}} + \psi_k \frac{\beta_{1,k}x_1 - (1 + \tau_k^d)Y_{1,k}}{Y_{2,k}} \\ \Rightarrow \tau_j^m - \tau_k^m &= \frac{1}{1 - \sigma_j} \frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}} - \frac{1}{1 - \sigma_k} \frac{\beta_{1,k}x_1 - (1 + \tau_k^d)Y_{1,k}}{Y_{2,k}}. \end{aligned}$$

The optimal policies align with Proposition 3 under the GIIA Supply System, where the parameter $1/\eta_j = -\psi_j$. Furthermore, this formula is the same as (A.66) that we derived directly using the Lagrange method.

C.2.3 Isomorphism Between the Generalized Multi-Sector Krugman Model, Multi-Sector Armington, and Eaton-Kortum model with EES and the Endogenous Technology Model

Kucheryavyy, Lyn, and Rodríguez-Clare (2023) proves the isomorphic gravity equations for models with varying returns to scale, including the generalized multi-sector Krugman model and multi-sector Armington and EK model with external economies of scale (EES). We also revisit Bai, Jin, and Lu (2025), who extend Eaton and Kortum (2001) with endogenous technology to a multi-sector setting. We provide proofs demonstrating that these models are isomorphic to the generalized multi-sector Krugman model. Consequently, the optimal policies align with those derived in Appendix C.2.2.

Generalized Multi-sector Krugman model Each sector contains a continuum of differentiated varieties. Preferences are structured as follows: Cobb-Douglas across sectors in country n with weights $\beta_{n,j}$, CES across country bundles within a sector j with an elasticity of substitution ϕ_j , and CES across varieties within a country bundle with elasticity σ_j . The corresponding demand function for a representative variety in sector j produced in country i and consumed in country n is:

$$q_{ni,j} = p_{ni,j}^{-\sigma_j} P_{ni,j}^{\sigma_j-1} \left(\frac{P_{ni,j}}{P_{n,j}} \right)^{1-\phi_j} \beta_{n,j} x_n,$$

where $p_{ni,j}$ is the price of the variety. $P_{ni,j} = M_{i,j}^{1/(1-\sigma_j)} p_{ni,j}$ is the price index in country n of country i varieties of sector j . $M_{i,j}$ is the measure of firms or the measure of varieties produced in country i sector j . $P_{n,j} = (\sum_i P_{ni,j}^{1-\phi_j})^{1/(1-\phi_j)}$ is the sector price index in country n .

Let $A_{i,j}$ represent the exogenous productivity in sector j of country i , which is common across all firms in that sector. Similarly, let $F_{i,j}$ denote the fixed cost (in terms of labor) associated with producing any variety in sector j of country i . Under monopolistic competition, the price index in country n for varieties from sector j in country i can be expressed as:

$$P_{ni,j} = M_{i,j}^{1/(1-\sigma_j)} (\bar{\sigma}_j w_i (1 + \tau_{ni,j}) d_{ni,j} / A_{i,j}),$$

where $\bar{\sigma}_j \equiv \sigma_j / (\sigma_j - 1)$ is the markup. When $n \neq i$, $\tau_{ni,j}$ denotes the import tariff imposed by country n on sector j imports from country i . When $n = i$, $\tau_{nn,j}$ represents the domestic tax imposed by country n on sector j . Denoting the revenue of the representative firm from sector j in country i

by $R_{i,j} \equiv \sum_n \frac{1}{1+\tau_{ni,j}} p_{ni,j} q_{ni,j}$, the corresponding profit is then given by $R_{i,j}/\sigma_j$.

$$\begin{aligned} \frac{R_{i,j}}{\sigma_j} &= \frac{1}{\sigma_j} \sum_n \frac{1}{1+\tau_{ni,j}} \left(\frac{p_{ni,j}}{P_{ni,j}}\right)^{1-\sigma_j} \left(\frac{P_{ni,j}}{P_{n,j}}\right)^{1-\phi_j} \beta_{n,j} x_n = \frac{M_{i,j}^{-1}}{\sigma_j} \sum_n \frac{1}{1+\tau_{ni,j}} \left(\frac{P_{ni,j}}{P_{n,j}}\right)^{1-\phi_j} \beta_{n,j} x_n = \frac{M_{i,j}^{-1}}{\sigma_j} \sum_n \frac{\pi_{ni,j}}{1+\tau_{ni,j}} \beta_{n,j} x_n \\ &= \frac{M_{i,j}^{-1}}{\sigma_j} w_i L_{i,j}. \end{aligned}$$

where sector-level trade shares are

$$\pi_{ni,j} = \left(\frac{P_{ni,j}}{P_{n,j}}\right)^{1-\phi_j} = \frac{A_{i,j}^{\phi_j-1} M_{i,j}^{(\phi_j-1)/(\sigma_j-1)} (w_i(1+\tau_{ni,j})d_{ni,j})^{1-\phi_j}}{\sum_l A_{l,j}^{\phi_j-1} M_{l,j}^{(\phi_j-1)/(\sigma_j-1)} (w_l(1+\tau_{nl,j})d_{nl,j})^{1-\phi_j}}.$$

Free entry then implies,

$$\frac{R_{i,j}}{\sigma_j} - F_{i,j} w_i = 0 \quad \Rightarrow \quad M_{i,j} = \frac{L_{i,j}}{\sigma_j F_{i,j}}.$$

Total labor $L_{i,j}$ must be consistent with the amounts produced for each market $L_{i,j}^p$, plus the fixed cost of entry, and labor markets must clear, $\sum_{j=1}^J L_{i,j} = \bar{L}_i$.

$$\begin{aligned} L_{i,j} &= M_{i,j} \sum_n q_{ni,j} d_{ni,j} / A_{i,j} + M_{i,j} F_{i,j} = M_{i,j} \sum_n p_{ni,j}^{-\sigma_j} P_{ni,j}^{\sigma_j-1} \left(\frac{P_{ni,j}}{P_{n,j}}\right)^{1-\phi_j} d_{ni,j} \beta_{n,j} x_n / A_{i,j} + \frac{1}{\sigma_j} \sum_n \frac{1}{1+\tau_{ni,j}} \pi_{ni,j} \beta_{n,j} x_n / w_i \\ &= \frac{1}{\sigma_j} \sum_n \frac{1}{1+\tau_{ni,j}} \left(\frac{P_{ni,j}}{P_{n,j}}\right)^{1-\phi_j} \beta_{n,j} x_n / w_i + \frac{1}{\sigma_j} \sum_n \frac{1}{1+\tau_{ni,j}} \pi_{ni,j} \beta_{n,j} x_n / w_i \\ &= \frac{1}{\sigma_j} \sum_n \frac{1}{1+\tau_{ni,j}} \pi_{ni,j} \beta_{n,j} x_n / w_i + \frac{1}{\sigma_j} \sum_n \frac{1}{1+\tau_{ni,j}} \pi_{ni,j} \beta_{n,j} x_n / w_i = \sum_n \frac{1}{1+\tau_{ni,j}} \pi_{ni,j} \beta_{n,j} x_n / w_i = L_{i,j}. \\ L_{i,j}^p &= \frac{\sigma_j-1}{\sigma_j} L_{i,j}. \end{aligned} \tag{A.75}$$

(A.75) implies that production labor maintains a constant share of total labor.

Trade shares are then given by

$$\pi_{ni,j} = \left(\frac{P_{ni,j}}{P_{n,j}}\right)^{1-\phi_j} = \frac{A_{i,j}^{\phi_j-1} F_{i,j}^{-(\phi_j-1)/(\sigma_j-1)} L_{i,j}^{(\phi_j-1)/(\sigma_j-1)} (w_i(1+\tau_{ni,j})d_{ni,j})^{1-\phi_j}}{\sum_l A_{l,j}^{\phi_j-1} F_{l,j}^{-(\phi_j-1)/(\sigma_j-1)} L_{l,j}^{(\phi_j-1)/(\sigma_j-1)} (w_l(1+\tau_{nl,j})d_{nl,j})^{1-\phi_j}},$$

with the price index given by

$$P_{n,j} = \sigma_j^{\frac{1}{\sigma_j-1}} \bar{\sigma}_j \left[\sum_l A_{l,j}^{\phi_j-1} F_{l,j}^{-(\phi_j-1)/(\sigma_j-1)} L_{l,j}^{(\phi_j-1)/(\sigma_j-1)} (w_l(1+\tau_{nl,j})d_{nl,j})^{1-\phi_j} \right]^{\frac{1}{1-\phi_j}}.$$

In the generalized Krugman model, the trade and scale elasticities are $\epsilon_j = \phi_j - 1$ and $\psi_j = 1/(\sigma_j - 1)$. If we set $\phi_j = \sigma_j$ for all j , the model is just the standard Krugman model, where preferences are CES across country bundles within a sector with an elasticity of substitution σ_j , and CES across varieties within a country bundle with the same elasticity σ_j .

Multi-sector Armington and EK model with External Economies of Scale Each sector contains a continuum of differentiated varieties. Preferences are structured as follows: Cobb-Douglas across sectors in country n with weights $\beta_{n,j}$, CES across country bundles within sector j with an elasticity of substitution σ_j . The corresponding demand function in sector j produced in country i and consumed in country n is:

$$q_{ni,j} = p_{ni,j}^{-\sigma_j} P_{n,j}^{\sigma_j-1} \beta_{n,j} x_n,$$

where $P_{n,j} = (\sum_i P_{ni,j}^{1-\sigma_j})^{1/(1-\sigma_j)}$ is sector j 's price index in country n .

In multi-sector Armington model, labor productivity in sector j country i is given by $A_{i,j} L_{i,j}^{\psi_j}$, where $A_{i,j}$ is an exogenous productivity parameter in sector j in country i , and $L_{i,j}^{\psi_j}$ captures EES in sector j .

Thus, the unit cost of good in sector j country i for delivery to market n is

$$c_{ni,j} = \frac{w_i(1 + \tau_{ni,j})d_{ni,j}}{A_{i,j} L_{i,j}^{\psi_j}}.$$

When $n \neq i$, $\tau_{ni,j}$ denotes the import tariff imposed by country n on sector j imports from country i . When $n = i$, $\tau_{ni,j}$ represents the domestic tax imposed by country n on sector j . All markets are perfectly competitive, indicating $p_{ni,j} = c_{ni,j}$. Total labor $L_{i,j}$ must be consistent with the amounts produced for each market, and labor markets must clear, $\sum_{j=1}^J L_{i,j} = \bar{L}_i$.

$$\begin{aligned} L_{i,j} &= \sum_n q_{ni,j} d_{ni,j} \frac{1}{A_{i,j} L_{i,j}^{\psi_j}} = \sum_n p_{ni,j}^{-\sigma_j} P_{n,j}^{\sigma_j-1} d_{ni,j} \beta_{n,j} x_n \frac{1}{A_{i,j} L_{i,j}^{\psi_j}} = \sum_n \frac{1}{1 + \tau_{ni,j}} p_{ni,j}^{1-\sigma_j} P_{n,j}^{\sigma_j-1} \beta_{n,j} x_n \frac{1}{w_i} \quad (\text{A.76}) \\ &= \sum_n \frac{\pi_{ni,j} \beta_{n,j} x_n}{1 + \tau_{ni,j}} \frac{1}{w_i}, \end{aligned}$$

where the sector-level trade shares are given by

$$\pi_{ni,j} = \left(\frac{p_{ni,j}}{P_{n,j}} \right)^{1-\sigma_j} = \frac{A_{i,j}^{\sigma_j-1} L_{i,j}^{\psi_j(\sigma_j-1)} (w_i(1 + \tau_{ni,j})d_{ni,j})^{1-\sigma_j}}{\sum_l A_{l,j}^{\sigma_j-1} L_{l,j}^{\psi_j(\sigma_j-1)} (w_l(1 + \tau_{nl,j})d_{nl,j})^{1-\sigma_j}}.$$

We can demonstrate that the multi-sector Armington model with EES is isomorphic to the generalized Krugman model, where the trade elasticity is given by $\epsilon_j = \sigma_j - 1$.

In the multi-sector EK model with EES, labor productivity in sector j of country i is given by $A_{i,j}L_{i,j}^{\psi_j}$, where the exogenous productivity parameter $A_{i,j}$ are independently drawn from a Fréchet distribution with shape parameter θ_j and scale parameter $\bar{T}_{i,j}$.

Similarly, the goods market clearing condition satisfies

$$w_i L_{i,j} = \sum_n \frac{\pi_{ni,j} \beta_{n,j} x_n}{1 + \tau_{ni,j}},$$

where the sector-level trade shares are

$$\pi_{ni,j} = \left(\frac{p_{ni,j}}{P_{n,j}} \right)^{-\theta_j} = \frac{\bar{T}_{i,j} L_{i,j}^{\psi_j \theta_j} (w_i (1 + \tau_{ni,j}) d_{ni,j})^{-\theta_j}}{\sum_l \bar{T}_{l,j} L_{l,j}^{\psi_j \theta_j} (w_l (1 + \tau_{nl,j}) d_{nl,j})^{-\theta_j}}.$$

We can demonstrate that the multi-sector EK model with EES is isomorphic to the generalized Krugman model, where the trade elasticity is given by $\epsilon_j = \theta_j$. Therefore, the optimal policies in these two models are identical to those in the generalized Krugman model.

Endogenous Technology We revisit the steady-state optimal policies in [Bai, Jin, and Lu \(2025\)](#) (BJL hereafter), an example of optimal trade policies in a multi-sector, multi-country GE model under productivity-driven supply curves. This framework extends [Eaton and Kortum \(2001\)](#) with endogenous technology to a multi-sector setting.

In BJL, total labor in each sector is divided into two components: production labor and researchers. Labor moves freely across sectors, and technology endogenously depends on the number of researchers. In steady-state equilibrium, the ratio of production labor to researchers in each sector remains constant, implying that sectoral technology can be expressed as a function of sectoral labor. Let r_j denote the fraction of researchers in each sector. Let $T_{n,j}$ denote the technology level of sector j in country n , which depends on an efficiency parameter $\rho_{n,j}$ and the endogenous researchers in the sector. Specifically, $T_{n,j} = \rho_{n,j} L_{n,j}^r = \rho_{n,j} r_j L_{n,j}$.

More generally, there may be constant, decreasing, or increasing returns to scale in innovation. The change in innovation effort further influences innovation efficiency. Specifically, the technology of the sector j in country n is a function of the sector's researchers, i.e.,

$$T_{n,j} = \rho_{n,j} (r_j L_{n,j})^{\nu_j}, \tag{A.77}$$

where ν_j governs the strength of economies of scale. The baseline B JL represents a special case of constant returns to scale where $\nu_j = 1$.

Therefore, the price in sector j of country n satisfies

$$p_{n,j} = T_{n,j}^{-\frac{1}{\theta_j}} w_n = w_n / (\bar{T}_{n,j} L_{n,j}^{\frac{\nu_j}{\theta_j}}),$$

where $\bar{T}_{n,j} = (\rho_{n,j} r^{\nu_j})^{-1/\theta_j}$ and θ_j is the trade elasticity in sector j .

Firms engage in Bertrand competition, where the lowest-cost producer of each good in each market captures the entire market by charging a markup just enough to keep the second-lowest-cost producer out of the market. In equilibrium, markups follow a Pareto distribution with parameter θ_j . Since all firms selling in the market charge a markup drawn from the same distribution, the total profits of firms earned by firms in the market are a constant share of that market's total sales. Specifically, firms (both domestic and foreign) that sell in the market earn profits equal to $1/(1 + \theta_j)$ of total sales. The remaining $\theta_j/(1 + \theta_j)$ share of total sales goes to production labor. Therefore, we can express the total sales of the sector j in country n as $\frac{1+\theta_j}{\theta_j} w_n L_{n,j}^p$, where $L_{n,j}^p$ can be replaced by $(1 - r_j) L_{n,j}$.

Goods market clearing conditions are

$$Y_{1,j} - \beta_{2,j} \frac{1}{1 + \tau_j^x} \pi_{21,j} x_2 = \beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} x_1, \quad (\text{A.78})$$

$$Y_{2,j} - \beta_{2,j} \pi_{22,j} x_2 = \beta_{1,j} \frac{1}{1 + \tau_j^m} \pi_{12,j} x_1, \quad (\text{A.79})$$

where $x_2 = \sum_{j=1}^J Y_{2,j}$, and $Y_{n,j} = \frac{1+\theta_j}{\theta_j} w_n (1 - r_j) L_{n,j}$, $(1 + \theta_j)/\theta_j$ captures firms' markup.

Home's expenditure is

$$x_1 = \sum_{j=1}^J Y_{1,j} + \sum_{j=1}^J \beta_{2,j} \frac{\tau_j^x}{1 + \tau_j^x} \pi_{21,j} x_2 + \sum_{j=1}^J \beta_{1,j} \frac{\tau_j^m}{1 + \tau_j^m} \pi_{12,j} x_1 + \sum_{j=1}^J \beta_{1,j} \frac{\tau_j^d}{1 + \tau_j^d} \pi_{11,j} x_1. \quad (\text{A.80})$$

Trade shares and prices satisfy equations (A.5) - (A.10), where the trade and scale elasticities are $\epsilon_j = \theta_j$ and $\psi_j = \nu_j/\theta_j$. Thus, the expressions for trade shares and sector price indexes in this endogenous technology model are the same as in the generalized Krugman models. The optimal policies in this model are identical to those in generalized Krugman model. When $\nu = 1$, the model reverts to the B JL framework, where the trade elasticity and scale elasticity are given by $\epsilon = \theta$ and

$\psi = 1/\theta$, respectively, and are constant across sectors. □

C.3 Differential Scale Elasticities and Imperfectly Mobile Labor

In this section, we extend the generalized multi-sector Krugman model with imperfectly mobile labor to analyze alternative supply-side specifications in a unified framework. This example combines features of the previous two examples: imperfectly mobile labor across sectors, as in Example 1, and the increasing returns to scale, as in Example 2.

In Section C.3.1, we formulate the Home government's problem and the world market equilibrium based on the original equilibrium conditions of this model. We solve $\{p_{n,j}\}$ as functions of wages and labor $\{w_{n,j}, L_{n,j}\}$ using the model-specific supply function $S(\{p_{n,j}, w_{n,j}, L_{n,j}\})$. We then directly take first-order conditions with respect to $\{w_{n,j}, L_{n,j}\}$ and impose multipliers on the labor market specifications Ω to derive optimal policies in this model. In Section C.3.2, we show that this model is a GIIA Supply System and employ matrix operations to establish a connection with Proposition 3.

C.3.1 Method 1: Model Specific Proof

In the model with differential scale elasticities and imperfectly mobile labor, due to the increasing returns to scale, prices are affected by both wages and labor. The supply-side assumption S satisfies

$$p_{n,j} = \frac{w_{n,j}}{\bar{T}_{n,j} L_{n,j}^{\psi_j}}, \quad (\text{A.81})$$

where $\bar{T}_{n,j}$ denotes the exogenous technology level beyond the economies of scale in sector j of country n and ψ_j is the scale elasticity in sector j . In the following market equilibrium constraints, we substitute $p_{n,j}$ with $w_{n,j}$ and $L_{n,j}$.

Home government solves the following problem:

$$\max_{\{\tau_j^d, \tau_j^x, \tau_j^m, x_1, w_{1,j}, w_{2,j}, L_{1,j}, L_{2,j}\}} \frac{x_1}{P_1},$$

subject to world market equilibrium characterized by the following constraints:

$$x_1 = \sum_{j=1}^J w_{1,j} L_{1,j} + \sum_{j=1}^J \beta_{2,j} \frac{\tau_j^x}{1 + \tau_j^x} \pi_{21,j} x_2 + \sum_{j=1}^J \beta_{1,j} \frac{\tau_j^m}{1 + \tau_j^m} \pi_{12,j} x_1 + \sum_{j=1}^J \beta_{1,j} \frac{\tau_j^d}{1 + \tau_j^d} \pi_{11,j} x_1, \quad (\gamma_x) \quad (\text{A.82})$$

$$\beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} x_1 = w_{1,j} L_{1,j} - \beta_{2,j} \frac{1}{1 + \tau_j^x} \pi_{21,j} x_2, \quad (\gamma_{1,j}, \quad J) \quad (\text{A.83})$$

$$\beta_{1,j} \frac{1}{1 + \tau_j^m} \pi_{12,j} x_1 = w_{2,j} L_{2,j} - \beta_{2,j} \pi_{22,j} x_2, \quad (\gamma_{2,j}, \quad J) \quad (\text{A.84})$$

The labor market specifications are

$$L_{1,j} = A_{1,j} \left(\frac{w_{1,j}}{W_1} \right)^{\kappa-1} \bar{L}_1, \quad (\gamma_{L1,j}, \quad J) \quad (\text{A.85})$$

$$L_{2,j} = A_{2,j} \left(\frac{w_{2,j}}{W_2} \right)^{\kappa-1} \bar{L}_2, \quad (\gamma_{L2,j}, \quad J) \quad (\text{A.86})$$

where $W_n = [\sum_{j=1}^J A_{n,j} w_{n,j}^\kappa]^\frac{1}{\kappa}$ and $\{x_2, P_1, P_2\}$ and trade shares are given by

$$x_2 = \sum_{j=1}^J w_{2,j} L_{2,j}, \quad (\text{A.87})$$

$$\pi_{11,j} = \frac{\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_{1,j} (1 + \tau_j^d))^{-\epsilon_j}}{\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_{1,j} (1 + \tau_j^d))^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} (w_{2,j} (1 + \tau_j^m) d_{12,j})^{-\epsilon_j}}, \quad (\text{A.88})$$

$$\pi_{12,j} = \frac{\bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} (w_{2,j} (1 + \tau_j^m) d_{12,j})^{-\epsilon_j}}{\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_{1,j} (1 + \tau_j^d))^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} (w_{2,j} (1 + \tau_j^m) d_{12,j})^{-\epsilon_j}}, \quad (\text{A.89})$$

$$\pi_{21,j} = \frac{\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_{1,j} (1 + \tau_j^x) d_{21,j})^{-\epsilon_j}}{\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_{1,j} (1 + \tau_j^x) d_{21,j})^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} (w_{2,j})^{-\epsilon_j}}, \quad (\text{A.90})$$

$$\pi_{22,j} = \frac{\bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} (w_{2,j})^{-\epsilon_j}}{\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_{1,j} (1 + \tau_j^x) d_{21,j})^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} (w_{2,j})^{-\epsilon_j}}, \quad (\text{A.91})$$

$$P_1 = \prod_{j=1}^J \left[\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_{1,j} (1 + \tau_j^d))^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} (w_{2,j} (1 + \tau_j^m) d_{12,j})^{-\epsilon_j} \right]^{-\frac{\beta_{1,j}}{\epsilon_j}} = 1, \quad (\text{A.92})$$

$$P_2 = \prod_{j=1}^J \left[\bar{T}_{1,j}^{\epsilon_j} L_{1,j}^{\epsilon_j \psi_j} (w_{1,j} (1 + \tau_j^x) d_{21,j})^{-\epsilon_j} + \bar{T}_{2,j}^{\epsilon_j} L_{2,j}^{\epsilon_j \psi_j} w_{2,j}^{-\epsilon_j} \right]^{-\frac{\beta_{2,j}}{\epsilon_j}}. \quad (\text{A.93})$$

One of the equilibrium conditions in (A.82), (A.83) and (A.84) is redundant, according to Walras' law, thus, we drop the expenditure constraint in optimization and set $\gamma_x = 0$. Or, equivalently, if we use Home's budget constraint, one of the market-clearing conditions is redundant.

The FOC over $\tau_j^d, \tau_j^m, \tau_j^x$ are the same as in Appendix B. We obtain the optimal policies:

$$1 + \tau_j^m = \gamma_{2,j}, \quad 1 + \tau_j^d = \gamma_{1,j}. \quad (\text{A.94})$$

$$1 + \tau_j^x = \frac{1 + \tau_j^d}{1 + \tau_j^m} \left(1 + \frac{1}{\epsilon_j \pi_{22,j}} \right). \quad (\text{A.95})$$

FOC over x_1

$$1 - \sum_{j=1}^J \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} - \sum_{j=1}^J \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \pi_{12,j} = 0.$$

FOC over $w_{1,j}$

$$\begin{aligned} & -\beta_{1,j} x_1 \pi_{11,j} + \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j \pi_{11,j} \pi_{12,j} x_1 + \gamma_{1,j} \beta_{2,j} \frac{1}{1 + \tau_j^x} \epsilon_j \pi_{21,j} \pi_{22,j} x_2 - \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \epsilon_j \pi_{12,j} \pi_{11,j} x_1 \\ & - \gamma_{2,j} \beta_{2,j} \epsilon_j \pi_{22,j} \pi_{21,j} x_2 + \gamma_{1,j} w_{1,j} L_{1,j} + \gamma_{L1,j} (\kappa - 1) L_{1,j} - \sum_{s=1}^J \gamma_{L1,s} (\kappa - 1) L_{1,s} \frac{w_{1,j} L_{1,j}}{W_1 \bar{L}_1} = 0. \end{aligned} \quad (\text{A.96})$$

FOC over $L_{1,j}$

$$\begin{aligned} & \psi_j \beta_{1,j} x_1 \pi_{11,j} - \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j \psi_j \pi_{11,j} \pi_{12,j} x_1 - \gamma_{1,j} \beta_{2,j} \frac{1}{1 + \tau_j^x} \epsilon_j \psi_j \pi_{21,j} \pi_{22,j} x_2 \\ & + \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \epsilon_j \psi_j \pi_{12,j} \pi_{11,j} x_1 + \gamma_{2,j} \beta_{2,j} \epsilon_j \psi_j \pi_{22,j} \pi_{21,j} x_2 + \gamma_{1,j} w_{1,j} L_{1,j} - \gamma_{L1,j} L_{1,j} = 0. \end{aligned} \quad (\text{A.97})$$

FOC over $w_{2,j}$

$$\begin{aligned} & -\beta_{1,j} x_1 \pi_{12,j} - \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j \pi_{11,j} \pi_{12,j} x_1 - \gamma_{1,j} \beta_{2,j} \frac{1}{1 + \tau_j^x} \epsilon_j \pi_{21,j} \pi_{22,j} x_2 + \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \epsilon_j \pi_{12,j} \pi_{11,j} x_1 \\ & + \gamma_{2,j} \beta_{2,j} \epsilon_j \pi_{22,j} \pi_{21,j} x_2 + (\Phi_2 + \gamma_{2,j}) w_{2,j} L_{2,j} + \gamma_{L2,j} (\kappa - 1) L_{2,j} - \sum_{s=1}^J \gamma_{L2,s} (\kappa - 1) L_{2,s} \frac{w_{2,j} L_{2,j}}{W_2 \bar{L}_2} = 0, \end{aligned} \quad (\text{A.98})$$

where $\Phi_2 = -\sum_{s=1}^J (\gamma_{1,s} \beta_{2,s} \frac{1}{1 + \tau_s^x} \pi_{21,s} + \gamma_{2,s} \beta_{2,s} \pi_{22,s})$.

FOC over $L_{2,j}$

$$\begin{aligned} & \psi_j \beta_{1,j} x_1 \pi_{12,j} + \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j \psi_j \pi_{11,j} \pi_{12,j} x_1 + \gamma_{1,j} \beta_{2,j} \frac{1}{1 + \tau_j^x} \epsilon_j \psi_j \pi_{21,j} \pi_{22,j} x_2 \\ & - \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \epsilon_j \psi_j \pi_{12,j} \pi_{11,j} x_1 - \gamma_{2,j} \beta_{2,j} \epsilon_j \psi_j \pi_{22,j} \pi_{21,j} x_2 + (\Phi_2 + \gamma_{2,j}) w_{2,j} L_{2,j} - \gamma_{L2,j} L_{2,j} = 0. \end{aligned} \quad (\text{A.99})$$

Combining FOCs over $w_{1,j}$ (A.96) and $L_{1,j}$ (A.97), we get

$$\begin{aligned}
& (\psi_j + 1)\gamma_{1,j}w_{1,j}L_{1,j} + \gamma_{L1,j}(\psi_j(\kappa - 1) - 1)L_{1,j} - \psi_j \sum_{s=1}^J \gamma_{L1,s}(\kappa - 1)L_{1,s} \frac{w_{1,j}L_{1,j}}{W_1 \bar{L}_1} = 0. \\
\Rightarrow & (\psi_j + 1)\gamma_{1,j} + \gamma_{L1,j}(\psi_j(\kappa - 1) - 1) \frac{1}{w_{1,j}} - \psi_j \sum_{s=1}^J \gamma_{L1,s}(\kappa - 1)L_{1,s} \frac{1}{W_1 \bar{L}_1} = 0. \tag{A.100}
\end{aligned}$$

Combining FOCs over $L_{1,j}$ (A.97), τ_j^x (A.95), and τ_j^d, τ_j^m (A.94), we get

$$\begin{aligned}
& \psi_j \beta_{1,j} x_1 \pi_{11,j} - \beta_{2,j} \left(\frac{1 + \tau_j^d}{1 + \tau_j^x} - (1 + \tau_j^m) \right) \epsilon_j \psi_j \pi_{21,j} \pi_{22,j} x_2 + \gamma_{1,j} w_{1,j} L_{1,j} - \gamma_{L1,j} L_{1,j} = 0 \\
\Rightarrow & \psi_j \beta_{1,j} x_1 \pi_{11,j} + \beta_{2,j} \frac{1 + \tau_j^d}{1 + \tau_j^x} \psi_j \pi_{21,j} x_2 + \gamma_{1,j} w_{1,j} L_{1,j} - \gamma_{L1,j} L_{1,j} = 0 \\
\Rightarrow & \gamma_{1,j} w_{1,j} L_{1,j} - \gamma_{L1,j} L_{1,j} + \psi_j (1 + \tau_j^d) w_{1,j} L_{1,j} = 0 \\
\Rightarrow & (\psi_j + 1)(1 + \tau_j^d) = \gamma_{L1,j} \frac{1}{w_{1,j}}.
\end{aligned}$$

Plugging back to (A.100),

$$\begin{aligned}
& (\psi_j + 1)(1 + \tau_j^d) + (\psi_j(\kappa - 1) - 1)(\psi_j + 1)(1 + \tau_j^d) - \psi_j \sum_{s=1}^J \gamma_{L1,s}(\kappa - 1)L_{1,s} \frac{1}{W_1 \bar{L}_1} = 0 \\
\Rightarrow & \psi_j(\kappa - 1)(\psi_j + 1)(1 + \tau_j^d) - \psi_j \sum_{s=1}^J \gamma_{L1,s}(\kappa - 1)L_{1,s} \frac{1}{W_1 \bar{L}_1} = 0 \\
\Rightarrow & (\psi_j + 1)(1 + \tau_j^d) = \sum_{s=1}^J \gamma_{L1,s} L_{1,s} \frac{1}{W_1 \bar{L}_1}.
\end{aligned}$$

Hence, for any sectors j and k , the optimal domestic taxes satisfy

$$\frac{1 + \tau_j^d}{1 + \tau_k^d} = \frac{1 + \psi_k}{1 + \psi_j}.$$

Combining FOCs over $L_{2,j}$ (A.99), τ_j^x (A.95), and τ_j^d, τ_j^m (A.94), we get

$$\begin{aligned}
& -(\Phi_2 + \gamma_{2,j})w_{2,j}L_{2,j} + \gamma_{L2,j}L_{2,j} = \psi_j\beta_{1,j}\pi_{12,j}x_1 + \left[\frac{1 + \tau_j^d}{1 + \tau_j^x} - (1 + \tau_j^m) \right] \beta_{2,j}\epsilon_j\psi_j\pi_{21,j}\pi_{22,j}x_2 \\
& \Rightarrow -(\Phi_2 + 1 + \tau_j^m) + \gamma_{L2,j}\frac{1}{w_{2,j}} = \psi_j\frac{\beta_{1,j}\pi_{12,j}x_1}{Y_{2,j}} - \frac{1 + \tau_j^d}{1 + \tau_j^x}\beta_{2,j}\psi_j\pi_{21,j}\frac{x_2}{Y_{2,j}} \\
& \Rightarrow -(\Phi_2 + 1 + \tau_j^m) + \gamma_{L2,j}\frac{1}{w_{2,j}} = \psi_j\frac{\beta_{1,j}\pi_{12,j}x_1}{Y_{2,j}} - \psi_j(1 + \tau_j^d)\frac{Y_{1,j}}{Y_{2,j}} + \psi_j\frac{\beta_{1,j}\pi_{11,j}x_1}{Y_{2,j}} \\
& \Rightarrow -(\Phi_2 + 1 + \tau_j^m) + \gamma_{L2,j}\frac{1}{w_{2,j}} = \psi_j\frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}}.
\end{aligned}$$

Combining FOCs over $w_{2,j}$ (A.98) and $L_{2,j}$ (A.99), we get

$$\begin{aligned}
& (\psi_j + 1)(\Phi_2 + \gamma_{2,j})w_{2,j}L_{2,j} + \gamma_{L2,j}(\psi_j(\kappa - 1) - 1)L_{2,j} - \psi_j \sum_{s=1}^J \gamma_{L2,s}(\kappa - 1)L_{2,s} \frac{w_{2,j}L_{2,j}}{W_2\bar{L}_2} = 0. \\
& \Rightarrow (\psi_j + 1)(\Phi_2 + 1 + \tau_j^m) + \gamma_{L2,j}(\psi_j(\kappa - 1) - 1)\frac{1}{w_{2,j}} - \psi_j \sum_{s=1}^J \gamma_{L2,s}(\kappa - 1)L_{2,s} \frac{1}{W_2\bar{L}_2} = 0 \\
& \Rightarrow \psi_j\kappa(\Phi_2 + 1 + \tau_j^m) + (\psi_j(\kappa - 1) - 1)\left(\psi_j\frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}}\right) - \psi_j \sum_{s=1}^J \gamma_{L2,s}(\kappa - 1)L_{2,s} \frac{1}{W_2\bar{L}_2} = 0 \\
& \Rightarrow \Phi_2 + 1 + \tau_j^m + \left(\psi_j\frac{\kappa - 1}{\kappa} - \frac{1}{\kappa}\right)\frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}} - \sum_{s=1}^J \gamma_{L2,s}\frac{\kappa - 1}{\kappa}L_{2,s}\frac{1}{W_2\bar{L}_2} = 0. \tag{A.101}
\end{aligned}$$

Finally, we take the difference of (A.101) between sectors j and k and derive the formula for optimal tariffs

$$\tau_j^m - \tau_k^m = \left(\frac{1}{\kappa} - \psi_j\frac{\kappa - 1}{\kappa}\right)\frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}} - \left(\frac{1}{\kappa} - \psi_k\frac{\kappa - 1}{\kappa}\right)\frac{\beta_{1,k}x_1 - (1 + \tau_k^d)Y_{1,k}}{Y_{2,k}}, \tag{A.102}$$

where $Y_{n,j} = w_{n,j}L_{n,j}$.

C.3.2 Method 2: Applying Proposition 3

We first prove that the model with differential scale elasticities and imperfectly mobile labor satisfies GIIA Supply System, then apply the optimal policy formulas from Proposition 3 using the structural parameters from this model.

In this extended model, goods market clearing conditions, expenditures, trade shares and prices satisfy equations (A.2) - (A.10). The labor market specifications are given by equations (A.85) and (A.86). As $\kappa \rightarrow \infty$, the model converges to one with increasing returns to scale and perfect labor

mobility, similar to the generalized Krugman model.

We begin by deriving $w_{n,j}$ as a function of $p_{n,j}$ and W_n , from the supply-side assumption S (A.81).

$$\begin{aligned} p_{n,j} &= \bar{T}_{n,j}^{-1} \bar{L}_{n,j}^{-\psi_j} w_{n,j} = \bar{T}_{n,j}^{-1} w_{n,j}^{1-\psi_j(\kappa-1)} W_n^{\psi_j(\kappa-1)} \bar{L}_n^{-\psi_j} \\ \Rightarrow w_{n,j} &= \left(p_{n,j} \bar{T}_{n,j} W_n^{-\psi_j(\kappa-1)} \bar{L}_n^{\psi_j} \right)^{\frac{1}{1-\psi_j(\kappa-1)}}. \end{aligned} \quad (\text{A.103})$$

Since the sectoral income is $Y_{n,j} = w_{n,j} L_{n,j}$, the supply curves in country n are given by

$$Y_{n,j} = A_{n,j} w_{n,j}^\kappa W_n^{1-\kappa} \bar{L}_n = A_{n,j} (p_{n,j} \bar{T}_{n,j})^{\frac{\kappa}{1-\psi_j(\kappa-1)}} W_n^{1-\frac{\kappa}{1-\psi_j(\kappa-1)}} \bar{L}_n^{\frac{\psi_j \kappa}{1-\psi_j(\kappa-1)}+1}. \quad (\text{A.104})$$

Plugging (A.103) into $W_n = [\sum_{j=1}^J A_{n,j} w_{n,j}^\kappa]^{\frac{1}{\kappa}}$, we derive W_n as a function of $p_{n,j}$.

$$W_n^\kappa = \sum_{j=1}^J A_{n,j} \left(p_{n,j} \bar{T}_{n,j} W_n^{-\psi_j(\kappa-1)} \bar{L}_n^{\psi_j} \right)^{\frac{\kappa}{1-\psi_j(\kappa-1)}}. \quad (\text{A.105})$$

Using the implicit function theorem, we obtain $\frac{\partial \ln W_n}{\partial \ln p_{n,j}}$.

$$\begin{aligned} \kappa \frac{\partial \ln W_n}{\partial \ln p_{n,j}} W_n^\kappa &= \frac{\kappa}{1-\psi_j(\kappa-1)} A_{n,j} \left(p_{n,j} \bar{T}_{n,j} W_n^{-\psi_j(\kappa-1)} \bar{L}_n^{\psi_j} \right)^{\frac{\kappa}{1-\psi_j(\kappa-1)}} \\ &\quad - \sum_{s=1}^J \psi_s(\kappa-1) \frac{\kappa}{1-\psi_s(\kappa-1)} A_{n,s} \left(p_{n,s} \bar{T}_{n,s} W_n^{-\psi_s(\kappa-1)} \bar{L}_n^{\psi_s} \right)^{\frac{\kappa}{1-\psi_s(\kappa-1)}} \frac{\partial \ln W_n}{\partial \ln p_{n,j}} \\ \Rightarrow \frac{\partial \ln W_n}{\partial \ln p_{n,j}} &= \frac{\frac{1}{1-\psi_j(\kappa-1)} A_{n,j} \left(p_{n,j} \bar{T}_{n,j} W_n^{-\psi_j(\kappa-1)} \bar{L}_n^{\psi_j} \right)^{\frac{\kappa}{1-\psi_j(\kappa-1)}}}{\sum_{s=1}^J \frac{1}{1-\psi_s(\kappa-1)} A_{n,s} \left(p_{n,s} \bar{T}_{n,s} W_n^{-\psi_s(\kappa-1)} \bar{L}_n^{\psi_s} \right)^{\frac{\kappa}{1-\psi_s(\kappa-1)}}} \\ &= \frac{\frac{1}{1-\psi_j(\kappa-1)} Y_{n,j} W_n^{\kappa-1} / \bar{L}_n}{\sum_{s=1}^J \frac{1}{1-\psi_s(\kappa-1)} Y_{n,s} W_n^{\kappa-1} / \bar{L}_n} = \frac{\frac{1}{1-\psi_j(\kappa-1)} Y_{n,j}}{\sum_{s=1}^J \frac{1}{1-\psi_s(\kappa-1)} Y_{n,s}}. \end{aligned} \quad (\text{A.106})$$

We then derive scaled partial supply elasticities $\frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,j}} \frac{Y_{n,s}}{Y_{n,j}}$ in Λ_n by incorporating (A.106) in (A.104).

$$\begin{aligned} \frac{\partial \ln Y_{n,k}}{\partial \ln p_{n,j}} \frac{Y_{n,k}}{Y_{n,j}} &= I_{k=j} \left(\frac{\kappa}{1-\psi_j(\kappa-1)} \right) + \left(1 - \frac{\kappa}{1-\psi_k(\kappa-1)} \right) \frac{Y_{n,k}}{Y_{n,j}} \frac{\partial \ln W_n}{\partial \ln p_{n,j}} \\ &= I_{k=j} \left(\frac{\kappa}{1-\psi_j(\kappa-1)} \right) + \left(1 - \frac{\kappa}{1-\psi_k(\kappa-1)} \right) \frac{\frac{1}{1-\psi_j(\kappa-1)} Y_{n,k}}{\sum_{s=1}^J \frac{1}{1-\psi_s(\kappa-1)} Y_{n,s}} \\ &= I_{k=j} \left(\frac{\kappa}{1-\psi_j(\kappa-1)} \right) + \frac{\kappa}{1-\psi_j(\kappa-1)} \frac{1 - \frac{\kappa}{1-\psi_k(\kappa-1)} Y_{n,k}}{\sum_{s=1}^J \frac{\kappa}{1-\psi_s(\kappa-1)} Y_{n,s}}, \end{aligned}$$

where $I_{s=j}(\frac{\kappa}{1-\psi_j(\kappa-1)})$ is an indicator function that equals $\frac{\kappa}{1-\psi_j(\kappa-1)}$ when $s = j$, and 0 otherwise.

The scaled partial supply elasticity matrix Λ_n of country n takes the form of

$$\Lambda_n = \begin{pmatrix} \frac{\kappa}{1-\psi_1(\kappa-1)} \left(1 + \frac{(1-\frac{\kappa}{1-\psi_1(\kappa-1)})Y_{n,1}}{\sum_{s=1}^J \frac{\kappa}{1-\psi_s(\kappa-1)}Y_{n,s}}\right) & \cdots & \frac{\kappa}{1-\psi_1(\kappa-1)} \frac{(1-\frac{\kappa}{1-\psi_j(\kappa-1)})Y_{n,j}}{\sum_{s=1}^J \frac{\kappa}{1-\psi_s(\kappa-1)}Y_{n,s}} & \cdots & \frac{\kappa}{1-\psi_1(\kappa-1)} \frac{(1-\frac{\kappa}{1-\psi_J(\kappa-1)})Y_{n,J}}{\sum_{s=1}^J \frac{\kappa}{1-\psi_s(\kappa-1)}Y_{n,s}} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \frac{\kappa}{1-\psi_j(\kappa-1)} \frac{(1-\frac{\kappa}{1-\psi_1(\kappa-1)})Y_{n,1}}{\sum_{s=1}^J \frac{\kappa}{1-\psi_s(\kappa-1)}Y_{n,s}} & \cdots & \frac{\kappa}{1-\psi_j(\kappa-1)} \left(1 + \frac{(1-\frac{\kappa}{1-\psi_j(\kappa-1)})Y_{n,j}}{\sum_{s=1}^J \frac{\kappa}{1-\psi_s(\kappa-1)}Y_{n,s}}\right) & \cdots & \frac{\kappa}{1-\psi_j(\kappa-1)} \frac{(1-\frac{\kappa}{1-\psi_J(\kappa-1)})Y_{n,J}}{\sum_{s=1}^J \frac{\kappa}{1-\psi_s(\kappa-1)}Y_{n,s}} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \frac{\kappa}{1-\psi_J(\kappa-1)} \frac{(1-\frac{\kappa}{1-\psi_1(\kappa-1)})Y_{n,1}}{\sum_{s=1}^J \frac{\kappa}{1-\psi_s(\kappa-1)}Y_{n,s}} & \cdots & \frac{\kappa}{1-\psi_J(\kappa-1)} \frac{(1-\frac{\kappa}{1-\psi_j(\kappa-1)})Y_{n,j}}{\sum_{s=1}^J \frac{\kappa}{1-\psi_s(\kappa-1)}Y_{n,s}} & \cdots & \frac{\kappa}{1-\psi_J(\kappa-1)} \left(1 + \frac{(1-\frac{\kappa}{1-\psi_J(\kappa-1)})Y_{n,J}}{\sum_{s=1}^J \frac{\kappa}{1-\psi_s(\kappa-1)}Y_{n,s}}\right) \end{pmatrix}. \quad (\text{A.107})$$

If $\sum_{j=1}^J \frac{\kappa Y_{n,j}}{1-\psi_j(\kappa-1)} \neq 0$, this matrix is invertible since

$$\det(\Lambda_n) = \left(\prod_j \frac{\kappa}{1-\psi_j(\kappa-1)} \right) \frac{\sum_j Y_{n,j}}{\sum_j \frac{\kappa}{1-\psi_j(\kappa-1)} Y_{n,j}} \neq 0.$$

The formula for Λ_n shows that this model satisfies the GIIA Supply System in Definition 4. The scaled partial elasticities of a sector's income with respect to price (scaled by the two sectors' size and elasticity parameter) satisfy $\frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,j}} \frac{Y_{n,s}}{Y_{n,j}} \frac{1}{\eta_j} - \frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,i}} \frac{Y_{n,s}}{Y_{n,i}} \frac{1}{\eta_i} = 0$ for $s \neq i$ or j , and $\frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,j}} \frac{Y_{n,s}}{Y_{n,j}} \frac{1}{\eta_j} - \frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,i}} \frac{Y_{n,s}}{Y_{n,i}} \frac{1}{\eta_i} = 1$ for $s = j$, where

$$\frac{\partial \ln Y_{n,k}}{\partial \ln p_{n,j}} \frac{Y_{n,k}}{Y_{n,j}} \frac{1}{\eta_j} = I_{k=j}(1) + \frac{(1-\frac{\kappa}{1-\psi_k(\kappa-1)})Y_{n,k}}{\sum_{s=1}^J \frac{\kappa}{1-\psi_s(\kappa-1)}Y_{n,s}},$$

and

$$\frac{1}{\eta_j} = \frac{1}{\kappa} - \psi_j \frac{\kappa-1}{\kappa}.$$

When $\kappa \rightarrow \infty$, $\frac{1}{\eta_j} \rightarrow -\psi_j$, the model converges to the generalized Krugman model, and the scaled partial supply elasticity matrix Λ_n of country n simplifies to (A.70).

Proof of optimal domestic taxes According to Proposition 2, optimal domestic taxes satisfy

$$(\Lambda_1 - I)(1 + \tau^d) = 0.$$

Consider two rows, j th and k th of (A.107)

$$\frac{\kappa}{1 - \psi_j(\kappa - 1)}(1 + \tau_j^d) + \frac{\kappa}{1 - \psi_j(\kappa - 1)} \sum_{s=1}^J \frac{(1 - \frac{\kappa}{1 - \psi_s(\kappa - 1)})Y_{n,s}}{\sum_{i=1}^J \frac{\kappa}{1 - \psi_i(\kappa - 1)}Y_{n,i}}(1 + \tau_s^d) = (1 + \tau_j^d), \quad (\text{A.108})$$

$$\frac{\kappa}{1 - \psi_k(\kappa - 1)}(1 + \tau_k^d) + \frac{\kappa}{1 - \psi_k(\kappa - 1)} \sum_{s=1}^J \frac{(1 - \frac{\kappa}{1 - \psi_s(\kappa - 1)})Y_{n,s}}{\sum_{i=1}^J \frac{\kappa}{1 - \psi_i(\kappa - 1)}Y_{n,i}}(1 + \tau_s^d) = (1 + \tau_k^d). \quad (\text{A.109})$$

Multiplying (A.108) and (A.109) by $(\frac{1}{\kappa} - \psi_j \frac{\kappa-1}{\kappa})$ and $(\frac{1}{\kappa} - \psi_k \frac{\kappa-1}{\kappa})$ respectively, and subtracting (A.109) from (A.108), we have

$$\begin{aligned} (1 + \tau_j^d) - (1 + \tau_k^d) &= (1 + \tau_j^d)(\frac{1}{\kappa} - \psi_j \frac{\kappa-1}{\kappa}) - (1 + \tau_k^d)(\frac{1}{\kappa} - \psi_k \frac{\kappa-1}{\kappa}) \\ \Rightarrow \frac{1 + \tau_j^d}{1 + \tau_k^d} &= \frac{1 + \psi_k}{1 + \psi_j} = \frac{\frac{\eta_j}{\eta_j - 1}}{\frac{\eta_k}{\eta_k - 1}}, \end{aligned}$$

where ψ_j is the scale elasticity in sector j .

Proof of optimal import tariffs According to Proposition 2, the optimal tariffs satisfy

$$\sum_{k=1}^J \tau_k^m \frac{\partial \ln(Y_{2,k})}{\partial \ln(p_{2,j})} \frac{Y_{2,k}}{Y_{2,j}} = \frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}}, \forall j$$

Consider two rows, j th and k th of (A.107),

$$\frac{\kappa}{1 - \psi_j(\kappa - 1)}\tau_j^m + \frac{\kappa}{1 - \psi_j(\kappa - 1)} \sum_{s=1}^J \frac{(1 - \frac{\kappa}{1 - \psi_s(\kappa - 1)})Y_{n,s}}{\sum_{i=1}^J \frac{\kappa}{1 - \psi_i(\kappa - 1)}Y_{n,i}}\tau_s^m = \frac{M_{1,j}}{Y_{2,j}}, \quad (\text{A.110})$$

$$\frac{\kappa}{1 - \psi_k(\kappa - 1)}\tau_k^m + \frac{\kappa}{1 - \psi_k(\kappa - 1)} \sum_{s=1}^J \frac{(1 - \frac{\kappa}{1 - \psi_s(\kappa - 1)})Y_{n,s}}{\sum_{i=1}^J \frac{\kappa}{1 - \psi_i(\kappa - 1)}Y_{n,i}}\tau_s^m = \frac{M_{1,k}}{Y_{2,k}}, \quad (\text{A.111})$$

Multiplying (A.110) and (A.111) by $(\frac{1}{\kappa} - \psi_j \frac{\kappa-1}{\kappa})$ and $(\frac{1}{\kappa} - \psi_k \frac{\kappa-1}{\kappa})$ respectively, and subtracting (A.111) from (A.110), we have

$$\tau_j^m - \tau_k^m = (\frac{1}{\kappa} - \psi_j \frac{\kappa-1}{\kappa}) \frac{\beta_{1,j}x_1 - (1 + \tau_j^d)Y_{1,j}}{Y_{2,j}} - (\frac{1}{\kappa} - \psi_k \frac{\kappa-1}{\kappa}) \frac{\beta_{1,k}x_1 - (1 + \tau_k^d)Y_{1,k}}{Y_{2,k}}.$$

The optimal policies align with Proposition 3 under the GIA Supply System, where the parameter $\frac{1}{\eta_j} = \frac{1}{\kappa} - \psi_j \frac{\kappa-1}{\kappa}$. Furthermore, this formula is the same as (A.102) that we derived directly using the Lagrange method. $\square$

C.4 Proof for Lemma 2: CES Labor Market

In this section, we first demonstrate that the labor market specification in a CES labor market is equivalent to that in one-factor Ricardo-Roy model. Next, we prove that it satisfies the GIIA Supply System and derive the formula for Home's optimal policies. In Sections C.4.1 and C.4.2, we discuss two special cases: perfectly mobile labor across sectors and immobile/sector-specific labor.

Here, total labor is a CES aggregator over labor in different sectors, $\bar{L}_n = [\sum_{j=1}^J \alpha_j^{-\frac{1}{\kappa-1}} L_{n,j}^{\frac{\kappa}{\kappa-1}}]^{\frac{\kappa-1}{\kappa}}$, where $\{\alpha_j\}$ are constants and $\kappa - 1 \geq 0$ captures the elasticity of substitution across labor in different sectors. This implies a labor supply curve as

$$L_{n,j} = \alpha_j \left(\frac{w_{n,j}}{W_n} \right)^{\kappa-1} \bar{L}_n, \quad \text{with } W_n = \left[\sum_{j=1}^J \alpha_j w_{n,j}^\kappa \right]^{\frac{1}{\kappa}}. \quad (\text{A.112})$$

When $\kappa \rightarrow \infty$, $\bar{L}_n = \sum_{j=1}^J L_{n,j}$ and labor is perfectly substitutable across sectors, which is widely adopted in the literature. For example, Costinot, Donaldson, and Komunjer (2012) extends Eaton and Kortum (2002) to multi-sector. In this case, the equilibrium wages are equalized across sectors, leading to $\Omega = \{(w_{n,j}, L_{n,j}) : \sum_{j=1}^J L_{n,j}(w) = \bar{L}_n, w_{n,j} = w_n\}$. When $\kappa = 1$, the model is the same as the specific-factor model with fixed labor in each sector, and thus $\Omega = \{(w_{n,j}, L_{n,j}) : L_{n,j} = \alpha_j \bar{L}_n, w_{n,j} = w_{n,j}(\mathbf{L})\}$. The optimal policies are in Sections C.4.1 and C.4.2 for details.

In general, we can rewrite the labor supply curve (A.112) and obtain the equilibrium labor market specification as

$$\Omega = \left\{ (w_{n,j}, L_{n,j}) : \frac{w_{n,j} L_{n,j}}{W_n \bar{L}_n} = \alpha_j \left(\frac{w_{n,j}}{W_n} \right)^\kappa, W_n = \left[\sum_{j=1}^J \alpha_j w_{n,j}^\kappa \right]^{\frac{1}{\kappa}} \right\}. \quad (\text{A.113})$$

The labor market specification (A.113) is equivalent to the one-factor Ricardo-Roy model, and it further implies a supply system,

$$Y_{n,j} = \alpha_j \left(\frac{w_{n,j}}{W_n} \right)^\kappa Y_n,$$

where the income in sector j , $Y_{n,j} = w_{n,j} L_{n,j}$. Let the share of sector j income in country n as $m_{n,j} = Y_{n,j}/Y_n$. Due to the constant returns to scale, prices are proportional to wages, which satisfy $p_{n,j} = \frac{w_{n,j}}{\bar{T}_{n,j}}$. The supply curves are

$$Y_{n,j} = \alpha_j (p_{n,j} \bar{T}_{n,j})^\kappa \left[\sum_{s=1}^J \alpha_s (p_{n,s} \bar{T}_{n,s})^\kappa \right]^{\frac{1-\kappa}{\kappa}} \bar{L}_n.$$

The corresponding element of the scaled partial supply elasticity matrix Λ_n is given by

$$\frac{\partial \ln(Y_{n,s})}{\partial \ln(p_{n,j})} \frac{Y_{n,s}}{Y_{n,j}} = I_{s=j}(\kappa) - (\kappa - 1)m_{n,s}.$$

The scaled partial supply elasticity matrix Λ_n takes the form of

$$\Lambda_n = \begin{pmatrix} \kappa - (\kappa - 1)m_{n,1} & \dots & -(\kappa - 1)m_{n,j} & \dots & -(\kappa - 1)m_{n,J} \\ \dots & \dots & \dots & \dots & \dots \\ -(\kappa - 1)m_{n,1} & \dots & \kappa - (\kappa - 1)m_{n,j} & \dots & -(\kappa - 1)m_{n,J} \\ \dots & \dots & \dots & \dots & \dots \\ -(\kappa - 1)m_{n,1} & \dots & -(\kappa - 1)m_{n,j} & \dots & \kappa - (\kappa - 1)m_{n,J} \end{pmatrix}. \quad (\text{A.114})$$

Therefore, this model satisfies the requirement of the GIA Supply System. The partial elasticities of a sector's income with respect to price (scaled by the two sectors' size and elasticity parameter) satisfy $\frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,j}} \frac{Y_{n,s}}{Y_{n,j}} \frac{1}{\eta_j} - \frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,i}} \frac{Y_{n,s}}{Y_{n,i}} \frac{1}{\eta_i} = 0$ for $s \neq i$ or j , and $\frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,j}} \frac{Y_{n,s}}{Y_{n,j}} \frac{1}{\eta_j} - \frac{\partial \ln Y_{n,s}}{\partial \ln p_{n,i}} \frac{Y_{n,s}}{Y_{n,i}} \frac{1}{\eta_i} = 1$ for $s = j$, where the parameter $\eta_j = \kappa$.

Consider the j th and k th rows of (12) in Proposition 2.

$$\begin{aligned} -(\kappa - 1) \sum_{s=1}^J m_{1,s} \tau_s^d + (\kappa - 1) \tau_j^d &= 0, \\ -(\kappa - 1) \sum_{s=1}^J m_{1,s} \tau_s^d + (\kappa - 1) \tau_k^d &= 0. \end{aligned}$$

The difference of the two implies $\tau_k^d = \tau_j^d$. Intuitively, without domestic frictions in the model, the home government has no incentives to use differential domestic taxes.

Consider the j th and k th rows of (13) in Proposition 2.

$$\begin{aligned} -(\kappa - 1) \sum_{s=1}^J m_{2,s} (\tau_s^m + \Phi_2 + 1) + \kappa (\tau_j^m + \Phi_2 + 1) &= \frac{M_{1,j}}{Y_{2,j}}, \\ -(\kappa - 1) \sum_{s=1}^J m_{2,s} (\tau_s^m + \Phi_2 + 1) + \kappa (\tau_k^m + \Phi_2 + 1) &= \frac{M_{1,k}}{Y_{2,k}}. \end{aligned}$$

The difference of the two implies,

$$\begin{aligned}\kappa(\tau_j^m - \tau_k^m) &= \frac{M_{1,j}}{Y_{2,j}} - \frac{M_{1,k}}{Y_{2,k}} \\ \Rightarrow \tau_j^m - \tau_k^m &= \frac{1}{\kappa} \left[\frac{\beta_{1,j}x_1 - Y_{1,j}}{Y_{2,j}} - \frac{\beta_{1,k}x_1 - Y_{1,k}}{Y_{2,k}} \right].\end{aligned}$$

This optimal tariff formula equation nests the case with perfectly mobile labor across sectors ($\kappa \rightarrow \infty$). In such a case, the optimal import tariffs can be set to zero, $\tau_j^m = 0$ for any j . This is because under a completely elastic relative Foreign supply, the Home has no market power to change relative foreign supply prices across sectors and, thus, no incentives to use heterogeneous tariffs across sectors. In contrast, with imperfect substitute labor, including the sector-specific labor model, where $\kappa < \infty$, imposing a tariff decreases the demand that Foreign sector faces and depresses the sector wage and price. The larger Home's net import share in the Foreign sectoral income, the greater the impact of tariffs on Foreign prices, and the higher the tariff.

To sum up, the optimal policies align with Proposition 3 under GIIA Supply System, where the parameter $\eta_j = \kappa$. Next, we discuss two special cases: perfectly mobile labor across sectors and immobile/sector-specific labor.

C.4.1 Perfectly Mobile Labor Across Sectors

With perfectly mobile labor across sectors ($\kappa \rightarrow \infty$), the labor market specifications are

$$\begin{aligned}\sum_{j=1}^J L_{1,j} &= \bar{L}_1, \\ \sum_{j=1}^J L_{2,j} &= \bar{L}_2.\end{aligned}$$

In this case, we can sum over goods markets across sectors. Now, the goods market clearing conditions become:

$$\begin{aligned}\sum_{j=1}^J \beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} x_1 &= w_1 \bar{L}_1 - \sum_{j=1}^J \beta_{2,j} \frac{1}{1 + \tau_j^x} \pi_{21,j} x_2, \quad (\gamma_1) \\ \sum_{j=1}^J \beta_{1,j} \frac{1}{1 + \tau_j^m} \pi_{12,j} x_1 &= w_2 \bar{L}_2 - \sum_{j=1}^J \beta_{2,j} \pi_{22,j} x_2. \quad (\gamma_2)\end{aligned}$$

It is important to note that when technology $T_{n,j}$ is endogenous and depends on $L_{n,j}$, as demon-

strated in Appendix C.2.3, we cannot simply aggregate the goods market conditions.

Since the multipliers γ_1 and γ_2 are not sector-specific, the optimal policy following equations (A.14) becomes $1 + \tau_j^m = \gamma_2 = 1 + \tau^m$ and $1 + \tau_j^d = \gamma_1 = 1 + \tau^d$. Thus, domestic taxes and tariffs are uniform across sectors. Furthermore, since the labor market specification is homogeneous of degree zero on wages, tax neutrality holds. Both the uniform tariff and domestic tax can be normalized to zero.

C.4.2 Immobile and Sector-Specific Labor

In this case, labor is fixed in each sector, $L_{n,j} = \bar{L}_{n,j}$, which is (A.112) with $\kappa = 1$. The goods market clearing conditions are sector-specific:

$$\begin{aligned}\beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} x_1 &= w_{1,j} \bar{L}_{1,j} - \beta_{2,j} \frac{1}{1 + \tau_j^x} \pi_{21,j} x_2, \quad (\gamma_{1,j}, \quad I) \\ \beta_{1,j} \frac{1}{1 + \tau_j^m} \pi_{12,j} x_1 &= w_{2,j} \bar{L}_{2,j} - \beta_{2,j} \pi_{22,j} x_2. \quad (\gamma_{2,j}, \quad I)\end{aligned}$$

In this case, the element of matrix Λ_n at row j and column k is given by $\frac{\partial \ln(Y_{n,k})}{\partial \ln(p_{n,j})} \frac{Y_{n,k}}{Y_{n,j}}$, which is 1 if $j = k$ and 0 if $j \neq k$. Therefore, $\Lambda_n = I$, which satisfies the definition of GIIA Supply System.

Proof of optimal domestic taxes Equation (12) shows $(\Lambda_1 - I)(\mathbf{1} + \tau^d) = 0$, which is satisfied identically and therefore does not determine the domestic-tax vector. Proposition 3, which requires $\eta_j \neq 1$, does not apply here.

Proof of optimal import tariffs Since $\Lambda_2 = I$, consider the j th and k th rows of (13) in Proposition 2.

$$\begin{aligned}\tau_j^m + 1 + \Phi_2 &= \frac{M_{1,j}}{Y_{2,j}}, \\ \tau_k^m + 1 + \Phi_2 &= \frac{M_{1,k}}{Y_{2,k}}.\end{aligned}$$

The difference of the two implies

$$\tau_j^m - \tau_k^m = \frac{\beta_{1,j} x_1 - (1 + \tau_j^d) Y_{1,j}}{Y_{2,j}} - \frac{\beta_{1,k} x_1 - (1 + \tau_k^d) Y_{1,k}}{Y_{2,k}}.$$

Thus, Home's import tariff imposed on sector j increases with the share of Home's net imports in Foreign's income, relative to sector k .

D Non-GIIA Supply System: Multi-factor Ricardo-Roy Model

In this section, we explain why multi-factor Ricardo-Roy model does not satisfy GIIA Supply System. In this model, Home government's problem and world market equilibrium are the same as those in one-factor Ricardo-Roy model shown in Section C.1, except for the labor market specifications, which satisfy

$$w_{n,j}L_{n,j} = \sum_g \frac{A_{n,j,g}w_{n,j}^\kappa}{W_{n,g}^\kappa} W_{n,g}\bar{L}_{n,g}, \quad W_{n,g} = \left[\sum_{s=1}^J A_{n,s,g} (w_{n,s})^\kappa \right]^{\frac{1}{\kappa}}. \quad (\text{A.115})$$

Since there are constant returns to scale, prices are proportional to wages, which satisfy $p_{n,j} = \frac{w_{n,j}}{\bar{T}_{n,j}}$. We can derive supply curves by combining the labor market specification Ω (Eq. A.115) with the supply-side assumption $\mathbb{S}$ (Eq. A.27) and the definition of income $Y_{n,j} = w_{n,j}L_{n,j}$. Eliminating the wage and labor $\{w_{n,j}, L_{n,j}\}$ yields the following supply curves:

$$Y_{n,j} = (p_{n,j}\bar{T}_{n,j})^\kappa \sum_g A_{n,j,g} \left[\sum_{s=1}^J A_{n,s,g} (p_{n,s}\bar{T}_{n,s})^\kappa \right]^{\frac{1-\kappa}{\kappa}} \bar{L}_{n,g}.$$

The corresponding element of the scaled partial supply elasticity matrix Λ_n is given by

$$\frac{\partial \ln(Y_{n,s})}{\partial \ln(p_{n,j})} \frac{Y_{n,s}}{Y_{n,j}} = I_{s=j}(\kappa) - (\kappa - 1) \frac{\sum_g \lambda_{n,s,g} \lambda_{n,j,g} W_{n,g} \bar{L}_{n,g}}{Y_{n,j}},$$

where $I_{s=j}(\kappa)$ is an indicator function that equals κ when $s = j$ and 0 otherwise. $\lambda_{n,j,g} = Y_{n,j,g} / (W_{n,g} \bar{L}_{n,g})$ is the share of type- g workers that enter sector j .

The scaled partial supply elasticity matrix Λ_n of country n , with its j th-row- i th-column entry as

$\frac{\partial \ln Y_{n,i}}{\partial \ln p_{n,j}} \frac{Y_{n,i}}{Y_{n,j}}$, takes the form of

$$\Lambda_n = \begin{pmatrix} \kappa - (\kappa - 1) \frac{\sum_g \lambda_{n,1,g} \lambda_{n,1,g} W_{n,g} \bar{L}_{n,g}}{Y_{n,1}} & \dots & -(\kappa - 1) \frac{\sum_g \lambda_{n,1,g} \lambda_{n,j,g} W_{n,g} \bar{L}_{n,g}}{Y_{n,1}} & \dots & -(\kappa - 1) \frac{\sum_g \lambda_{n,1,g} \lambda_{n,J,g} W_{n,g} \bar{L}_{n,g}}{Y_{n,1}} \\ \dots & \dots & \dots & \dots & \dots \\ -(\kappa - 1) \frac{\sum_g \lambda_{n,j,g} \lambda_{n,1,g} W_{n,g} \bar{L}_{n,g}}{Y_{n,j}} & \dots & \kappa - (\kappa - 1) \frac{\sum_g \lambda_{n,j,g} \lambda_{n,j,g} W_{n,g} \bar{L}_{n,g}}{Y_{n,j}} & \dots & -(\kappa - 1) \frac{\sum_g \lambda_{n,j,g} \lambda_{n,J,g} W_{n,g} \bar{L}_{n,g}}{Y_{n,j}} \\ \dots & \dots & \dots & \dots & \dots \\ -(\kappa - 1) \frac{\sum_g \lambda_{n,J,g} \lambda_{n,1,g} W_{n,g} \bar{L}_{n,g}}{Y_{n,J}} & \dots & -(\kappa - 1) \frac{\sum_g \lambda_{n,J,g} \lambda_{n,j,g} W_{n,g} \bar{L}_{n,g}}{Y_{n,J}} & \dots & \kappa - (\kappa - 1) \frac{\sum_g \lambda_{n,J,g} \lambda_{n,J,g} W_{n,g} \bar{L}_{n,g}}{Y_{n,J}} \end{pmatrix}. \quad (\text{A.116})$$

Foreign sector i 's supply, represented by $Y_{2,i}$, is affected by the sector j 's price $p_{2,j}$ through the expression $-(\kappa - 1) \sum_g \frac{\lambda_{2,j,g} W_{2,g} \bar{L}_{2,g}}{Y_{2,j}} \lambda_{2,i,g}$, and affected by sector J 's price $p_{2,J}$, $-(\kappa - 1) \sum_g \frac{\lambda_{2,J,g} W_{2,g} \bar{L}_{2,g}}{Y_{2,J}} \lambda_{2,i,g}$. In general, these cross-elasticities are influenced by asymmetric endogenous variables and cannot be canceled out. Hence, this model does not satisfy the GIIA Supply System as defined in Definition 4. The relative optimal tariffs between two sectors are also influenced by other sectors. Optimal policies still satisfy the general formula as in Proposition 2.

Proof of optimal domestic taxes According to Proposition 2, optimal domestic taxes satisfy

$$(\Lambda_1 - I)(1 + \tau^d) = 0,$$

where

$$\Lambda_1 - I = (\kappa - 1) \begin{pmatrix} 1 - \frac{\sum_g \lambda_{1,1,g} \lambda_{1,1,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,1}} & \dots & -\frac{\sum_g \lambda_{1,1,g} \lambda_{1,j,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,1}} & \dots & -\frac{\sum_g \lambda_{1,1,g} \lambda_{1,J,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,1}} \\ \dots & \dots & \dots & \dots & \dots \\ -\frac{\sum_g \lambda_{1,j,g} \lambda_{1,1,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,j}} & \dots & 1 - \frac{\sum_g \lambda_{1,j,g} \lambda_{1,j,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,j}} & \dots & -\frac{\sum_g \lambda_{1,j,g} \lambda_{1,J,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,j}} \\ \dots & \dots & \dots & \dots & \dots \\ -\frac{\sum_g \lambda_{1,J,g} \lambda_{1,1,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,J}} & \dots & -\frac{\sum_g \lambda_{1,J,g} \lambda_{1,j,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,J}} & \dots & 1 - \frac{\sum_g \lambda_{1,J,g} \lambda_{1,J,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,J}} \end{pmatrix}. \quad (\text{A.117})$$

Due to tax neutrality, the domestic tax can only be determined at a relative level. We define a $(J - 1) * (J - 1)$ matrix M that satisfies

$$M = (\kappa - 1) \begin{pmatrix} 1 - \frac{\sum_g \lambda_{1,1,g} \lambda_{1,1,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,1}} & \dots & -\frac{\sum_g \lambda_{1,1,g} \lambda_{1,j,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,1}} & \dots & -\frac{\sum_g \lambda_{1,1,g} \lambda_{1,J-1,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,1}} \\ \dots & \dots & \dots & \dots & \dots \\ -\frac{\sum_g \lambda_{1,j,g} \lambda_{1,1,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,j}} & \dots & 1 - \frac{\sum_g \lambda_{1,j,g} \lambda_{1,j,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,j}} & \dots & -\frac{\sum_g \lambda_{1,j,g} \lambda_{1,J-1,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,j}} \\ \dots & \dots & \dots & \dots & \dots \\ -\frac{\sum_g \lambda_{1,J-1,g} \lambda_{1,1,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,J-1}} & \dots & -\frac{\sum_g \lambda_{1,J-1,g} \lambda_{1,j,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,J-1}} & \dots & 1 - \frac{\sum_g \lambda_{1,J-1,g} \lambda_{1,J-1,g} W_{1,g} \bar{L}_{1,g}}{Y_{1,J-1}} \end{pmatrix}.$$

Using this matrix, we can rewrite the system of equations (A.117) as

$$M \cdot \begin{pmatrix} \tau_1^d - \tau_j^d \\ \dots \\ \tau_j^d - \tau_j^d \\ \dots \\ \tau_{j-1}^d - \tau_j^d \end{pmatrix} = \begin{pmatrix} 0 \\ \dots \\ 0 \\ \dots \\ 0 \end{pmatrix}.$$

As $|M| \neq 0$, $\tau_j^d - \tau_j^d = 0$, and thus domestic taxes are uniform across all sectors. $\square$

E Proof for Proposition 1 and 4: General Formula for Multi-Country Optimal Policies

In this section, we use the Lagrange multiplier method to study the optimization problem constrained to world equilibrium conditions shown in Appendix A, where γ_x is the multiplier on Home's expenditure, and $\gamma_{1,j}$ and $\gamma_{n,j}$ are the multipliers on the goods market clearing conditions that determine prices. According to the Walras' law, one of the equilibrium conditions is redundant; thus, we drop the expenditure constraint in optimization and set $\gamma_x = 0$. Or, equivalently, if we use Home's budget constraint, one of the market-clearing conditions is redundant. In addition, we normalize the price index of Home as the numeraire.

FOC over x_1

$$1 - \sum_{j=1}^J \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} - \sum_{i \neq 1}^N \sum_{j=1}^J \gamma_{i,j} \beta_{1,j} \frac{1}{1 + \tau_{i,j}^m} \pi_{1i,j} = 0.$$

FOC over import tariff $\tau_{n,j}^m$, $n \neq 1$

$$- \beta_{1,j} x_1 \frac{\pi_{1n,j}}{1 + \tau_{n,j}^m} - \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \frac{\partial \pi_{11,j}}{\partial \tau_{n,j}^m} x_1 + \gamma_{n,j} \beta_{1,j} \frac{\pi_{1n,j}}{(1 + \tau_{n,j}^m)^2} x_1 - \sum_{i \neq 1}^N \gamma_{i,j} \beta_{1,j} \frac{1}{1 + \tau_{i,j}^m} \frac{\partial \pi_{1i,j}}{\partial \tau_{n,j}^m} x_1 = 0.$$

Plugging into derivatives

$$- \beta_{1,j} x_1 \pi_{1n,j} - \gamma_{1,j} \frac{\beta_{1,j} \epsilon_j \pi_{11,j} \pi_{1n,j} x_1}{1 + \tau_j^d} + \gamma_{n,j} \frac{\beta_{1,j} \pi_{1n,j} x_1}{1 + \tau_{n,j}^m} - \sum_{i \neq 1}^N \gamma_{i,j} \frac{\beta_{1,j} \epsilon_j \pi_{1i,j} \pi_{1n,j} x_1}{1 + \tau_{i,j}^m} + \gamma_{n,j} \frac{\beta_{1,j} \epsilon_j \pi_{1n,j} x_1}{1 + \tau_{n,j}^m} = 0 \quad (\text{A.118})$$

$$\Rightarrow -1 - \gamma_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j \pi_{11,j} + \gamma_{n,j} \frac{1}{1 + \tau_{n,j}^m} - \sum_{i \neq 1}^N \gamma_{i,j} \frac{1}{1 + \tau_{i,j}^m} \epsilon_j \pi_{1i,j} + \gamma_{n,j} \frac{1}{1 + \tau_{n,j}^m} \epsilon_j = 0$$

$$\Rightarrow -1 - \gamma_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j \pi_{11,j} + \gamma_{n,j} \frac{1 + \epsilon_j}{1 + \tau_{n,j}^m} - \sum_{i \neq 1}^N \gamma_{i,j} \frac{1}{1 + \tau_{i,j}^m} \epsilon_j \pi_{1i,j} = 0. \quad (\text{A.119})$$

Thus, $\gamma_{n,j} \frac{1}{1 + \tau_{n,j}^m}$ is the same across all countries in sector j .

FOC over domestic tax τ_j^d

$$- \beta_{1,j} x_1 \frac{\pi_{11,j}}{1 + \tau_j^d} - \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \frac{\partial \pi_{11,j}}{\partial \tau_j^d} x_1 + \gamma_{1,j} \beta_{1,j} \frac{\pi_{11,j}}{(1 + \tau_j^d)^2} x_1 - \sum_{i \neq 1}^N \gamma_{i,j} \beta_{1,j} \frac{1}{1 + \tau_{i,j}^m} \frac{\partial \pi_{1i,j}}{\partial \tau_j^d} x_1 = 0.$$

Plugging into derivatives

$$- \beta_{1,j} x_1 \pi_{11,j} + \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j \pi_{11,j} (1 - \pi_{11,j}) x_1 + \gamma_{1,j} \beta_{1,j} \frac{\pi_{11,j}}{1 + \tau_j^d} x_1 - \sum_{i \neq 1}^N \gamma_{i,j} \beta_{1,j} \frac{1}{1 + \tau_{i,j}^m} \epsilon_j \pi_{1i,j} \pi_{11,j} x_1 = 0 \quad (\text{A.120})$$

$$\Rightarrow -1 + \gamma_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j (1 - \pi_{11,j}) + \gamma_{1,j} \frac{1}{1 + \tau_j^d} - \sum_{i \neq 1}^N \gamma_{i,j} \frac{1}{1 + \tau_{i,j}^m} \epsilon_j \pi_{1i,j} = 0.$$

Combining with (A.119),

$$\begin{aligned}\gamma_{1,j} \frac{1}{1 + \tau_j^d} &= \gamma_{n,j} \frac{1}{1 + \tau_{n,j}^m} \\ \Rightarrow 1 + \tau_j^d &= \gamma_{1,j}, \quad 1 + \tau_{n,j}^m = \gamma_{n,j}.\end{aligned}\tag{A.121}$$

FOC over export tax $\tau_{n,j}^x$, $n \neq 1$

$$- \gamma_{1,j} \beta_{n,j} \frac{1}{(1 + \tau_{n,j}^x)^2} \pi_{n1,j} x_n + \gamma_{1,j} \beta_{n,j} \frac{1}{1 + \tau_{n,j}^x} \frac{\partial \pi_{n1,j}}{\partial \tau_{n,j}^x} x_n + \sum_{i \neq 1}^N \gamma_{i,j} \beta_{n,j} \frac{\partial \pi_{ni,j}}{\partial \tau_{n,j}^x} x_n = 0.$$

Plugging into derivatives

$$- \gamma_{1,j} \beta_{n,j} \frac{1}{1 + \tau_{n,j}^x} \pi_{n1,j} x_n - \gamma_{1,j} \beta_{n,j} \frac{1}{1 + \tau_{n,j}^x} \epsilon_j \pi_{n1,j} (1 - \pi_{n1,j}) x_n + \sum_{i \neq 1}^N \gamma_{i,j} \beta_{n,j} \epsilon_j \pi_{ni,j} \pi_{n1,j} x_n = 0\tag{A.122}$$

$$\begin{aligned}\Rightarrow - \gamma_{1,j} \frac{1}{1 + \tau_{n,j}^x} - \gamma_{1,j} \frac{1}{1 + \tau_{n,j}^x} \epsilon_j (1 - \pi_{n1,j}) + \sum_{i \neq 1}^N \gamma_{i,j} \epsilon_j \pi_{ni,j} &= 0 \\ \Rightarrow \frac{1}{1 + \tau_{n,j}^x} &= \frac{\sum_{i \neq 1}^N \gamma_{i,j} \epsilon_j \pi_{ni,j}}{\gamma_{1,j} (1 + \epsilon_j (1 - \pi_{n1,j}))} = \frac{1}{1 + \tau_j^d} \frac{\sum_{i \neq 1}^N (1 + \tau_{i,j}^m) \epsilon_j \pi_{ni,j}}{1 + \epsilon_j (1 - \pi_{n1,j})}.\end{aligned}\tag{A.123}$$

FOC over $p_{1,j}$

$$\begin{aligned}- x_1 \beta_{1,j} \frac{\pi_{11,j}}{p_{1,j}} - \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \frac{\partial \pi_{11,j}}{\partial p_{1,j}} x_1 + \sum_{k=1}^J \gamma_{1,k} \frac{\partial E_{11,k}}{\partial p_{1,j}} - \sum_{i \neq 1}^N \gamma_{i,j} \beta_{1,j} \frac{1}{1 + \tau_{i,j}^m} \frac{\partial \pi_{1i,j}}{\partial p_{1,j}} x_1 + \sum_{k=1}^J \sum_{i \neq 1}^N \gamma_{i,k} \frac{\partial E_{1i,k}}{\partial p_{1,j}} &= 0 \\ \Rightarrow - x_1 \beta_{1,j} \pi_{11,j} + \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \epsilon_j \pi_{11,j} (1 - \pi_{11,j}) x_1 + \sum_{k=1}^J \gamma_{1,k} \frac{\partial Y_{1,k}}{\partial \ln p_{1,j}} + \gamma_{1,j} \sum_{i \neq 1}^N \beta_{i,j} \frac{1}{1 + \tau_{i,j}^x} \epsilon_j \pi_{1i,j} (1 - \pi_{1i,j}) x_i \\ - \sum_{i \neq 1}^N \gamma_{i,j} \beta_{1,j} \frac{1}{1 + \tau_{i,j}^m} \epsilon_j \pi_{1i,j} \pi_{11,j} x_1 - \sum_{i \neq 1}^N \gamma_{i,j} \sum_{n \neq 1}^N \beta_{n,j} \epsilon_j \pi_{ni,j} \pi_{n1,j} x_n &= 0,\end{aligned}\tag{A.124}$$

where $\frac{\partial E_{11,k}}{\partial \ln p_{1,j}} = \frac{\partial Y_{1,k}}{\partial \ln p_{1,j}} + I_{k=j} \left[\sum_{i \neq 1}^N \beta_{i,k} \frac{x_i}{1 + \tau_{i,k}^x} \epsilon_k \pi_{i1,k} (1 - \pi_{i1,k}) \right] - \beta_{1,k} \frac{\pi_{11,k}}{1 + \tau_k^d} \frac{\partial \sum_{s=1}^J Y_{1,s}}{\partial \ln p_{1,j}}.$

Combining (A.124) with (A.121) and (A.122), we derive the formula for domestic taxes

$$\begin{aligned}
& -x_1\beta_{1,j}\pi_{11,j} + \sum_{k=1}^J \gamma_{1,k} \frac{\partial Y_{1,k}}{\partial \ln p_{1,j}} - \gamma_{1,j} \sum_{n \neq 1}^N \beta_{n,j} \frac{1}{1 + \tau_{n,j}^x} \pi_{n1,j} x_n = 0 \\
\Rightarrow & -x_1\beta_{1,j}\pi_{11,j} + \sum_{k=1}^J \gamma_{1,k} \frac{\partial Y_{1,k}}{\partial \ln p_{1,j}} - \gamma_{1,j} Y_{1,j} + \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} x_1 = 0 \\
\Rightarrow & \sum_{k=1}^J \gamma_{1,k} \frac{\partial Y_{1,k}}{\partial \ln p_{1,j}} = \gamma_{1,j} Y_{1,j} \\
\Rightarrow & \sum_{k=1}^J (1 + \tau_k^d) \frac{\partial \ln Y_{1,k}}{\partial \ln p_{1,j}} \frac{Y_{1,k}}{Y_{1,j}} = 1 + \tau_j^d. \tag{A.125}
\end{aligned}$$

FOC over $p_{n,j}$, $n \neq 1$

$$\begin{aligned}
& -x_1\beta_{1,j} \frac{\pi_{1n,j}}{p_{n,j}} - \gamma_{1,j}\beta_{1,j} \frac{1}{1 + \tau_j^d} \frac{\partial \pi_{11,j}}{\partial p_{n,j}} x_1 + \sum_{k=1}^J \gamma_{1,k} \frac{\partial E_{11,k}}{\partial p_{n,j}} - \sum_{i \neq 1}^N \gamma_{i,j}\beta_{1,j} \frac{1}{1 + \tau_{i,j}^m} \frac{\partial \pi_{1i,j}}{\partial p_{n,j}} x_1 + \sum_{k=1}^J \sum_{i \neq 1}^N \gamma_{i,k} \frac{\partial E_{1i,k}}{\partial p_{n,j}} = 0 \\
\Rightarrow & \sum_{k=1}^J \sum_{i \neq 1}^N \gamma_{i,k} \frac{\partial E_{1i,k}}{\partial p_{n,j}} p_{n,j} = x_1\beta_{1,j}\pi_{1n,j} - \sum_{k=1}^J \gamma_{1,k} \frac{\partial E_{11,k}}{\partial p_{n,j}} p_{n,j} \\
\Rightarrow & \sum_{k=1}^J \sum_{i \neq 1}^N (1 + \tau_{i,k}^m) \frac{\partial E_{1i,k}}{\partial \ln p_{n,j}} = x_1\beta_{1,j}\pi_{1n,j} - \sum_{k=1}^J (1 + \tau_k^d) \frac{\partial E_{11,k}}{\partial \ln p_{n,j}}. \tag{A.126}
\end{aligned}$$

We can rewrite (A.126) into matrix and determine the $(N-1) \times J$ tariffs

$$\begin{pmatrix} \frac{\partial E_{12,1}}{\partial \ln p_{2,1}} \frac{1}{Y_{2,1}} & \cdots & \frac{\partial E_{12,J}}{\partial \ln p_{2,1}} \frac{1}{Y_{2,1}} & \vdots & \frac{\partial E_{13,1}}{\partial \ln p_{2,1}} \frac{1}{Y_{2,1}} & \cdots & \frac{\partial E_{1N,J}}{\partial \ln p_{2,1}} \frac{1}{Y_{2,1}} \\ \cdots & \cdots & \cdots & \vdots & \cdots & \cdots & \cdots \\ \frac{\partial E_{12,1}}{\partial \ln p_{2,J}} \frac{1}{Y_{2,J}} & \cdots & \frac{\partial E_{12,J}}{\partial \ln p_{2,J}} \frac{1}{Y_{2,J}} & \vdots & \frac{\partial E_{13,1}}{\partial \ln p_{2,J}} \frac{1}{Y_{2,J}} & \cdots & \frac{\partial E_{1N,J}}{\partial \ln p_{2,J}} \frac{1}{Y_{2,J}} \\ \frac{\partial E_{12,1}}{\partial \ln p_{3,1}} \frac{1}{Y_{3,1}} & \cdots & \frac{\partial E_{12,J}}{\partial \ln p_{3,1}} \frac{1}{Y_{3,1}} & \vdots & \frac{\partial E_{13,1}}{\partial \ln p_{3,1}} \frac{1}{Y_{3,1}} & \cdots & \frac{\partial E_{1N,J}}{\partial \ln p_{3,1}} \frac{1}{Y_{3,1}} \\ \cdots & \cdots & \cdots & \vdots & \cdots & \cdots & \cdots \\ \frac{\partial E_{12,1}}{\partial \ln p_{N,J}} \frac{1}{Y_{N,J}} & \cdots & \frac{\partial E_{12,J}}{\partial \ln p_{N,J}} \frac{1}{Y_{N,J}} & \vdots & \frac{\partial E_{13,1}}{\partial \ln p_{N,J}} \frac{1}{Y_{N,J}} & \cdots & \frac{\partial E_{1N,J}}{\partial \ln p_{N,J}} \frac{1}{Y_{N,J}} \end{pmatrix} \begin{pmatrix} 1 + \tau_{2,1}^m \\ \cdots \\ 1 + \tau_{2,J}^m \\ 1 + \tau_{3,1}^m \\ \cdots \\ 1 + \tau_{N,J}^m \end{pmatrix} = \begin{pmatrix} \beta_{1,1}\pi_{12,1}x_1 \frac{1}{Y_{2,1}} - \sum_{k=1}^J (1 + \tau_k^d) \frac{\partial E_{11,k}}{\partial \ln p_{2,1}} \frac{1}{Y_{2,1}} \\ \cdots \\ \beta_{1,J}\pi_{12,J}x_1 \frac{1}{Y_{2,J}} - \sum_{k=1}^J (1 + \tau_k^d) \frac{\partial E_{11,k}}{\partial \ln p_{2,J}} \frac{1}{Y_{2,J}} \\ \beta_{1,1}\pi_{13,1}x_1 \frac{1}{Y_{3,1}} - \sum_{k=1}^J (1 + \tau_k^d) \frac{\partial E_{11,k}}{\partial \ln p_{3,1}} \frac{1}{Y_{3,1}} \\ \cdots \\ \beta_{1,J}\pi_{1N,J}x_1 \frac{1}{Y_{N,J}} - \sum_{k=1}^J (1 + \tau_k^d) \frac{\partial E_{11,k}}{\partial \ln p_{N,J}} \frac{1}{Y_{N,J}} \end{pmatrix}, \tag{A.127}$$

where the semi-export-supply-elasticities are

$$\begin{aligned}\frac{\partial E_{11,k}}{\partial \ln p_{n,j}} &= -I_{k=j} \left[\sum_{i \neq 1}^N \beta_{i,k} \frac{x_i}{1 + \tau_{i,k}^x} \epsilon_k \pi_{i1,k} \pi_{in,k} \right] - \beta_{n,k} \frac{\pi_{n1,k}}{1 + \tau_{n,k}^x} \frac{\partial \sum_{s=1}^J Y_{n,s}}{\partial \ln p_{n,j}}, \quad (n \neq 1) \\ \frac{\partial E_{1n,k}}{\partial \ln p_{n,j}} &= \frac{\partial Y_{n,k}}{\partial \ln p_{n,j}} + I_{k=j} \left[\sum_{s \neq 1}^N \beta_{s,k} x_s \epsilon_k (1 - \pi_{sn,k}) \pi_{sn,k} \right] - \beta_{n,k} \pi_{nn,k} \frac{\partial \sum_{s=1}^J Y_{n,s}}{\partial \ln p_{n,j}}, \quad (n \neq 1) \\ \frac{\partial E_{1i,k}}{\partial \ln p_{n,j}} &= -I_{k=j} \left[\sum_{s \neq 1}^N \beta_{s,k} x_s \epsilon_k \pi_{si,k} \pi_{sn,k} \right] - \beta_{n,k} \pi_{ni,k} \frac{\partial \sum_{s=1}^J Y_{n,s}}{\partial \ln p_{n,j}}, \quad (i \neq n, i \neq 1).\end{aligned}$$

Plugging derivatives into (A.126),

$$\begin{aligned}& \sum_{k=1}^J (1 + \tau_{n,k}^m) \frac{\partial Y_{n,k}}{\partial \ln p_{n,j}} + (1 + \tau_{n,j}^m) \sum_{s \neq 1} \beta_{s,j} x_s \epsilon_j (1 - \pi_{sn,j}) \pi_{sn,j} - \sum_{i \neq 1, n} (1 + \tau_{i,j}^m) \sum_{s \neq 1} \beta_{s,j} x_s \epsilon_j \pi_{si,j} \pi_{sn,j} \\ & - \frac{\partial \sum_{s=1}^J Y_{n,s}}{\partial \ln p_{n,j}} \sum_{k=1}^J \beta_{n,k} \left[(1 + \tau_{n,k}^m) \pi_{nn,k} + \sum_{i \neq 1, n} (1 + \tau_{i,k}^m) \pi_{ni,k} \right] \\ & = x_1 \beta_{1,j} \pi_{1n,j} + (1 + \tau_j^d) \sum_{i \neq 1} \beta_{i,j} \frac{x_i}{1 + \tau_{i,j}^x} \epsilon_j \pi_{i1,j} \pi_{in,j} + \frac{\partial \sum_{s=1}^J Y_{n,s}}{\partial \ln p_{n,j}} \sum_{k=1}^J (1 + \tau_k^d) \beta_{n,k} \frac{\pi_{n1,k}}{1 + \tau_{n,k}^x} \\ & \Rightarrow \sum_{k=1}^J (\Phi_n + 1 + \tau_{n,k}^m) \frac{\partial Y_{n,k}}{\partial \ln p_{n,j}} + (1 + \tau_{n,j}^m) \sum_{s \neq 1} \beta_{s,j} x_s \epsilon_j (1 - \pi_{sn,j}) \pi_{sn,j} - \sum_{i \neq 1, n} (1 + \tau_{i,j}^m) \sum_{s \neq 1} \beta_{s,j} x_s \epsilon_j \pi_{si,j} \pi_{sn,j} \\ & = x_1 \beta_{1,j} \pi_{1n,j} + (1 + \tau_j^d) \sum_{i \neq 1} \beta_{i,j} \frac{x_i}{1 + \tau_{i,j}^x} \epsilon_j \pi_{i1,j} \pi_{in,j} \\ & \Rightarrow \sum_{k=1}^J (\Phi_n + 1 + \tau_{n,k}^m) \frac{\partial Y_{n,k}}{\partial \ln p_{n,j}} = x_1 \beta_{1,j} \pi_{1n,j} + \sum_{s \neq 1} \beta_{s,j} x_s \pi_{sn,j} \left[(\epsilon_j + 1) \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} - \epsilon_j (1 + \tau_{n,j}^m) \right] \quad (\text{A.128})\end{aligned}$$

where $\Phi_n = -\sum_{i \neq 1}^N \sum_{k=1}^J \beta_{n,k} (1 + \tau_{i,k}^m) \pi_{ni,k} - \sum_{k=1}^J \beta_{n,k} (1 + \tau_k^d) \frac{\pi_{n1,k}}{1 + \tau_{n,k}^x}$.

This equation provides a supply-system representation of the import tariff. Its left-hand side captures how a change in $p_{n,j}$ affects production across all sectors of country n . Φ_n is the country-level multilateral expenditure-feedback, which is endogenous but common across sectors for a given country n . It summarizes how changes in country n 's aggregate expenditure affect demand across all origins.

□

F Proof for Proposition 5: Multi-Country Optimal Tariffs under GIIA Supply System

In this section, we first present the formula for the optimal domestic taxes and tariffs within a GIIA Supply System with η_j . Then, we prove Proposition 5. Lastly, we present the formula for the optimal policies within a GIIA Supply System with constant returns to scale.

Similar to the proof in Section B.1, the optimal domestic taxes satisfy

$$(1 + \tau_j^d) - (1 + \tau_k^d) = (1 + \tau_j^d) \frac{1}{\eta_j} - (1 + \tau_k^d) \frac{1}{\eta_k} \Rightarrow \frac{1 + \tau_j^d}{1 + \tau_k^d} = \frac{\frac{\eta_j}{\eta_j - 1}}{\frac{\eta_k}{\eta_k - 1}}.$$

According to the tax neutrality shown in Lemma 1, we can set $1 + \tau_j^d = (1 + \bar{\tau}^d) \frac{\eta_j}{\eta_j - 1}, \forall j$. The uniform shifter $\bar{\tau}^d$ can be assigned any arbitrary value, and one of the trade policies can be normalized, without changing the real allocations and welfare impacts of optimal policies.

(A.128) can be rewritten as follows:

$$\begin{aligned} \sum_{k=1}^J (\Phi_n + 1 + \tau_{n,k}^m) \frac{\partial \ln Y_{n,k}}{\partial \ln p_{n,j}} \frac{Y_{n,k}}{Y_{n,j}} \frac{1}{\eta_j} &= \frac{1}{\eta_j} \frac{1}{Y_{n,j}} \left\{ x_1 \beta_{1,j} \pi_{1n,j} + \sum_{s \neq 1} \beta_{s,j} x_s \pi_{sn,j} \left[(\epsilon_j + 1) \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} - \epsilon_j (1 + \tau_{n,j}^m) \right] \right\} \\ \Rightarrow \tau_{n,j}^m - \tau_{n,k}^m &= \frac{1}{\eta_j} \frac{1}{Y_{n,j}} \left\{ x_1 \beta_{1,j} \pi_{1n,j} + \sum_{s \neq 1} \beta_{s,j} x_s \pi_{sn,j} \left[(\epsilon_j + 1) \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} - \epsilon_j (1 + \tau_{n,j}^m) \right] \right\} \\ &\quad - \frac{1}{\eta_k} \frac{1}{Y_{n,k}} \left\{ x_1 \beta_{1,k} \pi_{1n,k} + \sum_{s \neq 1} \beta_{s,k} x_s \pi_{sn,k} \left[(\epsilon_k + 1) \frac{1 + \tau_k^d}{1 + \tau_{s,k}^x} - \epsilon_k (1 + \tau_{n,k}^m) \right] \right\} \end{aligned} \quad (\text{A.129})$$

$$\text{or } \tau_{n,j}^m = \frac{1}{\eta_j} \frac{1}{Y_{n,j}} \left\{ x_1 \beta_{1,j} \pi_{1n,j} + \sum_{s \neq 1} \beta_{s,j} x_s \pi_{sn,j} \left[(\epsilon_j + 1) \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} - \epsilon_j (1 + \tau_{n,j}^m) \right] \right\} + \tau_n^m, \quad (\text{A.130})$$

where τ_n^m is endogenous but common across sectors. The GIIA restriction implies that the jk th element of Λ_n , after scaling by $1/\eta_j$, can be written as

$$\frac{\Lambda_{n,jk}}{\eta_j} \equiv \frac{\partial \ln Y_{n,k}}{\partial \ln p_{n,j}} \frac{Y_{n,k}}{Y_{n,j}} \frac{1}{\eta_j} = I_{k=j} + a_{n,k}. \quad (\text{A.131})$$

Hence, the optimal tariff satisfies

$$\tau_{n,j}^m = \frac{1}{\eta_j} \frac{1}{Y_{n,j}} \left\{ x_1 \beta_{1,j} \pi_{1n,j} + \sum_{s \neq 1} \beta_{s,j} x_s \pi_{sn,j} \left[(\epsilon_j + 1) \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} - \epsilon_j (1 + \tau_{n,j}^m) \right] \right\} - (\Phi_n + 1) - \sum_{k=1}^J (\Phi_n + 1 + \tau_{n,k}^m) a_{n,k},$$

and

$$\tau_n^m = -(\Phi_n + 1) - \sum_{k=1}^J (\Phi_n + 1 + \tau_{n,k}^m) a_{n,k}.$$

Equivalently, for any reference sector s ,

$$\tau_n^m = -(\Phi_n + 1) - \sum_{k=1}^J (\Phi_n + 1 + \tau_{n,k}^m) \left[\frac{\partial \ln Y_{n,k}}{\partial \ln p_{n,s}} \frac{Y_{n,k}}{Y_{n,s}} \frac{1}{\eta_s} - I_{k=s} \right].$$

In the two-country case,

$$\sum_{s \neq 1} \beta_{s,j} x_s \pi_{sn,j} \left[(\epsilon_j + 1) \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} - \epsilon_j (1 + \tau_{n,j}^m) \right] = - \frac{1 + \tau_j^d}{1 + \tau_{n,j}^x} \beta_{n,j} x_n \pi_{n1,j}.$$

Therefore,

$$\begin{aligned} \tau_{n,j}^m &= \frac{1}{\eta_j} \frac{1}{Y_{n,j}} \left[x_1 \beta_{1,j} \pi_{1n,j} - \frac{1 + \tau_j^d}{1 + \tau_{n,j}^x} \beta_{n,j} x_n \pi_{n1,j} \right] + \tau_n^m \\ &= \frac{1}{\eta_j} \frac{1}{Y_{n,j}} \left[x_1 \beta_{1,j} - (1 + \tau_j^d) Y_{1,j} \right] + \tau_n^m \\ &= \frac{1}{\eta_j} \frac{M_{1,j}}{Y_{n,j}} + \tau_n^m, \end{aligned} \tag{A.132}$$

which is consistent with Equation (17) in Proposition 3.

F.1 Optimal tariffs and average net import share

Next, we prove the first point in Proposition 5. By summing up $n \neq 1$ in (A.130), we derive:

$$\eta_j \sum_{n \neq 1}^N (\tau_{n,j}^m - \tau_n^m) Y_{n,j} = x_1 \beta_{1,j} \sum_{n \neq 1}^N \pi_{1n,j} + \sum_{s \neq 1}^N \beta_{s,j} x_s \sum_{n \neq 1}^N \pi_{sn,j} \left[(\epsilon_j + 1) \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} - \epsilon_j (1 + \tau_{n,j}^m) \right]. \tag{A.133}$$

For each foreign country $s \neq 1$, the export-tax FOC (A.123) implies

$$\frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} = \frac{\epsilon_j \sum_{n \neq 1}^N (1 + \tau_{n,j}^m) \pi_{sn,j}}{1 + \epsilon_j (1 - \pi_{s1,j})}. \tag{A.134}$$

Using $\sum_{n \neq 1}^N \pi_{sn,j} = 1 - \pi_{s1,j}$, we therefore have

$$\begin{aligned}
\sum_{n \neq 1}^N \pi_{sn,j} \left[(\epsilon_j + 1) \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} - \epsilon_j (1 + \tau_{n,j}^m) \right] &= (\epsilon_j + 1) (1 - \pi_{s1,j}) \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} - \epsilon_j \sum_{n \neq 1}^N (1 + \tau_{n,j}^m) \pi_{sn,j} \\
&= (\epsilon_j + 1) (1 - \pi_{s1,j}) \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} - [1 + \epsilon_j (1 - \pi_{s1,j})] \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} \\
&= - \pi_{s1,j} \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x}.
\end{aligned} \tag{A.135}$$

Substituting (A.135) into (A.133) yields

$$\eta_j \sum_{n \neq 1}^N (\tau_{n,j}^m - \tau_n^m) Y_{n,j} = x_1 \beta_{1,j} \sum_{n \neq 1}^N \pi_{1n,j} - \sum_{n \neq 1}^N \beta_{n,j} x_n \pi_{n1,j} \frac{1 + \tau_j^d}{1 + \tau_{n,j}^x}$$

Using $\sum_{n \neq 1}^N \pi_{1n,j} = 1 - \pi_{11,j}$ and the goods-market-clearing condition for Home production,

$$(1 + \tau_j^d) Y_{1,j} = \beta_{1,j} x_1 \pi_{11,j} + \sum_{n \neq 1}^N \beta_{n,j} x_n \pi_{n1,j} \frac{1 + \tau_j^d}{1 + \tau_{n,j}^x},$$

(A.133) becomes

$$\begin{aligned}
\eta_j \sum_{n \neq 1}^N (\tau_{n,j}^m - \tau_n^m) Y_{n,j} &= x_1 \beta_{1,j} - (1 + \tau_j^d) Y_{1,j} = M_{1,j} \\
\Rightarrow \sum_{n \neq 1}^N (\tau_{n,j}^m - \tau_n^m) Y_{n,j} &= \frac{1}{\eta_j} M_{1,j}.
\end{aligned} \tag{A.136}$$

F.2 Identical foreign countries

We then prove the second point in Proposition 5. When foreign countries are identical, the optimal import tariffs and export taxes are identical across all countries within a sector j , i.e., $\tau_{n,j}^m = \tau_{i,j}^m$ and $\tau_{n,j}^x = \tau_{i,j}^x, \forall n \neq i > 1, \forall j$. Trade shares between any two foreign countries within a sector are equal. Furthermore, $\beta_{n,j} = \beta_{i,j}$ and $x_n = x_i, \forall n \neq i > 1, \forall j$. The optimal export tax in (A.123) can be expressed as

$$\begin{aligned}
\frac{1 + \tau_j^d}{1 + \tau_{n,j}^x} &= (1 + \tau_{n,j}^m) \frac{\sum_{i \neq 1}^N \epsilon_j \pi_{ni,j}}{1 + \epsilon_j (1 - \pi_{n1,j})} = (1 + \tau_{n,j}^m) \frac{\epsilon_j (1 - \pi_{n1,j})}{1 + \epsilon_j (1 - \pi_{n1,j})} \\
\Rightarrow 1 + \tau_{n,j}^x &= \frac{1 + \tau_j^d}{1 + \tau_{n,j}^m} \left(1 + \frac{1}{\epsilon_j (1 - \pi_{n1,j})} \right).
\end{aligned} \tag{A.137}$$

To derive the optimal import tariff, we can first reduce the term from (A.130) to

$$\sum_{s \neq 1}^N \beta_{s,j} x_s \pi_{sn,j} \left[(\epsilon_j + 1) \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} - \epsilon_j (1 + \tau_{n,j}^m) \right] = \beta_{n,j} x_n (1 - \pi_{n1,j}) \left[(\epsilon_j + 1) \frac{1 + \tau_j^d}{1 + \tau_{n,j}^x} - \epsilon_j (1 + \tau_{n,j}^m) \right].$$

Combining (A.137), it follows that

$$\sum_{s \neq 1}^N \beta_{s,j} x_s \pi_{sn,j} \left[(\epsilon_j + 1) \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} - \epsilon_j (1 + \tau_{n,j}^m) \right] = -\beta_{n,j} x_n \pi_{n1,j} \frac{1 + \tau_j^d}{1 + \tau_{n,j}^x}. \quad (\text{A.138})$$

Substituting (A.138) into (A.130) gives

$$\begin{aligned} \tau_{n,j}^m &= \frac{1}{\eta_j} \frac{1}{Y_{n,j}} \left[\pi_{1n,j} \beta_{1,j} x_1 - \frac{1 + \tau_j^d}{1 + \tau_{n,j}^x} \pi_{n1,j} \beta_{n,j} x_n \right] + \tau_n^m \\ &= \frac{1}{\eta_j} \frac{M_{1n,j}}{Y_{n,j}} + \tau_n^m, \end{aligned} \quad (\text{A.139})$$

where

$$M_{1n,j} \equiv \pi_{1n,j} \beta_{1,j} x_1 - \frac{1 + \tau_j^d}{1 + \tau_{n,j}^x} \pi_{n1,j} \beta_{n,j} x_n, \quad (\text{A.140})$$

and

$$\tau_n^m = \tau_g^m = \bar{\tau}^m$$

Taking the difference between sectors j and k eliminates τ_n^m and yields

$$\tau_{n,j}^m - \tau_{n,k}^m = \frac{1}{\eta_j} \frac{M_{1n,j}}{Y_{n,j}} - \frac{1}{\eta_k} \frac{M_{1n,k}}{Y_{n,k}}. \quad (\text{A.141})$$

F.3 No trade between foreign countries

When foreign countries do not trade with each other, the trade shares between any two foreign countries are zero, i.e., $\pi_{ni,j} = 0$ and $\pi_{nm,j} = 1 - \pi_{n1,j}, \forall n \neq i, n, i \neq 1, \forall j$. The optimal export tax in (A.123) can be written as

$$\begin{aligned} \frac{1 + \tau_j^d}{1 + \tau_{n,j}^x} &= \frac{(1 + \tau_{n,j}^m) \epsilon_j \pi_{nm,j}}{1 + \epsilon_j (1 - \pi_{n1,j})} \\ \Rightarrow 1 + \tau_{n,j}^x &= \frac{1 + \tau_j^d}{1 + \tau_{n,j}^m} \left(1 + \frac{1}{\epsilon_j (1 - \pi_{n1,j})} \right). \end{aligned} \quad (\text{A.142})$$

Because $\pi_{sn,j} = 0$ for $s \neq n$, the term in (A.130) reduces to

$$\begin{aligned} \sum_{s \neq 1}^N \beta_{s,j} x_s \pi_{sn,j} \left[(\epsilon_j + 1) \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} - \epsilon_j (1 + \tau_{n,j}^m) \right] &= \beta_{n,j} x_n \pi_{nn,j} \left[(\epsilon_j + 1) \frac{1 + \tau_j^d}{1 + \tau_{n,j}^x} - \epsilon_j (1 + \tau_{n,j}^m) \right] \\ &= -\beta_{n,j} x_n \pi_{n1,j} \frac{1 + \tau_j^d}{1 + \tau_{n,j}^x}, \end{aligned} \quad (\text{A.143})$$

where the last equality follows from (A.142) and $\pi_{nn,j} = 1 - \pi_{n1,j}$.

Substituting (A.143) into (A.130) gives

$$\begin{aligned} \tau_{n,j}^m &= \frac{1}{\eta_j} \frac{1}{Y_{n,j}} \left[\pi_{1n,j} \beta_{1,j} x_1 - \frac{1 + \tau_j^d}{1 + \tau_{n,j}^x} \pi_{n1,j} \beta_{n,j} x_n \right] + \tau_n^m \\ &= \frac{1}{\eta_j} \frac{M_{1n,j}}{Y_{n,j}} + \tau_n^m. \end{aligned} \quad (\text{A.144})$$

Therefore, for any sectors j and k ,

$$\tau_{n,j}^m - \tau_{n,k}^m = \frac{1}{\eta_j} \frac{M_{1n,j}}{Y_{n,j}} - \frac{1}{\eta_k} \frac{M_{1n,k}}{Y_{n,k}}. \quad (\text{A.145})$$

F.4 GIIA Supply System with constant returns to scale

Uniform optimal domestic taxes In a multi-country case, consider a GIIA Supply System exhibiting constant returns to scale with $\eta_j = \kappa$ for any j (see proof in Section C.1). We integrate Definition 4 with Equation (A.125):

$$\sum_{k=1}^J (1 + \tau_k^d) \frac{\partial \ln Y_{1,k}}{\partial \ln p_{1,j}} \frac{Y_{1,k}}{Y_{1,j}} = 1 + \tau_j^d \quad \Rightarrow \quad \tau_j^d - \tau_i^d = 0.$$

Thus, the optimal domestic tax is uniform across sectors. Due to the tax neutrality in Lemma 1, one domestic tax can be normalized to zero. Therefore, all domestic taxes are zero.

Optimal tariffs Within a GIIA Supply System with constant returns to scale, (A.130) becomes

$$\tau_{n,j}^m = \frac{1}{\kappa} \frac{1}{Y_{n,j}} \left\{ x_1 \beta_{1,j} \pi_{1n,j} + \sum_{s \neq 1} \beta_{s,j} x_s \pi_{sn,j} \left[(\epsilon_j + 1) \frac{1 + \tau_j^d}{1 + \tau_{s,j}^x} - \epsilon_j (1 + \tau_{n,j}^m) \right] \right\} + \tau_n^m \quad (\text{A.146})$$

where

$$\begin{aligned}\Phi_n &= -\sum_{i \neq 1}^N \sum_{k=1}^J \beta_{n,k} (1 + \tau_{i,k}^m) \pi_{ni,k} - \sum_{k=1}^J \beta_{n,k} \frac{\pi_{n1,k}}{1 + \tau_{n,k}^x}, \\ \tau_n^m &= \frac{\kappa - 1}{\kappa} \sum_{k=1}^J (\Phi_n + 1 + \tau_{n,k}^m) \lambda_{n,k} - (\Phi_n + 1) = \frac{\kappa - 1}{\kappa} \sum_{k=1}^J (1 + \tau_{n,k}^m) \lambda_{n,k} - \frac{1}{\kappa} \Phi_n - 1.\end{aligned}$$

The semi-export-supply-elasticities are

$$\begin{aligned}\frac{\partial E_{11,k}}{\partial \ln p_{n,j}} &= -I_{k=j} \left[\sum_{i \neq 1}^N \beta_{i,k} \frac{x_i}{1 + \tau_{i,k}^x} \epsilon_k \pi_{i1,k} \pi_{in,k} \right] - \beta_{n,k} \frac{\pi_{n1,k}}{1 + \tau_{n,k}^x} \frac{\partial \sum_{s=1}^J Y_{n,s}}{\partial \ln p_{n,j}}, \quad (n \neq 1) \\ &= -I_{k=j} \left[\sum_{i \neq 1}^N \beta_{i,k} \frac{x_i}{1 + \tau_{i,k}^x} \epsilon_k \pi_{i1,k} \pi_{in,k} \right] - \beta_{n,k} \frac{\pi_{n1,k}}{1 + \tau_{n,k}^x} Y_{n,j}, \\ \frac{\partial E_{1n,k}}{\partial \ln p_{n,j}} &= \frac{\partial Y_{n,k}}{\partial \ln p_{n,j}} + I_{k=j} \left[\sum_{s \neq 1}^N \beta_{s,k} x_s \epsilon_k (1 - \pi_{sn,k}) \pi_{sn,k} \right] - \beta_{n,k} \pi_{nn,k} \frac{\partial \sum_{s=1}^J Y_{n,s}}{\partial \ln p_{n,j}}, \quad (n \neq 1) \\ &= I_{k=j} [\kappa Y_{n,k}] - (\kappa - 1) \lambda_{n,k} Y_{n,j} + I_{k=j} \left[\sum_{s \neq 1}^N \beta_{s,k} x_s \epsilon_k (1 - \pi_{sn,k}) \pi_{sn,k} \right] - \beta_{n,k} \pi_{nn,k} Y_{n,j}, \\ \frac{\partial E_{1i,k}}{\partial \ln p_{n,j}} &= -I_{k=j} \left[\sum_{s \neq 1}^N \beta_{s,k} x_s \epsilon_k \pi_{si,k} \pi_{sn,k} \right] - \beta_{n,k} \pi_{ni,k} \frac{\partial \sum_{s=1}^J Y_{n,s}}{\partial \ln p_{n,j}}, \quad (i \neq n \neq 1) \\ &= -I_{k=j} \left[\sum_{s \neq 1}^N \beta_{s,k} x_s \epsilon_k \pi_{si,k} \pi_{sn,k} \right] - \beta_{n,k} \pi_{ni,k} Y_{n,j}.\end{aligned}$$

The optimal tariffs depend on the LHS matrix, dE . The diagonal blocks of dE are

$$\begin{aligned}\frac{\partial E_{1n}}{\partial \ln p_n} \circ \frac{1}{Y_n} &= \begin{pmatrix} \frac{\partial E_{1n,1}}{\partial \ln p_{n,1}} \frac{1}{Y_{n,1}} & \dots & \frac{\partial E_{1n,J}}{\partial \ln p_{n,J}} \frac{1}{Y_{n,J}} \\ \vdots & \ddots & \vdots \\ \frac{\partial E_{1n,1}}{\partial \ln p_{n,J}} \frac{1}{Y_{n,J}} & \dots & \frac{\partial E_{1n,J}}{\partial \ln p_{n,J}} \frac{1}{Y_{n,J}} \end{pmatrix} \\ &= \begin{pmatrix} \kappa - (\kappa - 1) \lambda_{n,1} + \epsilon_1 \sum_{h \neq 1}^N (1 - \pi_{hn,1}) s_{hn,1} - \beta_{n,1} \pi_{nn,1} & \dots & -(\kappa - 1) \lambda_{n,J} - \beta_{n,J} \pi_{nn,J} \\ \vdots & & \vdots \\ -(\kappa - 1) \lambda_{n,1} - \beta_{n,1} \pi_{nn,1} & \dots & \kappa - (\kappa - 1) \lambda_{n,J} + \epsilon_J \sum_{h \neq 1}^N (1 - \pi_{hn,J}) s_{hn,J} - \beta_{n,J} \pi_{nn,J} \end{pmatrix},\end{aligned}$$

where we define $s_{hn,j} = \frac{\pi_{hn,j} \beta_{h,j} x_h}{Y_{n,j}}$ as the share of the exporter n 's income from an importer h .

The off-diagonal blocks of dE are

$$\begin{aligned} \frac{\partial E_{1i}}{\partial \ln p_n} \circ \frac{1}{Y_n} &= \begin{pmatrix} \frac{\partial E_{1i,1}}{\partial \ln p_{n,1}} \frac{1}{Y_{n,1}} & \cdots & \frac{\partial E_{1i,J}}{\partial \ln p_{n,1}} \frac{1}{Y_{n,1}} \\ \cdots & & \cdots \\ \frac{\partial E_{1i,1}}{\partial \ln p_{n,J}} \frac{1}{Y_{n,J}} & \cdots & \frac{\partial E_{1i,J}}{\partial \ln p_{n,J}} \frac{1}{Y_{n,J}} \end{pmatrix} \\ &= \begin{pmatrix} -\epsilon_1 \sum_{h \neq 1}^N \pi_{hi,1} s_{hm,1} - \beta_{n,1} \pi_{ni,1} & \cdots & -\beta_{n,J} \pi_{ni,J} \\ \vdots & & \vdots \\ -\beta_{n,1} \pi_{ni,1} & \cdots & -\epsilon_J \sum_{h \neq 1}^N \pi_{hi,J} s_{hm,J} - \beta_{n,J} \pi_{ni,J} \end{pmatrix}. \end{aligned}$$

G Proof for Lemma 3: Convergence & Divergence

We first establish the result for an arbitrary number of foreign countries. Suppose that no foreign country sources sector- j goods from Home, so that

$$d_{g1,j} \rightarrow \infty, \quad g \neq 1.$$

It follows that $\pi_{g1,j} = 0$ and $\sum_{i \neq 1} \pi_{gi,j} = 1$ for $g \neq 1$.

At these zero shares, the export-tax first-order conditions reduce to identities and impose no additional restrictions. The FOC over $p_{n,j}$ becomes

$$\begin{aligned} \sum_{k=1}^J (\Phi_n + 1 + \tau_{n,k}^m) \frac{\partial Y_{n,k}}{\partial \ln p_{n,j}} + (1 + \tau_{n,j}^m) \sum_{s \neq 1} \beta_{s,j} x_s \epsilon_j (1 - \pi_{sn,j}) \pi_{sn,j} - \sum_{i \neq 1, n} (1 + \tau_{i,j}^m) \sum_{s \neq 1} \beta_{s,j} x_s \epsilon_j \pi_{si,j} \pi_{sn,j} \\ = x_1 \beta_{1,j} \pi_{1n,j} \end{aligned} \quad (\text{A.147})$$

Under a GIIA Supply System with sector-specific parameter η_j , the row of the sector- j tariff block associated with foreign country n is

$$\begin{aligned} \left[\eta_j + \frac{\epsilon_j}{Y_{n,j}} \sum_{g \neq 1} \beta_{g,j} x_g \pi_{gn,j} (1 - \pi_{gn,j}) \right] (1 + \tau_{n,j}^m) - \frac{\epsilon_j}{Y_{n,j}} \sum_{i \neq 1, n} \left(\sum_{g \neq 1} \beta_{g,j} x_g \pi_{gn,j} \pi_{gi,j} \right) (1 + \tau_{i,j}^m) \\ = \beta_{1,j} \pi_{1n,j} \frac{x_1}{Y_{n,j}} + \eta_j (1 + \tau_n^m). \end{aligned} \quad (\text{A.148})$$

Using $1 - \pi_{gn,j} = \sum_{i \neq 1,n} \pi_{gi,j}$, equation (A.148) becomes

$$\eta_j(1 + \tau_{n,j}^m) + \frac{\epsilon_j}{Y_{n,j}} \sum_{g \neq 1} \beta_{g,j} x_g \pi_{gn,j} \sum_{i \neq 1,n} \pi_{gi,j} (\tau_{n,j}^m - \tau_{i,j}^m) = \beta_{1,j} \pi_{1n,j} \frac{x_1}{Y_{n,j}} + \eta_j(1 + \tau_n^m). \quad (\text{A.149})$$

Since $M_{1n,j} = \beta_{1,j} \pi_{1n,j} x_1$, rearranging (A.149) gives

$$\tau_{n,j}^m = \frac{1}{\eta_j Y_{n,j}} \left[M_{1n,j} + \epsilon_j \sum_{g \neq 1} \beta_{g,j} x_g \pi_{gn,j} \sum_{i \neq 1,n} \pi_{gi,j} (\tau_{i,j}^m - \tau_{n,j}^m) \right] + \tau_n^m = \frac{M_{1n,j} + \mathcal{N}_{n,j}}{\eta_j Y_{n,j}} + \tau_n^m. \quad (\text{A.150})$$

where

$$\mathcal{N}_{n,j} = \epsilon_j \left(\sum_{g \neq 1} \sum_{i \neq 1,n} \beta_{g,j} x_g \pi_{gn,j} \pi_{gi,j} \right) \left(\sum_{i \neq 1,n} \omega_{in,j} \tau_{i,j}^m - \tau_{n,j}^m \right). \quad (\text{A.151})$$

$$\omega_{in,j} \equiv \frac{\sum_{g \neq 1} \beta_{g,j} x_g \pi_{gn,j} \pi_{gi,j}}{\sum_{\tilde{g} \neq 1} \sum_{\tilde{i} \neq 1,n} \beta_{\tilde{g},j} x_{\tilde{g}} \pi_{\tilde{g}n,j} \pi_{\tilde{g}\tilde{i},j}}, \quad \sum_{i \neq 1,n} \omega_{in,j} = 1. \quad (\text{A.152})$$

The network components add up to zero. Indeed,

$$\sum_{n \neq 1} \mathcal{N}_{n,j} = \epsilon_j \sum_{g \neq 1} \beta_{g,j} x_g \sum_{n \neq 1} \pi_{gn,j} \left(\sum_{i \neq 1} \pi_{gi,j} \tau_{i,j}^m - \tau_{n,j}^m \right) = 0,$$

where the last equality follows from $\sum_{n \neq 1} \pi_{gn,j} = 1$. Normalizing the nonnegative coefficients on $\tau_{i,j}^m$ gives the weights in (A.152), from which (39) follow immediately.

For three countries, the general expression reduces to the following two-country tariff block.

$$\begin{bmatrix} \eta_j + \frac{\epsilon_j}{Y_{2,j}} \sum_{g \neq 1} \beta_{g,j} x_g \pi_{g2,j} \pi_{g3,j} & -\frac{\epsilon_j}{Y_{2,j}} \sum_{g \neq 1} \beta_{g,j} x_g \pi_{g2,j} \pi_{g3,j} \\ -\frac{\epsilon_j}{Y_{3,j}} \sum_{g \neq 1} \beta_{g,j} x_g \pi_{g2,j} \pi_{g3,j} & \eta_j + \frac{\epsilon_j}{Y_{3,j}} \sum_{g \neq 1} \beta_{g,j} x_g \pi_{g2,j} \pi_{g3,j} \end{bmatrix} \begin{bmatrix} 1 + \tau_{2,j}^m \\ 1 + \tau_{3,j}^m \end{bmatrix} = \begin{bmatrix} \beta_{1,j} \pi_{12,j} \frac{x_1}{Y_{2,j}} + \eta_j(1 + \tau_2^m) \\ \beta_{1,j} \pi_{13,j} \frac{x_1}{Y_{3,j}} + \eta_j(1 + \tau_3^m) \end{bmatrix}. \quad (\text{A.153})$$

Expanding the first row of (A.153) gives

$$\begin{aligned} & \eta_j(1 + \tau_{2,j}^m) + \frac{\epsilon_j}{Y_{2,j}} \left(\sum_{g \neq 1} \beta_{g,j} x_g \pi_{g2,j} \pi_{g3,j} \right) (\tau_{2,j}^m - \tau_{3,j}^m) = \beta_{1,j} \pi_{12,j} \frac{x_1}{Y_{2,j}} + \eta_j(1 + \tau_2^m), \\ \Rightarrow \quad & \tau_{2,j}^m = \frac{1}{\eta_j} \frac{\beta_{1,j} x_1 \pi_{12,j}}{Y_{2,j}} + \tau_2^m - \frac{\epsilon_j}{\eta_j Y_{2,j}} \left(\sum_{g \neq 1} \beta_{g,j} x_g \pi_{g2,j} \pi_{g3,j} \right) (\tau_{2,j}^m - \tau_{3,j}^m). \end{aligned} \quad (\text{A.154})$$

Similarly, the second row yields

$$\begin{aligned} \eta_j(1 + \tau_{3,j}^m) + \frac{\epsilon_j}{Y_{3,j}} \left(\sum_{g \neq 1} \beta_{g,j} x_g \pi_{g2,j} \pi_{g3,j} \right) (\tau_{3,j}^m - \tau_{2,j}^m) &= \beta_{1,j} \pi_{13,j} \frac{x_1}{Y_{3,j}} + \eta_j(1 + \tau_3^m), \\ \Rightarrow \tau_{3,j}^m &= \frac{1}{\eta_j} \frac{\beta_{1,j} x_1 \pi_{13,j}}{Y_{3,j}} + \tau_3^m + \frac{\epsilon_j}{\eta_j Y_{3,j}} \left(\sum_{g \neq 1} \beta_{g,j} x_g \pi_{g2,j} \pi_{g3,j} \right) (\tau_{2,j}^m - \tau_{3,j}^m). \end{aligned} \quad (\text{A.155})$$

Subtracting (A.155) from (A.154) gives

$$\tau_{2,j}^m - \tau_{3,j}^m = \frac{1}{\eta_j} \left(\frac{\beta_{1,j} x_1 \pi_{12,j}}{Y_{2,j}} - \frac{\beta_{1,j} x_1 \pi_{13,j}}{Y_{3,j}} \right) + \tau_2^m - \tau_3^m - \frac{\epsilon_j}{\eta_j} \left(\sum_{g \neq 1} \beta_{g,j} x_g \pi_{g2,j} \pi_{g3,j} \right) \left(\frac{1}{Y_{2,j}} + \frac{1}{Y_{3,j}} \right) (\tau_{2,j}^m - \tau_{3,j}^m). \quad (\text{A.156})$$

In the absence of trade network, the optimal tariff depends only on the bilateral net import share, i.e., $\tau_{n,j}^m = \frac{1}{\eta_j} \frac{\beta_{1,j} x_1 \pi_{1n,j}}{Y_{n,j}} + \tau_n^m$. Thus, when $\eta_j > 0$, such as the Ricardo-Roy case, where $\eta_j = \kappa$, the additional terms in (A.154) and (A.155) operate against the tariff gap. If $\tau_{2,j}^m > \tau_{3,j}^m$, introducing the trade network lowers the tariff on country 2 and raises the tariff on country 3. The opposite occurs when $\tau_{2,j}^m < \tau_{3,j}^m$. Therefore, the tariff gap between countries 2 and 3 is compressed within sector j , inducing cross-country tariff convergence.

When $\eta_j < 0$, such as the Krugman case, where $\eta_j = -1/\psi_j$, the signs of the additional terms are reversed. If $\tau_{2,j}^m > \tau_{3,j}^m$, the network raises the tariff on country 2 relative to its bilateral benchmark and lowers the tariff on country 3 relative to its bilateral benchmark. Thus, the network corrections operate in the direction of tariff divergence. The directions are again reversed when $\tau_{2,j}^m < \tau_{3,j}^m$. $\square$

H Numerical Examples: Heterogeneity and Convergence in Optimal Policies

Our analysis has shown that optimal tariffs can vary between sectors and countries. Meanwhile, in the GIIA Supply System with positive supply parameters $\eta_j = \kappa$, interdependence works as a counterforce that moves tariffs closer. This section presents numerical examples to highlight how technologies and trade costs affect tariff heterogeneity and convergence across countries and sectors. The welfare implications of these examples are summarized in Table A-1.

Table A-1: Welfare Implications: Optimal versus Alternative Policies

Parameters		Optimal	Endogenous bilateral share	Pre-tax bilateral share
		(1)	(2)	(3)
Case 1	$N = 2, J = 2, d = \begin{pmatrix} 1 & 1.2 \\ 1.2 & 1 \end{pmatrix}$	4.85	4.85	0.52
Case 1a	$N = 2, J = 10, d = \begin{pmatrix} 1 & 1.2 \\ 1.2 & 1 \end{pmatrix}$	2.89	2.89	1.54
Case 2	$N = 3, J = 10, d = \begin{pmatrix} 1 & 1.2 & 1.2 \\ 1.2 & 1 & 1.2 \\ 1.2 & 1.2 & 1 \end{pmatrix}$	2.67	2.61	0.96
Case 3	$N = 3, J = 10, d = \begin{pmatrix} 1 & 1.2 & 1.2 \\ 1.2 & 1 & 10 \\ 1.2 & 10 & 1 \end{pmatrix}$	4.84	4.84	-1.61
Case 4	$N = 3, J = 10, d = \begin{pmatrix} 1 & 1.01 & 2 \\ 2 & 1 & 1.01 \\ 1.01 & 1.01 & 1 \end{pmatrix}$	3.31	2.51	-5.09

Note: This table summarizes welfare gains for the Home country following policy implementation, measured as the percentage change of the real consumption (x_1/P_1) relative to real consumption before policies. Column (1) reports results under optimal policies, Column (2) applies policies from equations (36) and (37) using endogenous bilateral shares, and Column (3) uses the same policy formulas but uses pre-tax bilateral trade shares and net import shares. All values are expressed in percentages. $\epsilon = 4, \kappa = 1.1, \bar{L}_n = 1, \forall n, A_{n,j} = 1, \forall n, j. T_1 = [50, \dots, 1], T_2 = [1, \dots, 50]$. If there are three countries, $T_3 = T_1 \circ r$, where r represents values ranging from 2 to 0.5, comprising 10 elements.

Tariff heterogeneity We begin with the simplest case with two countries and two sectors, with a focus on sector heterogeneity. Table A-1, Case 1 presents the technologies and trade costs for this case with $T_1 = [50, 1]$ and $T_2 = [1, 50]$. Tariffs across the two sectors can be very different: we normalize the tariff in sector 1 to zero and find that the optimal tariff on sector 2 is 51.1%. In this case, formulas (36) and (37), which incorporate endogenous bilateral trade shares, exactly capture the home government's incentives to exploit its monopsony power. In contrast, substituting pre-tax bilateral trade shares and net import shares for endogenous bilateral trade shares would greatly reduce the welfare gain from trade policies, from 4.85% (Column 1) to 0.52% (Column 3). This occurs because pre-tax net import shares fail to account for the endogenous elasticities and import patterns across sectors that are influenced by trade policies.

The variation in optimal tariffs across sectors depends on the heterogeneity of Home's buyer power, which arises from comparative advantages between countries. To illustrate this point, we

expand the model from two to ten sectors while maintaining the same technology ranges for both countries (Case 1a of Table A-1). This expansion reduces sectoral differences as foreign goods become more similar across sectors, comparative advantages weaken, and the home country's buyer power becomes more uniform across sectors. Although optimal tariffs still range from 0 (sector 1) to 51.1% (sector 10) as in the two-sector case, the share of each sector declines, resulting in lower cross-sectoral variation in tariffs. This small variation is associated with a reduction in welfare gains from optimal policies by 40%, from 4.85% to 2.89%.

Interdependency When the home country trades with more than one partner, its optimal tariffs are also shaped by the interdependency across foreign countries. Case 2 of Table A-1 illustrates this with a three-country example using the technologies from Figure 1, where country 3 has a comparative advantage in sector 1, relative to Home; however, the trade cost between countries is symmetric in this case. The optimal tariffs in this multi-country setting have two distinct features. First, the bilateral trade share formulas (36) and (37) break down with more than two countries. Ignoring cross-country interdependency and applying only bilateral trade formulas would reduce welfare gains from the optimal 2.67% to 2.61%. Nevertheless, using the endogenous trade shares and prices still outperforms using pre-tax bilateral trade shares and net import shares, which would further reduce the welfare gain to 0.96%.

Second, expanding from two to more countries weakens Home's market power by providing alternative trading partners for foreign countries. When country 2 can redirect trade toward country 3, its self-elasticity becomes larger, reducing Home's ability to use tariffs to exploit its market power. As a result, welfare gains decline when an additional country is added, with underlying parameter patterns remaining comparable across cases. The lower welfare outcome in Case 2 (three countries) compared to Case 1a (two countries) illustrates this interdependency effect.

When foreign countries have a weaker network, the home country can impose higher and more differential tariffs on the foreign country-sector and achieve greater welfare gains. Case 3 provides an example when the interdependence between countries 2 and 3 is weaker than Case 2, allowing Home to extract higher welfare from its policies. Specifically, we shut down trade flows between country 3 and country 2 by setting an extremely high trade cost ($d_{23,j} = d_{32,j} = 10$). This restriction forces countries 2 and 3 to export exclusively to country 1 (Home), reducing their export supply elasticity in response to Home's tariffs. As a result, Home gains greater monopsony power, leading to welfare gains of 4.84%, much higher than those in Case 2. In addition, using formulas (36) and

(37) with endogenous bilateral trade shares generates welfare gains of 4.84%, identical to the optimal one, as proved in Proposition 5. In contrast, using only the pre-tax bilateral trade shares and net import shares fails to capture the general equilibrium effects that endogenously change countries' self- and cross-elasticities. This results in a welfare loss of -1.61% for the home country.

Case 4 illustrates how country asymmetries and trade network matter for Home's optimal policies. According to Proposition 5, when foreign countries are identical or do not trade with each other, optimal policies depend solely on bilateral trade shares. Hence, if foreign countries are asymmetric and engage in large bilateral trade with each other, Home's policies based purely on bilateral trade shares tend to deviate significantly from optimal levels, resulting in lower welfare gains or even losses.

Case 4 presents such a scenario. We construct an asymmetric trade network when country 1 imports almost exclusively from country 2 by imposing high export costs from country 3 to country 1 ($d_{13,j} = 2$), while country 1 exports almost exclusively to country 3 due to high export costs from country 1 to country 2 ($d_{21,j} = 2$). This structure creates highly asymmetric trading relationships between Home and the two foreign countries. Moreover, countries 2 and 3 face low bilateral trade costs ($d_{23,j} = d_{32,j} = 1.01$) and thus engage in extensive trade with each other.

The trade network creates important policy spillovers that bilateral formulas fail to capture. Without accounting for these trade network effects, Home would impose high tariffs on imports from country 2 due to its large imports from that country. However, such high tariffs would reduce country 2's prices and, given the low bilateral trade costs between countries 2 and 3, encourage country 3 to increase its imports from country 2. This increased demand from country 3 reduces country 2's export supply to Home.

Policies that overlook these network effects can generate substantial welfare costs. Simply applying bilateral share formulas generates the outcome in Figure 1b, where Home imposes relatively low tariffs on sector 10 of country 3. In contrast, the optimal policies internalize the large trade between countries 2 and 3, leading to policy convergence as shown in Figure 1a. The difference leads to large welfare impacts. Using endogenous bilateral trade shares yields 24% lower welfare gain compared to the optimal policy (2.51% versus 3.31%). The large general equilibrium effect further implies that using pre-tax bilateral trade shares and net import shares actually causes welfare losses rather than gains.

I Nash Optimal Policies

In this section, all countries are allowed to impose policies on other countries. We derive a country's optimal policies given the policies of other countries.

As proven in Appendix F4, under the multi-country setup that satisfies the GIIA Supply System with constant returns to scale, a country implementing optimal policies will not impose any domestic taxes. Therefore, domestic taxes will not be included in the setup in this section.

We consider a Nash equilibrium where all countries implement optimal policies given other countries' policies. Country i 's policy is defined as $\Gamma = \{1 + \tau_{ni,j}^x, 1 + \tau_{in,j}^m\}$, where $\tau_{ni,j}^x$ denotes the export tax imposed by country i on country n in sector j , and $\tau_{in,j}^m$ represents the import tariff imposed by country i on country n in sector j . Given the policies of other countries, country i 's government chooses $\{\tau_{ni,j}^x, \tau_{in,j}^m\}$ to maximize domestic consumers' consumption, $\max x_i/P_i$, subject to world market equilibrium and the policies of other countries. The world converges to a Nash equilibrium, where each country adopts its optimal policy given the policies of other countries.

Take country 1 as an example. Country 1's government solves the following problem:

$$\max_{\{\tau_{n1,j}^x, \tau_{1n,j}^m, x_1, x_n, p_{1,j}, p_{n,j}\}} \frac{x_1}{P_1},$$

subject to world market equilibrium characterized by the following constraints:

$$\beta_{1,j} \pi_{11,j} x_1 = Y_{1,j} - \sum_{i \neq 1}^N \beta_{i,j} \frac{1}{1 + \tau_{i1,j}^x} \frac{1}{1 + \tau_{1i,j}^m} \pi_{i1,j} x_i, \quad (\gamma_{1,j}) \quad (\text{A.157})$$

$$\beta_{1,j} \frac{1}{1 + \tau_{1n,j}^m} \frac{1}{1 + \tau_{n1,j}^x} \pi_{1n,j} x_1 = Y_{n,j} - \sum_{i \neq 1}^N \beta_{i,j} \frac{1}{1 + \tau_{in,j}^m} \frac{1}{1 + \tau_{ni,j}^x} \pi_{in,j} x_i, \quad (\gamma_{n,j}) \quad (\text{A.158})$$

$$x_1 = \sum_{j=1}^J Y_{1,j} + \sum_{i \neq 1}^N \sum_{j=1}^J \left[\beta_{i,j} \frac{\tau_{i1,j}^x}{1 + \tau_{i1,j}^x} \frac{1}{1 + \tau_{1i,j}^m} \pi_{i1,j} x_i + \beta_{1,j} \frac{\tau_{1i,j}^m}{1 + \tau_{1i,j}^m} \pi_{1i,j} x_1 \right], \quad (\gamma_{x1}) \quad (\text{A.159})$$

$$x_n = \sum_{j=1}^J Y_{n,j} + \sum_{i \neq n}^N \sum_{j=1}^J \left[\beta_{i,j} \frac{\tau_{in,j}^x}{1 + \tau_{in,j}^x} \frac{1}{1 + \tau_{ni,j}^m} \pi_{in,j} x_i + \beta_{n,j} \frac{\tau_{ni,j}^m}{1 + \tau_{ni,j}^m} \pi_{ni,j} x_n \right], \quad (\gamma_{xn}) \quad (\text{A.160})$$

and the labor market specifications

$$L_{1,j} = A_{1,j} \left(\frac{w_{1,j}}{W_1} \right)^{\kappa-1} \bar{L}_1,$$

$$L_{n,j} = A_{n,j} \left(\frac{w_{n,j}}{W_n} \right)^{\kappa-1} \bar{L}_n,$$

where $\{P_1, P_n\}$ and trade shares are given by

$$\begin{aligned}
P_1 &= \prod_{j=1}^J \left[p_{1,j}^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j}(1 + \tau_{1i,j}^x)(1 + \tau_{1i,j}^m)d_{1i,j})^{-\epsilon_j} \right]^{-\frac{\beta_{1,j}}{\epsilon_j}}, \\
P_n &= \prod_{j=1}^J \left[(p_{1,j}(1 + \tau_{n1,j}^x)(1 + \tau_{n1,j}^m)d_{n1,j})^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j}(1 + \tau_{ni,j}^x)(1 + \tau_{ni,j}^m)d_{ni,j})^{-\epsilon_j} \right]^{-\frac{\beta_{n,j}}{\epsilon_j}}, \\
\pi_{11,j} &= \frac{(p_{1,j})^{-\epsilon_j}}{(p_{1,j})^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j}(1 + \tau_{1i,j}^m)(1 + \tau_{1i,j}^x)d_{1i,j})^{-\epsilon_j}}, \\
\pi_{1n,j} &= \frac{(p_{n,j}(1 + \tau_{1n,j}^m)(1 + \tau_{1n,j}^x)d_{1n,j})^{-\epsilon_j}}{(p_{1,j})^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j}(1 + \tau_{1i,j}^m)(1 + \tau_{1i,j}^x)d_{1i,j})^{-\epsilon_j}}, \\
\pi_{n1,j} &= \frac{(p_{1,j}(1 + \tau_{n1,j}^x)(1 + \tau_{n1,j}^m)d_{n1,j})^{-\epsilon_j}}{(p_{1,j}(1 + \tau_{n1,j}^x)(1 + \tau_{n1,j}^m)d_{n1,j})^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j}(1 + \tau_{ni,j}^x)(1 + \tau_{ni,j}^m)d_{ni,j})^{-\epsilon_j}}, \\
\pi_{nm,j} &= \frac{(p_{m,j}(1 + \tau_{nm,j}^x)(1 + \tau_{nm,j}^m)d_{nm,j})^{-\epsilon_j}}{(p_{1,j}(1 + \tau_{n1,j}^x)(1 + \tau_{n1,j}^m)d_{n1,j})^{-\epsilon_j} + \sum_{i \neq 1}^N (p_{i,j}(1 + \tau_{ni,j}^x)(1 + \tau_{ni,j}^m)d_{ni,j})^{-\epsilon_j}}.
\end{aligned}$$

According to the Walras' law, one of the equilibrium conditions in equations (A.157)-(A.160) is redundant; thus, we drop country 1's expenditure constraint in optimization and set $\gamma_{x1} = 0$.

Without loss of generality, we normalize country 1's final goods price $P_1 = 1$. We define $E_{11,j} = Y_{1,j} - \sum_{i \neq 1}^N \beta_{i,j} \frac{1}{1 + \tau_{1i,j}^x} \frac{1}{1 + \tau_{1i,j}^m} \pi_{i1,j} x_i$ as the supply of country 1 on its domestic market, and $E_{1n,j} = Y_{n,j} - \sum_{i \neq 1}^N \beta_{i,j} \frac{1}{1 + \tau_{in,j}^x} \frac{1}{1 + \tau_{in,j}^m} \pi_{in,j} x_i$ as the export supply of country n on country 1's market.

We apply the Lagrange multiplier method to derive the formula for each country's optimal policies. We write each goods-market clearing constraint as residual supply minus country 1's demand equal to zero. The following proposition presents the optimal policies for country 1, given the policies of other countries.

Proposition 6 (Nash Optimal Policies and Multipliers). *Given all other countries' policies, the optimal policy of country 1's government satisfies, for any country $n \neq 1$ and sector $\forall j$,*

$$\text{Import tariff: } 1 + \tau_{1n,j}^m = \frac{(\gamma_{xn} + \gamma_{n,j}) \frac{1}{1 + \tau_{1n,j}^x} - \gamma_{xn}}{1/P_1}, \quad (\text{A.161})$$

Export tax:

$$\frac{1}{1 + \tau_{n1,j}^x} = \frac{\sum_{i \neq 1, n}^N \frac{\gamma_{xi} + \gamma_{i,j}}{1 + \tau_{ni,j}^x} \frac{1}{1 + \tau_{ni,j}^m} \epsilon_j \pi_{ni,j} + (\gamma_{xn} + \gamma_{n,j}) \epsilon_j \pi_{nn,j} + \sum_{i \neq 1, n}^N \frac{\gamma_{xn} - \gamma_{xi}}{1 + \tau_{ni,j}^m} \epsilon_j \pi_{ni,j} - \gamma_{xn} \frac{1}{1 + \tau_{n1,j}^m} \epsilon_j (1 - \pi_{n1,j})}{\frac{1}{P_1} \frac{1}{1 + \tau_{n1,j}^m} (1 + \epsilon_j (1 - \pi_{n1,j}))}, \quad (\text{A.162})$$

where $\gamma_{n,j}$ is the multiplier on country n 's goods-market clearing condition written as residual supply minus country 1's demand equal to zero, and γ_{xn} is the multiplier on country n 's expenditure. These multipliers satisfy the FOCs with respect to prices and FOCs with respect to expenditures.

Tax neutrality The proof of tax neutrality for the country i follows a similar approach to that presented in Lemma 1. When a country implements its optimal policies, it can normalize a tariff it imposes on a specific country and sector to zero, given the policies of other countries. Since there are N countries in total and each country can set one tariff to zero while implementing its optimal policies, a total of N tariffs can be normalized to zero.

Country 1's optimal conditions

FOC over x_1

$$\frac{1}{P_1} - \sum_{j=1}^J \gamma_{1,j} \beta_{1,j} \pi_{11,j} - \sum_{i \neq 1}^N \sum_{j=1}^J \gamma_{i,j} \beta_{1,j} \frac{1}{1 + \tau_{1i,j}^m} \frac{1}{1 + \tau_{1i,j}^x} \pi_{1i,j} + \sum_{i \neq 1}^N \gamma_{xi} \sum_{j=1}^J \beta_{1,j} \frac{\tau_{1i,j}^x}{1 + \tau_{1i,j}^x} \frac{1}{1 + \tau_{1i,j}^m} \pi_{1i,j} = 0. \quad (\text{A.163})$$

FOC over $x_n, n \neq 1$

$$-\gamma_{xn} - \sum_{j=1}^J \gamma_{n,j} \beta_{n,j} \pi_{nn,j} - \sum_{i \neq n}^N \sum_{j=1}^J \gamma_{i,j} \beta_{n,j} \frac{1}{1 + \tau_{ni,j}^m} \frac{1}{1 + \tau_{ni,j}^x} \pi_{ni,j} + \sum_{i \neq n}^N \gamma_{xi} \sum_{j=1}^J \beta_{n,j} \frac{\tau_{ni,j}^x}{1 + \tau_{ni,j}^x} \frac{1}{1 + \tau_{ni,j}^m} \pi_{ni,j} = 0. \quad (\text{A.164})$$

FOC over import tariff $\tau_{1n,j}^m, n \neq 1$

$$\begin{aligned} & -\frac{1}{P_1} \beta_{1,j} x_1 \frac{\pi_{1n,j}}{1 + \tau_{1n,j}^m} - \gamma_{1,j} \beta_{1,j} \frac{\partial \pi_{11,j}}{\partial \tau_{1n,j}^m} x_1 + \gamma_{n,j} \beta_{1,j} \frac{\pi_{1n,j}}{(1 + \tau_{1n,j}^m)^2} \frac{1}{1 + \tau_{1n,j}^x} x_1 - \sum_{i \neq 1}^N \gamma_{i,j} \beta_{1,j} \frac{1}{1 + \tau_{1i,j}^m} \frac{1}{1 + \tau_{1i,j}^x} \frac{\partial \pi_{1i,j}}{\partial \tau_{1n,j}^m} x_1 \\ & - \gamma_{xn} \beta_{1,j} \frac{\tau_{1n,j}^x}{1 + \tau_{1n,j}^x} \frac{1}{(1 + \tau_{1n,j}^m)^2} \pi_{1n,j} x_1 + \sum_{i \neq 1}^N \gamma_{xi} \beta_{1,j} \frac{\tau_{1i,j}^x}{1 + \tau_{1i,j}^x} \frac{1}{1 + \tau_{1i,j}^m} \frac{\partial \pi_{1i,j}}{\partial \tau_{1n,j}^m} x_1 = 0. \end{aligned}$$

Plugging into derivatives

$$\begin{aligned}
& -\frac{1}{P_1}\beta_{1,j}x_1\pi_{1n,j} - \gamma_{1,j}\beta_{1,j}\epsilon_j\pi_{11,j}\pi_{1n,j}x_1 + (\gamma_{n,j} + \gamma_{xn})\beta_{1,j}\frac{\pi_{1n,j}}{1 + \tau_{1n,j}^m}\frac{1 + \epsilon_j}{1 + \tau_{1n,j}^x}x_1 - \sum_{i \neq 1}^N (\gamma_{i,j} + \gamma_{xi})\frac{\beta_{1,j}\pi_{1n,j}}{1 + \tau_{1i,j}^m}\frac{\epsilon_j\pi_{1i,j}x_1}{1 + \tau_{1i,j}^x} \\
& - \gamma_{xn}\beta_{1,j}\frac{1}{1 + \tau_{1n,j}^m}\pi_{1n,j}x_1 + \sum_{i \neq 1} \gamma_{xi}\beta_{1,j}\frac{1}{1 + \tau_{1i,j}^m}\epsilon_j\pi_{1i,j}\pi_{1n,j}x_1 - \gamma_{xn}\beta_{1,j}\frac{1}{1 + \tau_{1n,j}^m}\epsilon_j\pi_{1n,j}x_1 = 0 \quad (\text{A.165}) \\
\Rightarrow & -\frac{1}{P_1} - \gamma_{1,j}\epsilon_j\pi_{11,j} + \frac{\gamma_{n,j} + \gamma_{xn}}{1 + \tau_{1n,j}^m}\frac{1 + \epsilon_j}{1 + \tau_{1n,j}^x} - \sum_{i \neq 1}^N \frac{\gamma_{i,j} + \gamma_{xi}}{1 + \tau_{1i,j}^m}\frac{\epsilon_j\pi_{1i,j}}{1 + \tau_{1i,j}^x} - \gamma_{xn}\frac{1 + \epsilon_j}{1 + \tau_{1n,j}^m} + \sum_{i \neq 1} \gamma_{xi}\frac{\pi_{1i,j}}{1 + \tau_{1i,j}^m}\epsilon_j = 0.
\end{aligned}$$

Thus, $-(\gamma_{n,j} + \gamma_{xn})\frac{1}{1 + \tau_{1n,j}^m}\frac{1}{1 + \tau_{1n,j}^x} + \gamma_{xn}\frac{1}{1 + \tau_{1n,j}^m}$ is the same across all countries in sector j . Replacing each $-(\gamma_{i,j} + \gamma_{xi})\frac{1}{1 + \tau_{1i,j}^m}\frac{1}{1 + \tau_{1i,j}^x} + \gamma_{xi}\frac{1}{1 + \tau_{1i,j}^m}$, we derive

$$\begin{aligned}
& -\frac{1}{P_1} - \gamma_{1,j}\epsilon_j\pi_{11,j} + \frac{\gamma_{n,j} + \gamma_{xn}}{1 + \tau_{1n,j}^m}\frac{1 + \epsilon_j}{1 + \tau_{1n,j}^x} - \gamma_{xn}\frac{1 + \epsilon_j}{1 + \tau_{1n,j}^m} - \frac{\gamma_{n,j} + \gamma_{xn}}{1 + \tau_{1n,j}^m}\frac{\epsilon_j(1 - \pi_{11,j})}{1 + \tau_{1n,j}^x} + \gamma_{xn}\frac{\epsilon_j}{1 + \tau_{1n,j}^m}(1 - \pi_{11,j}) = 0 \\
\Rightarrow & -\frac{1}{P_1} - \gamma_{1,j}\epsilon_j\pi_{11,j} + (\gamma_{n,j} + \gamma_{xn})\frac{1 + \epsilon_j\pi_{11,j}}{1 + \tau_{1n,j}^m}\frac{1}{1 + \tau_{1n,j}^x} - \gamma_{xn}\frac{1 + \epsilon_j\pi_{11,j}}{1 + \tau_{1n,j}^m} = 0.
\end{aligned}$$

Combining FOCs over x_1 and $\tau_{1n,j}^m$, we derive

$$\frac{1}{P_1} + \sum_{j=1}^J (-(\gamma_{n,j} + \gamma_{xn})\frac{1}{1 + \tau_{1n,j}^x} + \gamma_{xn})\beta_{1,j}\frac{1}{1 + \tau_{1n,j}^m} = 0,$$

or

$$\sum_{j=1}^J \beta_{1,j} \frac{(\frac{1}{P_1} - \gamma_{1,j})\epsilon_j\pi_{11,j}}{1 + \epsilon_j\pi_{11,j}} = 0. \quad (\text{A.166})$$

FOC over export tax $\tau_{n1,j}^x$, $n \neq 1$

$$\begin{aligned}
& \gamma_{1,j}\beta_{n,j}\frac{1}{(1 + \tau_{n1,j}^x)^2}\frac{1}{1 + \tau_{n1,j}^m}\pi_{n1,j}x_n - \gamma_{1,j}\beta_{n,j}\frac{1}{1 + \tau_{n1,j}^x}\frac{1}{1 + \tau_{n1,j}^m}\frac{\partial \pi_{n1,j}}{\partial \tau_{n1,j}^x}x_n - \sum_{i \neq 1}^N \gamma_{i,j}\beta_{n,j}\frac{1}{1 + \tau_{ni,j}^x}\frac{1}{1 + \tau_{ni,j}^m}\frac{\partial \pi_{ni,j}}{\partial \tau_{n1,j}^x}x_n \\
& + \sum_{i \neq 1,n}^N \gamma_{xi}\beta_{n,j}\frac{\tau_{ni,j}^x}{1 + \tau_{ni,j}^x}\frac{1}{1 + \tau_{ni,j}^m}\frac{\partial \pi_{ni,j}}{\partial \tau_{n1,j}^x}x_n + \gamma_{xn}\sum_{i \neq n}^N \beta_{n,j}\frac{\tau_{ni,j}^m}{1 + \tau_{ni,j}^m}\frac{\partial \pi_{ni,j}}{\partial \tau_{n1,j}^x}x_n = 0.
\end{aligned}$$

Plugging into derivatives

$$\begin{aligned}
& \gamma_{1,j}\beta_{n,j}\frac{1}{1+\tau_{n1,j}^x}\frac{1}{1+\tau_{n1,j}^m}\pi_{n1,j}(1+\epsilon_j(1-\pi_{n1,j}))x_n - \sum_{i \neq 1}^N \gamma_{i,j}\beta_{n,j}\frac{1}{1+\tau_{ni,j}^x}\frac{1}{1+\tau_{ni,j}^m}\epsilon_j\pi_{ni,j}\pi_{n1,j}x_n \\
& + \sum_{i \neq 1,n}^N \gamma_{xi}\beta_{n,j}\frac{\tau_{ni,j}^x}{1+\tau_{ni,j}^x}\frac{1}{1+\tau_{ni,j}^m}\epsilon_j\pi_{ni,j}\pi_{n1,j}x_n + \gamma_{xn}\sum_{i \neq n}^N \beta_{n,j}\frac{\tau_{ni,j}^m}{1+\tau_{ni,j}^m}\epsilon_j\pi_{ni,j}\pi_{n1,j}x_n - \gamma_{xn}\beta_{n,j}\frac{\tau_{n1,j}^m}{1+\tau_{n1,j}^m}\epsilon_j\pi_{n1,j}x_n = 0
\end{aligned} \tag{A.167}$$

$$\begin{aligned}
\Rightarrow & \gamma_{1,j}\frac{1}{1+\tau_{n1,j}^x}\frac{1}{1+\tau_{n1,j}^m}(1+\epsilon_j(1-\pi_{n1,j})) - \sum_{i \neq 1,n}^N (\gamma_{i,j} + \gamma_{xi})\frac{1}{1+\tau_{ni,j}^x}\frac{1}{1+\tau_{ni,j}^m}\epsilon_j\pi_{ni,j} - (\gamma_{n,j} + \gamma_{xn})\epsilon_j\pi_{nn,j} \\
& + \sum_{i \neq 1,n}^N (\gamma_{xi} - \gamma_{xn})\frac{1}{1+\tau_{ni,j}^m}\epsilon_j\pi_{ni,j} + \gamma_{xn}\frac{1}{1+\tau_{n1,j}^m}\epsilon_j(1-\pi_{n1,j}) = 0.
\end{aligned}$$

FOC over $p_{1,j}$

$$\begin{aligned}
& -\frac{1}{p_1}x_1\beta_{1,j}\frac{\pi_{11,j}}{p_{1,j}} - \gamma_{1,j}\beta_{1,j}\frac{\partial \pi_{11,j}}{\partial p_{1,j}}x_1 + \sum_{k=1}^J \gamma_{1,k}\frac{\partial E_{11,k}}{\partial p_{1,j}} - \sum_{i \neq 1}^N \gamma_{i,j}\beta_{1,j}\frac{1}{1+\tau_{1i,j}^x}\frac{1}{1+\tau_{1i,j}^m}\frac{\partial \pi_{1i,j}}{\partial p_{1,j}}x_1 + \sum_{k=1}^J \sum_{i \neq 1}^N \gamma_{i,k}\frac{\partial E_{1i,k}}{\partial p_{1,j}} \\
& + \sum_{n \neq 1}^N \gamma_{xn}\sum_{i \neq n}^N \left[\beta_{i,j}\frac{\tau_{in,j}^x}{1+\tau_{in,j}^x}\frac{1}{1+\tau_{in,j}^m}\frac{\partial \pi_{in,j}}{\partial p_{1,j}}x_i + \beta_{n,j}\frac{\tau_{ni,j}^m}{1+\tau_{ni,j}^m}\frac{\partial \pi_{ni,j}}{\partial p_{1,j}}x_n \right] = 0 \\
\Rightarrow & -\frac{1}{p_1}x_1\beta_{1,j}\pi_{11,j} + \gamma_{1,j}\beta_{1,j}\epsilon_j\pi_{11,j}(1-\pi_{11,j})x_1 + \sum_{k=1}^J \gamma_{1,k}\frac{\partial Y_{1,k}}{\partial p_{1,j}}p_{1,j} + \gamma_{1,j}\sum_{i \neq 1}^N \beta_{i,j}\frac{1}{1+\tau_{1i,j}^x}\frac{1}{1+\tau_{1i,j}^m}\epsilon_j\pi_{1i,j}(1-\pi_{1i,j})x_i \\
& - \sum_{i \neq 1}^N \gamma_{i,j}\beta_{1,j}\frac{1}{1+\tau_{1i,j}^m}\frac{1}{1+\tau_{1i,j}^x}\epsilon_j\pi_{1i,j}\pi_{11,j}x_1 - \sum_{i \neq 1}^N \gamma_{i,j}\beta_{n,j}\sum_{n \neq 1}^N \frac{1}{1+\tau_{ni,j}^m}\frac{1}{1+\tau_{ni,j}^x}\epsilon_j\pi_{ni,j}\pi_{n1,j}x_n \\
& + \sum_{n \neq 1}^N \gamma_{xn}\sum_{i \neq n}^N \left[\beta_{i,j}\frac{\tau_{in,j}^x}{1+\tau_{in,j}^x}\frac{1}{1+\tau_{in,j}^m}\epsilon_j\pi_{in,j}\pi_{i1,j}x_i + \beta_{n,j}\frac{\tau_{ni,j}^m}{1+\tau_{ni,j}^m}\epsilon_j\pi_{ni,j}\pi_{n1,j}x_n \right] - \sum_{n \neq 1}^N \gamma_{xn}\beta_{n,j}\frac{\tau_{n1,j}^m}{1+\tau_{n1,j}^m}\epsilon_j\pi_{n1,j}x_n = 0.
\end{aligned} \tag{A.168}$$

Combining (A.168) with (A.165) and (A.167),

$$\begin{aligned}
& -\frac{1}{P_1}x_1\beta_{1,j} + \sum_{k=1}^J \gamma_{1,k} \frac{\partial Y_{1,k}}{\partial p_{1,j}} p_{1,j} - \gamma_{1,j} \sum_{n \neq 1}^N \frac{\beta_{n,j}}{1 + \tau_{n1,j}^x} \frac{\pi_{n1,j} x_n}{1 + \tau_{n1,j}^m} + \sum_{n \neq 1}^N \left(\frac{\gamma_{n,j}}{1 + \tau_{1n,j}^x} - \gamma_{xn} \frac{\tau_{1n,j}^x}{1 + \tau_{1n,j}^x} \right) \frac{\beta_{1,j} \pi_{1n,j} x_1}{1 + \tau_{1n,j}^m} = 0 \\
\Rightarrow & -\frac{1}{P_1}x_1\beta_{1,j} + \sum_{k=1}^J \gamma_{1,k} \frac{\partial Y_{1,k}}{\partial p_{1,j}} p_{1,j} + \gamma_{1,j} \beta_{1,j} \pi_{11,j} x_1 - \gamma_{1,j} Y_{1,j} + \left(\frac{\gamma_{n,j}}{1 + \tau_{1n,j}^x} - \gamma_{xn} \frac{\tau_{1n,j}^x}{1 + \tau_{1n,j}^x} \right) \frac{\beta_{1,j}}{1 + \tau_{1n,j}^m} (1 - \pi_{11,j}) x_1 = 0 \\
\Rightarrow & -\frac{1}{P_1}x_1\beta_{1,j} + \sum_{k=1}^J \gamma_{1,k} \frac{\partial L_{1,k}}{\partial \ln p_{1,j}} + \gamma_{1,j} \beta_{1,j} \pi_{11,j} x_1 - \beta_{1,j} \frac{-\frac{1}{P_1} - \gamma_{1,j} \epsilon_j \pi_{11,j}}{1 + \epsilon_j \pi_{11,j}} (1 - \pi_{11,j}) x_1 = 0 \\
\Rightarrow & \sum_{k=1}^J \gamma_{1,k} \frac{\partial L_{1,k}}{\partial \ln p_{1,j}} - \beta_{1,j} \frac{\frac{1}{P_1} - \gamma_{1,j}}{1 + \epsilon_j \pi_{11,j}} (1 + \epsilon_j) x_1 = 0 \\
\Rightarrow & \gamma_{1,j} (\kappa - 1) Y_{1,j} - \sum_{k=1}^J \gamma_{1,k} (\kappa - 1) \frac{Y_{1,k}}{W_1 L_1} Y_{1,j} - \beta_{1,j} \frac{\frac{1}{P_1} - \gamma_{1,j}}{1 + \epsilon_j \pi_{11,j}} (1 + \epsilon_j) x_1 = 0 \\
\Rightarrow & \sum_{j=1}^J \beta_{1,j} \frac{\frac{1}{P_1} - \gamma_{1,j}}{1 + \epsilon_j \pi_{11,j}} (1 + \epsilon_j) x_1 = 0. \tag{A.169}
\end{aligned}$$

Using (A.166), we solve for $\gamma_{1,j} = \frac{1}{P_1}$. Thus, the optimal import tariff is

$$\begin{aligned}
& -\frac{1}{P_1} + (\gamma_{n,j} + \gamma_{xn}) \frac{1}{1 + \tau_{1n,j}^m} \frac{1}{1 + \tau_{1n,j}^x} - \gamma_{xn} \frac{1}{1 + \tau_{1n,j}^m} = 0 \\
\Rightarrow & 1 + \tau_{1n,j}^m = \frac{(\gamma_{xn} + \gamma_{n,j}) \frac{1}{1 + \tau_{1n,j}^x} - \gamma_{xn}}{1/P_1}. \tag{A.170}
\end{aligned}$$

Plugging (A.170) into (A.167), the optimal export tax is

$$\frac{1}{1 + \tau_{n1,j}^x} = \frac{\sum_{i \neq 1,n}^N \frac{\gamma_{xi} + \gamma_{ij}}{1 + \tau_{ni,j}^x} \frac{1}{1 + \tau_{ni,j}^m} \epsilon_j \pi_{ni,j} + (\gamma_{xn} + \gamma_{n,j}) \epsilon_j \pi_{nn,j} + \sum_{i \neq 1,n}^N \frac{\gamma_{xn} - \gamma_{xi}}{1 + \tau_{ni,j}^m} \epsilon_j \pi_{ni,j} - \gamma_{xn} \frac{1}{1 + \tau_{n1,j}^m} \epsilon_j (1 - \pi_{n1,j})}{\frac{1}{P_1} \frac{1}{1 + \tau_{n1,j}^m} (1 + \epsilon_j (1 - \pi_{n1,j}))}. \tag{A.171}$$

where $\gamma_{n,j}$ is the shadow value of country n 's residual supply to country 1 and γ_{xn} reflects how changes in country n 's expenditure affect country 1's utility. When $\gamma_{xn} = 0$ for $n \neq 1$, and other countries do not impose policies, (A.170) and (A.171) are equivalent to (A.121) and (A.123) under country 1's unilateral policies.

FOC over $p_{n,j}$, $n \neq 1$

$$\begin{aligned}
& -\frac{1}{P_1} x_1 \beta_{1,j} \frac{\pi_{1n,j}}{p_{n,j}} - \frac{1}{P_1} \beta_{1,j} \frac{\partial \pi_{11,j}}{\partial p_{n,j}} x_1 - \sum_{k=1}^J \gamma_{1,k} \frac{\partial E_{11,k}}{\partial p_{n,j}} - \sum_{i \neq 1}^N \gamma_{i,j} \beta_{1,j} \frac{1}{1 + \tau_{1i,j}^m} \frac{1}{1 + \tau_{1i,j}^x} \frac{\partial \pi_{1i,j}}{\partial p_{n,j}} x_1 + \sum_{k=1}^J \sum_{i \neq 1}^N \gamma_{i,k} \frac{\partial E_{1i,k}}{\partial p_{n,j}} \\
& + \gamma_{xn} \sum_{k=1}^J \frac{\partial Y_{n,k}}{\partial p_{n,j}} p_{n,j} + \sum_{h \neq 1}^N \gamma_{xh} \sum_{i \neq h}^N \left[\beta_{i,j} \frac{\tau_{ih,j}^x}{1 + \tau_{ih,j}^x} \frac{1}{1 + \tau_{ih,j}^m} \frac{\partial \pi_{ih,j}}{\partial p_{n,j}} x_i + \beta_{h,j} \frac{\tau_{hi,j}^m}{1 + \tau_{hi,j}^m} \frac{\partial \pi_{hi,j}}{\partial p_{n,j}} x_h \right] = 0 \\
\Rightarrow & -\frac{1}{P_1} x_1 \beta_{1,j} \pi_{1n,j} - \frac{1}{P_1} \beta_{1,j} \epsilon_j \pi_{11,j} \pi_{1n,j} x_1 + \sum_{k=1}^J (\gamma_{xn} + \gamma_{n,k}) \frac{\partial Y_{n,k}}{\partial p_{n,j}} p_{n,j} - \gamma_{1,j} \sum_{i \neq 1}^N \beta_{i,j} \frac{1}{1 + \tau_{1i,j}^x} \frac{1}{1 + \tau_{1i,j}^m} \epsilon_j \pi_{1i,j} \pi_{in,j} x_i \\
& - \sum_{i \neq 1}^N \gamma_{i,j} \beta_{1,j} \frac{1}{1 + \tau_{1i,j}^m} \frac{1}{1 + \tau_{1i,j}^x} \epsilon_j \pi_{1i,j} \pi_{1n,j} x_1 + \gamma_{n,j} \beta_{1,j} \frac{1}{1 + \tau_{1n,j}^m} \frac{1}{1 + \tau_{1n,j}^x} \epsilon_j \pi_{1n,j} x_1 - \sum_{i \neq 1}^N \gamma_{i,j} \sum_{h \neq 1}^N \frac{\beta_{h,j} x_h}{1 + \tau_{hi,j}^m} \frac{\epsilon_j \pi_{hi,j} \pi_{hn,j}}{1 + \tau_{hi,j}^x} \\
& + \gamma_{n,j} \sum_{h \neq 1}^N \beta_{h,j} \frac{1}{1 + \tau_{hn,j}^m} \frac{1}{1 + \tau_{hn,j}^x} \epsilon_j \pi_{hn,j} x_h + \sum_{h \neq 1}^N \gamma_{xh} \sum_{i \neq h}^N \beta_{i,j} \frac{\tau_{ih,j}^x}{1 + \tau_{ih,j}^x} \frac{1}{1 + \tau_{ih,j}^m} \epsilon_j \pi_{ih,j} \pi_{in,j} x_i \\
& + \sum_{h \neq 1}^N \gamma_{xh} \sum_{i \neq h}^N \beta_{h,j} \frac{\tau_{hi,j}^m}{1 + \tau_{hi,j}^m} \epsilon_j \pi_{hi,j} \pi_{hn,j} x_h - \gamma_{xn} \sum_{i \neq n}^N \beta_{i,j} \frac{\tau_{in,j}^x}{1 + \tau_{in,j}^x} \frac{1}{1 + \tau_{in,j}^m} \epsilon_j \pi_{in,j} x_i - \sum_{h \neq 1}^N \gamma_{xh} \beta_{h,j} \frac{\tau_{hn,j}^m}{1 + \tau_{hn,j}^m} \epsilon_j \pi_{hn,j} x_h = 0 \\
\Rightarrow & -\frac{1}{P_1} x_1 \beta_{1,j} \pi_{1n,j} + \sum_{k=1}^J (\gamma_{xn} + \gamma_{n,k}) \frac{\partial Y_{n,k}}{\partial \ln p_{n,j}} - \gamma_{1,j} \sum_{i \neq 1}^N \frac{\beta_{i,j}}{1 + \tau_{1i,j}^x} \frac{\epsilon_j \pi_{1i,j} \pi_{in,j} x_i}{1 + \tau_{1i,j}^m} - \sum_{i \neq 1}^N \gamma_{i,j} \sum_{h \neq 1}^N \frac{\beta_{h,j}}{1 + \tau_{hi,j}^m} \frac{\epsilon_j \pi_{hi,j} \pi_{hn,j} x_h}{1 + \tau_{hi,j}^x} \\
& + \gamma_{n,j} \sum_{h \neq 1}^N \beta_{h,j} \frac{1}{1 + \tau_{hn,j}^m} \frac{1}{1 + \tau_{hn,j}^x} \epsilon_j \pi_{hn,j} x_h + \sum_{h \neq 1}^N \gamma_{xh} \sum_{i \neq \{h,n\}}^N \beta_{i,j} \frac{\tau_{ih,j}^x}{1 + \tau_{ih,j}^x} \frac{1}{1 + \tau_{ih,j}^m} \epsilon_j \pi_{ih,j} \pi_{in,j} x_i \\
& - \gamma_{xn} \sum_{i \neq n}^N \beta_{i,j} \frac{\tau_{in,j}^x}{1 + \tau_{in,j}^x} \frac{1}{1 + \tau_{in,j}^m} \epsilon_j \pi_{in,j} x_i + \sum_{h \neq 1}^N \gamma_{xh} \sum_{i \neq h}^N \beta_{h,j} \frac{\tau_{hi,j}^m}{1 + \tau_{hi,j}^m} \epsilon_j \pi_{hi,j} \pi_{hn,j} x_h - \sum_{h \neq 1}^N \gamma_{xh} \beta_{h,j} \frac{\tau_{hn,j}^m}{1 + \tau_{hn,j}^m} \epsilon_j \pi_{hn,j} x_h = 0.
\end{aligned}
\tag{A.172}$$

J Quantifying the Consequences of Optimal Policies

In this section, we compute the counterfactual equilibrium under unilateral and Nash optimal policies using the exact hat method and the formula for the optimal policies in the multi-country case satisfying GIIA Supply System. We first present the counterfactual systems for unilateral and Nash policies in the constant returns to scale, where $\eta_j = \kappa$. We then extend the unilateral-policy system to allow for economies of scale. Variables without 'prime' are observed variables from data (trade matrix $\pi_{ni,j}$; sectoral income $w_{n,j} L_{n,j}$; sectoral income share $\lambda_{n,j}$), and variables with 'prime' are counterfactuals after implementing optimal policies. Variables with 'hat' denote the ratios of prime variables to the observed ones.

J.1 Unilateral optimal policies

Given Home's policies, the counterfactual equilibrium satisfies²⁸

$$\hat{p}_{1,j} = \hat{w}_{1,j}, \quad (A.173)$$

$$\hat{p}_{n,j} = \hat{w}_{n,j}, \quad n \neq 1 \quad (A.174)$$

$$Y'_{1,j} = w'_{1,j} L'_{1,j} = \hat{w}_{1,j} \hat{L}_{1,j} w_{1,j} L_{1,j}, \quad (A.175)$$

$$Y'_{n,j} = w'_{n,j} L'_{n,j} = \hat{w}_{n,j} \hat{L}_{n,j} w_{n,j} L_{n,j}, \quad n \neq 1 \quad (A.176)$$

$$\hat{p}_n = \prod_{j=1}^J \left[\pi_{n1,j} \left(\hat{p}_{1,j} (1 + \tau_{n,j}^{x'}) \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{ni,j} \hat{p}_{i,j}^{-\epsilon_j} \right]^{-\beta_{n,j}/\epsilon_j}, \quad n \neq 1 \quad (A.177)$$

$$\hat{p}_1 = \prod_{j=1}^J \left[\pi_{11,j} \hat{p}_{1,j}^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{1i,j} (\hat{p}_{i,j} (1 + \tau_{i,j}^{m'}))^{-\epsilon_j} \right]^{-\beta_{1,j}/\epsilon_j}, \quad (A.178)$$

$$\pi'_{11,j} = \frac{\pi_{11,j} \hat{p}_{1,j}^{-\epsilon_j}}{\pi_{11,j} \hat{p}_{1,j}^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{1i,j} \left(\hat{p}_{i,j} (1 + \tau_{i,j}^{m'}) \right)^{-\epsilon_j}}, \quad (A.179)$$

$$\pi'_{1n,j} = \frac{\pi_{1n,j} \left(\hat{p}_{n,j} (1 + \tau_{n,j}^{m'}) \right)^{-\epsilon_j}}{\pi_{11,j} \hat{p}_{1,j}^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{1i,j} \left(\hat{p}_{i,j} (1 + \tau_{i,j}^{m'}) \right)^{-\epsilon_j}}, \quad n \neq 1 \quad (A.180)$$

$$\pi'_{n1,j} = \frac{\pi_{n1,j} \left(\hat{p}_{1,j} (1 + \tau_{n,j}^{x'}) \right)^{-\epsilon_j}}{\pi_{n1,j} \left(\hat{p}_{1,j} (1 + \tau_{n,j}^{x'}) \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{ni,j} \hat{p}_{i,j}^{-\epsilon_j}}, \quad n \neq 1 \quad (A.181)$$

$$\pi'_{nm,j} = \frac{\pi_{nm,j} \hat{p}_{m,j}^{-\epsilon_j}}{\pi_{n1,j} \left(\hat{p}_{1,j} (1 + \tau_{n,j}^{x'}) \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{ni,j} \hat{p}_{i,j}^{-\epsilon_j}}, \quad n \neq 1, m \neq 1 \quad (A.182)$$

$$x'_1 = \sum_{j=1}^J Y'_{1,j} + \sum_{i \neq 1}^N \sum_{j=1}^J \frac{\tau_{i,j}^{x'}}{1 + \tau_{i,j}^{x'}} \pi'_{i1,j} \beta_{i,j} x'_i + \sum_{i \neq 1}^N \sum_{j=1}^J \frac{\tau_{i,j}^{m'}}{1 + \tau_{i,j}^{m'}} \pi'_{1i,j} \beta_{1,j} x'_1, \quad (A.183)$$

$$x'_n = \sum_{j=1}^J Y'_{n,j}, \quad n \neq 1 \quad (A.184)$$

$$\pi'_{11,j} \beta_{1,j} x'_1 = Y'_{1,j} - \sum_{i \neq 1}^N \frac{1}{1 + \tau_{i,j}^{x'}} \pi'_{i1,j} \beta_{i,j} x'_i, \quad (A.185)$$

$$\frac{1}{1 + \tau_{n,j}^{m'}} \pi'_{1n,j} \beta_{1,j} x'_1 = Y'_{n,j} - \sum_{i \neq 1}^N \pi'_{in,j} \beta_{i,j} x'_i, \quad n \neq 1. \quad (A.186)$$

²⁸As we have proven in Appendix F.4, in the setup we use, a country implementing optimal policies will not impose any domestic taxes. Therefore, domestic taxes will not be included in the setup in this section.

The labor market specifications become

$$\lambda'_{n,j} = \frac{\lambda_{n,j} \hat{w}_{n,j}^\kappa}{\sum_{k=1} \lambda_{n,k} \hat{w}_{n,k}^\kappa}, \quad W'_n = W_n \left(\sum_{k=1} \lambda_{n,k} \hat{w}_{n,k}^\kappa \right)^{\frac{1}{\kappa}}.$$

Thus,

$$\begin{aligned} W'_n \bar{L}_n &= W_n \bar{L}_n \left(\sum_{k=1} \lambda_{n,k} \hat{w}_{n,k}^\kappa \right)^{\frac{1}{\kappa}} \Rightarrow \frac{w'_{n,j} L'_{n,j}}{\lambda'_{n,j}} = \frac{w_{n,j} L_{n,j}}{\lambda_{n,j}} \left(\sum_{k=1} \lambda_{n,k} \hat{w}_{n,k}^\kappa \right)^{\frac{1}{\kappa}} \\ \Rightarrow \hat{L}_{n,j} &= \frac{\hat{w}_{n,j}^{\kappa-1}}{\left(\sum_{k=1} \lambda_{n,k} \hat{w}_{n,k}^\kappa \right)^{\frac{\kappa-1}{\kappa}}}. \end{aligned} \quad (\text{A.187})$$

The equations for optimal export taxes and import tariffs are derived from (A.123) and (A.126),

$$\frac{1}{1 + \tau'_{n,j}} = \frac{\sum_{i \neq 1}^N (1 + \tau'_{i,j}) \epsilon_j \pi'_{ni,j}}{1 + \epsilon_j (1 - \pi'_{n1,j})}, \quad n \neq 1, \forall j \quad (\text{A.188})$$

$$\sum_{k=1}^J \sum_{i \neq 1}^N (1 + \tau'_{i,k}) \frac{\partial E'_{1i,k}}{\partial \ln p'_{n,j}} = x'_1 \beta_{1,j} \pi'_{1n,j} - \sum_{k=1}^J \frac{\partial E'_{11,k}}{\partial \ln p'_{n,j}}, \quad n \neq 1, \forall j \quad (\text{A.189})$$

where

$$\begin{aligned} \frac{\partial E'_{11,k}}{\partial \ln p'_{n,j}} &= -I_{k=j} \left[\sum_{i \neq 1}^N \beta_{i,k} \frac{x'_i}{1 + \tau'_{i,k}} \epsilon_k \pi'_{i1,k} \pi'_{in,k} \right] - \beta_{n,k} \frac{\pi'_{n1,k}}{1 + \tau'_{n,k}} Y'_{n,j}, \quad (n \neq 1) \\ \frac{\partial E'_{1n,k}}{\partial \ln p'_{n,j}} &= I_{k=j} [\kappa Y'_{n,k}] - (\kappa - 1) \lambda'_{n,k} Y'_{n,j} + I_{k=j} \left[\sum_{s \neq 1}^N \beta_{s,k} x'_s \epsilon_k (1 - \pi'_{sn,k}) \pi'_{sn,k} \right] - \beta_{n,k} \pi'_{nn,k} Y'_{n,j}, \\ \frac{\partial E'_{1i,k}}{\partial \ln p'_{n,j}} &= -I_{k=j} \left[\sum_{s \neq 1}^N \beta_{s,k} x'_s \epsilon_k \pi'_{si,k} \pi'_{sn,k} \right] - \beta_{n,k} \pi'_{ni,k} Y'_{n,j}. \quad (i \neq n) \end{aligned}$$

We use equations (A.173)-(A.189) to determine variables $\{\hat{w}_{n,j}, \hat{L}_{n,j}, \hat{p}_{n,j}, \hat{P}_n, x'_n, \pi'_{ni,j}, 1 + \tau'_{n,j}, 1 + \tau'_{n,j}\}$. Due to tax neutrality, we normalize one export tax to zero. We can then calculate the change in welfare for each country ($\hat{C}_n = \frac{\hat{x}_n}{\hat{P}_n}$).

J.2 Nash optimal policies

Given policies of all countries, the counterfactual equilibrium satisfies

$$\hat{p}_{1,j} = \hat{w}_{1,j}, \quad (A.190)$$

$$\hat{p}_{n,j} = \hat{w}_{n,j}, \quad n \neq 1 \quad (A.191)$$

$$Y'_{1,j} = w'_{1,j} L'_{1,j} = \hat{w}_{1,j} \hat{L}_{1,j} w_{1,j} L_{1,j}, \quad (A.192)$$

$$Y'_{n,j} = w'_{n,j} L'_{n,j} = \hat{w}_{n,j} \hat{L}_{n,j} w_{n,j} L_{n,j}, \quad n \neq 1 \quad (A.193)$$

$$\hat{P}_n = \prod_{j=1}^J \left[\pi_{n1,j} \left(\hat{p}_{1,j} (1 + \tau_{n1,j}^{x'}) (1 + \tau_{n1,j}^{m'}) \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{ni,j} \left(\hat{p}_{i,j} (1 + \tau_{ni,j}^{x'}) (1 + \tau_{ni,j}^{m'}) \right)^{-\epsilon_j} \right]^{-\beta_{n,j}/\epsilon_j}, \quad n \neq 1 \quad (A.194)$$

$$\hat{P}_1 = \prod_{j=1}^J \left[\pi_{11,j} \hat{p}_{1,j}^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{1i,j} \left(\hat{p}_{i,j} (1 + \tau_{1i,j}^{m'}) (1 + \tau_{1i,j}^{x'}) \right)^{-\epsilon_j} \right]^{-\beta_{1,j}/\epsilon_j}, \quad (A.195)$$

$$\pi'_{11,j} = \frac{\pi_{11,j} \hat{p}_{1,j}^{-\epsilon_j}}{\pi_{11,j} \hat{p}_{1,j}^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{1i,j} \left(\hat{p}_{i,j} (1 + \tau_{1i,j}^{m'}) (1 + \tau_{1i,j}^{x'}) \right)^{-\epsilon_j}}, \quad (A.196)$$

$$\pi'_{1n,j} = \frac{\pi_{1n,j} \left(\hat{p}_{n,j} (1 + \tau_{1n,j}^{m'}) (1 + \tau_{1n,j}^{x'}) \right)^{-\epsilon_j}}{\pi_{11,j} \hat{p}_{1,j}^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{1i,j} \left(\hat{p}_{i,j} (1 + \tau_{1i,j}^{m'}) (1 + \tau_{1i,j}^{x'}) \right)^{-\epsilon_j}}, \quad n \neq 1 \quad (A.197)$$

$$\pi'_{n1,j} = \frac{\pi_{n1,j} \left(\hat{p}_{1,j} (1 + \tau_{n1,j}^{x'}) (1 + \tau_{n1,j}^{m'}) \right)^{-\epsilon_j}}{\pi_{n1,j} \left(\hat{p}_{1,j} (1 + \tau_{n1,j}^{x'}) (1 + \tau_{n1,j}^{m'}) \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{ni,j} \left(\hat{p}_{i,j} (1 + \tau_{ni,j}^{x'}) (1 + \tau_{ni,j}^{m'}) \right)^{-\epsilon_j}}, \quad n \neq 1 \quad (A.198)$$

$$\pi'_{nm,j} = \frac{\pi_{nm,j} \left(\hat{p}_{mj} (1 + \tau_{nm,j}^{x'}) (1 + \tau_{nm,j}^{m'}) \right)^{-\epsilon_j}}{\pi_{n1,j} \left(\hat{p}_{1,j} (1 + \tau_{n1,j}^{x'}) (1 + \tau_{n1,j}^{m'}) \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{ni,j} \left(\hat{p}_{i,j} (1 + \tau_{ni,j}^{x'}) (1 + \tau_{ni,j}^{m'}) \right)^{-\epsilon_j}}, \quad n \neq 1 \quad (A.199)$$

$$x'_1 = \sum_{j=1}^J Y'_{1,j} + \sum_{i \neq 1}^N \sum_{j=1}^J \frac{\tau_{i1,j}^{x'}}{1 + \tau_{i1,j}^{x'}} \frac{1}{1 + \tau_{i1,j}^{m'}} \pi'_{i1,j} \beta_{i,j} x'_i + \sum_{i \neq 1}^N \sum_{j=1}^J \frac{\tau_{i1,j}^{m'}}{1 + \tau_{i1,j}^{m'}} \pi'_{i1,j} \beta_{1,j} x'_1, \quad (A.200)$$

$$x'_n = \sum_{j=1}^J Y'_{n,j} + \sum_{i \neq n}^N \sum_{j=1}^J \frac{\tau_{in,j}^{x'}}{1 + \tau_{in,j}^{x'}} \frac{1}{1 + \tau_{in,j}^{m'}} \pi'_{in,j} \beta_{i,j} x'_i + \sum_{i \neq n}^N \sum_{j=1}^J \frac{\tau_{in,j}^{m'}}{1 + \tau_{in,j}^{m'}} \pi'_{in,j} \beta_{n,j} x'_n, \quad n \neq 1 \quad (A.201)$$

$$\pi'_{11,j} \beta_{1,j} x'_1 = Y'_{1,j} - \sum_{i \neq 1}^N \frac{1}{1 + \tau_{i1,j}^{x'}} \frac{1}{1 + \tau_{i1,j}^{m'}} \pi'_{i1,j} \beta_{i,j} x'_i, \quad (A.202)$$

$$\frac{1}{1 + \tau_{1n,j}^{m'}} \frac{1}{1 + \tau_{1n,j}^{x'}} \pi'_{1n,j} \beta_{1,j} x'_1 = Y'_{n,j} - \sum_{i \neq 1}^N \frac{1}{1 + \tau_{in,j}^{m'}} \frac{1}{1 + \tau_{in,j}^{x'}} \pi'_{in,j} \beta_{i,j} x'_i, \quad n \neq 1. \quad (A.203)$$

The labor market specifications become

$$\lambda'_{n,j} = \frac{\lambda_{n,j} \hat{w}_{n,j}^\kappa}{\sum_{k=1}^J \lambda_{n,k} \hat{w}_{n,k}^\kappa}, \quad W'_n = W_n \left(\sum_{k=1}^J \lambda_{n,k} \hat{w}_{n,k}^\kappa \right)^{\frac{1}{\kappa}}.$$

Thus,

$$\begin{aligned} W'_n \bar{L}_n &= W_n \bar{L}_n \left(\sum_{k=1} \lambda_{n,k} \hat{w}_{n,k}^\kappa \right)^{\frac{1}{\kappa}} \Rightarrow \frac{w'_{n,j} L'_{n,j}}{\lambda'_{n,j}} = \frac{w_{n,j} L_{n,j}}{\lambda_{n,j}} \left(\sum_{k=1} \lambda_{n,k} \hat{w}_{n,k}^\kappa \right)^{\frac{1}{\kappa}} \\ \Rightarrow \hat{L}_{n,j} &= \frac{\hat{w}_{n,j}^{\kappa-1}}{\left(\sum_{k=1} \lambda_{n,k} \hat{w}_{n,k}^\kappa \right)^{\frac{\kappa-1}{\kappa}}}. \end{aligned} \quad (\text{A.204})$$

Given the policies of other countries, country 1's unilateral optimal policies are determined by the FOC equations (A.163), (A.164), (A.170), (A.171), and (A.172). Similarly, in this counterfactual equilibrium, we solve the following FOC equations:

FOC over x'_1

$$\frac{1}{P'_1} - \sum_{j=1}^J \gamma_{1,j} \beta_{1,j} \pi'_{11,j} - \sum_{i \neq 1}^N \sum_{j=1}^J \gamma_{i,j} \beta_{1,j} \frac{1}{1 + \tau'_{1i,j}} \frac{1}{1 + \tau'_{1i,j}} \pi'_{1i,j} + \sum_{i \neq 1}^N \gamma_{xi} \sum_{j=1}^J \beta_{1,j} \frac{\tau'_{1i,j}}{1 + \tau'_{1i,j}} \frac{1}{1 + \tau'_{1i,j}} \pi'_{1i,j} = 0. \quad (\text{A.205})$$

FOC over $x'_n, n \neq 1$

$$-\gamma_{xn} - \sum_{j=1}^J \gamma_{n,j} \beta_{n,j} \pi'_{nn,j} - \sum_{i \neq n}^N \sum_{j=1}^J \gamma_{i,j} \beta_{n,j} \frac{1}{1 + \tau'_{ni,j}} \frac{1}{1 + \tau'_{ni,j}} \pi'_{ni,j} + \sum_{i \neq n}^N \gamma_{xi} \sum_{j=1}^J \beta_{n,j} \frac{\tau'_{ni,j}}{1 + \tau'_{ni,j}} \frac{1}{1 + \tau'_{ni,j}} \pi'_{ni,j} = 0. \quad (\text{A.206})$$

FOC over $\tau'_{1n,j}, n \neq 1$

$$1 + \tau'_{1n,j} = \frac{(\gamma_{xn} + \gamma_{n,j}) \frac{1}{1 + \tau'_{1n,j}} - \gamma_{xn}}{1/P'_1}. \quad (\text{A.207})$$

FOC over $\tau'_{n1,j}, n \neq 1$

$$\frac{1}{1 + \tau'_{n1,j}} = \frac{\sum_{i \neq 1, n}^N \frac{\gamma_{xi} + \gamma_{ij}}{1 + \tau'_{ni,j}} \frac{1}{1 + \tau'_{ni,j}} \epsilon_j \pi'_{ni,j} + (\gamma_{xn} + \gamma_{n,j}) \epsilon_j \pi'_{nn,j} + \sum_{i \neq 1, n}^N \frac{\gamma_{xn} - \gamma_{xi}}{1 + \tau'_{ni,j}} \epsilon_j \pi'_{ni,j} - \gamma_{xn} \frac{1}{1 + \tau'_{n1,j}} \epsilon_j (1 - \pi'_{n1,j})}{\frac{1}{P'_1} \frac{1}{1 + \tau'_{n1,j}} (1 + \epsilon_j (1 - \pi'_{n1,j}))}. \quad (\text{A.208})$$

FOC over $p'_{n,j}$, $n \neq 1$

$$\begin{aligned}
& -\frac{1}{P'_1} x'_1 \beta_{1,j} \pi'_{1n,j} + \sum_{k=1}^J (\gamma_{xn} + \gamma_{n,k}) \frac{\partial Y'_{n,k}}{\partial \ln p'_{n,j}} - \gamma_{1,j} \sum_{i \neq 1}^N \frac{\beta_{i,j}}{1 + \tau'_{i1,j}} \frac{\epsilon_j \pi'_{i1,j} \pi'_{in,j} x'_i}{1 + \tau'_{i1,j}} - \sum_{i \neq 1}^N \gamma_{ij} \sum_{h \neq 1}^N \frac{\beta_{h,j}}{1 + \tau'_{hi,j}} \frac{\epsilon_j \pi'_{hi,j} \pi'_{hn,j} x'_h}{1 + \tau'_{hi,j}} \\
& + \gamma_{n,j} \sum_{h \neq 1}^N \beta_{h,j} \frac{1}{1 + \tau'_{hn,j}} \frac{1}{1 + \tau'_{hn,j}} \epsilon_j \pi'_{hn,j} x'_h + \sum_{h \neq 1}^N \gamma_{xh} \sum_{i \neq \{h,n\}}^N \beta_{i,j} \frac{\tau'_{ih,j}}{1 + \tau'_{ih,j}} \frac{1}{1 + \tau'_{ih,j}} \epsilon_j \pi'_{ih,j} \pi'_{in,j} x'_i \\
& - \gamma_{xn} \sum_{i \neq n}^N \beta_{i,j} \frac{\tau'_{in,j}}{1 + \tau'_{in,j}} \frac{1}{1 + \tau'_{in,j}} \epsilon_j \pi'_{in,j} x'_i + \sum_{h \neq 1}^N \gamma_{xh} \sum_{i \neq h}^N \beta_{h,j} \frac{\tau'_{hi,j}}{1 + \tau'_{hi,j}} \epsilon_j \pi'_{hi,j} \pi'_{hn,j} x'_h - \sum_{h \neq 1}^N \gamma_{xh} \beta_{h,j} \frac{\tau'_{hn,j}}{1 + \tau'_{hn,j}} \epsilon_j \pi'_{hn,j} x'_h = 0,
\end{aligned} \tag{A.209}$$

where $\frac{\partial Y'_{n,k}}{\partial \ln p'_{n,j}} = I_{k=j} [\kappa Y'_{n,j}] - (\kappa - 1) \lambda'_{n,k} Y'_{n,j}$. Without loss of generality, we normalize country 1's final goods price $P'_1 = 1$.

Due to tax neutrality, when a country implements its optimal policies, it can normalize a export tax it imposes on a specific country and sector to zero, given the policies of other countries. Since there are N countries in total and each country can set one export tax to zero while implementing its optimal policies, a total of N export taxes can be normalized to zero. To derive the equilibrium variables, we follow several steps. First, we let country 1 impose unilateral optimal policies and use equations (A.190)–(A.209) to determine the variables $\{\hat{w}_{n,j}, \hat{L}_{n,j}, \hat{p}_{n,j}, \hat{P}_n, x'_n, \pi'_{nm,j}, 1 + \tau'_{li,j}, 1 + \tau'_{il,j}\}$ and multipliers $\{\gamma_{xi}, \gamma_{n,j}\}$, $\forall i \neq 1, \forall n, m, j$, given that the policies of other countries are initially set to zero. Next, we update country 1's policies. Similarly, country 2 imposes unilateral optimal policies, given the policies of all other countries. We repeat these steps until each variable remains unchanged when any country imposes optimal policies. Finally we calculate the change in welfare for each country ($\hat{C}_n = \frac{\hat{x}_n}{\hat{P}_n}$).

J.3 Extension with Economies of Scale

We next compute the counterfactual equilibrium under unilateral optimal policies in the model with differential scale elasticities and imperfectly mobile labor. In this model, labor is imperfectly mobile across sectors within each country. Therefore, country n has sector-specific wages $w_{n,j}$. Given

Home's policies, the counterfactual equilibrium satisfies

$$\hat{p}_{n,j} = \hat{w}_{n,j} \hat{L}_{n,j}^{-\psi_j}, \quad (\text{A.210})$$

$$Y'_{n,j} = w'_{n,j} L'_{n,j} = \hat{w}_{n,j} \hat{L}_{n,j} Y_{n,j}. \quad (\text{A.211})$$

$$\hat{P}_1 = \prod_{j=1}^J \left[\pi_{11,j} \left(\hat{p}_{1,j} (1 + \tau_j^{d'}) \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{1i,j} \left(\hat{p}_{i,j} (1 + \tau_{i,j}^{m'}) \right)^{-\epsilon_j} \right]^{-\beta_{1,j}/\epsilon_j}, \quad (\text{A.212})$$

$$\hat{P}_n = \prod_{j=1}^J \left[\pi_{n1,j} \left(\hat{p}_{1,j} (1 + \tau_{n,j}^{x'}) \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{ni,j} \hat{p}_{i,j}^{-\epsilon_j} \right]^{-\beta_{n,j}/\epsilon_j}, \quad n \neq 1. \quad (\text{A.213})$$

$$\pi'_{11,j} = \frac{\pi_{11,j} \left(\hat{p}_{1,j} (1 + \tau_j^{d'}) \right)^{-\epsilon_j}}{\pi_{11,j} \left(\hat{p}_{1,j} (1 + \tau_j^{d'}) \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{1i,j} \left(\hat{p}_{i,j} (1 + \tau_{i,j}^{m'}) \right)^{-\epsilon_j}}, \quad (\text{A.214})$$

$$\pi'_{1n,j} = \frac{\pi_{1n,j} \left(\hat{p}_{n,j} (1 + \tau_{n,j}^{m'}) \right)^{-\epsilon_j}}{\pi_{11,j} \left(\hat{p}_{1,j} (1 + \tau_j^{d'}) \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{1i,j} \left(\hat{p}_{i,j} (1 + \tau_{i,j}^{m'}) \right)^{-\epsilon_j}}, \quad n \neq 1, \quad (\text{A.215})$$

$$\pi'_{n1,j} = \frac{\pi_{n1,j} \left(\hat{p}_{1,j} (1 + \tau_{n,j}^{x'}) \right)^{-\epsilon_j}}{\pi_{n1,j} \left(\hat{p}_{1,j} (1 + \tau_{n,j}^{x'}) \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{ni,j} \hat{p}_{i,j}^{-\epsilon_j}}, \quad n \neq 1, \quad (\text{A.216})$$

$$\pi'_{nm,j} = \frac{\pi_{nm,j} \hat{p}_{m,j}^{-\epsilon_j}}{\pi_{n1,j} \left(\hat{p}_{1,j} (1 + \tau_{n,j}^{x'}) \right)^{-\epsilon_j} + \sum_{i \neq 1}^N \pi_{ni,j} \hat{p}_{i,j}^{-\epsilon_j}}, \quad n \neq 1, \quad m \neq 1. \quad (\text{A.217})$$

$$x'_1 = \sum_{j=1}^J Y'_{1,j} + \sum_{i \neq 1}^N \sum_{j=1}^J \frac{\tau_{i,j}^{x'}}{1 + \tau_{i,j}^{x'}} \pi'_{i1,j} \beta_{i,j} x'_i + \sum_{i \neq 1}^N \sum_{j=1}^J \frac{\tau_{i,j}^{m'}}{1 + \tau_{i,j}^{m'}} \pi'_{1i,j} \beta_{1,j} x'_1 + \sum_{j=1}^J \frac{\tau_j^{d'}}{1 + \tau_j^{d'}} \pi'_{11,j} \beta_{1,j} x'_1. \quad (\text{A.218})$$

$$x'_n = \sum_{j=1}^J Y'_{n,j}, \quad n \neq 1. \quad (\text{A.219})$$

$$\frac{\pi'_{11,j} \beta_{1,j} x'_1}{1 + \tau_j^{d'}} = Y'_{1,j} - \sum_{i \neq 1}^N \frac{\pi'_{i1,j} \beta_{i,j} x'_i}{1 + \tau_{i,j}^{x'}}, \quad (\text{A.220})$$

$$\frac{\pi'_{1n,j} \beta_{1,j} x'_1}{1 + \tau_{n,j}^{m'}} = Y'_{n,j} - \sum_{i \neq 1}^N \pi'_{in,j} \beta_{i,j} x'_i, \quad n \neq 1. \quad (\text{A.221})$$

The labor market specifications become

$$\lambda'_{n,j} = \frac{\lambda_{n,j} \hat{w}_{n,j}^\kappa}{\sum_{k=1}^J \lambda_{n,k} \hat{w}_{n,k}^\kappa}, \quad W'_n = W_n \left(\sum_{k=1}^J \lambda_{n,k} \hat{w}_{n,k}^\kappa \right)^{1/\kappa}.$$

Thus,

$$W'_n \bar{L}_n = W_n \bar{L}_n \left(\sum_{k=1}^J \lambda_{n,k} \hat{w}_{n,k}^\kappa \right)^{1/\kappa}$$

$$\Rightarrow \hat{L}_{n,j} = \frac{\hat{w}_{n,j}^{\kappa-1}}{\left(\sum_{k=1}^J \lambda_{n,k} \hat{w}_{n,k}^\kappa \right)^{(\kappa-1)/\kappa}}. \quad (\text{A.222})$$

This model satisfies a GIIA Supply System with

$$\frac{1}{\eta_j} = \frac{1}{\kappa} - \psi_j \frac{\kappa-1}{\kappa}, \quad \eta_j = \frac{\kappa}{1 - \psi_j(\kappa-1)}. \quad (\text{A.223})$$

The optimal domestic taxes therefore satisfy

$$\frac{1 + \tau_j^{d'}}{1 + \tau_k^{d'}} = \frac{1 + \psi_k}{1 + \psi_j} = \frac{\eta_j/(\eta_j - 1)}{\eta_k/(\eta_k - 1)}. \quad (\text{A.224})$$

The optimal export taxes satisfy

$$\frac{1}{1 + \tau_{n,j}^{x'}} = \frac{\epsilon_j \sum_{i \neq 1}^N (1 + \tau_{i,j}^{m'}) \pi'_{ni,j}}{(1 + \tau_j^{d'}) \left[1 + \epsilon_j (1 - \pi'_{n1,j}) \right]}, \quad n \neq 1. \quad (\text{A.225})$$

The optimal import tariffs satisfy

$$\sum_{k=1}^J \sum_{i \neq 1}^N (1 + \tau_{i,k}^{m'}) \frac{\partial E'_{1i,k}}{\partial \ln p'_{n,j}} = x'_1 \beta_{1,j} \pi'_{1n,j} - \sum_{k=1}^J (1 + \tau_k^{d'}) \frac{\partial E'_{11,k}}{\partial \ln p'_{n,j}}, \quad n \neq 1. \quad (\text{A.226})$$

To express the semi-export-supply elasticities, define

$$D'_n \equiv \sum_{\ell=1}^J \eta_\ell Y'_{n,\ell}, \quad (\text{A.227})$$

$$\frac{\partial Y'_{n,k}}{\partial \ln p'_{n,j}} = Y'_{n,j} \left[I_{k=j} \eta_j + \eta_j \frac{(1 - \eta_k) Y'_{n,k}}{D'_n} \right], \quad (\text{A.228})$$

$$\frac{\partial \sum_{k=1}^J Y'_{n,k}}{\partial \ln p'_{n,j}} = x'_n \frac{\eta_j Y'_{n,j}}{D'_n}, \quad n \neq 1. \quad (\text{A.229})$$

Table A-2: List of Manufacturing Sectors

Sector Code	Short Name	Description
1	Food	Food, Beverages and Tobacco
2	Textiles	Textiles, wearing apparel and leather products
3	Wood	Wood and products of wood and cork; articles of straw and plaiting materials
4	Paper	Paper and paper products, Printing and reproduction of recorded media
5	Oil	Coke and refined petroleum products
6	Chemicals	Chemicals and chemical products
7	Rubber	Rubber and plastics
8	Mineral	Other non-metallic mineral
9	Metals	Basic metals and fabricated metal
10	Computer	Computer, electronic, optical products and electrical equipment
11	Machinery	Machinery and equipment n.e.c.
12	Vehicles	Vehicles and other transport equipment
13	Furniture	Furniture and other manufacturing

Then,

$$\frac{\partial E'_{11,k}}{\partial \ln p'_{n,j}} = -I_{k=j} \sum_{s \neq 1}^N \frac{\beta_{s,k} x'_s}{1 + \tau_{s,k}^{x'}} \epsilon_k \pi'_{s1,k} \pi'_{sn,k} - \frac{\beta_{n,k} \pi'_{n1,k}}{1 + \tau_{n,k}^{x'}} x'_n \frac{\eta_j Y'_{n,j}}{D'_n}, \quad (\text{A.230})$$

$$\frac{\partial E'_{1n,k}}{\partial \ln p'_{n,j}} = Y'_{n,j} \left[I_{k=j} \eta_j + \eta_j \frac{(1 - \eta_k) Y'_{n,k}}{D'_n} \right] - \beta_{n,k} \pi'_{nn,k} x'_n \frac{\eta_j Y'_{n,j}}{D'_n} + I_{k=j} \sum_{s \neq 1}^N \beta_{s,k} x'_s \epsilon_k (1 - \pi'_{sn,k}) \pi'_{sn,k}, \quad (\text{A.231})$$

$$\frac{\partial E'_{1i,k}}{\partial \ln p'_{n,j}} = -\beta_{n,k} \pi'_{ni,k} x'_n \frac{\eta_j Y'_{n,j}}{D'_n} - I_{k=j} \sum_{s \neq 1}^N \beta_{s,k} x'_s \epsilon_k \pi'_{si,k} \pi'_{sn,k}, \quad i \neq 1, n. \quad (\text{A.232})$$

Equations (A.210)–(A.232) determine $\left\{ \hat{w}_{n,j}, \hat{L}_{n,j}, \hat{p}_{n,j}, \hat{P}_n, x'_n, \pi'_{ni,j}, 1 + \tau_j^{d'}, 1 + \tau_{n,j}^{m'}, 1 + \tau_{n,j}^{x'} \right\}$. We use the Home price index as the numeraire, $\hat{P}_1 = 1$, and, by tax neutrality, normalize one domestic tax and one export tax to zero. Welfare changes are then given by $\hat{C}_n = \frac{\hat{x}_n}{\hat{P}_n}$.

K Data and Additional Quantitative Results

Table A-2 reports the 13 manufacturing sectors used in the quantitative analysis.

Table A-3: U.S. Nash Policies across Sectors

Sector	Tariff mean	Tariff s.d.	Export-tax mean	Export-tax s.d.
Oil	20.84	1.81	4.26	2.26
Paper	21.26	1.57	3.57	2.01
Chemicals	21.25	1.39	4.12	1.86
Machinery	22.74	5.45	4.45	3.80
Food	21.45	0.64	3.14	0.69
Vehicles	22.12	1.98	3.33	1.74
Rubber	21.80	1.53	3.12	1.24
Mineral	21.80	0.65	2.74	0.57
Wood	21.93	1.77	2.58	1.59
Metals	22.14	1.25	2.58	0.91
Computer	23.15	3.05	2.54	0.92
Furniture	22.99	1.82	1.93	1.46
Textiles	24.49	2.33	0.40	1.44
Average	22.15	1.94	2.98	1.58

Note: This table summarizes the U.S. policies (mean and standard deviation) for 13 sectors under Nash optimal policies. The sectors are ranked in ascending order based on the U.S. sectoral net imports relative to world sectoral income before the policy. The last row represents the average of the entire column.

Table A-4: China's Nash Policies across Sectors

Sector	Tariff mean	Tariff s.d.	Export-tax mean	Export-tax s.d.
Textiles	17.73	1.66	8.89	3.07
Computer	20.92	1.69	5.25	1.90
Furniture	22.29	0.88	3.07	0.90
Mineral	22.22	0.86	2.58	1.18
Rubber	22.51	0.73	2.32	0.53
Machinery	23.11	1.33	2.05	1.39
Wood	23.02	1.43	1.65	1.37
Food	23.05	1.05	1.50	0.91
Paper	23.32	1.17	1.23	1.04
Metals	23.52	1.75	1.17	1.50
Oil	22.98	0.68	1.54	0.46
Vehicles	23.61	0.91	0.98	0.69
Chemicals	23.76	0.94	0.94	0.68
Average	22.47	1.16	2.55	1.20

Note: This table summarizes China's policies (mean and standard deviation) for 13 sectors under Nash optimal policies. The sectors are ranked in ascending order based on China's sectoral net imports relative to world sectoral income before the policy. The last row represents the average of the entire column.

Table A-5: U.S. Nash Policies across Countries

Country	Tariff mean	Tariff s.d.	<i>Slope</i>	Export-tax mean	Export-tax s.d.
NLD	21.10	0.77	0.22	3.61	1.21
AUS	21.14	0.78	0.19	3.38	1.05
FRA	21.46	1.22	0.49	3.24	1.19
GBR	21.62	0.60	0.28	3.06	1.02
BRA	21.04	0.76	0.59	3.33	0.98
KOR	21.33	0.78	0.43	3.27	0.97
ROW	21.93	0.92	0.60	2.76	1.04
TUR	21.96	0.70	0.35	2.78	0.94
ESP	21.63	0.58	0.45	2.81	0.70
CHE	21.81	0.93	0.20	3.06	0.94
DEU	21.72	0.72	0.64	2.87	0.75
JPN	21.86	0.70	0.42	2.71	0.83
IND	22.47	0.71	0.72	2.17	0.78
CHN	22.54	0.59	0.77	2.09	0.59
ITA	21.93	0.69	0.61	2.65	0.79
CAN	25.81	2.34	0.23	2.77	3.17
RUS	22.16	0.38	0.09	2.35	0.50
IDN	23.00	1.81	0.44	1.73	1.70
MEX	24.34	8.51	0.64	6.00	6.32
Average	22.15	1.29	0.44	2.98	1.34

Note: This table summarizes the U.S. import tariffs (mean, standard deviation, and tariff-import slope) and export taxes (mean and standard deviation) for the other 19 countries under Nash optimal policies. The column "*Slope*" represents tariff-import slope, which is the regression coefficient of U.S. optimal tariffs on each country's sectors with respect to the after-tax U.S. net import share relative to that country-sector income in different sectors, normalized by κ . Countries are ranked in ascending order based on the U.S. total net import share relative to each country's total income before the policy. The last row represents the average of the entire column.

Table A-6: China's Nash Policies across Countries

Country	Tariff mean	Tariff s.d.	<i>Slope</i>	Export-tax mean	Export-tax s.d.
NLD	21.92	1.87	0.23	3.12	2.61
TUR	21.24	1.31	0.19	3.21	1.69
RUS	22.06	2.54	0.12	2.73	3.73
MEX	21.63	1.39	0.08	3.26	1.80
ESP	22.16	1.43	0.57	2.63	1.89
IND	21.21	0.80	0.42	3.27	1.06
U.S.	22.17	1.69	0.42	2.75	2.47
CAN	22.76	2.47	0.24	2.45	3.33
IDN	22.51	1.91	0.25	2.54	2.40
BRA	22.64	1.08	0.58	2.09	1.30
ITA	22.30	1.27	0.68	2.36	1.53
JPN	22.61	2.27	0.39	2.69	3.47
GBR	22.70	1.82	0.42	2.38	2.39
FRA	22.72	1.75	0.36	2.42	2.31
ROW	23.11	1.44	0.58	2.13	1.74
CHE	23.10	1.64	0.45	1.76	1.62
AUS	23.47	3.29	0.28	2.36	4.68
DEU	23.10	1.81	0.31	2.13	2.33
KOR	23.45	1.82	0.41	2.20	2.32
Average	22.47	1.77	0.37	2.55	2.35

Note: This table summarizes China's import tariffs (mean, standard deviation, and tariff-import slope) and export taxes (mean and standard deviation) for the other 19 countries under Nash optimal policies. The column "*Slope*" represents tariff-import slope, which is the regression coefficient of China's optimal tariffs on each country's sectors with respect to China's after-tax net import share relative to that country-sector income in different sectors, normalized by κ . Countries are ranked in ascending order based on China's total net import share relative to each country's total income before the policy. The last row represents the average of the entire column.

L Alternative Proof for Proposition 2 and 3

As shown in Appendix B, one of the equilibrium conditions in (A.1), (A.2) and (A.3) is redundant according to Walras' law. In this section, we use Home's budget constraint and eliminate one of the goods market clearing conditions by setting $\gamma_{2,J} = 0$. We then prove the optimal policies under the general framework and our defined GIIA Supply System, as shown in Propositions 2 and 3. The Home government's maximization problem and the world market equilibrium constraints in the two-country case are detailed in Appendix B.

FOC over x_1

$$1 - \sum_{j=1}^J \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \pi_{11,j} - \sum_{j=1}^J \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \pi_{12,j} - \gamma_x \left(1 - \sum_{j=1}^J \beta_{1,j} \frac{\tau_j^m}{1 + \tau_j^m} \pi_{12,j} - \sum_{j=1}^J \beta_{1,j} \frac{\tau_j^d}{1 + \tau_j^d} \pi_{11,j} \right) = 0.$$

FOC over export tax τ_j^x

$$\gamma_{1,j} \beta_{2,j} \frac{\pi_{21,j} x_2}{(1 + \tau_j^x)^2} - \gamma_{1,j} \beta_{2,j} \frac{1}{1 + \tau_j^x} \frac{\partial \pi_{21,j}}{\partial \tau_j^x} x_2 - \gamma_{2,j} \beta_{2,j} \frac{\partial \pi_{22,j}}{\partial \tau_j^x} x_2 + \gamma_x \beta_{2,j} \frac{\pi_{21,j} x_2}{(1 + \tau_j^x)^2} + \gamma_x \beta_{2,j} \frac{\tau_j^x}{1 + \tau_j^x} \frac{\partial \pi_{21,j}}{\partial \tau_j^x} x_2 = 0.$$

After plugging in derivatives and simplifying, the FOC over τ_j^x becomes

$$1 + \tau_j^x = \frac{(\gamma_x + \gamma_{1,j}) (\epsilon_j \pi_{22,j} + 1)}{(\gamma_x + \gamma_{2,j}) \epsilon_j \pi_{22,j}}. \quad (\text{A.233})$$

FOC over import tariff τ_j^m

$$\begin{aligned} & -\beta_{1,j} x_1 \frac{\pi_{12,j}}{1 + \tau_j^m} - \gamma_{1,j} \beta_{1,j} \frac{1}{1 + \tau_j^d} \frac{\partial \pi_{11,j}}{\partial \tau_j^m} x_1 + \gamma_{2,j} \beta_{1,j} \frac{\pi_{12,j}}{(1 + \tau_j^m)^2} x_1 - \gamma_{2,j} \beta_{1,j} \frac{1}{1 + \tau_j^m} \frac{\partial \pi_{12,j}}{\partial \tau_j^m} x_1 \\ & + \gamma_x \beta_{1,j} \frac{1}{(1 + \tau_j^m)^2} \pi_{12,j} x_1 + \gamma_x \beta_{1,j} \frac{\tau_j^m}{1 + \tau_j^m} \frac{\partial \pi_{12,j}}{\partial \tau_j^m} x_1 + \gamma_x \beta_{1,j} \frac{\tau_j^d}{1 + \tau_j^d} \frac{\partial \pi_{11,j}}{\partial \tau_j^m} x_1 = 0. \end{aligned}$$

Plugging in derivatives and simplifying, the FOC over τ_j^m becomes

$$1 + \tau_j^m = \frac{(\gamma_x + \gamma_{2,j}) (1 + \epsilon_j \pi_{11,j})}{(\gamma_x + \gamma_{1,j}) \frac{1}{1 + \tau_j^d} \epsilon_j \pi_{11,j} + 1}. \quad (\text{A.234})$$

FOC over domestic tax τ_j^d

$$\begin{aligned}
& -\beta_{1,j}x_1 \frac{\pi_{11,j}}{1+\tau_j^d} + \gamma_{1,j} \frac{1}{(1+\tau_j^d)^2} \beta_{1,j} \pi_{11,j} x_1 - \gamma_{1,j} \frac{1}{1+\tau_j^d} \beta_{1,j} \frac{\partial \pi_{11,j}}{\partial \tau_j^d} x_1 - \gamma_{2,j} \beta_{1,j} \frac{1}{1+\tau_j^m} \frac{\partial \pi_{12,j}}{\partial \tau_j^d} x_1 \\
& + \gamma_x \beta_{1,j} \frac{\tau_j^m}{1+\tau_j^m} \frac{\partial \pi_{12,j}}{\partial \tau_j^d} x_1 + \gamma_x \beta_{1,j} \frac{1}{(1+\tau_j^d)^2} \pi_{11,j} x_1 + \gamma_x \beta_{1,j} \frac{\tau_j^d}{1+\tau_j^d} \frac{\partial \pi_{11,j}}{\partial \tau_j^d} x_1 = 0.
\end{aligned}$$

Plugging in derivatives and simplifying, the FOC over τ_j^d becomes

$$1 + \tau_j^d = \frac{(\gamma_x + \gamma_{1,j}) (1 + \epsilon_j \pi_{12,j})}{(\gamma_x + \gamma_{2,j}) \frac{1}{1+\tau_j^m} \epsilon_j \pi_{12,j} + 1}. \quad (\text{A.235})$$

Combining the optimal tariff (A.234) and domestic tax (A.235), we can get

$$1 + \tau_j^m = \gamma_x + \gamma_{2,j}, \quad 1 + \tau_j^d = \gamma_x + \gamma_{1,j}. \quad (\text{A.236})$$

Combining the above two equations with equation (A.233), we get

$$1 + \tau_j^x = \frac{1 + \tau_j^d}{1 + \tau_j^m} \left(1 + \frac{1}{\epsilon_j \pi_{22,j}} \right). \quad (\text{A.237})$$

When $\gamma_x = 0$, the optimal policy formulas are identical to those in Appendix B.

FOC over $p_{1,j}$

$$\begin{aligned}
& -\beta_{1,j}x_1 \pi_{11,j} + \gamma_{1,j} \beta_{1,j} \frac{1}{1+\tau_j^d} \epsilon_j \pi_{11,j} \pi_{12,j} x_1 + \gamma_{1,j} \beta_{2,j} \frac{1}{1+\tau_j^x} \epsilon_j \pi_{21,j} \pi_{22,j} x_2 - \gamma_{2,j} \beta_{1,j} \frac{1}{1+\tau_j^m} \epsilon_j \pi_{12,j} \pi_{11,j} x_1 \\
& - \gamma_{2,j} \beta_{2,j} \epsilon_j \pi_{22,j} \pi_{21,j} x_2 - \gamma_x \beta_{2,j} \frac{\tau_j^x}{1+\tau_j^x} \epsilon_j \pi_{21,j} \pi_{22,j} x_2 + \gamma_x \beta_{1,j} \frac{\tau_j^m}{1+\tau_j^m} \epsilon_j \pi_{12,j} \pi_{11,j} x_1 - \gamma_x \beta_{1,j} \frac{\tau_j^d}{1+\tau_j^d} \epsilon_j \pi_{11,j} \pi_{12,j} x_1 \\
& + \sum_{k=1}^J (\gamma_x + \gamma_{1,k}) \frac{\partial Y_{1,k}}{\partial p_{1,j}} p_{1,j} = 0.
\end{aligned} \quad (\text{A.238})$$

FOC over $p_{2,j}$

$$\begin{aligned}
& -\beta_{1,j}x_1\pi_{12,j} - \gamma_{1,j}\beta_{1,j}\frac{1}{1+\tau_j^d}\epsilon_j\pi_{11,j}\pi_{12,j}x_1 - \gamma_{1,j}\beta_{2,j}\frac{1}{1+\tau_j^x}\epsilon_j\pi_{21,j}\pi_{22,j}x_2 + \gamma_{2,j}\beta_{1,j}\frac{1}{1+\tau_j^m}\epsilon_j\pi_{12,j}\pi_{11,j}x_1 \\
& + \gamma_{2,j}\beta_{2,j}\epsilon_j\pi_{22,j}\pi_{21,j}x_2 + \gamma_x\beta_{2,j}\frac{\tau_j^x}{1+\tau_j^x}\epsilon_j\pi_{21,j}\pi_{22,j}x_2 - \gamma_x\beta_{1,j}\frac{\tau_j^m}{1+\tau_j^m}\epsilon_j\pi_{12,j}\pi_{11,j}x_1 + \gamma_x\beta_{1,j}\frac{\tau_j^d}{1+\tau_j^d}\epsilon_j\pi_{12,j}\pi_{11,j}x_1 \\
& + \sum_{k=1}^J(\Phi_2 + \gamma_x + \gamma_{2,k})\frac{\partial Y_{2,k}}{\partial p_{2,j}}p_{2,j} = 0,
\end{aligned} \tag{A.239}$$

where $\Phi_2 = \sum_{s=1}^J \gamma_x\beta_{2,s}\frac{\tau_s^x}{1+\tau_s^x}\pi_{21,s} - \sum_{s=1}^J(\gamma_{1,s}\beta_{2,s}\frac{1}{1+\tau_s^x}\pi_{21,s} + \gamma_{2,s}\beta_{2,s}\pi_{22,s}) - \gamma_x$.

L.1 Proof of optimal domestic taxes

Combining FOCs over $p_{1,j}$ (A.238), τ_j^x (A.237), and τ_j^d, τ_j^m (A.236), we get

$$\begin{aligned}
& -\beta_{1,j}x_1\pi_{11,j} + \beta_{2,j}\left(\frac{1+\tau_j^d}{1+\tau_j^x} - (1+\tau_j^m)\right)\epsilon_j\pi_{21,j}\pi_{22,j}x_2 + \sum_{k=1}^J(1+\tau_k^d)\frac{\partial Y_{1,k}}{\partial p_{1,j}}p_{1,j} = 0 \\
\Rightarrow & -\beta_{1,j}x_1\pi_{11,j} - \beta_{2,j}\frac{1+\tau_j^d}{1+\tau_j^x}\pi_{21,j}x_2 + \sum_{k=1}^J(1+\tau_k^d)\frac{\partial Y_{1,k}}{\partial p_{1,j}}p_{1,j} = 0 \\
\Rightarrow & \sum_{k=1}^J(1+\tau_k^d)\frac{\partial Y_{1,k}}{\partial p_{1,j}}p_{1,j} - (1+\tau_j^d)Y_{1,j} = 0 \\
\Rightarrow & \sum_{k=1}^J(1+\tau_k^d)\frac{\partial \ln(Y_{1,k})}{\partial \ln(p_{1,j})}\frac{Y_{1,k}}{Y_{1,j}}\frac{1}{\eta_j} - (1+\tau_j^d)\frac{1}{\eta_j} = 0.
\end{aligned} \tag{A.240}$$

Hence, domestic taxes $\boldsymbol{\tau}^d = [\tau_1^d, \dots, \tau_j^d, \dots, \tau_J^d]'$ for J sectors satisfy

$$(\Lambda_1 - I)(1 + \boldsymbol{\tau}^d) = 0,$$

where element of matrix Λ_n at row j and column k is given by $\frac{\partial \ln(Y_{n,k})}{\partial \ln(p_{n,j})}\frac{Y_{n,k}}{Y_{n,j}}$, and

$$\Lambda_1 - I = \begin{pmatrix} \frac{\partial \ln(Y_{1,1})}{\partial \ln(p_{1,1})}\frac{Y_{1,1}}{Y_{1,1}} - 1 & \dots & \frac{\partial \ln(Y_{1,j})}{\partial \ln(p_{1,1})}\frac{Y_{1,j}}{Y_{1,1}} & \dots & \frac{\partial \ln(Y_{1,J})}{\partial \ln(p_{1,1})}\frac{Y_{1,J}}{Y_{1,1}} \\ \dots & \dots & \dots & \dots & \dots \\ \frac{\partial \ln(Y_{1,1})}{\partial \ln(p_{1,j})}\frac{Y_{1,1}}{Y_{1,j}} & \dots & \frac{\partial \ln(Y_{1,j})}{\partial \ln(p_{1,j})}\frac{Y_{1,j}}{Y_{1,j}} - 1 & \dots & \frac{\partial \ln(Y_{1,J})}{\partial \ln(p_{1,j})}\frac{Y_{1,J}}{Y_{1,j}} \\ \dots & \dots & \dots & \dots & \dots \\ \frac{\partial \ln(Y_{1,1})}{\partial \ln(p_{1,J})}\frac{Y_{1,1}}{Y_{1,J}} & \dots & \frac{\partial \ln(Y_{1,j})}{\partial \ln(p_{1,J})}\frac{Y_{1,j}}{Y_{1,J}} & \dots & \frac{\partial \ln(Y_{1,J})}{\partial \ln(p_{1,J})}\frac{Y_{1,J}}{Y_{1,J}} - 1 \end{pmatrix}.$$

Divide equation (A.240) by η_j . For any sector k and j , optimal domestic taxes satisfy

$$\sum_{s=1}^J \frac{\partial \ln Y_{1,s}}{\partial \ln p_{1,j}} \frac{Y_{1,s}}{Y_{1,j}} (1 + \tau_s^d) \frac{1}{\eta_j} = (1 + \tau_j^d) \frac{1}{\eta_j}, \quad (\text{A.241})$$

$$\sum_{s=1}^J \frac{\partial \ln Y_{1,s}}{\partial \ln p_{1,k}} \frac{Y_{1,s}}{Y_{1,k}} (1 + \tau_s^d) \frac{1}{\eta_k} = (1 + \tau_k^d) \frac{1}{\eta_k}. \quad (\text{A.242})$$

where η_j is the partial supply elasticity parameter in sector j . Using the definition of GIIA Supply System in Definition 4, and subtracting (A.242) from (A.241), we have

$$\begin{aligned} (1 + \tau_j^d) - (1 + \tau_k^d) &= (1 + \tau_j^d) \frac{1}{\eta_j} - (1 + \tau_k^d) \frac{1}{\eta_k} \\ \Rightarrow \frac{1 + \tau_j^d}{1 + \tau_k^d} &= \frac{\frac{\eta_j}{\eta_j - 1}}{\frac{\eta_k}{\eta_k - 1}}. \end{aligned}$$

Due to the tax neutrality shown in Lemma 1, we can set $1 + \tau_j^d = (1 + \bar{\tau}^d) \frac{\eta_j}{\eta_j - 1}, \forall j$. The uniform shifter $\bar{\tau}^d$ can be assigned any arbitrary value, and one of the trade policies can be normalized, without changing the real allocations and welfare impacts of optimal policies.

L.2 Proof of optimal import tariffs

Combining FOC over $p_{2,j}$ (A.239), τ_j^x (A.237), and τ_j^d, τ_j^m (A.236), we get

$$\begin{aligned} \sum_{k=1}^J (\Phi_2 + 1 + \tau_k^m) \left[\frac{\partial \ln Y_{2,k}}{\partial \ln p_{2,j}} Y_{2,k} \right] &= \beta_{1,j} \pi_{12,j} x_1 + \left[\frac{1 + \tau_j^d}{1 + \tau_j^x} - (1 + \tau_j^m) \right] \beta_{2,j} \epsilon_j \pi_{21,j} \pi_{22,j} x_2 \\ \Rightarrow \sum_{k=1}^J (\Phi_2 + 1 + \tau_k^m) \left[\frac{\partial \ln Y_{2,k}}{\partial \ln p_{2,j}} \frac{Y_{2,k}}{Y_{2,j}} \right] &= \frac{\beta_{1,j} \pi_{12,j} x_1}{Y_{2,j}} - \frac{1 + \tau_j^d}{1 + \tau_j^x} \beta_{2,j} \pi_{21,j} \frac{x_2}{Y_{2,j}} \\ \Rightarrow \sum_{k=1}^J (\Phi_2 + 1 + \tau_k^m) \left[\frac{\partial \ln Y_{2,k}}{\partial \ln p_{2,j}} \frac{Y_{2,k}}{Y_{2,j}} \right] &= \frac{\beta_{1,j} \pi_{12,j} x_1}{Y_{2,j}} - (1 + \tau_j^d) \frac{Y_{1,j}}{Y_{2,j}} + \frac{\beta_{1,j} \pi_{11,j} x_1}{Y_{2,j}}, \end{aligned} \quad (\text{A.243})$$

where Φ_2 is common across all sectors. The above equation becomes

$$\sum_{k=1}^J (\Phi_2 + 1 + \tau_k^m) \left[\frac{\partial \ln Y_{2,k}}{\partial \ln p_{2,j}} \frac{Y_{2,k}}{Y_{2,j}} \right] = \frac{\beta_{1,j} x_1 - (1 + \tau_j^d) Y_{1,j}}{Y_{2,j}}. \quad (\text{A.244})$$

Let $\tau^m = [\tau_1^m, \dots, \tau_j^m, \dots, \tau_J^m]'$ denote the import tariff vector. We can write equation (A.244) in

matrix form as $\Lambda_2(\tau^m + \Phi_2 \mathbf{1} + \mathbf{1}) = \left(\frac{M_{1,j}}{Y_{2,j}} \right)_{j=1}^J$, where Λ_2 is given by

$$\Lambda_2 = \begin{pmatrix} \frac{\partial \ln(Y_{2,1})}{\partial \ln(p_{2,1})} \frac{Y_{2,1}}{Y_{2,1}} & \cdots & \frac{\partial \ln(Y_{2,j})}{\partial \ln(p_{2,1})} \frac{Y_{2,j}}{Y_{2,1}} & \cdots & \frac{\partial \ln(Y_{2,J})}{\partial \ln(p_{2,1})} \frac{Y_{2,J}}{Y_{2,1}} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \frac{\partial \ln(Y_{2,1})}{\partial \ln(p_{2,j})} \frac{Y_{2,1}}{Y_{2,j}} & \cdots & \frac{\partial \ln(Y_{2,j})}{\partial \ln(p_{2,j})} \frac{Y_{2,j}}{Y_{2,j}} & \cdots & \frac{\partial \ln(Y_{2,J})}{\partial \ln(p_{2,j})} \frac{Y_{2,J}}{Y_{2,j}} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \frac{\partial \ln(Y_{2,1})}{\partial \ln(p_{2,J})} \frac{Y_{2,1}}{Y_{2,J}} & \cdots & \frac{\partial \ln(Y_{2,j})}{\partial \ln(p_{2,J})} \frac{Y_{2,j}}{Y_{2,J}} & \cdots & \frac{\partial \ln(Y_{2,J})}{\partial \ln(p_{2,J})} \frac{Y_{2,J}}{Y_{2,J}} \end{pmatrix}. \quad (\text{A.245})$$

Divide equation (A.244) by η_j . For any sector k and j , optimal import tariffs satisfy

$$\sum_{s=1}^J \frac{\partial \ln Y_{2,s}}{\partial \ln p_{2,j}} \frac{Y_{2,s}}{Y_{2,j}} (\Phi_2 + 1 + \tau_s^m) \frac{1}{\eta_j} = \frac{1}{\eta_j} \frac{M_{1,j}}{Y_{2,j}}, \quad (\text{A.246})$$

$$\sum_{s=1}^J \frac{\partial \ln Y_{2,s}}{\partial \ln p_{2,k}} \frac{Y_{2,s}}{Y_{2,k}} (\Phi_2 + 1 + \tau_s^m) \frac{1}{\eta_k} = \frac{1}{\eta_k} \frac{M_{1,k}}{Y_{2,k}}. \quad (\text{A.247})$$

Using the definition of GIIA Supply System in Definition 4, and subtracting (A.247) from (A.246), we have

$$\begin{aligned} & \left(\frac{\partial \ln Y_{2,j}}{\partial \ln p_{2,j}} \frac{1}{\eta_j} - \frac{\partial \ln Y_{2,j}}{\partial \ln p_{2,k}} \frac{Y_{2,j}}{Y_{2,k}} \frac{1}{\eta_k} \right) (\Phi_2 + 1 + \tau_j^m) + \left(\frac{\partial \ln Y_{2,k}}{\partial \ln p_{2,j}} \frac{Y_{2,k}}{Y_{2,j}} \frac{1}{\eta_j} - \frac{\partial \ln Y_{2,k}}{\partial \ln p_{2,k}} \frac{1}{\eta_k} \right) (\Phi_2 + 1 + \tau_k^m) \\ &= \frac{1}{\eta_j} \frac{M_{1,j}}{Y_{2,j}} - \frac{1}{\eta_k} \frac{M_{1,k}}{Y_{2,k}} \\ \Rightarrow \tau_j^m - \tau_k^m &= \frac{1}{\eta_j} \frac{M_{1,j}}{Y_{2,j}} - \frac{1}{\eta_k} \frac{M_{1,k}}{Y_{2,k}}, \end{aligned}$$

Since this relationship holds for every pair of sectors, there exists a sector-common endogenous term $\bar{\tau}^m$ such that

$$1 + \tau_j^m = \frac{1}{\eta_j} \frac{M_{1,j}}{Y_{2,j}} + (1 + \bar{\tau}^m), \quad \forall j.$$